

PRIME DNA

The Floor of the Weil Form

The edge of the minimizer, the echo lattice,
and one inequality for the Riemann Hypothesis

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Abstract

Let $\lambda(a)$ be the floor of the odd-sector Weil form over test functions supported in $[-a, a]$. Weil's criterion, in Yoshida's odd-test form, says that $\lambda(a) \geq 0$ for every a is equivalent to the Riemann Hypothesis; the floor is certified positive to $a = 0.8$ (Zhu) and numerically decays doubly exponentially in a . We study the floor as a function of the window. Two results are exact. First, the archimedean part of the form on a window is one half of the Dirichlet form of the logarithmic Laplacian plus a bounded form, with an explicit confining wall $\frac{1}{2} \log(1/(a - |y|))$ at the edge; by the boundary theory of the logarithmic Laplacian, a bounded minimizer is bounded by $C_0(\log(a/\delta))^{-1/2}$ at distance δ from the edge, and numerically it behaves like $C(a)(\log(a/\delta) + \beta)^{-1/2}$ with an amplitude $C(a)$ that we measure on meshes graded to 10^{-12} of the edge. Consequently the entry of a prime power into the window produces no kink in λ but a soft kink, λ' continuous with an increment of order $C^2/\log(1/\varepsilon)$ at scale ε , which we verify over four decades. Second, between entries the floor obeys the sliding-window identity

$$a \lambda'(a) = - \left[D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f) \right],$$

proved for the K -mode form and, for the exact form, as a two-sided Dini-derivative statement holding at every point of differentiability, where f is the minimizer, D_∞ is the archimedean dilation form, whose symbol tends to 1, $D_P(f) = - \sum_n 2\Lambda(n)n^{-1/2} \log n g'_f(\log n)$ is the Euler product differentiated against the derivative of the autocorrelation of f at the prime-power positions, and P, M are the polar moments. We evaluate its three terms in ball arithmetic on the K -mode minimizer and confirm the identity independently on the Fourier side; three order-one quantities cancel to the size of λ . The floor is continuous and non-increasing, and under a bounded-variation hypothesis on near-minimizers it is locally Lipschitz; every minimizer computed, on meshes graded to 10^{-12} of the edge and with up to 160 modes, has total variation exactly $4 \sup |f|$, being unimodal on each half window, so the regularity clause of the conjecture below reduces to that hypothesis. A third statement is numerical: on seven supports, with and without primes, $\lambda'(a) = -2C(a)^2$ to a part in a thousand (an eighth, at an entry, to 1.7%), the Hadamard formula one expects for this window problem, so that $\Phi' = 2C^2/\lambda$ with $\Phi = -\log \lambda$. We prove this boundary law under the edge law with two derivatives, as a Pohozaev identity whose boundary term is the overlap of the leak of the dilation generator with the singular force outside the edge; and for the archimedean form, without primes, we prove the edge law itself from an exact Wiener–Hopf representation of the critical point: the edge of its transform is

$A_0/(t\sigma_-(t))$ with σ_- the outer factor of the symbol, the profile is $|A_0|(\Psi_\infty(1/\delta) - \lambda - \gamma)^{-1/2}$ with Euler's constant γ , verified to a part in a thousand with nothing fitted, and the law $\Phi' = \pi T_{\text{eff}}$ stated below is exact for that form; with the prime 2 inside the window the constants are unchanged, the edge is accompanied by its echo, and the law of the zeros is a theorem under RH for $a_2 < a < a_3$. At every support below $a_\infty = 0.843$, where the walks that the entries generate inside the window are confined, the echoes form a finite tree, the symbol of the edge is the Schur complement of the tree's adjacency, a finite continued fraction along each chain, the constants of the edge are unchanged and the law of the zeros is a theorem under RH; beyond a_∞ the echo set is dense, the representation with the window's symbol remains exact, and the band that governs the onset of the law is the one the edge sees, the spectral measure of the confined echo lattice at the edge's coupling, whose mass lies within a few horizons at every support computed while its support reaches the spectral radius of the edge's component, several horizons at $a \geq 2$ and growing: the law holds from five horizons on inside the region where the symbol of the window still changes sign. At every support the edge law and the boundary law are theorems under the boundedness hypothesis: above its periodic band the window's symbol factors expand in the lattice of the entries with coefficients indexed by the walks of the echo tree, a walk of k steps cannot return to within $\frac{1}{2}e^{-2ak}$ of its start, so the echoes that accumulate at an edge are long chains of small total weight, and the chains from the far edge across the window are the leading correction to the edge amplitude, found in the data at their predicted frequencies; under that hypothesis the Hypothesis reduces to the single inequality below. By the explicit formula, the defect of the zeros above a height against their mean density is the prime sum at the entries, the oscillating part of the symbol, equal to twice the square of the echoes' amplitudes averaged over the tail and verified on the zeros to a few parts in a million; so the two sampling hypotheses reduce to one statement about the minimizer alone, that the sharp tail law holds at a bounded multiple of the horizon. The derivative of the floor is a property of the ground state alone: the pencil $(-Q', Q)$ has eigenvalues of size $6 \times 10^{10} T^*$ at $a = 1$ while the minimizer's direction gives $3.7 T^*$, so no inequality between the two forms can hold. With the horizon $T^*(a) = 2\pi e^{2a}$, finite-mode upper bounds on the floor from height 17 to height 130 (down to $\lambda \approx 10^{-93}$, with up to 230 modes and rigorous enclosures of the truncated matrix's lowest eigenvalue beyond $a = 1$) give Φ'/T^* rising from 3.2 to 3.9 on coarse grids, and the Hellmann–Feynman derivative resolves a sawtooth at every prime-power entry: a rise into the entry, of 10 to 45% over the last 0.003, as the form without the entering prime heads toward indefiniteness, a drop across it, and a relaxation after, with peaks 3.3 at the entry of 2 and 4.1–4.3 at the entries of 3, 4, 5, 7. We state the bounded-relay conjecture, λ locally absolutely continuous with $\lambda' \geq -cT^*\lambda$ almost everywhere, which implies the Riemann Hypothesis by Grönwall's inequality from Zhu's certified support, and the decay law $\Phi'/T^* \rightarrow c_0$; the data exclude the raw phase-space count, disfavour the logarithmic law and the derivative of Zhu's even-sector Landau–Widom law, and favour a constant near 3.8. Evaluated on the first 6000 zeros of ζ , the sum of $|F(\gamma)|^2$ over the zeros returns the floor of the K -mode minimizer to 3×10^{-4} or better at $a = 0.6, 0.8, 1.0$, down to $\lambda = 10^{-26}$, and the zero-side dilation sum returns its derivative to 3×10^{-3} or better: the explicit formula closes on the actual zeros for both, a third and independent measurement of the identity after the matrices and the Fourier symbol. The zeros below the horizon carry at most 10^{-5} of the floor, the first zero 10^{-24} of it at $a = 1$, and the floor is made by the zeros between one and a half and three horizons. The fraction of the floor carried by the zeros above height T is $\Phi'/(\pi T)$ from a few horizons on, on every minimizer computed (edge-FEM to 1% over two decades of height, K -mode to 1–3% down to $\lambda = 10^{-93}$): the rate of decay of the floor is π times the effective height of the zeros that carry it, $\Phi' = \pi T_{\text{eff}}$, and the two conjectures say that this height is tied to the horizon. Under the Riemann Hypothesis the floor is a sampling constant for the zeros above the horizon, and a prolate-type count bounds Φ below by about half of what is observed. Nothing in this paper proves the Riemann Hypothesis. The last section states what is missing as one inequality: under the boundedness hypothesis, with the boundary law and the analytic tail law now theorems below a_∞ and no zeros involved in either, the Hypothesis follows from $2|A_0(a)|^2 \leq cT^*(a)\lambda(a)$ for all a , the square of the edge amplitude of the minimizer against

its energy times the horizon; the inequality contains the Hypothesis, since it fails wherever the floor is negative, and in prime-side form it is a statement about the weighted multiplicative relations among the prime powers below e^{2a} , the moments of the spectral measure of the echo lattice at the edge. Taken on the plane, the zero side is the quadratic functional $4S(z)^2$ of the minimizer's transform at every zero wherever it lies, and the exponential rate of the negative part of the floor is the zeros' maximal distance from the line: the edge of the minimizer reads the plane, growing exponentially if the Hypothesis fails and decaying faster than any exponential if it holds with the conjecture, with no third behaviour. The main result is a theorem: under the boundedness hypothesis the Riemann Hypothesis follows from that one inequality, every other link being proved; the inequality is where the paper rests, its remainder in the dilation identity, the prime part, having no fixed sign. Theorems, computations with their precision, and conjectures are distinguished throughout.

Standalone research paper. Companion to Primes, Folds, and the One Dot and to Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas. Revised after an independent referee report; the report's corrections are incorporated and acknowledged in Section 10.

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Status conventions

A **Theorem**, **Proposition** or **Lemma** without citation is proved in the text. A statement with a citation is taken from the literature and its hypotheses are stated. A **Computation** is a numerical result with its method, precision and cross-checks; a finite-mode minimum is a minimum over a subspace and hence an upper bound on the floor, and every such value is labelled by its mode count K ; derivatives of finite-mode floors are Hellmann–Feynman derivatives of the K -mode form and are called so, never “exact”. A **Conjecture** is a precise statement with the evidence for it. A **Heuristic** or **Remark** makes no mathematical claim. The hypothesis that a minimizer is bounded, needed for the edge law, is written (H_∞) ; the hypothesis that near-minimizers have bounded sup norm times total variation, which gives Lipschitz continuity of the floor, is written (H_{BV}) ; both are carried explicitly wherever they are used. A step of a proof marked *sketch density* is written at the density of a research sketch, with its estimates stated but not isolated as lemmas; the verification notes at the end of Section 9.2 list every such step.

Main theorem

Theorem (the reduction). *Assume the boundedness hypothesis (H_∞) of Section 3, that the minimizer of the odd Weil form on the window $[-a, a]$ is bounded, and the variation hypothesis (H_{BV}) of Section 5, under which the floor is Lipschitz (Proposition 5.6); the second is used only for the absolute continuity of the floor. Both are removable by the representation of the transform: the first is a theorem for the prime-free form (Proposition 8.6) and at every support granting the steps of Theorem 9.12 marked sketch density, the second is Proposition 9.15 under the local boundedness of one height. Let $\lambda(a)$ be the floor of the form, $T^*(a) = 2\pi e^{2a}$ the horizon, $A_0(a)$ the edge amplitude of the minimizer, and $a_Z = 0.8$ the support to which the floor is certified positive. If for some constant c and some locally integrable $e \geq 0$*

$$2|A_0(a)|^2 \leq (cT^*(a) + e(a))\lambda(a) \quad \text{for almost every } a \geq a_Z,$$

then $\lambda(a) > 0$ for every $a > 0$ and the Riemann Hypothesis holds (Theorem 9.1, Corollary 9.13, Proposition 9.23).

Every other link is a theorem: the edge law of the minimizer and the boundary law $\lambda' = -2|A_0|^2$ at almost every support, every support that is not an entry and satisfies a Diophantine condition that holds almost everywhere (Theorem 9.12), the analytic tail law, the Wiener–Hopf representation of the transform with the window’s symbol, the echo tree and its band. The inequality itself is not proved. It contains the Hypothesis, since it fails wherever the floor is negative; in the dilation identity $a\lambda' = -[D_\infty + D_P + 2P^2 + 2PM]$ it asks for a sign of the remainder against $T^*\lambda$, and the obstruction is the prime part D_P , which has no fixed sign. This is where the paper rests. In prime-side form the inequality is a statement about the weighted multiplicative relations among the prime powers below e^{2a} , the moments of the spectral measure of the echo lattice at the edge (Proposition 9.26); and since the window reads an off-line zero only at second order in its distance from the line (Theorem 9.27), the Hypothesis is a matter of structure, not of detection at any finite scale.

1 The form, the floor and the horizon

For $f \in L^2(\mathbb{R})$ real with compact support write $F = \hat{f}$, $F(t) = \int f(x)e^{itx} dx$, and $g = f \star \tilde{f}$, $g(u) = \int f(x)f(x-u) dx$, so that $\hat{g} = |F|^2$. The Weil form in the normalization used throughout is

$$Q(f) = W_\infty(g) - \sum_{n \geq 2} \Lambda(n)n^{-1/2}(g(\log n) + g(-\log n)) + (\hat{g}(i/2) + \hat{g}(-i/2)), \quad (1)$$

$$W_\infty(g) = -(\gamma + \log \pi)g(0) + 2 \int_0^\infty \frac{g(0)e^{-2x} - g(x)e^{-x/2}}{1 - e^{-2x}} dx. \quad (2)$$

By the explicit formula [1], $Q(f) = \sum_\rho \hat{g}((\rho - \frac{1}{2})/i)$ over the nontrivial zeros; on the geometric side, in Fourier form,

$$Q(f) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \Psi(t) dt + \text{polar}, \quad \Psi(t) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi - \sum_n \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n), \quad (3)$$

where the polar term is $2F(i/2)^2$ for even f and $-2P(f)^2$, $P(f) = \int f(x) \sinh(x/2) dx$, for odd f ; the multiplier $\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ is that of W_∞ by the integral representation of ψ . This is the normalization of Zhu [6] and of Checkpoints 15–20 of [22]; Computation 2.1 confirms the identification numerically.

Definition 1.1. For $a > 0$ the *odd floor* is

$$\lambda(a) = \inf\{Q(f) : f \text{ odd, } \operatorname{supp} f \subset [-a, a], \|f\|_2 = 1, Q(f) < \infty\},$$

$\Phi(a) = -\log \lambda(a)$ wherever $\lambda(a) > 0$, and the *horizon* is $T^*(a) = 2\pi e^{2a}$. $N(T)$ denotes the number of zeros with $0 < \gamma \leq T$.

Since g is supported in $[-2a, 2a]$, only the prime powers with $\log n < 2a$ enter $Q(f)$; the *entries* are $a_n = \frac{1}{2} \log n$. The form Q itself does not depend on a , so λ is non-increasing.

Theorem 1.2 (Weil [1]; Yoshida [2], Prop. 1, as recorded in Suzuki [4]). *The Riemann Hypothesis holds if and only if $Q(f) \geq 0$ for every compactly supported smooth f ; and $Q(f) \geq 0$ for every odd such f already implies it. Hence $\text{RH} \iff \lambda(a) \geq 0$ for all $a > 0$.*

Theorem 1.3 (Zhu [6], Theorem 6.2). *For $a = 0.8$, in the odd sector, $8.206 \times 10^{-15} \leq \lambda(0.8) \leq 2.347 \times 10^{-14}$, certified by interval arithmetic (and in the even sector $8.9 \times 10^{-18} \leq \lambda^{\text{even}}(0.8) \leq 2.523 \times 10^{-16}$). In particular $\lambda(a) > 0$ for $0 < a \leq 0.8$.*

The horizon is the height at which the zeros become too dense for the window. A function of exponential type a has zeros of upper density at most a/π (Jensen's formula; see Boas [12]); the zeros of ζ have density $(2\pi)^{-1} \log(t/2\pi)$ at height t , which equals a/π exactly at $t = T^*(a)$. Zhu's resolution threshold, $2\pi e^{(4+o(1))e^a}$ for a one-stroke positivity certificate at half-width a [6], is a different and far larger scale. Everything below is organized around the pair $(\lambda(a), T^*(a))$.

Lemma 1.4 (existence, monotonicity, continuity, differentiability a.e.). *For every $a > 0$ the infimum defining $\lambda(a)$ is attained, the infimum over odd $C_c^\infty(-a, a)$ functions is the same number, λ is non-increasing and continuous, and λ is differentiable at almost every a .*

Proof sketch. By Proposition 3.1 and the discussion after it, Q on $L_{\text{odd}}^2(-a, a)$ is one half of the Dirichlet form of L_Δ plus a form bounded on L^2 ; the form domain $\{f : \int \log(2 + |t|)|F|^2 < \infty\}$ embeds compactly into $L^2(-a, a)$ [8, Thm. 2.1 ff.], so Q is bounded below and lower semicontinuous on the unit sphere and a minimizing sequence has a convergent subsequence. Density of C_c^∞ in the form domain gives the second claim. Monotonicity is inclusion of the admissible sets, and a monotone function is differentiable almost everywhere. Continuity: for $a < a'$ and a minimizer f' at a' , the dilate $f'_{a/a'}$ (notation of Section 5) is admissible at a , so $0 \leq \lambda(a) - \lambda(a') \leq Q(f'_{a/a'}) - Q(f')$. Minimizers at supports in $[a, a+1]$ have form norm bounded in terms of a and $\lambda(a)$ alone (the wall is positive, the prime and polar terms are bounded on the unit sphere), and on a form-bounded set $Q(f_b) - Q(f) \rightarrow 0$ uniformly as $b \rightarrow 1$: in the archimedean term because $|\Psi_\infty(s/b) - \Psi_\infty(s)| \leq w_* |\log b|$ with $w_* = \sup_t |t\Psi'_\infty(t)| < \infty$, in the prime terms because $|g_f(u) - g_f(u')| \leq \|f\| \|f(\cdot - u) - f(\cdot - u')\|$ and $\|f(\cdot - h) - f\|^2 \leq h^2 T^2 \|f\|^2 + 4 \int_{|t|>T} |F|^2$ is uniformly small on a form-bounded set, in the polar term because P is a bounded functional. Letting $a' \downarrow a$ gives right continuity; left continuity is $\lambda(a') \leq Q(f_{a'/a}) \rightarrow Q(f) = \lambda(a)$ as $a' \uparrow a$, for a minimizer f at a . In particular the floor cannot jump. $\square$

2 Three engines

All computations use the orthonormal odd Legendre basis on the window, $f(y) = a^{-1/2} \sum_{i < K} c_i N_i P_{2i+1}(y/a)$, $N_i = \sqrt{(4i+3)/2}$, in which the matrix $Q_K(a)$ of (1) is assembled from the exact cross-correlations $g_{ij}(x) = \int \phi_i(u) \phi_j(u-x) du$ by Gauss–Legendre rules exact for the polynomial products. The K -mode floor $\lambda_K(a)$ is the lowest eigenvalue of $Q_K(a)$; it is an upper bound on $\lambda(a)$, and λ_1 below denotes the second eigenvalue. Three implementations are used, all in the repository accompanying this paper [24].

1. `rh_weil_odd.py`: mpmath, 50–60 digits, $K \leq 72$, 80 outer quadrature nodes. Reproduces the floors and the failure points of [21] to 4–6 digits. Its outer rule is exact only to polynomial degree 159, which costs 10^{-4} in relative accuracy at $K = 72$.
2. `rh_weil_arb.py`: ball arithmetic (python-flint 0.9.0 over FLINT/Arb), $K \leq 230$, 500–800 bits, with certified Gauss–Legendre rules, $2K + 8$ inner and $2K + 100$ outer nodes, and a rigorous enclosure of the lowest eigenvalue of the assembled matrix. The Legendre recurrence is evaluated with a midpoint reset at each degree, since its ball version doubles the radius per degree; the enclosure therefore covers the assembly and the eigenvalue computation but not the polynomial evaluations, which are 500-bit floating point, nor the truncation of the outer rule. It reproduces the mpmath floors to twenty digits where both are converged and runs a 180-mode point in six minutes.
3. `rh_edge_fem.py`: odd-extended hat functions on a mesh of the half window graded geometrically to $10^{-12}a$ (350–490 nodes; the “two-stage” mesh has ratio 0.9 from $0.05a$ to $10^{-4}a$ and $\frac{1}{2}$ below), exact cross-correlations, closed-form archimedean tails, double precision. Built to read the minimizer at the edge, where a Legendre truncation resolves only $\delta \gtrsim a/K^2$. Its amplitudes carry a mesh bias of 10^{-3} (prime-free) to 10^{-2} (full form).

Computation 2.1. $\lambda_{40}(0.8) = 1.5935 \times 10^{-14}$ and $\lambda_{64}(0.8) = 1.5659 \times 10^{-14}$, both inside Zhu’s certified odd interval of Theorem 1.3, which fixes the normalization and shows the truncation error at that support. At $a = 0.3$, where no prime is inside the window, $\lambda_{64}(0.3) = 0.2225569$ and the edge-FEM floor 0.2225566 agree to 1.3×10^{-6} ; at $a_3 = \frac{1}{2} \log 3$ the 64-mode profile and the FEM profile agree to 10^{-4} for $|y| \leq 0.99a$.

3 The archimedean wall and the edge law

Proposition 3.1 (decomposition on the window). *Let $J_\Gamma(s) = e^{-|s|/2}/(1-e^{-2|s|})$, $\mathcal{J}(x) = \int_x^\infty J_\Gamma = \operatorname{artanh}(e^{-x/2}) + \arctan(e^{-x/2})$, and $c_1 = -(\gamma + \log \pi) - \frac{\pi}{2} - 3 \log 2$. For f supported in $[-a, a]$,*

$$W_\infty(g_f) = c_1 \|f\|^2 + \frac{1}{2} \iint_{[-a, a]^2} J_\Gamma(u-v) |f(u) - f(v)|^2 du dv + \int_{-a}^a V_a(y) |f(y)|^2 dy, \quad V_a(y) = \mathcal{J}(a-y) + \mathcal{J}(a+y),$$

and $V_a(y) = \frac{1}{2} \log \frac{1}{a-|y|} + \kappa_a + o(1)$ as $|y| \rightarrow a$, with $\kappa_a = \frac{1}{2} \log 4 + \frac{\pi}{4} + \mathcal{J}(2a)$.

Proof. With $g(0) - g(x) = \frac{1}{2} \int |f(u) - f(u-x)|^2 du$, the integral in (2) is $\frac{1}{2} \iint_{\mathbb{R}^2} J_\Gamma(u-v) |f(u) - f(v)|^2$, and $2 \int_0^\infty (e^{-2x} - e^{-x/2})/(1 - e^{-2x}) dx = \psi(\frac{1}{4}) + \gamma = -\pi/2 - 3 \log 2$ by the integral representation of ψ . Splitting the double integral according to whether v lies in the window gives the three terms, the two cross-regions contributing $\frac{1}{2} \int V_a |f|^2$ each. The substitution $w = e^{-s/2}$ gives $\int_x^\infty J_\Gamma = \int_0^{e^{-x/2}} 2 dw/(1 - w^4) = \mathcal{J}(x)$, and $\operatorname{artanh}(e^{-x/2}) = \frac{1}{2} \log(4/x) + o(1)$ gives the constant; at $a = a_3$, $\kappa_a = 2.660623$, and the direct evaluation of $V_a - \frac{1}{2} \log(1/\delta)$ at $\delta = 10^{-9}$ gives 2.6606231. $\square$

The symbol of the whole-line form $\frac{1}{2} \iint J_\Gamma |f(u) - f(v)|^2$ is $\sigma(\xi) = \int J_\Gamma(s)(1 - \cos s\xi) ds = \log |\xi| + O(1)$, so on the window the archimedean form is one half of the Dirichlet form $\int \log(\xi^2) |\hat{f}|^2$ of the logarithmic Laplacian L_Δ of Chen–Weth [8] plus a bounded quadratic form; the prime terms are bounded translation operators and the polar term has rank one. Write $Q(f) = \frac{1}{2} \mathcal{E}_L(f) + \langle Bf, f \rangle$ with B bounded and self-adjoint on $L^2(-a, a)$ (its kernel is bounded on the window). The weak Euler–Lagrange equation of a minimizer is then

$$L_\Delta f = 2(\lambda - B)f \quad \text{in } (-a, a), \quad f = 0 \text{ outside}, \quad (4)$$

tested against $C_c^\infty(-a, a)$.

Hypothesis (H_∞). The minimizer f is bounded. *This is not known to follow from the form domain, which embeds in no L^p , $p > 2$; an L^∞ bound of De Giorgi–Moser type for L_Δ with a bounded lower-order perturbation would supply it. Numerically $\max |f| = 1.37$ at a_3 and 1.61 at $a = 0.3$.*

Theorem 3.2 (edge law; Chen–Weth [8], Thm. 1.11; Hernández-Santamaría–López Ríos–Saldaña [9], Thm. 1.1). *Assume (H_∞). Then f is continuous on $\mathbb{R}$ and there is C_0 with*

$$|f(y)| \leq C_0 \left(\log \frac{1}{a-|y|} \right)^{-1/2} \quad (|y| < a, \quad a - |y| < 0.1).$$

In particular $f(\pm a) = 0$.

Proof sketch. By (4) and (H_∞), $f \in H(\Omega) \cap L^\infty(\mathbb{R})$ is a weak solution of $L_\Delta f = h$ with $h = 2(\lambda - B)f \in L^\infty(\Omega)$, $\Omega = (-a, a)$, which is a Lipschitz domain with the uniform exterior sphere condition. [8, Thm. 1.11] gives continuity and the rate $(\log(1/\delta))^{-\tau}$ for every $\tau < \frac{1}{2}$; [9, Thm. 1.1] gives $\tau = \frac{1}{2}$, with $\ell(r) = -1/\ln \min\{0.1, r\}$. $\square$

The exponent $\frac{1}{2}$ is sharp for the torsion function of a small interval [9, Thm. 1.2], and a Hopf-type lemma gives the rate from below for nonnegative supersolutions [9, Thm. 1.4]; the minimizer here is odd, so the lemma does not apply to it directly and the lower rate is a numerical statement (Computation 3.4).

Heuristic 3.3. In $t = \log(1/\delta)$, $\delta = a - y$, equation (4) near the edge has the shape $th(t) - \frac{1}{2} \int_0^t h(\tau) d\tau = O(h) + O(1)$, the coefficient t being $\frac{1}{2}$ from the wall and $\frac{1}{2}$ from the interior kernel, and the nonlocal term coming from the long-range part of the kernel, which in the variable t is the constant $\frac{1}{2}$ on one side; its solutions are $h = Ct^{-1/2}(1 + O(1/t))$. We have not derived the equation in this form.

Checkpoint 17 of [22] applied Theorem 3.2 to the forced solutions of the same operator and left open the existence of the normalized trace $\lim \sqrt{\log(1/\delta)} f(a - \delta)$; the following computation is the numerical answer for the ground state.

Computation 3.4 (the edge amplitude). On every mesh and with or without primes, $1/f(a - \delta)^2$ is linear in $t = \log(a/\delta)$ with relative residual below 10^{-4} over $14 \leq t \leq 20$; the local exponent $-d \log f/d \log t$ descends toward $\frac{1}{2}$ exactly as $\frac{1}{2}t/(t + \beta)$ does. Writing $f(a - \delta) = C(t + \beta)^{-1/2}(1 + o(1))$ on the two-stage mesh:

form	a	λ (FEM)	C	β	mesh dependence of C
prime-free	0.3	0.2225566	1.3089	-1.49	10^{-3}
full	$a_3 = 0.5493$	1.4914×10^{-5}	0.02412	-1.87	10^{-2}

The six K -mode endpoint values of [21] at a_3 ($f_K(a_3) = 0.009471, \dots, 0.008207$ for $K = 24, \dots, 64$) are this law read at the truncation's resolution: with the fitted C, β , the effective $t_K = C^2/f_K^2 - \beta$ equals $2 \log K + 2.0$ within 0.3 for all six, with and without primes, so a degree- $2K$ expansion resolves $\delta \approx e^{-2}a/K^2$ at the edge and $f_K(a)^2 = C^2/(2 \log K + \beta')$ with $\beta' = 2 + \beta$.

4 The relay is a soft kink

At an entry a_n the new term $-2\Lambda(n)n^{-1/2}g(\log n)$ vanishes and grows with a .

Proposition 4.1 (K -mode kink). *For the K -mode form, λ_K is continuous at a_n and $\lambda'_K(a_n^+) - \lambda'_K(a_n^-) = 4\Lambda(n)n^{-1/2}f_K(a_n)^2$, where f_K is the normalized K -mode minimizer at entry.*

Proof. For $a = a_n + \varepsilon$ and odd f , $g(\log n) = \int_{a-2\varepsilon}^a f(y)f(y - \log n) dy = - \int_0^{2\varepsilon} f(a - \delta)f(a - 2\varepsilon + \delta) d\delta$, which is $-2\varepsilon f(a)^2 + O(\varepsilon^2)$ for a function continuous at the edge; the K -mode minimizer is a polynomial. Hellmann–Feynman with the minimizer frozen gives the jump. $\square$

Computation 4.2 (the K -mode kink). With 24 modes at the entry of 3, the measured jump is 0.948 of the prediction 2.276×10^{-4} at step 5×10^{-4} and 0.9998 at step 10^{-5} (`rh_kink_steps.py` in [24]); the deficit at coarser steps is the reorganization of the minimizer within the scale gap/ $\|Q'_K\| \approx 3 \times 10^{-5}$, the ratio of the spectral gap $\lambda_1 - \lambda_K$ to the norm of the entering term's derivative.

For the exact form, under (H_∞) , Theorem 3.2 gives $f(a_n) = 0$ and the proposition's jump is a quantity of the truncation. What replaces it follows from the fitted profile of Computation 3.4: with $f(a - \delta) = C(\log(a/\delta) + \beta)^{-1/2}$, the product integral is

$$-2\Lambda(n)n^{-1/2}g(\log n) = \frac{4\Lambda(n)C^2}{\sqrt{n}} \frac{\varepsilon}{\log(a/\varepsilon) + \beta + 1 - \log 2} \left(1 + O(\log^{-2}(1/\varepsilon))\right), \quad (5)$$

so λ' is continuous across the entry and its increment at scale ε is $4\Lambda(n)n^{-1/2}C^2/(\log(a/\varepsilon) + \beta + 1 - \log 2)$: a soft kink, vanishing only logarithmically. (An earlier version omitted the constant $1 - \log 2$ in the denominator, which is an 8% effect at $\varepsilon = 10^{-3}$.)

Computation 4.3 (the soft kink). On the edge FEM (134-node mesh, for which $C = 0.0241$, $\beta = -1.93$) at the entry of 3, with $D(\varepsilon) = [\lambda(a_3 + \varepsilon) - 2\lambda(a_3) + \lambda(a_3 - \varepsilon)]/\varepsilon$, all entries $\times 10^{-4}$:

ε	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
measured $D(\varepsilon)$	2.45	1.90	1.48	1.21	1.01
first-order energy of the entering term on the a_3 minimizer	3.42	2.13	1.59	1.27	1.06
law (5)	3.15	2.11	1.59	1.27	1.06

The law reproduces the first-order energy from $\varepsilon = 10^{-4}$ down, as it must, since it is that quantity evaluated on the fitted profile; the measured increment sits 3–5% below both, which is the second-order reorganization of the minimizer, and at 10^{-3} the smooth curvature $\lambda''\varepsilon \approx 0.9 \times 10^{-4}$ is still inside D . The increment falls by about 22% per decade, as the law does. The 24-mode form gives the constant 2.28×10^{-4} .

At the resolution of the grids of Section 6 (steps 0.004 to 0.025) the entries beyond $n = 3$ leave no visible trace in λ'_K . The Hellmann–Feynman derivative resolves what those steps average over.

Computation 4.4 (the relay resolved). `rh_dilation.py` gives $\lambda'_K(a) = \langle c, Q'_K(a)c \rangle$ at $a_n + \text{offset}$ for the first five entries ($K = 48, 48, 64, 80, 96$ for $n = 2, 3, 4, 5, 7$; ball arithmetic, central differences of the matrix at step 10^{-30}). Entries are Φ'/T^* ; data in `relay_scan.csv`.

n	a_n	-0.01	-0.003	-0.001	-0.0001	+0.0001	+0.001	+0.01
2	0.3466	2.42	2.96	3.16	3.27	2.80	2.59	2.30
3	0.5493	2.31	3.02	3.65	4.08	3.41	3.17	3.08
4	0.6931	3.61	3.32	3.70	4.08	3.73	3.63	3.51
5	0.8047	3.96	2.98	3.32	4.32	3.63	3.48	3.51
7	0.9730	3.18	3.66	3.06	4.12	3.59	3.54	3.84

Over the last 0.003 before each entry the ratio rises by 10%, 35%, 23%, 45%, 13% ($n = 2, 3, 4, 5, 7$), monotonically for $n = 2, 3$ and after a dip at -0.003 for $n = 4, 5, 7$; across the entry, between offsets ∓ 0.0001 , it drops by 14%, 16%, 9%, 16%, 13%, which at this scale is the K -mode kink of Proposition 4.1 and the soft kink (5) nearly coincide, since $1/K^2 \approx 10^{-4}$; then it relaxes. The peaks at the entries are 3.27, 4.08, 4.08, 4.32, 4.12 and the dips between them reach 2.3. The values are stable in K : at $a_3 - 0.001$, $\Phi'/T^* = 3.6404, 3.6485, 3.6505$ for $K = 32, 48, 64$; at $a_5 - 0.003$, 3.001 at $K = 64$ against 2.98 at $K = 80$. The rise is the form without the entering prime power heading toward its failure point a_{fail} , the deleted-form failure of [21] (0.5574 for $n = 3$): λ_K falls roughly linearly there ($\lambda'_K \approx -1.14 \times 10^{-3}$ at $a = 0.549$, reaching zero at 0.562 by linear extrapolation) while λ'_K barely moves, so $\Phi' = -\lambda'_K/\lambda_K$ grows like $1/(a_{\text{fail}} - a)$ until the entry rescues the floor. Each value near a_3 was checked against a finite difference of λ_K with step 2×10^{-4} to four digits. The coarse grids average this sawtooth into the plateau of Computation 6.1.

The relay is therefore the sawtooth described in [21], measured on the K -mode form: not a kick at the entry but a rise into it, cut off by the entering prime power, whose first-order action at the entry is the soft kink.

5 The sliding-window identity

For f supported in $[-a, a]$ and $b > 0$ let $f_b(x) = b^{-1/2} f(x/b)$, supported in $[-ab, ab]$ with $\|f_b\| = \|f\|$. Define, with $F = \hat{f}$ and $g = g_f$,

$$D_\infty(f) = \frac{1}{2\pi} \int |F(t)|^2 t \partial_t \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) dt, \quad D_P(f) = - \sum_{\log n < 2a} \frac{2\Lambda(n)}{\sqrt{n}} \log n g'(\log n),$$

$P(f) = \int f(x) \sinh(x/2) dx$, $M(f) = \int f(x) x \cosh(x/2) dx$, and

$$\mathcal{D}(f) = D_\infty(f) + D_P(f) + 2P(f)^2 + 2P(f)M(f). \quad (6)$$

Lemma 5.1 (dilation derivative of the form). *Let a lie strictly between two consecutive entries and let f be supported in $[-a, a]$ with $Q(f) < \infty$ and g_f differentiable at every $\log n < 2a$. Then $b \mapsto Q(f_b)$ is differentiable at $b = 1$ and $\frac{d}{db} Q(f_b)|_{b=1} = -\mathcal{D}(f)$.*

Proof. $\hat{f}_b(t) = \sqrt{b} F(bt)$ exactly, jumps included. In (1), $g_{f_b}(u) = g_f(u/b)$, so the prime terms are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, differentiable at $b = 1$ with derivative $2\Lambda(n)n^{-1/2} \log n g'_f(\log n) =$ the n -th term of $-D_P(f)$; for b near 1 no new prime enters. The archimedean term is $\frac{1}{2\pi} \int b |F(bt)|^2 \Psi_\infty(t) dt = \frac{1}{2\pi} \int |F(s)|^2 \Psi_\infty(s/b) ds$ with $\Psi_\infty = \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$; since $t\Psi'_\infty(t)$ is bounded on $\mathbb{R}$, differentiation under the integral is justified by dominated convergence and gives $-\frac{1}{2\pi} \int |F|^2 t\Psi'_\infty(t) dt = -D_\infty(f)$ at $b = 1$. For the polar term, $P(f_b) = \sqrt{b} \int f(y) \sinh(by/2) dy$ gives $\frac{d}{db} P(f_b)|_{b=1} = \frac{1}{2}P(f) + \frac{1}{2}M(f)$, so $\frac{d}{db}(-2P(f_b)^2)|_{b=1} = -2P^2 - 2PM$. $\square$

Theorem 5.2 (sliding-window identity, K -mode form). *Let a lie strictly between two consecutive entries and let $\lambda_K(a)$ be simple with normalized eigenvector c , f the corresponding function. Then λ_K is differentiable at a and*

$$a \lambda'_K(a) = -\mathcal{D}(f). \quad (7)$$

Proof. The basis functions are a dilation family, $\phi_i^{(ab)} = (\phi_i^{(a)})_b$, so $Q_K(ab) = [Q((\phi_i^{(a)})_b, (\phi_j^{(a)})_b)]_{ij}$ is real-analytic in b near 1 (each entry is a finite sum of terms as in Lemma 5.1, with g_{ij} polynomial), Hellmann–Feynman for a simple eigenvalue gives $\frac{d}{db} \lambda_K(ab)|_{b=1} = \langle c, \frac{d}{db} Q_K(ab)|_{b=1} c \rangle = \frac{d}{db} Q(f_b)|_{b=1}$ by bilinearity, and Lemma 5.1 evaluates it. $\square$

Proposition 5.3 (exact form: two-sided Dini derivatives). *Let a lie strictly between two consecutive entries and let f be a minimizer at support a (Lemma 1.4) with g_f differentiable at every $\log n < 2a$. Then*

$$a D^+ \lambda(a) \leq -\mathcal{D}(f) \leq a D_- \lambda(a),$$

where D^+, D_- are the upper-right and lower-left Dini derivatives; hence $a \lambda'(a) = -\mathcal{D}(f)$ at every point of differentiability of λ , which by Lemma 1.4 is almost every a .

Proof. For every $b > 0$, f_b is admissible at support ab , so $\lambda(ab) \leq Q(f_b)$, with equality at $b = 1$. By Lemma 5.1, $Q(f_b) = \lambda(a) - (b-1)\mathcal{D}(f) + o(b-1)$. For $b > 1$, $(\lambda(ab) - \lambda(a))/(b-1) \leq -\mathcal{D}(f) + o(1)$; for $b < 1$ the division reverses the inequality. Since $\frac{d}{db} = a \frac{d}{da}$ at $b = 1$, the two bounds follow. $\square$

Remark 5.4. The hypothesis on g'_f is implied by $f' \in L^1$, which the edge law $C(\log(1/\delta))^{-1/2}$ suggests but which is not proved; it is the only condition beyond existence. The form-domain condition $\int \log(2 + |t|)|F|^2 < \infty$ does not give $\int |t||F|^2 < \infty$, so the prime part of $\mathcal{D}$ is to be read in x -space, as defined.

The hypothesis is natural in the class of functions of bounded variation, and in that class the dilation derivative is bounded, which makes the floor Lipschitz.

Lemma 5.5 (dilation of a function of bounded variation). *Let $f \in L^\infty(\mathbb{R})$ have bounded variation $\text{Var } f$ and support in $[-a, a]$. Then g_f is Lipschitz with constant $\|f\|_\infty \text{Var } f$; the map $b \mapsto Q(f_b)$ is locally Lipschitz on $(0, \infty)$, entries included, with $b \frac{d}{db} Q(f_b) = -\mathcal{D}(f_b)$ at every b at which g_f is differentiable at the points $(\log n)/b$; and for every b*

$$|\mathcal{D}(f_b)| \leq w_* \|f\|^2 + \frac{\|f\|_\infty \text{Var } f}{b} \sum_{\log n < 2ab} \frac{2\Lambda(n) \log n}{\sqrt{n}} + (2\sigma(ab)^2 + 2\sigma(ab)\mu(ab)) \|f\|^2,$$

where $w_* = \sup_t t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) = 2.0807$, attained at $t = 0.520$, $\sigma(a)^2 = \int_{-a}^a \sinh^2(x/2) dx$ and $\mu(a)^2 = \int_{-a}^a x^2 \cosh^2(x/2) dx$. If f is continuous, $g_f \in C^1$ and $b \mapsto Q(f_b)$ is C^1 .

Proof. For f of bounded variation, $\int |f(y) - f(y-h)| dy \leq |h| \text{Var } f$, so $|g_f(u) - g_f(u')| \leq \|f\|_\infty \int |f(x-u) - f(x-u')| dx \leq \|f\|_\infty \text{Var } f |u-u'|$. The prime terms of $Q(f_b)$ are $-2\Lambda(n)n^{-1/2}g_f(\log n/b)$, locally Lipschitz in b , with $b \frac{d}{db} g_f(\log n/b) = -(\log n/b) g'_f(\log n/b) = -\log n g'_{f_b}(\log n)$ whenever g_f is differentiable at $\log n/b$; a term with $\log n = 2ab$ enters with the value $g_f(2a) = 0$ and continuously, since g_f vanishes on $[2a, \infty)$. The archimedean and polar terms are C^1 in b for every b , with the derivatives computed in Lemma 5.1. The bound uses $|t \partial_t \text{Re } \psi| \leq w_*$, $|g'_{f_b}| \leq \|f_b\|_\infty \text{Var } f_b = b^{-1} \|f\|_\infty \text{Var } f$, and $|P(f_b)| \leq \sigma(ab) \|f\|$, $|M(f_b)| \leq \mu(ab) \|f\|$ by Cauchy-Schwarz. If f is continuous, $g'_f(u) = -\int f(z+u) d(Df)(z)$ with Df the finite derivative measure of f , which is continuous in u by dominated convergence. $\square$

Proposition 5.6 (the floor is Lipschitz under a variation bound). *Let $0 < a_0 < a_1$ and $M > 0$, and assume*

(H_{BV}) for every $a \in [a_0, a_1]$ and $\varepsilon > 0$ there is $f \in L^\infty$ of bounded variation, supported in $[-a, a]$, with $\|f\| = 1$, $Q(f) \leq \lambda(a) + \varepsilon$ and $\|f\|_\infty \text{Var } f \leq M$.

Then for $a_0 \leq a < a' \leq a_1$,

$$0 \leq \lambda(a) - \lambda(a') \leq L \log \frac{a'}{a} \leq \frac{L}{a_0} (a' - a), \quad L = w_* + \frac{a_1}{a_0} M \sum_{\log n < 2a_1} \frac{2\Lambda(n) \log n}{\sqrt{n}} + 2\sigma(a_1)^2 + 2\sigma(a_1)\mu(a_1).$$

In particular λ is Lipschitz on $[a_0, a_1]$, hence absolutely continuous there, with $\lambda' \geq -L/a_0$ almost everywhere.

Proof. The left inequality is Lemma 1.4. For the right one take f' as in (H_{BV}) at a' and put $b_0 = a/a'$. The dilate f'_{b_0} is supported in $[-a, a]$ with unit norm, so $\lambda(a) \leq Q(f'_{b_0})$. By Lemma 5.5, $b \mapsto Q(f'_b)$ is Lipschitz on $[b_0, 1]$ with $|b \frac{d}{db} Q(f'_b)| \leq L$ almost everywhere (the prime sum runs over $\log n < 2a'b \leq 2a_1$, and $b^{-1} \leq a_1/a_0$), so $Q(f'_{b_0}) \leq Q(f') + L \log(1/b_0) \leq \lambda(a') + \varepsilon + L \log(a'/a)$. Let $\varepsilon \rightarrow 0$. $\square$

Remark 5.7. (H_{BV}) concerns near-minimizers, not the minimizer, and it follows from a bound on the K -mode minimizers uniform in K : $\lambda(a) = \inf_K \lambda_K(a)$ by Lemma 1.4 (the odd Legendre truncations of an odd $C_c^\infty(-a, a)$ function converge to it in the form norm, their jumps at $\pm a$ being of vanishing size and costing $O(\text{jump}^2)$ in a norm that tolerates jumps), so the K -mode minimizers are near-minimizers of the exact problem, and they are polynomials, hence of bounded variation. Under (H_{BV}) on every compact subinterval of $[a_0, \infty)$ the regularity clause of Conjecture 7.1 holds

and the conjecture is the inequality $\lambda' \geq -cT^*\lambda$ a.e. alone. The trivial bound $\lambda' \geq -L/a$ also shows what the conjecture asks: on $[0.6, 1]$ with $M = 9$ (Computation 5.8) it gives $L = 126$ and $\lambda' \geq -211$, while λ' is -4×10^{-5} at $a = 0.6$ and -3×10^{-24} at $a = 1$; the conjecture is that $\mathcal{D}(f)$, whose three terms are each of order one, cancels to $O(T^*\lambda)$ on the minimizer, as in (8).

Computation 5.8 (bounded variation of the minimizers). `rh_bv_check.py`. On the edge-FEM minimizers of Computation 8.11 (486 nodes, edge resolved to $10^{-12}a$, $\|f\| = 1$): the supremum, the total variation over the window, and the increments of the Hardy integral $H(\delta_0) = \int_{\delta_0}^{a/10} f(a - \delta)^2 d\delta/\delta$ between $\delta_0 = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}$ (finiteness of $H(0)$ is the $H^{1/2}$ condition at the edge).

a	primes	$\sup f $	$\text{Var } f$	$\text{Var } f / (4 \sup f)$	$H(10^{-3})$	increments per three decades
0.3	no	1.70138	6.80553	1.0000	2.64	1.63, 0.93, 0.62
0.4	no	1.47286	5.89146	1.0000	2.13	1.26, 0.69, 0.46
0.5	no	1.31669	5.26678	1.0000	1.83	1.01, 0.58, 0.39
0.6	no	1.20136	4.80545	1.0000	1.66	0.86, 0.52, 0.35
0.4	yes	1.55514	6.22057	1.0000	0.56	0.23, 0.12, 0.08
0.45	yes	1.54288	6.17152	1.0000	0.17	0.049, 0.027, 0.018
0.5	yes	1.49876	5.99842	1.0006	0.037	0.0054, 0.0030, 0.0020
a_3	yes	1.47916	5.91666	1.0000	0.0060	0.0005, 0.0003, 0.0002

On every minimizer $\text{Var } f = 4 \sup |f|$ to four digits (1.0006 at $a = 0.5$ with primes, where the half-window profile has a small interior ripple): each half-window profile rises from 0 at the centre to its maximum and falls monotonically to the edge. Mesh refinement at $a = 0.5$ without primes (124, 237, 471, 486 nodes; $\delta_{\min} = 6 \times 10^{-9}$ and then 7×10^{-13} ; 100, 200, 400, 200 uniform intervals on the inner 90% of the half window; the last mesh with the two-stage grading) gives $\sup |f| = 1.316720, 1.316696, 1.316694, 1.316694$ and $\text{Var } f = 5.266880, 5.266785, 5.266777, 5.266777$. The Hardy increments decrease like those of $C^2 \log(\log(a/\delta_0) + \beta)$ with the C, β of Computation 8.11 (at $a = 0.3$: 1.63, 0.93, 0.62 against 1.66, 0.83, 0.55), i.e. $H(\delta_0) \rightarrow \infty$ doubly logarithmically as the edge law predicts: the minimizers are not in $H^{1/2}$, and $L^\infty \cap BV$ is the right class. On the K -mode minimizers (ball-arithmetic engine, two steps of inverse iteration, evaluated on 44,000 points of the half window graded to $10^{-12}a$; the variation includes the jump $f_K(a)$ at the edge):

a	K	12	24	48	72	96	128
0.6	$\sup f_K $	1.461834	1.461771	1.461786	1.461780	1.461771	
	$\text{Var } f_K$	5.847337	5.847085	5.847145	5.847121	5.847085	
0.8	$\sup f_K $	1.407373	1.421807	1.421859	1.421851 [†]	1.421844	1.421842
	$\text{Var } f_K$	5.629491	5.687230	5.687435	5.687404 [†]	5.687377	5.687368

([†]: $K = 64$.) At $a = 1.0$ (600 bits): $\sup |f_K| = 1.392228, 1.399706, 1.399722, 1.399716, 1.399714$ and $\text{Var } f_K = 5.568912, 5.598825, 5.598888, 5.598865, 5.598858$ for $K = 24, 48, 80, 120, 160$. In every case $\text{Var } f_K / (4 \sup |f_K|) = 1.0000$, and $\|f_K\|_\infty \text{Var } f_K$ converges, to 8.55, 8.09, 7.84 at $a = 0.6, 0.8, 1.0$: (H_{BV}) holds on these supports with $M = 9$, in the form of Remark 5.7.

Since $t \partial_t \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) \rightarrow 1$, $D_\infty(f) = \|f\|^2 + O(\int |F|^2 t^{-2})$; on a minimizer $\|f\| = 1$ while λ is 10^{-90} at $a = 1.5$, so (7) forces

$$D_P(f) = -D_\infty(f) - 2P^2 - 2PM + O(aT^*\lambda) : \quad (8)$$

the Euler product, differentiated and applied to the derivative of the minimizer's autocorrelation at the prime-power positions, cancels the archimedean dilation form to ninety digits. Together with the explicit formula on the same function, $\sum 2\Lambda(n)n^{-1/2}g(\log n) = W_\infty(g) - 2P^2 - \lambda$, both the values and the log-weighted derivatives of g_f at the prime powers are pinned by the archimedean side.

Computation 5.9 (the identity on the K -mode minimizer). `rh_dilation.py` differentiates the archimedean matrix A , the prime matrices Π_n and the polar matrix $ss^\top$ of $Q_K(a)$ by central differences with step 10^{-30} in ball arithmetic and contracts them with the eigenvector; every entry is a ball with the radius shown, and the last column is the sum, which is $a\lambda'_K$ by Theorem 5.2.

a	K	λ_K	$a\langle c, A'c \rangle$	$a\langle c, \Pi'c \rangle$	$-2a\langle c, (ss^\top)'c \rangle$	sum = $a\lambda'_K$	radius
0.6	48	5.9659×10^{-7}	-1.014313	+1.086974	-0.072687	-2.493408×10^{-5}	4×10^{-20}
0.8	64	1.5659×10^{-14}	-1.016605	+1.103966	-0.087361	$-1.295854 \times 10^{-12}$	4×10^{-27}
1.0	80	1.4820×10^{-26}	-1.018057	+1.114966	-0.096909	$-2.516852 \times 10^{-24}$	3×10^{-39}
1.25	140	3.4161×10^{-51}	-1.019178	+1.123585	-0.104407	$-1.233712 \times 10^{-48}$	2×10^{-63}

The three columns are $-D_\infty$, $-D_P$ and $-(2P^2 + 2PM)$ evaluated through the matrices; that their sum is $a\lambda'_K$ is Theorem 5.2, not an independent check. The independent check is on the Fourier side: at $K = 12$, $a = 0.6$ (`rh_fourier_check.py`), with the same coefficient vector, D_∞ evaluated as $(1/2\pi) \int |F|^2 t \partial_t \Re \psi(\frac{1}{4} + \frac{it}{2}) dt$ with F in closed form through spherical Bessel functions and the Parseval tail, D_P from $g'(\log 2)$ and $g'(\log 3)$ of the exact autocorrelation, and P, M by quadrature, gives $D_\infty = 1.01430470102641$, $D_P = -1.08691690953774$, $2P^2 + 2PM = 0.0726373685293583$ and $-\mathcal{D}(f) = -2.51600180293 \times 10^{-5}$, against $a\langle c, Q'_K c \rangle = -2.51600180295 \times 10^{-5}$ from the matrices: twelve digits. (The eigenvector enters through one step of inverse iteration from a shift at relative distance 2^{-20} below λ_K , which fixes $a\langle c, Q'_K c \rangle$ to about seven digits: at $K = 12$ a second step moves it by 1×10^{-7} relatively, to $-2.51600203711 \times 10^{-5}$, which is the central difference of λ_K itself to twelve digits. The ball radii in the table bound the arithmetic only.) The archimedean dilation form sits at 1.014–1.019, the excess over $\|f\|^2$ being the finite- t correction; the prime dilation form grows with the primes present and cancels it; the polar term grows from 7% to 10%. The implied Φ'/T^* are 3.339, 3.324, 3.658, 3.774; at $a = 0.8$ this lies inside the sawtooth of the entry of 5 (Computation 4.4).

5.1 The derivative is a property of the ground state

Let $Q'_K(a)$ be the a -derivative of the K -mode matrix, a symmetric form on the K -mode space. Is there an inequality $-Q'_K \leq cT^* Q_K$ between the two forms, which would give the derivative bound for every f at once? Since a form inequality on the exact form would restrict to every K -mode subspace, the pencil $(-Q'_K, Q_K)$ answers this for the exact form too.

Computation 5.10 (the pencil). Largest and smallest eigenvalues of $Q_K^{-1}(-Q'_K)$ divided by T^* , computed from the midpoint matrices with an approximate QR eigenvalue routine at 500 bits (Q_K has condition number up to 10^{26} here, leaving over a hundred digits):

a	minimizer direction	largest	second, third	smallest
0.6	3.34	74.7	4.26, 3.69	-70.5
0.8	3.32	1.46×10^5	332, 6.08	-1.46×10^5
1.0	3.66	6.2×10^{10}	3.1×10^7 , 8.2×10^4	-6.2×10^{10}

The extreme eigenvalues come in $\pm$ pairs and grow explosively. They are near-null directions of Q_K whose dilation derivative is enormous relative to their value; heuristically they are the directions along which the minimizer rotates as a moves, and their size is of the order of the inverse of the scale gap/ $\|Q'_K\|$ of Computation 4.2. The ground state is neither extreme; it is a special interior direction in which the dilation derivative is anomalously small, by ten orders at $a = 1$. Hence the derivative bound of Section 7 is a statement about the ground-state eigenvector alone, $\lambda' = \langle f, Q'f \rangle$, and cannot be an inequality between forms. A proof must pass through the Euler–Lagrange equation (4) of the minimizer.

6 The floor to height 130

Computation 6.1 (the band). Three data sets (`floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`): $K = 40$, step 0.0125 on $[0.30, 1.20]$ with the entries $a_n \pm 0.004$ added; $K = 72$, step 0.025 on $[0.475, 1.05]$; and the ball-arithmetic engine at pairs $a \pm 0.005$ around the centres 1.06, 1.10, 1.15, 1.25, 1.33, 1.40, 1.45, 1.50 with $K = 110, 110, 110, 140, 140, 180, 180, 180$, plus $K = 180$ at 1.25 and $K = 230$ at 1.5. Φ' is a centered difference over the nearest admissible pair (± 0.025 for $K = 72$, ± 0.005 for arb), never across an entry; the pair at 1.10 straddles $a_9 = 1.0986$ and is excluded. Every value is a K -mode upper bound on the floor. Convergence: λ_{40} agrees with λ_{72} to 5% up to $a = 0.925$ and 26% at 1.0; λ_{140} with λ_{180} to 2% at 1.25; at 1.5, λ_{180} and λ_{230} differ by 39% and the ratio by 4%, more modes raising it, so the last ratio is a lower bound. For $a > 1$ each eigenvalue carries a rigorous enclosure of the assembled matrix’s lowest eigenvalue: radius 2×10^{-64} for the four points computed before the midpoint reset of Section 2, $\approx 10^{-143}$ at 500 bits after it, and $\approx 10^{-233}$ at 800 bits; the enclosures do not cover the polynomial evaluations or the outer rule.

a	K	T^*	λ_K	Φ'	Φ'/T^*
0.500	72	17.1	1.938×10^{-4}	55.2	3.23
0.650	72	23.1	1.837×10^{-8}	74.8	3.24
0.775	72	29.6	2.105×10^{-13}	107.5	3.63
0.900	72	38.0	7.176×10^{-20}	138.5	3.64
1.000	72	46.4	1.491×10^{-26}	172.6	3.72
1.060	110	52.4	2.97×10^{-31}	195.8	3.74
1.150	110	62.7	1.04×10^{-39}	240.3	3.83
1.250	180	76.5	3.59×10^{-51}	289.7	3.79
1.330	140	89.8	3.58×10^{-62}	339.2	3.78
1.400	180	103.3	1.73×10^{-73}	398.4	3.86
1.450	180	114.2	1.22×10^{-82}	443.5	3.88
1.500	230	126.2	2.97×10^{-93}	491.1	3.89

For the $K = 72$ rows λ_K is the grid value at the centre; for the arb rows it is the geometric mean of the pair. Means of Φ'/T^* : 3.23 on $[0.5, 0.7]$ (5 grid points), 3.61 on $[0.85, 1.05]$ (5), 3.79 on $[1.06, 1.25]$ (3), 3.85 on $[1.33, 1.5]$ (4). One-parameter least-squares fits of Φ'/T^* against a on the nine tabulated rows with $a \geq 0.85$, residual = rms of the relative deviations: $\Phi' = cT^*$ with $c = 3.79$, 2.1%; $\Phi' = cT^* \log T^*$, 7.5%; the raw count $\Phi \propto N(T^*)$, whose ratio is $\propto 4a - 1$, 18.4%; the derivative of Zhu’s even-sector law $\Phi \simeq C N(T^*)/\log N(T^*)$, whose ratio is $\propto 2a(L - 1)/L^2$ with $L = \log N(T^*) = 2a + \log(2a - 1)$ and tends to C/π , 8.1%. With Zhu’s own $C = 2\pi^2$ that derivative is 3.14 at $a = 1$ and 3.72 at $a = 1.5$, a 42% rise against the observed 7%. The ratio

λ_1/λ_K stays above 5.7 and grows overall to 10^6 at $a = 1$ (with local decreases of up to 18%), so the K -mode ground state is simple throughout.

At step 0.004 the entries beyond $n = 3$ show no net drop in Φ' , because the difference quotient across an entry averages the steep rise before it against the drop at it; Computation 4.4 resolves both. Over the fine scan, Φ'/T^* lies in $[2.3, 4.3]$, and the supremum over all measured points, the quantity Conjecture 7.1 bounds, is 4.32, at the entry of 5.

7 The conjectures

Conjecture 7.1 (bounded relay). *There are $a_0 > 0$ and $c > 0$ such that λ is locally absolutely continuous on $[a_0, \infty)$ and*

$$\lambda'(a) \geq -cT^*(a)\lambda(a) \quad \text{for almost every } a \geq a_0.$$

The statement is meaningful whatever the sign of λ ; absolute continuity is part of the conjecture: Lemma 1.4 gives continuity and differentiability a.e., and Proposition 5.6 gives Lipschitz continuity under (H_{BV}) , a bound on the sup norm times the total variation of near-minimizers, which Computation 5.8 supports on every minimizer computed ($\text{Var } f = 4 \sup |f| \approx 6$, converged in the mesh and in K). Where $\lambda > 0$ it reads $\Phi' \leq cT^*$.

Proposition 7.2. *Conjecture 7.1 with $a_0 \leq 0.8$ implies the Riemann Hypothesis.*

Proof. Put $u(a) = \lambda(a) \exp(\frac{c}{2}T^*(a))$, which is locally absolutely continuous with $u' = e^{cT^*/2}(\lambda' + cT^*\lambda) \geq 0$ a.e., using $\frac{d}{da}T^* = 2T^*$. Hence u is non-decreasing and $\lambda(a) \geq \lambda(a_0) \exp(-\frac{c}{2}(T^*(a) - T^*(a_0)))$ for $a \geq a_0$. With $a_0 = 0.8$, Theorem 1.3 gives $\lambda(0.8) \geq 8.206 \times 10^{-15} > 0$, so $\lambda > 0$ on $[0.8, \infty)$, and $\lambda > 0$ on $(0, 0.8]$ by the same theorem. Theorem 1.2 concludes. $\square$

Conjecture 7.3 (decay law). *For $a \geq a_0$, $\lambda(a) > 0$ and $\Phi'(a)/T^*(a)$ is bounded above and below by positive constants. In its sharp form, $\Phi'(a)/T^*(a) \rightarrow c_0$, i.e. $\lambda(a) = \exp(-(c_0/2 + o(1))2\pi e^{2a})$.*

Computation 6.1 supports the weak form from height 17 to 130 and, on the odd sector, favours a constant $c_0 \approx 3.8$ (residual 2.1%) over every law with a logarithmic factor; the derivative of Zhu's even-sector law, whose constant $C = 2\pi^2$ would give the asymptote 2π , predicts a rise six times larger than observed. Whether the ratio saturates or continues to creep up beyond height 130 is not decidable from these data. Conjecture 7.1 is the one that implies the Riemann Hypothesis; the constant is not its content.

By Theorem 5.2 and Proposition 5.3, Conjecture 7.1 is, at almost every a , the inequality

$$\mathcal{D}(f) \leq c a T^*(a) Q(f) \quad \text{on the minimizer } f, \tag{9}$$

and by Computation 5.10 it holds only there.

Remark 7.4 (what a proof cannot be). Positivity of Q_{24} on a window is destroyed by displacing $\log 2$ by 3×10^{-14} at $a = 0.8$ and by displacements of order $\exp(-cT^*(a))$ in general [21, 24], so no statement about the primes that is invariant under such displacements can imply positivity; and Checkpoint 20 of [22] shows that bounding the reflected prime terms separately from the polar term yields $-\infty$ on translated tests. A proof of Conjecture 7.1 must therefore use the exact multiplicative structure of Λ , and by Computation 5.10 it must use that f minimizes Q .

7.1 Under the Riemann Hypothesis

Let PW_a be the Paley–Wiener space of transforms $F = \hat{f}$ of $f \in L^2(-a, a)$, here with f odd. If all zeros are real, $Q(f) = \sum_{\gamma} |F(\gamma)|^2$ (the sum over γ with multiplicity, of both signs; it converges when $Q(f) < \infty$), and the dilation computation of Lemma 5.1 on the zero side, $Q(f_b) = \sum_{\gamma} b |F(b\gamma)|^2$, gives at a point of differentiability

$$a\lambda' = \lambda + \sum_{\gamma} \gamma \partial_t |F|^2(\gamma)$$

whenever the last sum converges absolutely, so Conjecture 7.1 reads $\sum_{\gamma} \gamma \partial_t |F|^2(\gamma) \geq -(1 + caT^*) \sum_{\gamma} |F(\gamma)|^2$: the minimizer is small at the zeros to the same order in its derivative as in its value. And

$$\lambda(a) = \inf \{ \sum_{\gamma} |F(\gamma)|^2 : F \in \text{PW}_a \text{ odd, } \|f\|_2 = 1 \}$$

is the lower sampling constant of the zeros for PW_a . The zeros have density $(2\pi)^{-1} \log(t/2\pi)$, which exceeds a/π exactly above $T^*(a)$, so only the zeros above the horizon can sample PW_a , and the floor measures how well a band-limited function can hide from them.

Both statements can be evaluated on the zeros themselves. The explicit formula is unconditional, and every zero used below lies on the line, so for the K -mode minimizer the sum over the zeros of $|F_K(\gamma)|^2$ must return $Q(f_K) = \lambda_K$ up to the contribution of the zeros above the last one used; this closes the loop on the object the form was built from.

Computation 7.5 (the floor and the identity on the zeros). `rh_zero_side.py`, with the first 6000 zeros ($\gamma \leq 6365.85$; `data/zeros_6000.txt`, from `mpmath.zetazero`, the first 400 to 36 digits and the rest to 21). For the K -mode minimizers of Computation 5.9, F_K is evaluated in closed form through spherical Bessel functions at 35 digits; the sums run over $\gamma > 0$ with the factor 2; the tail above γ_{6000} is the Riemann–von Mangoldt density integrated against $|F_K|^2$, with $|F_K(t)|^2$ replaced by its mean $2f_K(a)^2/t^2$ beyond $\gamma_{6000} + 60$, and the derivative tail is the corresponding $-2\times$ that; $a\lambda'_K$ is the matrix-side value of Computation 5.9, the sums run over $\gamma > 0$.

a	K	λ_K	$2 \sum F_K(\gamma) ^2 + \text{tail}$	ratio	$a\lambda'_K$	$\lambda_K + 2 \sum \gamma \partial_t F_K ^2 + \text{tail}$	ratio
0.6	48	5.96594×10^{-7}	5.96554×10^{-7}	0.99993	-2.49341×10^{-5}	-2.49310×10^{-5}	0.99988
0.8	64	1.56587×10^{-14}	1.56568×10^{-14}	0.99988	-1.29585×10^{-12}	-1.29685×10^{-12}	1.00077
1.0	80	1.48201×10^{-26}	1.48157×10^{-26}	0.99971	-2.51685×10^{-24}	-2.52433×10^{-24}	1.0030

At $a = 0.6$ the value ratio with $N = 10, 30, 100, 300, 1000, 3000, 6000$ zeros (tail included) is 0.977, 0.952, 0.973, 0.989, 0.9974, 0.99967, 0.99993; the residual is the fluctuation of the zeros about their mean density, of the size of the tail. The derivative sum is first-order sensitive to the positions of the zeros, by $4\gamma |F'_K(\gamma)|^2 \varepsilon$ per zero for an error ε in γ , while the value sum is not; with all zeros at 21 digits the derivative sum at $a = 1$ came out $+7 \times 10^{-22}$ against -2.5×10^{-24} , stable from $N = 30$ on, and the 36-digit zeros cure it. The residuals of the derivative sums, 1.2×10^{-4} , 8×10^{-4} , 3×10^{-3} , grow with a as this sensitivity does (the zeros beyond the 400th are still at 21 digits); the value sums are insensitive, and their residuals are the fluctuation of the zeros about their mean density in the tail. The explicit formula, which built the form from the primes and the archimedean symbol, returns on the actual zeros both the floor and its rate of change.

Where the floor sits among the zeros is the second reading of the same computation. At $a = 0.6$ ($T^* = 20.9$) one zero lies below the horizon and carries 1.0×10^{-5} of $2 \sum |F_K(\gamma)|^2$; at $a = 0.8$ ($T^* = 31.1$) four zeros lie below and carry 1.0×10^{-7} ; at $a = 1$ ($T^* = 46.4$) eight lie below and

carry 1.6×10^{-13} , the first zero alone 9×10^{-25} . The largest single terms are at $\gamma_7 = 40.9$ (15% of the floor), $\gamma_{13} = 59.3$ (19%) and $\gamma_{27} = 94.7$ (8%), at 2.0, 1.9 and 2.0 horizons, and at each support the mass sits between 1.5 and 3 horizons. So the minimizer hides from the zeros below the horizon to an accuracy far beyond the floor itself, and the floor is made by the zeros between the horizon and about three times the horizon, as the sampling picture says. This is the shadow in its most literal form: $2 \sum_{\gamma>0} |F_K(\gamma)|^2 = \lambda_K$ says that the worst test function is small at every zero, to 10^{-13} at $a = 1$, and the derivative column says that it is small there in its dilation derivative too.

The distribution has a law. Write $\vartheta(T)$ for the fraction of $2 \sum_{\gamma>0} |F(\gamma)|^2$ carried by the zeros above height T .

Computation 7.6 (the tail law). `rh_tailLaw.py`, same zeros. (i) The edge-FEM minimizers with primes of Computation 8.11 (486 nodes, edge resolved to $10^{-12}a$), with F of the odd piecewise-linear function in closed form, the value at the last node included as a jump, and λ' the matrix-side derivative of the same computation. The FEM floor is recovered from the zeros: $2 \sum_{\gamma \leq \gamma_{6000}} |F(\gamma)|^2 / \lambda_{\text{FEM}}$ plus the tail $\Phi' / (\pi \gamma_{6000})$ is 0.999995, 0.999997, 0.999999 at $a = 0.4, 0.45, 0.5$, a check of the FEM engine on the zeros independent of everything else. Then $T\vartheta(T)/T^*$ against $\Phi' / (\pi T^*)$, with $\Phi' = -\lambda' / \lambda_{\text{FEM}}$:

a	$\Phi' / (\pi T^*)$	$T/T^* = 2$	3	5	7	10	20	50	100
0.4	0.7213	0.674	0.732	0.742	0.718	0.721	0.728	0.724	0.724
0.45	0.8676	0.956	0.836	0.857	0.856	0.872	0.865	0.868	0.867
0.5	1.0811	1.105	1.267	1.085	1.103	1.060	1.081	1.083	1.081
a_3	1.2948	0.889	0.967	1.059	1.103	1.135	1.160	1.184	1.197

From seven horizons on, $T\vartheta(T)/T^*$ equals $\Phi' / (\pi T^*)$ to 1% at $a = 0.4, 0.45, 0.5$ (0.995–1.009 of it at $T/T^* = 7, 10, 20, 50, 100$), with no logarithmic drift over the decade and a half covered. At the entry a_3 the ratio rises from 0.69 at two horizons to 0.92 at a hundred: there λ' is still moving at the scale $1/T$ that a zero at height T probes (Computation 4.3), and the tail follows the derivative at that scale. The law is a property of the transform sampled by the density of zeros: at $a = 0.5$ the density integral $2 \int_T^\infty |F|^2 \rho$, $\rho(t) = \log(t/2\pi)/2\pi$, is 1.068, 1.060, 1.032, 1.020, 1.012, 1.006 times $-\lambda' / (\pi T)$ at $T/T^* = 5, 7, 10, 20, 50, 100$, the $O(1/\log T)$ correction of Proposition 7.8 below, while the zero sum is 1.004, 1.020, 0.981, 1.000, 1.002, 1.000 times it. (ii) The K -mode minimizers, per-zero shares from `data/zeros_6000.txt` for $a \leq 1.25$ and from the first 1500 zeros at 50 digits for $a = 1.5$ (`data/zeros_1500_50digits.txt`; in the plunge region $|F_K(\gamma)|$ is far below $|F'_K(\gamma)|$ times the error of a 21-digit zero), ϑ at fixed T/T^* :

a	K	$\Phi' / (\pi T^*)$	$\vartheta(2T^*)$	$\vartheta(2.5T^*)$	$\vartheta(3T^*)$	$\vartheta(5T^*)$	$\vartheta(10T^*)$	$T\vartheta/T_{\text{eff}}$ at 3	5	10	median
0.6	48	1.063	0.594	0.481	0.370	0.215	0.1065	1.045	1.011	1.002	2.39
0.8	64	1.058	0.596	0.480	0.415	0.226	0.1009	1.176	1.067	0.953	2.32
1.0	80	1.164	0.810	0.526	0.407	0.240	0.1170	1.048	1.032	1.005	2.65
1.25	140	1.201	0.781	0.528	0.433	0.249	0.1210	1.082	1.038	1.007	2.67
1.5	230	1.239	0.858	0.554	0.434	0.251	0.1230	1.052	1.015	0.993	2.74

($T_{\text{eff}} = \Phi' / \pi$; the median is the height, in horizons, below which half of the floor sits. At $a = 1.5$ the Hellmann–Feynman contraction failed, two inverse-iteration steps at 800 bits leaving the $K = 230$ eigenvector short of the 10^{-90} relative accuracy the contraction needs, and $\Phi' = 491.1$ is the central difference of the $K = 230$ floor between 1.495 and 1.505 from Computation 6.1; the tail law returns it, $T\vartheta/T^* = 1.233, 1.230$ at 7 and 10 horizons against $\Phi' / (\pi T^*) = 1.239$. The λ_K of the $a = 1.25$ row of Computation 5.9 was corrected on this occasion to 3.4161×10^{-51} , the value the zeros

return; its Φ'/T^* was already the correct one.) The tail law holds on the K -mode minimizers to 1–3% from five horizons on. Below that, the plunge region $[T^*, 3T^*]$ carries more than the law, $T\vartheta/T^*$ reaching 1.4–1.5 at $1.5T^*$; the median height of the floor's mass is 2.39, 2.32, 2.65, 2.67, 2.74 horizons at $a = 0.6, 0.8, 1.0, 1.25, 1.5$ (and 1.8, 2.0, 2.4 on the FEM minimizers at 0.4, 0.45, 0.5), rising with a and, at the last point, at $e = 2.72$ The curves $\vartheta(T/T^*)$ at $a = 0.6$ and 0.8 coincide to 0.01 at every point of the grid, those at 0.4, 0.45, 0.5 approach them from below as a grows, and those at 1.0, 1.25 and 1.5 lie above them in the plunge region ($\vartheta(2T^*) = 0.81, 0.78, 0.86$ against 0.59) and rejoin them in the tail. Measured in units of T_{eff} rather than T^* , that is as $T\vartheta(T)/T_{\text{eff}}$, all five agree to 5% from 2.5 horizons on (1.10–1.14 at 2.5, 1.05–1.18 at 3, 1.01–1.07 at 5, 0.99–1.01 at 10), across floors spanning 86 orders of magnitude

Conjecture 7.7 (tail law). *Assume RH. For every $a > 0$ that is not an entry, with f the minimizer at support a and λ' the derivative of the floor,*

$$\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\frac{\lambda'(a)}{\pi}; \quad \text{equivalently} \quad \Phi'(a) = \pi T_{\text{eff}}(a), \quad T_{\text{eff}}(a) = \lim_{T \rightarrow \infty} T\vartheta(T).$$

Under (H_∞) this is a theorem for $a_2 < a < a_3$ (Corollary 8.9), at every support below $a_\infty = 0.843$ where the echo set is finite (Corollary 8.15), and for the archimedean form at every support (Corollary 8.5).

The heuristic behind it is the edge law: a shape with $f(a - \delta) \sim C(\log(a/\delta))^{-1/2}$ has $|F(t)|^2 \approx 4C^2 \cos^2(ta)/(t^2 \log t)$ for large t , the zeros have density $\log(t/2\pi)/2\pi$, and the product integrated from T gives $C^2/(\pi T)$, which is $-\lambda'/(2\pi T)$ by Conjecture 8.1. The asymptotic expansion carries a ratio of logarithms, $\log(T/2\pi)/(\log(aT) + \beta)$, that would be 1.1 to 1.4 at the heights of Computation 7.6 and that the data do not show; the law is sharper than its derivation, which suggests a sum rule behind it. In the form $\Phi' = \pi T_{\text{eff}}$ the two conjectures of Section 7 read: Conjecture 7.1 is $T_{\text{eff}}(a) \leq (c/\pi)T^*(a)$, the effective height of the zeros that carry the floor is a bounded multiple of the horizon; Conjecture 7.3 is $T_{\text{eff}}/T^* \rightarrow c_0/\pi$, and the measured c_0 between 3.2 and 3.9 is T_{eff} between 1.0 and 1.24 horizons.

The limit in Conjecture 7.7 is a theorem once the edge law is assumed with two derivatives.

Proposition 7.8 (edge law implies tail law). *Assume RH. Let f be real, odd, supported in $[-a, a]$ and C^2 on $(-a, a)$, and suppose that for some $C > 0$, $\beta \in \mathbb{R}$ and $\delta_0 > 0$,*

$$f(a - \delta) = C(\log(a/\delta) + \beta)^{-1/2} + \psi(\delta), \quad 0 < \delta < \delta_0, \quad \psi^{(k)}(\delta) = o(\delta^{-k} \log^{-3/2}(1/\delta)) \quad (k = 0, 1, 2) \text{ as } \delta \rightarrow 0^+.$$

Then

$$\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = \frac{2C^2}{\pi}.$$

Proof. Step 1: the transform. Put $L(\delta) = \log(a/\delta) + \beta$, $\varphi(\delta) = CL(\delta)^{-1/2}$, so $\varphi'(\delta) = \frac{C}{2}L^{-3/2}\delta^{-1} > 0$ and $|\varphi''(\delta)| \leq C'L^{-3/2}\delta^{-2}$ on $(0, \delta_0)$. Let χ be smooth, 1 on $[0, \delta_0/2]$ and 0 beyond δ_0 ; put $f_+(x) = (\varphi + \psi)(a - x)\chi(a - x)$ near the right edge, $f_-(x) = -f_+(-x)$, and $f_0 = f - f_+ - f_-$, which is C^2 with compact support in $(-a, a)$, so $F_0(t) = O(t^{-2})$. With $u = (\varphi + \psi)\chi$, $u(0^+) = u(\delta_0) = 0$, one integration by parts gives $F_+(t) = e^{ita} \int_0^{\delta_0} u(\delta) e^{-it\delta} d\delta = e^{ita} (it)^{-1} \int_0^{\delta_0} u'(\delta) e^{-it\delta} d\delta$. In $u' = \varphi'\chi + \psi'\chi + (\varphi + \psi)\chi'$ the last term is smooth and supported away from 0, contributing $O(1/t)$. For $J(t) = \int_0^{\delta_0} \varphi'\chi e^{-it\delta} d\delta$ split at $\delta = 1/t$: on $[0, 1/t]$, $e^{-it\delta} = 1 + O(t\delta)$ gives $\varphi(1/t) + O(t \int_0^{1/t} \varphi'\delta d\delta) = \varphi(1/t) + O(L(1/t)^{-3/2})$; on $[1/t, \delta_0]$ one more integration by parts gives

$O(\varphi'(1/t)/t) + O(t^{-1} \int_{1/t}^{\delta_0} |(\varphi'\chi)'|) = O(L(1/t)^{-3/2})$. Hence $J(t) = CL(1/t)^{-1/2}(1 + O(1/\log t))$.

The same splitting applied to $\int \psi' \chi e^{-it\delta}$, with $\int_0^{1/t} |\psi'| = o(\log^{-1/2} t)$ and the hypothesis on ψ'' , gives $o(\log^{-1/2} t)$. Therefore $F_+(t) = e^{ita}(it)^{-1}CL(1/t)^{-1/2}(1 + O(1/\log t))$, and since f is real and odd, $F_-(t) = -\overline{F_+(t)} = e^{-ita}(it)^{-1}CL(1/t)^{-1/2}(1 + O(1/\log t))$. So

$$F(t) = \frac{2C \cos(ta)}{it\sqrt{L(1/t)}}(1 + O(1/\log t)) + O(t^{-2}), \quad |F(t)|^2 = \frac{4C^2 \cos^2(ta)}{t^2 L(1/t)}(1 + O(1/\log t)) + O(t^{-3}).$$

Step 2: the sum over the zeros. Under RH the zeros are real and $N(t) = \frac{t}{2\pi} \log \frac{t}{2\pi e} + \frac{7}{8} + S(t) + R(t)$ with R smooth, $R = O(1/t)$, $R' = O(t^{-2})$, and, by Littlewood [5, Ch. XIV, §14.13], $S(t) = O(\log t / \log \log t)$ and $S_1(t) = \int_0^t S = O(\log t / (\log \log t)^2)$. Put $w(t) = t^{-2}L(1/t)^{-1}$, $\rho(t) = \frac{1}{2\pi} \log \frac{t}{2\pi}$, so $\rho/L(1/t) \cdot t^2 w = 1 + O(1/\log t)$. Since $\sum_{|\gamma|>T} = 2 \sum_{\gamma>T}$ and $4C^2 \cos^2(ta)w = 2C^2 w(1 + \cos 2ta)$, the claim is $\sum_{\gamma>T} w(\gamma) = \frac{1}{2\pi T}(1 + o(1))$, $\sum_{\gamma>T} \cos(2\gamma a)w(\gamma) = o(1/T)$, and $\sum_{\gamma>T} (w(\gamma)/\log \gamma + \gamma^{-3}) = o(1/T)$. For a positive decreasing g , $\sum_{\gamma>T} g(\gamma) = \int_T^\infty g dN = \int_T^\infty g \rho dt - g(T)S(T) - \int_T^\infty S g' dt + \int_T^\infty g R' dt$, and with $|S| \ll \log t$ the last three terms are $O(g(T) \log T + \int_T^\infty |g'| \log t + \int_T^\infty g/t^2)$: for $g = w$ this is $O(\log T/T^2) = o(1/T)$ while $\int_T^\infty w \rho = \frac{1}{2\pi T}(1 + O(1/\log T))$; for $g = w/\log t$ and $g = t^{-3}$ everything is $o(1/T)$. For the oscillatory sum put $G(t) = \cos(2ta)w(t)$: $\int_T^\infty G \rho dt = O(w(T)\rho(T)/a) = o(1/T)$ by one integration by parts against $\sin(2ta)/2a$, $\int_T^\infty G R' = o(1/T)$, and

$$\int_T^\infty G dS = -G(T)S(T) + G'(T)S_1(T) + \int_T^\infty S_1 G'' dt,$$

where $|G(T)S(T)| = O(\log T w(T))$, $|G'(T)S_1(T)| = O(a \log T w(T))$, and $|G''| \leq 4a^2 w + O(w/t)$ gives $\int_T^\infty |S_1 G''| = O(a^2 \int_T^\infty \frac{\log t}{(\log \log t)^2 t^2 L(1/t)} dt) = O(\frac{a^2}{T(\log \log T)^2})$, all $o(1/T)$. Hence $\sum_{|\gamma|>T} |F(\gamma)|^2 = 4C^2 \cdot \frac{1}{2\pi T}(1 + o(1)) = \frac{2C^2}{\pi T}(1 + o(1))$. $\square$

With Conjecture 8.1, $-\lambda' = 2C^2$, the proposition is Conjecture 7.7; its hypothesis is the edge law of Theorem 3.2 strengthened to an asymptotic equality with two derivatives, which is what Computation 3.4 measures and (H_∞) does not give; for the prime-free form, Proposition 8.4 derives that strengthening, with its constants, from (H_∞) and the Wiener–Hopf representation. RH enters twice, to read Q as the sum over real zeros and through Littlewood’s bound on S_1 ; without the latter the sum over the zeros is still $O(1/T)$, only the constant is lost. The $O(1/\log T)$ in the proof is the correction seen in the density integral of Computation 7.6; that the zero sums show none of it at one percent is the sum rule still to be found. The Euler–Lagrange equation gives an exact identity for the tail at every height, in which the logarithms cancel.

Proposition 7.9 (tail identity). *Let f be a minimizer at support a and let K be the operator with symbol Ψ , so that $Q(f) = \langle f, Kf \rangle - 2P(f)^2$ and f satisfies $Kf = \lambda f + 2P(f)s$ on $(-a, a)$ with $s(y) = \sinh(y/2)\mathbf{1}_{[-a, a]}(y)$. Put $g_{\text{out}} = Kf \cdot \mathbf{1}_{|y|>a}$, the force outside the window, and let f_T be the low-pass of f at height T , $\widehat{f_T} = F\mathbf{1}_{|t|\leq T}$. Then for every $T > 0$*

$$\frac{1}{2\pi} \int_{|t|>T} |F(t)|^2 (\Psi(t) - \lambda) dt = \frac{2P(f)}{2\pi} \int_{|t|>T} \overline{F(t)} \Sigma(t) dt - \int_{|y|>a} f_T(y) g_{\text{out}}(y) dy, \quad \Sigma = \hat{s}.$$

Explicitly, for $|y| > a$, $g_{\text{out}}(y) = -\int J_\Gamma(y-z)f(z)dz - \sum_n \Lambda(n)n^{-1/2}(f(y-\log n) + f(y+\log n))$.

Proof. On the whole line $Kf = \lambda f + 2Ps + g_{\text{out}}$, since the window equation holds inside and g_{out} is by definition the value of Kf outside. In Fourier form, $(\Psi - \lambda)F = 2P\Sigma + G$ with $G = \widehat{g_{\text{out}}}$. Multiply by

$\bar{F}\mathbf{1}_{|t|>T}$ and integrate: the term with G is $\frac{1}{2\pi} \int_{|t|>T} \bar{F}G = \langle f - f_T, g_{\text{out}} \rangle$ by Parseval, and $\langle f, g_{\text{out}} \rangle = 0$ because the supports are disjoint. The formula for g_{out} is the kernel of K off the diagonal: from (2), $(K_\infty f)(y) = c_0 f(y) + \int_0^\infty [2f(y)e^{-2x} - (f(y-x) + f(y+x))e^{-x/2}] (1 - e^{-2x})^{-1} dx$, which for $f(y) = 0$ is $-\int J_\Gamma(y-z)f(z)dz$, and the prime part of Q is the operator $-\sum_n \Lambda(n)n^{-1/2}(\tau_{\log n} + \tau_{-\log n})$. Both sides converge: g_{out} is smooth away from $\pm a$, decays exponentially, and by the edge law grows like $-C(\log(1/\delta))^{1/2}$ at distance δ outside the edge, while f_T is bounded and $O(1/|y|)$. $\square$

The identity is the mechanism of the tail law. Just outside the edge, $g_{\text{out}}(a+\delta) = -CL(\delta)^{1/2}(1+o(1))$ with $L(\delta) = \log(a/\delta) + \beta$ (the kernel $1/(2|s|)$ acting on the edge profile $CL^{-1/2}$), and the low-pass of f leaks outside the edge as $f_T(a+\delta) = CL(1/T)^{-1/2}\pi^{-1}(\frac{\pi}{2} - \text{Si}(T\delta))(1+o(1))$; their overlap is $2C^2/(\pi T)$ times $\int_0^\infty (\frac{\pi}{2} - \text{Si}) = 1$, with $L^{-1/2}L^{1/2} = 1$: no logarithm, and it is the tail law, since the left side is $\int_{|t|>T} |F|^2 \rho(1+o(1))$. The remaining pieces, the overlap with the smooth part of g_{out} (the regular part of the kernel, the other edge, the reflected primes) and the polar integral, are each of order $CL(1/T)^{-1/2}/T$, larger than C^2/T when C is small, and must cancel.

Computation 7.10 (the tail identity on the FEM minimizers). `rh_sum_rule.py`, on the edge-FEM minimizers with primes at $a = 0.5$ and 0.45 (Computation 8.11; the prime 2 is the only one in the window). The left side is the frequency integral of the closed-form transform against $\Psi - \lambda$ from T to $400T^*$, with the tail beyond by Computation 7.6; g_{out} is the archimedean kernel applied to the piecewise-linear f , the $1/(2s)$ part of J_Γ integrated exactly on every segment and the smooth remainder by Gauss–Legendre, plus the reflected prime; f_T is the inverse sine transform of the closed-form F on $[0, T]$. At $a = 0.5$, $T/T^* = 5, 10, 20$: right side over left side 1.000, 0.990, 0.980; at $a = 0.45$, $T/T^* = 5, 20$: 1.011, 0.989. Beyond twenty horizons the grid in y no longer resolves the oscillation of the leak, whose period is $2\pi/T$, and the ratio drifts to 0.94 and 0.91 at 50 and 100 horizons at $a = 0.5$; the left side itself equals the law of Computation 7.6 to 0.1% there. The polar piece is 0.30, 0.30, 0.28 of the left side at $a = 0.5$, so it is not small. The overlap of f_T with the edge-law force $-CL(\delta)^{1/2}$ alone, on $\delta < a/10$ with the C, β of Computation 8.11, is 0.93, 1.09, 0.98 of the law $-\lambda'/(\pi T)$ at $a = 0.5$ and 1.07, 1.01 at $a = 0.45$, and the rest of the overlap plus the polar piece is +0.09, -0.10, -0.01 and -0.04, -0.02 of the law: the edge overlap is the law and the two order- $CL^{-1/2}/T$ pieces cancel, each to about ten percent. What remains of the sum rule is analytic: the overlap of the leak with $-CL^{1/2}$ as an asymptotic expansion in $1/L(1/T)$ (the leading term has no logarithm; the first correction is not yet derived and is at most ten percent at two units of L), and the cancellation of the smooth overlap against the polar piece, which is a statement about the reflected primes and the regular part of the kernel at the edge.

The first of these is settled by a symmetry.

Lemma 7.11 (the edge overlap has no first-order logarithmic correction). *Let $L(\delta) = \log(a/\delta) + \beta$, $\varphi(\delta) = CL(\delta)^{-1/2}$ the edge profile and $\gamma_e(\delta) = -CL(\delta)^{1/2}$ the edge force, both on $0 < \delta < \delta_0$, and let the leak of φ through the sharp low-pass at height T be $\ell_T(\delta) = \int_0^{\delta_0} \varphi(u) \frac{\sin(T(\delta+u))}{\pi(\delta+u)} du$. Then, with $L_T = L(1/T)$,*

$$-2 \int_0^{\delta_0} \ell_T(\delta) \gamma_e(\delta) d\delta = \frac{2C^2}{\pi T} J(L_T), \quad J(L) = \int_0^\infty \int_0^\infty \left[\frac{L + \log(1/v)}{L + \log(1/s)} \right]^{1/2} \frac{\sin(s+v)}{s+v} ds dv = 1 + O(L^{-2}),$$

the integrals taken in the Abel sense and cut where the model ends. The term of order $1/L$ vanishes: the kernel $\sin(s+v)/(s+v)$ is symmetric in (s, v) while $\log[(L + \log(1/v))/(L + \log(1/s))]$ is antisymmetric, so only even powers of it survive, and the leading one is $\frac{1}{8} \log^2(s/v)/L^2$, which integrates to $\pi^2/(24L^2)$ against $\int_0^1 \log^2 \frac{1-\sigma}{\sigma} d\sigma = \pi^2/3$ and $\int_0^\infty \sin w dw = 1$ after the substitution

$s = w\sigma$, $v = w(1-\sigma)$. Numerically $J(L) = 1.020, 1.012, 1.008, 1.005, 1.005, 1.003$ for $L = 3, \dots, 8$ (`rh_sum_rule.py`); the series $1 + \pi^2/(24L^2) - 1.898/L^3 + \dots$ is asymptotic and useful only from $L \approx 7$.

Proof. Substitute $u = s/T$, $\delta = v/T$ in the double integral, so that $\varphi(s/T)\gamma_e(v/T) = -C^2[L_T + \log(1/v)]^{1/2}[L_T + \log(1/s)]^{-1/2}$ and the kernel becomes $T \sin(s+v)/(\pi(s+v))$ with $du d\delta = T^{-2} ds dv$. Write the ratio as $\exp(u)$ with $u = \frac{1}{2}[\log(1+x_v) - \log(1+x_s)]$, $x = \log(1/\cdot)/L_T$; u changes sign under $s \leftrightarrow v$ and the kernel does not, so $\int \int e^u k = \int \int \cosh(u) k = \int \int (1 + u^2/2 + \dots)k$. With $s = w\sigma$, $v = w(1-\sigma)$, $ds dv = w dw d\sigma$ and $u = \frac{1}{2} \log \frac{1-\sigma}{\sigma} L_T^{-1} + O(L_T^{-2})$ is independent of w at leading order, which gives the constant and the second-order term. $\square$

So the edge overlap is the tail law with a correction of order $L(1/T)^{-2}$, one to two percent at the heights of Computation 7.6, which is what the zero sums show. What remains of the sum rule is the cancellation of the smooth overlap against the polar piece. At leading order both are $2CL_T^{-1/2}/(\pi T)$ times a constant: the polar piece is $2P\langle f, s_{>T} \rangle$ with the high-pass of $s = \sinh(y/2)\mathbf{1}_{[-a,a]}$ concentrated in the Gibbs layer of width $1/T$ at the edges, giving $4P \sinh(a/2) CL_T^{-1/2}/(\pi T)$, and the smooth overlap is $-2g_{\text{reg}}(a) CL_T^{-1/2}/(\pi T)$ with $g_{\text{reg}}(a)$ the regular part of g_{out} at the edge. They cancel if and only if $g_{\text{reg}}(a) = 2P \sinh(a/2)$, which is the value of Kf just inside the edge by the eigen-equation: the sum rule reduces to the statement that the regular part of Kf is continuous across the edge, the singularity $\mp CL^{1/2}$ being carried entirely by the outside. Computation 7.10 supports it to about twenty percent of $g_{\text{reg}}(a)$. It is a local statement, and it holds.

Lemma 7.12 (the regular part of Kf is continuous across the edge). *Let f satisfy the hypothesis of Proposition 7.8 and put $\varphi(\delta) = CL(\delta)^{-1/2}$. Then*

$$\lim_{\delta \rightarrow 0^+} \left[(K_\infty f)(a - \delta) - (K_\infty f)(a + \delta) - CL(\delta)^{1/2} \right] = 0.$$

Consequently, for the minimizer, $g_{\text{reg}}(a) := \lim_{\delta \rightarrow 0^+} [g_{\text{out}}(a + \delta) + CL(\delta)^{1/2}] = (Kf)(a^-) = 2P(f) \sinh(a/2)$, and the function $h = 2P(f)s + g_{\text{reg}}$ is continuous at $\pm a$.

Proof. Write $k(x) = e^{-x/2}/(1-e^{-2x}) = \frac{1}{2x} + \frac{1}{4} + O(x)$ and $m(x) = e^{-2x}/(1-e^{-2x}) = \frac{1}{2x} - \frac{1}{2} + O(x)$, so that by (2) $(K_\infty f)(y) = c_0 f(y) + \int_0^\infty [2f(y)m(x) - (f(y-x) + f(y+x))k(x)] dx$. Outside, $f(a+\delta) = 0$ gives $(K_\infty f)(a+\delta) = -\int_0^{2a} f(a-u)k(u+\delta) du$. Inside, with $f(a-\delta+x) = \varphi(\delta-x)\mathbf{1}_{x < \delta}$, split the integral at $x = \delta$: on $[0, \delta]$ it is $\int_0^\delta [2\varphi(\delta)(m-k) + (2\varphi(\delta) - \varphi(\delta+x) - \varphi(\delta-x))k] dx = O(\delta\varphi(\delta)) + O(\int_0^\delta x^2 |\varphi''| k) = o(1)$, since $|\varphi''| \leq C' L^{-3/2} \delta^{-2}$; on $[\delta, \infty)$ it is $2\varphi(\delta) \int_\delta^\infty m - \int_\delta^\infty \varphi(\delta+x)k(x) dx = \varphi(\delta) \log \frac{1}{1-e^{-2\delta}} - \int_{2\delta}^{2a} f(a-u)k(u-\delta) du$. Hence

$$(K_\infty f)(a-\delta) - (K_\infty f)(a+\delta) - CL(\delta)^{1/2} = \left[\varphi(\delta) \log \frac{1}{2\delta} - CL(\delta)^{1/2} \right] + \int_0^{2a} f(a-u) [k(u+\delta) - k(u-\delta)\mathbf{1}_{u > 2\delta}] du + o(1).$$

The bracket is $CL(\delta)^{-1/2} [\log \frac{1}{2\delta} - L(\delta)] = -C(\beta + \log 2a)L(\delta)^{-1/2} \rightarrow 0$. In the integral, the part over $u < 2\delta$ is $O(\varphi(2\delta))$, and over $u > 2\delta$ one has $k(u+\delta) - k(u-\delta) = -\delta/(u^2 - \delta^2) + O(\delta)$, whose contribution is $O(\delta \int_{2\delta}^\infty \varphi(u)u^{-2} du) + O(\delta) = O(\varphi(2\delta)) + O(\delta) \rightarrow 0$. The terms with ψ in place of φ are integrable against $1/u$ by the hypothesis on ψ and contribute the same continuous function on both sides. For the consequence, $g_{\text{out}} = K_\infty f + K_P f$ outside and $K_P f$ is continuous across the edge (f is continuous), so $g_{\text{reg}}(a) = (K_\infty f)(a^-) + (K_P f)(a) = (Kf)(a^-) = \lambda f(a) + 2P \sinh(a/2)$, and $f(a) = 0$. $\square$

Corollary 7.13 (the sum rule). *Under the hypotheses of Proposition 7.8, for the minimizer,*

$$\frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi - \lambda) dt = \frac{2C^2}{\pi T} \left(1 + O(L(1/T)^{-1})\right),$$

the edge overlap contributing $\frac{2C^2}{\pi T}(1 + O(L^{-2}))$ by Lemma 7.11 and the polar piece plus the smooth overlap contributing $O(C^2/(TL))$.

Proof. In the identity of Proposition 7.9 write $g_{\text{out}} = -CL^{1/2}\mathbf{1}_{\text{edge}} + g_{\text{reg}}$. The polar piece is $2P\langle f, s_{>T} \rangle$ and the smooth overlap is $-\langle f_T, g_{\text{reg}} \rangle = -\langle f, g_{\text{reg}} \rangle + \langle f, (g_{\text{reg}})_{>T} \rangle = \langle f, (g_{\text{reg}})_{>T} \rangle$, the low-pass being self-adjoint and the supports disjoint; their sum is $\langle f, h_{>T} \rangle$ with $h = 2Ps + g_{\text{reg}}$. Inside the window the eigen-equation reads $2P \sinh(y/2) = K_{\infty}f_{\text{int}} + K_P f + e_{\text{in}} - \lambda f$, where f_{int} is the interior part of f and e_{in} the bounded remainder of K_{∞} on the edge profile; outside, $g_{\text{reg}} = K_{\infty}f_{\text{int}} + K_P f + e_{\text{out}}$ with $e_{\text{out}} = (K_{\infty}\varphi) + CL^{1/2}$. The part $K_{\infty}f_{\text{int}} + K_P f$ is smooth across the edge, so h is a smooth function plus the edge terms $e_{\text{in}} - \lambda f$ and e_{out} , which are $O(C)$, continuous at the edge by Lemma 7.12, and of the form $C \times (\text{a function with an expansion in powers of } L^{-1/2})$. The high-pass of the smooth part against f is $O(T^{-N})$; the high-pass of the $O(C)$ edge terms against f , whose own edge is $CL^{-1/2}$, is $O(C^2/(TL))$ by the computation of Lemma 7.11 with $L^{-1/2}$ in place of $L^{1/2}$. $\square$

So the sum rule is the tail identity with two facts about the edge: the overlap of the leak with the singular force is log-free by symmetry, and the regular force is continuous across the edge by the eigen-equation. The data say that the $O(1/L)$ of the corollary is small in practice (the zero sums sit on the law to one percent from five horizons on, where $L(1/T) \approx 2$); its coefficient is not computed here. The corollary is, however, only the far-field form of something more robust.

Computation 7.14 (the tail law is analytic). `rh_sum_rule.py`. (i) On the FEM minimizers with primes, the edge law $CL(\delta)^{-1/2}$ with the C, β of Computation 8.11 is not yet accurate at the scales that the heights of Computation 7.6 probe: $f(a-\delta)^2 L(\delta)/C^2$ at $\delta = 1/(kT^*)$ is 1.48, 2.04, 3.20, 3.17 for $k = e$ at $a = 0.4, 0.45, 0.5, a_3, 1.20, 1.33, 1.54, 1.43$ at $k = 10$, and reaches 1.02–1.06 only at $k = 100$; yet the tail law holds at one percent from five horizons. (ii) On the prime-free critical points of Computation 8.11 ($a = 0.3, 0.4, 0.5, 0.6$; the form $W_{\infty} - 2P^2$, whose floor is $+0.223, -0.078, -0.322, -0.536$), with λ' the matrix-side derivative and no zero anywhere, $T \cdot \frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi_{\infty} - \lambda) dt$ over $-\lambda'/\pi$ is

a	$T/T^* = 3$	5	10	20	50	100
0.3	0.986	0.980	1.014	1.008	1.002	0.999
0.4	0.978	1.020	1.009	1.001	1.000	1.000
0.5	0.998	1.000	0.997	0.997	1.001	1.001
0.6	1.012	1.007	1.004	1.001	0.999	1.000

The law $T \frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi - \lambda) \rightarrow -\lambda'/\pi$ is therefore a property of critical points of window forms with the logarithmic symbol, holding to one percent from three horizons on whatever the sign of the floor, and the zeros enter only through (S_{κ}) : they sample this analytic tail at their density. (iii) The sampling constant $M(\kappa)$ of (S_{κ}) , the density integral over the zero sum above κT^* on the FEM minimizers with primes, is 8.9, 1.53, 1.39, 1.03, 1.01, 1.01, 1.06, 1.05 at $a = 0.5$, 3.6, 0.91, 1.14, 1.09, 1.08, 1.07, 1.07, 1.03 at $a = 0.45$ and 1.8, 0.73, 1.16, 1.01, 1.05, 1.05, 1.08, 1.03 at $a = 0.4$, for $\kappa = 1, 1.5, 2, e, 3, 4, 5, 10$: above the exhaustion height the zeros sample the tail with constant one to within ten percent, and below two horizons they do not.

Two things follow. The finite-height sum rule is analytic and has an exact form: the dilation derivative is itself an edge pairing, and the tail is its remainder at height T .

Proposition 7.15 (the dilation derivative as an edge pairing). *Let f be a critical point as in Proposition 7.9, $\mathcal{D}(f)$ the dilation form (6), and $(xf)' = f + xf'$ the dilation generator, an L^1 function supported in the window. For $T > 0$ let $((xf)')_T$ be its low-pass at height T and $B_T = \langle g_{\text{out}}, ((xf)')_T \rangle$. Then for every T*

$$\mathcal{D}(f) = \text{tail}(T) + \frac{1}{\pi} T |F(T)|^2 (\Psi(T) - \lambda) + \text{tail}_D(T) - 4P \langle s, ((xf)')_{>T} \rangle + 2B_T,$$

with $\text{tail}(T) = \frac{1}{2\pi} \int_{|t|>T} |F|^2 (\Psi - \lambda)$ and $\text{tail}_D(T) = \frac{1}{2\pi} \int_{|t|>T} |F|^2 t \Psi'$. In particular B_T converges, $\mathcal{D}(f) = 2B_\infty$, and by Theorem 5.2 (Proposition 5.3 for the exact form) and the boundary law (Conjecture 8.1, proved under the edge law with derivatives in Proposition 8.3)

$$-a\lambda' = 2 \lim_{T \rightarrow \infty} \langle g_{\text{out}}, ((xf)')_T \rangle, \quad B_\infty = aC^2.$$

Proof. Write $\mathcal{D} = \frac{1}{2\pi} \int |F|^2 t \Psi' + 2P^2 + 2PM$ and cut the integral at $|t| = T$. On $|t| < T$ integrate by parts: $\frac{1}{2\pi} \int_{|t|<T} |F|^2 t \Psi' = \frac{1}{\pi} T |F(T)|^2 \Psi(T) - \frac{1}{2\pi} \int_{|t|<T} \Psi |F|^2 - \frac{1}{2\pi} \int_{|t|<T} \Psi t (|F|^2)'$. The second term is $-(\lambda + 2P^2) + \frac{1}{2\pi} \int_{|t|>T} \Psi |F|^2$. In the third, $\Psi t (|F|^2)' = 2 \text{Re} [\overline{\Psi F} t F']$ and $\Psi F = \lambda F + 2P\Psi + G$ by the eigen-equation; since $tF' = -(\widehat{xf})'$, Parseval gives $-\frac{1}{2\pi} \int_{|t|<T} \Psi t (|F|^2)' = -\lambda [\frac{1}{\pi} T |F(T)|^2 - 1 + \frac{1}{2\pi} \int_{|t|>T} |F|^2] + 4P \langle s, ((xf)')_T \rangle + 2 \langle g_{\text{out}}, ((xf)')_T \rangle$. Collecting, and using $\langle s, (xf)' \rangle = -M/2$ (integration by parts, $f(\pm a) = 0$) to write $4P \langle s, ((xf)')_T \rangle = -2PM - 4P \langle s, ((xf)')_{>T} \rangle$, the terms $\pm\lambda$, $\pm 2P^2$ and $\pm 2PM$ cancel and the identity follows. As $T \rightarrow \infty$ every term but $2B_T$ tends to 0 (tail and tail_D since $|F|^2 \Psi$ is integrable; the boundary term since $t|F|^2 \Psi \rightarrow 0$; the polar term since $s_{>T} \rightarrow 0$ in L^2 and $(xf)' \in L^1 \cap L^\infty$ away from the edge with the edge singularity $\delta^{-1} L^{-3/2}$ integrable), so $B_T \rightarrow \mathcal{D}/2$. $\square$

Corollary 7.16 (the tail as the remainder of the pairing). $\text{tail}(T) = 2(B_\infty - B_T) + 4P \langle s, ((xf)')_{>T} \rangle - \frac{1}{\pi} T |F(T)|^2 (\Psi(T) - \lambda) - \text{tail}_D(T)$. *The polar term is not small: the high-pass of $(xf)'$ carries the edge singularity $\delta^{-1} L^{-3/2}$, whose Gibbs layer against the jump of s gives $-2aPC \sinh(a/2) L(1/T)^{-1/2} (1 + o(1))$, decaying only logarithmically, and $2(B_\infty - B_T)$ cancels it at that order, as in Corollary 7.13; the tail is the $O(1/T)$ remainder of that cancellation. The identity is the dilation identity in Fourier form and does not by itself make the law easier; its value is the exact expression of $-a\lambda'$ as an edge pairing.*

Computation 7.17 (the pairing identity on the prime-free critical point). `rh_cutoff_identity.py`, on the prime-free FEM critical point at $a = 0.5$, every term computed independently: $\mathcal{D} = D_\infty + 2P^2 + 2PM = 1.129084$ from the profile ($D_\infty = 1.020022$, $P = 0.134363$, $M = 0.271485$), against $-a\lambda' = 1.129075$ from the ball-arithmetic engine; $B_T = \langle g_{\text{out}}, ((xf)')_T \rangle$ from the archimedean kernel in x -space and the inverse sine transform of $-tS'(t)$, $B_T = 0.523, 0.549, 0.550, 0.551$ at $T/T^* = 3, 5, 10, 20$ against $\mathcal{D}/2 = 0.5645$; the right side of Proposition 7.15 over $\mathcal{D}$ is 1.010, 1.020, 1.020, 1.011, the two percent being the resolution of the leak of $(xf)'$, whose edge singularity is $\delta^{-1} L^{-3/2}$, on the y -grid; and $\text{tail}(T)$ is 0.997, 0.998, 0.992, 0.989 of $\mathcal{D}/(\pi a T)$. The polar term is $-0.047, -0.044, -0.039, -0.035$, five times the tail and decaying like $L(1/T)^{-1/2}$ as the corollary says, so $B_\infty - B_T$ is dominated by its cancellation; separating the $O(1/T)$ remainder needs B_T to a part in ten thousand, beyond this computation.

The sharpest form of the sum rule is local in t , and it is exact.

Computation 7.18 (the envelope law). The average of $t^2(\Psi(t) - \lambda)|F(t)|^2$ over a window of width $4\pi/a$ centred at T (two periods of the interference of the two edges), divided by $-\lambda'$, on the FEM critical points (`rh_sum_rule.py`):

	$T/T^* = 1$	2	3	5	10	20	50	100	300
$a = 0.5$, no primes	1.197	1.016	1.012	1.001	1.001	1.000	1.000	1.000	1.000
$a = 0.3$, no primes	-2.81	1.025	1.112	1.016	1.006	1.001	1.000	1.000	1.001
$a = 0.5$, primes	-6.30	2.244	1.270	1.046	1.006	1.021	1.005	1.011	1.008
$a = 0.4$, primes	0.782	1.876	1.035	1.145	1.013	0.957	1.032	0.993	0.975

Without primes the local envelope equals $-\lambda'$ to a part in a thousand from ten horizons on and to 0.1% from twenty, over two decades of height, with no logarithmic drift; with primes it fluctuates by a few percent about 1, the cosines of the symbol beating against the interference of the edges.

Conjecture 7.19 (Wiener–Hopf form of the edge). *Let f be a critical point on $[-a, a]$ and $\sigma_\infty = \Psi_\infty - \lambda$. For a real even symbol σ with finitely many real zeros $\pm t_j$ write $\tilde{\sigma} = \sigma \prod_j (t^2 + t_j^2)/(t^2 - t_j^2) > 0$ and $\tilde{\sigma} = \sigma_+ \sigma_-$ for its canonical factorization, $\sigma_\pm = \tilde{\sigma}^{1/2} e^{\pm i\theta}$ on the line with*

$$\theta(t) = \frac{1}{2} \mathcal{H}[\log \tilde{\sigma}](t) = \frac{t}{\pi} \int_0^\infty \frac{\log \tilde{\sigma}(\tau) - \log \tilde{\sigma}(t)}{t^2 - \tau^2} d\tau, \quad (10)$$

odd in t , so that σ_+ continues analytically without zeros to the upper half-plane and σ_- to the lower. (i) Without primes ($\sigma = \sigma_\infty$) there are a height $T_0(a)$ and a real constant κ such that, with $S(t) = \int_0^a f \sin(tx) dx$ ($F = 2iS$),

$$S(t) = -\frac{|A| \cos(ta + \theta(t) - \kappa/t)}{t \tilde{\sigma}(t)^{1/2}} (1 + O(t^{-2})), \quad |A|^2 = -\frac{1}{2} \lambda', \quad t > T_0; \quad (11)$$

equivalently $F(t) = A(t)e^{ita}/(t\sigma_-(t)) - \overline{A(t)}e^{-ita}/(t\sigma_+(t)) + O(t^{-3})$ with $A(t) = -i|A|e^{-i\kappa/t}$: the amplitude of each edge is purely imaginary up to the phase κ/t , and there is no free phase. (ii) With the entries n inside the window ($\log n < 2a$; $c_n = \Lambda(n)n^{-1/2}$) each edge is accompanied by its echoes, at $a - \log n, a - \log n - \log m, \dots$, as long as they stay inside the window. For a single entry with $a < \log n < 2a$ there is one echo; the symbol of the edge is $\sigma_{\text{eff}} = \sigma_\infty - c_n^2/\sigma_\infty$, and with $\rho(t) = c_n/\sigma_\infty(t)$, θ_{eff} the phase (10) of σ_{eff} ,

$$S(t) = -\frac{|A| |1 + \rho e^{-it \log n}| \cos(ta + \theta_{\text{eff}}(t) + \arg(1 + \rho e^{-it \log n}) - \kappa/t)}{t \tilde{\sigma}_{\text{eff}}(t)^{1/2}} (1 + O(t^{-2})). \quad (12)$$

To first order in ρ , (12) is (11) with the symbol $\sigma_a = \Psi_a - \lambda$ of the window, $\Psi_a = \Psi_\infty - \sum_{\log n < 2a} 2\Lambda(n)n^{-1/2} \cos(t \log n)$. In both cases the local envelope of Computation 7.18 is $2|A|^2 = -\lambda'$, exactly in (i) and up to $O(\rho^2)$ in (ii), and the nulls of S sit at $ta = \pi/2 + k\pi - \vartheta(t) + \kappa/t$ with ϑ the phase of the respective form.

The form comes from the two edges as two half-line problems, and the argument is the classical one for a Wiener–Hopf equation with a symbol of order zero; no cutoff is needed to see it. Write the eigen-equation on the whole line, $(K - \lambda)f = 2Ps 1_{[-a, a]} + g_{\text{out}}$ with $g_{\text{out}} = Kf$ outside the window as in Proposition 7.9, transform it, $\sigma F = 2P\Sigma + G$, and multiply by e^{-ita} . The function $\tilde{F} = e^{-ita}F = \int_{-2a}^0 f(a+u)e^{itu} du$ is analytic and bounded in the lower half-plane and $O(1/t)$ (a “−” function); $\tilde{G}_+ = \int_{u>0} g_{\text{out}}(a+u)e^{itu} du$ is analytic in the upper half-plane (a “+” function) and $O(t^{-1} \log^{1/2} t)$ by Lemma 7.12; and $\tilde{H} = e^{-ita}(2P\Sigma + \int_{x<-a} g_{\text{out}} e^{itx} dx)$ is a “−” function,

$\tilde{H} = -2iP \sinh(a/2)/(t-i0) + e^{-2ita}\Phi_-(t) + O(t^{-2})$, the first term the step of the polar term at the edge and $\Phi_- = O(t^{-1} \log^{1/2} t)$ the far edge, its polar step and its outside force. The primes outside the window drop out: for $\log n > 2a$ the shifted copies $f(x \mp \log n)$ lie entirely outside the window, so they are part of g_{out} , and their transforms $e^{\pm it \log n} F$ reproduce the term $-2\Lambda(n)n^{-1/2} \cos(t \log n) F$ of σF . So, exactly,

$$\sigma_a \tilde{F} = \tilde{H} + \tilde{G}_+, \quad (13)$$

with the symbol σ_a of the window, the partial copies of the entries inside the window (the part of $f(x - \log n)$ on $x > a$, of $f(x + \log n)$ on $x < -a$) staying in $\tilde{G}_+$ and $\tilde{H}$ on their sides.

Without primes. Divide (13) by σ_+ and split $\tilde{H}/\sigma_+$ additively into its parts analytic in the two half-planes, $[\cdot]_- + [\cdot]_+$. The function $E = \sigma_- \tilde{F} - [\tilde{H}/\sigma_+]_- = \tilde{G}_+/\sigma_+ + [\tilde{H}/\sigma_+]_+$ is entire and tends to zero in every direction, since $\sigma_{\pm} \sim (\log t)^{1/2}$, so $E \equiv 0$ and $\tilde{F} = [\tilde{H}/\sigma_+]_-/\sigma_-$. For the polar step, $[1/((t-i0)\sigma_+(t))]_- = 1/((t-i0)\sigma_+(0))$; for the far edge, $[e^{-2ita}\Phi_-/\sigma_+]_- = e^{-2ita}\Phi_-/\sigma_+ - \sum_j c_j/(t-z_j)$ by closing the Cauchy integral in the lower half-plane, the sum running over the singularities z_j of $1/\sigma_+ = \sigma_-/\sigma$ there, the complex zeros of σ , with $c_j \propto e^{-2iz_j a}$. Hence

$$\tilde{F}(t) = \frac{A(t)}{t\sigma_-(t)} + \frac{e^{-2ita}\Phi_-(t)}{\sigma(t)} + O(t^{-2}), \quad A(t) = -\frac{2iP \sinh(a/2)}{\sigma_+(0)} - \sum_j \frac{c_j t}{t-z_j} = A_0 + \frac{A_1}{t} + \dots,$$

the first term the right edge and the second the left one ($e^{-2ita}\Phi_-/\sigma = e^{-ita}F_L$). The symmetry $\tilde{F}(-t) = \overline{\tilde{F}(t)}$ of a real f , with $\sigma_-(-t) = \overline{\sigma_-(t)}$, gives $A(-t) = -\overline{A(t)}$: A_0 is purely imaginary and A_1 real, so $A(t) = A_0 e^{-i\kappa/t} (1 + O(t^{-2}))$ with $\kappa = A_1/|A_0|$, the first correction is a phase, and (11) follows. The far edge and the polar term fix $|A_0|$ (the sum over the c_j is the coupling of the two edges, which is the quantization of $\lambda(a)$), and $|A_0|^2 = -\lambda'/2$ is the boundary law $\lambda' = -2C^2$ read as a flux: the x -space asymptotics of $1/\sigma_-$ is the edge profile $L(\delta)^{-1/2}$, so $|A_0| = C$. All the logarithms of the edge profile are inside σ_- , which is why the profile of Computation 7.14(i) can be 1.5–3 times above its asymptotic law at scale $1/T$ while the tail law holds at that T to one percent; Proposition 7.8 and Corollary 7.13 are the far field of this structure. The real zeros $\pm t_j$ of σ are the frequencies of the standing waves $e^{\pm it_j x}$ that solve $(K - \lambda)u = 0$ on the line and make up the interior of the critical point; with them the homogeneous problem has solutions, E is a rational function with simple poles at $\pm t_j$ instead of zero, and the standing-wave amplitudes enter A_0 and κ (the $\arctan(t_j/t)$ of the zeros' factors are part of κ). Proposition 7.23 and Corollary 7.24 below make this rigorous, for every L^2 critical point of the prime-free form and without any hypothesis on the edge: the decay of the three functions needs only $f \in L^2$, and the factor of a symbol growing like $\log t$ is the outer function of $\tilde{\sigma}^{1/2}$.

With primes. An entry inside the window shifts the window onto itself: $(K_{\infty} - \lambda)f = 2Ps + \sum_n c_n (f(x - \log n) + f(x + \log n))$ on the window, K_{∞} the archimedean part of K . The right edge's singularity is thereby copied to $a - \log n$, the copy to $a - \log n - \log m$, and so on while the points stay inside the window: a finite set of echoes, a chain for each entry, and the mirror chains from the left edge. Take one entry with $a < \log n < 2a$, so that each chain has one link, and split f near a and near $a - \log n$ into an edge piece f_R and an echo piece f_e with a partition of unity. At the echo point $(K_{\infty} - \lambda)f_e = c_n f_R(x + \log n) + \text{smooth}$, so $F_e = (c_n/\sigma_{\infty})e^{-it \log n} F_R$ up to rapidly decaying terms. At the edge $(K_{\infty} - \lambda)f_R = 2Ps + c_n f_e(x - \log n) + \text{smooth}$ on $x < a$, and the echo shifted back is a two-sided singularity at a with transform $c_n e^{it \log n} F_e = (c_n^2/\sigma_{\infty})F_R$; moved to the left it makes the edge a half-line problem with the real even symbol $\sigma_{\text{eff}} = \sigma_{\infty} - c_n^2/\sigma_{\infty}$, to which everything above applies: $\tilde{F}_R = A(t)/(t\sigma_{\text{eff},-})$. Since f is odd, $F = 2i \text{Im}(F_R + F_e)$, which is (12). With the symbol $\sigma_a = \sigma_{\infty} - 2c_n \cos(t \log n)$ of the window instead one has $\sigma_a^{-1/2} = \sigma_{\infty}^{-1/2} (1 + \rho \cos(t \log n)) + O(\rho^2)$ and $\theta_a = \theta_{\infty} - \rho \sin(t \log n) + O(\rho^2)$, the same as (12) to first order in ρ : the Hilbert transform of the

cosine is the echo. The difference is at second order and it is the window's: for a constant σ_∞ the factorization $\sigma_\infty - 2c_n \cos \tau = (\alpha - \beta e^{-i\tau})(\alpha - \beta e^{i\tau})$, $\alpha^2 = \frac{1}{2}(\sigma_\infty + \sqrt{\sigma_\infty^2 - 4c_n^2})$, $\beta = c_n/\alpha$, expands $1/\sigma_{a,-}$ into the infinite chain $\alpha^{-1} \sum_k (c_n/\alpha^2)^k e^{-ik\tau}$, with α^2 the fixed point of $\sigma \mapsto \sigma_\infty - c_n^2/\sigma$: the edge of a half-line with all its echoes, while the window cuts the chain where it leaves $[-a, a]$, and σ_{eff} is the continued fraction $\sigma_\infty - c_n^2/(\sigma_\infty - c_n^2/(\dots))$ with as many levels as there are echoes inside. Part (iii) of the following computation measures exactly this.

Computation 7.20 (the Wiener–Hopf phase). On the prime-free critical point at $a = 0.5$ of Computation 7.14 (edge FEM, 486 nodes to $10^{-12}a$, $\lambda = -0.32224$, $\lambda' = -2.25815$) the symbol $\sigma = \Psi_\infty - \lambda$ has the single real zero $t_\lambda = 4.562$, and θ of (10) is computed by the trapezoidal rule on a grid of step 0.01 to 8000 and geometric to 10^{11} (self-test on $\log \frac{t^2+1}{t^2+4}$, whose phase is $\arctan \frac{1}{t} - \arctan \frac{2}{t}$: error 5×10^{-10}); `rh_wiener_hopf.py`, logs in `data/`. (i) *The nulls.* S has 319 nulls t_k between 2.8 and 120 horizons, $t_k a = \pi/2 + k\pi + r_k$. The shift r_k is 0.339, 0.294, 0.239, 0.199, 0.162, 0.141, 0.136 at 3.2, 5, 10, 20, 50, 100, 120 horizons, and $-\theta(t_k)$ is 0.259, 0.243, 0.214, 0.187, 0.157, 0.138, 0.134; the difference $r_k + \theta(t_k) = 0.080, 0.051, 0.025, 0.013, 0.005, 0.0023, 0.0022$ is κ/t_k with $\kappa = 4.34$ (fitted on $t \geq 10T^*$; against $t_\lambda = 4.56$, the $\arctan(t_\lambda/t)$ of the zero's factor), and the residual $r_k + \theta(t_k) - \kappa/t_k$ has root mean square 1.2×10^{-4} over all 319 nulls, with no trend. The alternatives fail: a $1/t$ shift alone leaves 0.19; the adiabatic value of the phase, $-\frac{\pi}{4} t \frac{d}{dt} \log \tilde{\sigma} \approx -\frac{\pi}{4 \log(t/2\pi)}$, fitted as $c/\log(t/2\pi)$ (best $c = 0.79$ against $\pi/4 = 0.785$), leaves 3.5×10^{-3} , thirty times the exact phase, its error being 0.029, 0.018, 0.010 at 3.2, 3.9, 5 horizons, where the Hilbert integral (10) differs from its local value by that much. A second mesh (471 nodes, coarser between $\delta = 10^{-6}$ and 10^{-3}) reproduces r_k to 2×10^{-3} up to 20 horizons and drifts beyond, where its mesh at scale $1/t$ shows. (ii) *Pointwise.* With $|A| = (-\lambda'/2)^{1/2} = 1.0626$ and the fitted κ , nothing else adjusted, the relative L^2 distance over one interference period $4\pi/a$ between S and $-|A| \cos(ta + \theta - \kappa/t)/(t\tilde{\sigma}^{1/2})$ is $4.5 \times 10^{-3}, 1.5 \times 10^{-3}, 3.4 \times 10^{-4}, 8 \times 10^{-5}, 3 \times 10^{-5}$ at 3, 5, 10, 20, 50 horizons, decaying like T^{-2} (2.7×10^{-4} at 100, the mesh); with $\kappa = 0$ it is 0.093, 0.052, 0.026, 0.013, 0.005, the phase κ/t itself, and with $\theta = 0$ it is 0.36, 0.29, 0.24, 0.20, 0.16. (iii) *With the prime 2* (the critical point of $W - 2P^2$ at $a = 0.5$, $\lambda = 1.9 \times 10^{-4}$, $\lambda' = -0.011236$; the entry 2 is the only one inside the window, with $a < \log 2 < 2a$, so each edge has one echo, at $\pm(a - \log 2) = \pm 0.193$). The window's symbol $\Psi_\infty - 0.980 \cos(t \log 2) - \lambda$ has three real zeros, 3.31, 6.95, 10.53, and its phase oscillates at the prime's frequency with amplitude about $\rho = 0.490/\sigma_\infty$: $-\theta_a(t_k) = 0.16, 0.29, 0.05, 0.13, 0.13, 0.21, 0.23, 0.07$ at 9.8, 15, 20, 30, 50, 70, 100, 120 horizons. The nulls follow it: $r_k + \theta_a(t_k) - \kappa/t_k$ has root mean square 0.010 over 3–120 horizons ($\kappa = 62.4$), against 0.25–0.32 for r_k , 0.11–0.20 for $r_k + \theta_a(t_k)$, 0.08–0.13 for the smooth law $c/\log(t/2\pi)$, and 0.070 for the phase of the prime-free symbol at the same λ plus κ/t ; so the wobble of the nulls is the Hilbert transform of the prime's cosine, and the form (11) with σ_a is within 0.14, 0.051, 0.024, 0.016, 0.010 of S at 3, 5, 10, 20, 50 horizons, a percent-level remainder that does not decay. The echo form (12) removes it: with $\sigma_{\text{eff}} = \sigma_\infty - 0.240/\sigma_\infty$ (real zeros 3.86, 6.29, 10.26) and $\kappa = 62.2$, the null residual is 1.1×10^{-3} in root mean square over the 204 nulls between 5 and 80 horizons (largest 3.6×10^{-3} ; the window's symbol leaves 9.8×10^{-3} there, no shift 0.32), rising to 0.02 above 80 horizons where the mesh at scale $1/t$ shows, and the pointwise distance is 0.010, 0.0021, 0.0007 at 10, 20, 50 horizons, decaying like T^{-2} (0.0099 at 100, the mesh; 0.15 and 0.044 at 3 and 5, where the interior of this critical point, with $\lambda \approx 0$ and three standing waves, is still felt). So the second-order remainder of the window's symbol is the echo that the window cuts off, and the few-percent modulation of the envelope in Computation 7.18 is its trace.

The derivation is a theorem for the prime-free form, for every L^2 critical point and with no hypothesis on the edge. We need two facts about the symbol.

Lemma 7.21 (the factor of a logarithmic symbol). *Let $\tilde{\sigma}$ be even, positive and C^2 on $\mathbb{R}$ with $c_1 \leq \tilde{\sigma}(t) \leq c_2 \log(e + |t|)$, $(\log \tilde{\sigma})' \in L^2$ and $(\log \tilde{\sigma})''$ bounded, and let θ be given by (10). Then $\sigma_+ = \tilde{\sigma}^{1/2} e^{i\theta}$ is the boundary value of the outer function of $\tilde{\sigma}^{1/2}$: it extends to an analytic function without zeros in the upper half-plane, continuous up to the line, with $c \leq |\sigma_+(z)| \leq C(\log(e + |z|))^{1/2}$ there; $\sigma_-(z) = \overline{\sigma_+(\bar{z})}$ is analytic without zeros in the lower half-plane with the same bounds; $\sigma_+ \sigma_- = \tilde{\sigma}$ on the line; θ is odd and C^1 with θ, θ' bounded, so that $1/\sigma_{\pm}$ are bounded with bounded derivative on the line.*

Proof. $w = \tilde{\sigma}^{1/2}$ has $\log w \in L^1(d\tau/(1 + \tau^2))$, so the outer function $O(z) = \exp \left[\frac{1}{\pi i} \int \left(\frac{1}{\tau - z} - \frac{\tau}{1 + \tau^2} \right) \log w(\tau) d\tau \right]$ is analytic and zero-free in the upper half-plane, $|O(t)| = w(t)$ on the line and $\log |O(x + iy)| = P_y * \log w(x)$ is the Poisson integral [16, Ch. II]. Its argument on the line is the conjugate function of $\log w$; for even w the term $\tau/(1 + \tau^2)$ drops out by parity and the two half-lines combine into (10), which is odd and vanishes at 0; so $O = \sigma_+$, the unimodular constant of the outer function being fixed by $\theta(0) = 0$. Since $\log w \geq \frac{1}{2} \log c_1$, $|O(z)| \geq c$. Since $\log w(\tau) \leq \frac{1}{2} \log \log(e + |\tau|) + C$ and $\log \log(e + |\tau|) \leq \log \log(e + |z|) + \log(1 + |\tau - x|/(e + |z|))$ for $z = x + iy$, the Poisson integral gives $\log |O(z)| \leq \frac{1}{2} \log \log(e + |z|) + C + \frac{1}{2} \int P_y(s) \log(1 + |s|/y) ds$, and the last integral equals $\int P_1(s) \log(1 + |s|) ds < \infty$. Finally $\theta' = \frac{1}{2} \mathcal{H}[(\log \tilde{\sigma})']$, and the Hilbert transform of a C^1 function in L^2 with bounded derivative is bounded (split the integral at $|\tau - t| = 1$). Continuity up to the line follows from that of $\log w$ and θ , and the statements for σ_- are the reflection $z \mapsto \bar{z}$. $\square$

Lemma 7.22 (the kernel of $1/\sigma_+$ away from the origin). *Let $\sigma = \Psi_{\infty} - \lambda$ with $\lambda > \Psi_{\infty}(0)$, so that σ has exactly one pair of simple real zeros $\pm t_{\lambda}$ (Ψ_{∞} is even and increasing on $t > 0$, as $\frac{d}{dt} \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) = -\frac{1}{2} \operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) > 0$), and let $\tilde{\sigma} = \sigma(t^2 + t_{\lambda}^2)/(t^2 - t_{\lambda}^2)$. Then $\tilde{\sigma}$ satisfies the hypotheses of Lemma 7.21; there is $\eta \in (0, \frac{1}{2})$ such that $\tilde{\sigma}$ is analytic and zero-free in the strip $-\eta \leq \operatorname{Im} z \leq 0$, so that $1/\sigma_+ = \sigma_-/\tilde{\sigma}$ is bounded and analytic there; and the kernel of $1/\sigma_+$,*

$$k_+(\xi) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-it\xi}}{\sigma_+(t)} dt \quad (\xi > 0, \text{ as an improper integral}),$$

is a real C^2 function on $(0, \infty)$ with $|k_+^{(j)}(\xi)| \leq C_{\delta} e^{-\eta\xi}$ on $\xi \geq \delta$, $j \leq 2$, for every $\delta > 0$; $1/\sigma_+$ is the transform of the distribution k_+ supported on $[0, \infty)$.

Proof. σ is real-analytic and even with $\sigma(t) = \log(t/2\pi) - \lambda + O(t^{-2})$ and derivatives $O(t^{-j})$ by Stirling's formula, and $\sigma < 0$ on $|t| < t_{\lambda}$, so $\tilde{\sigma} > 0$, $\tilde{\sigma} \asymp \log(e + |t|)$ and $(\log \tilde{\sigma})^{(j)} = O(t^{-j}/\log t)$: Lemma 7.21 applies. For $z = x - iy$, $0 \leq y \leq \eta < \frac{1}{2}$, both $\frac{1}{4} + \frac{iz}{2}$ and $\frac{1}{4} - \frac{iz}{2}$ have real part at least $\frac{1}{4} - \frac{\eta}{2} > 0$, so Stirling's formula holds uniformly and $\sigma(z) = \log(|x|/2\pi) - \lambda + O(1/|x|)$ there; hence $|\tilde{\sigma}(z)| \geq \frac{1}{2} \log |x|$ for $|x| \geq X_0$, and on the compact rectangle $|x| \leq X_0$ the continuous function $\tilde{\sigma}$, which has no zero on the real segment, has none for η small. So $1/\sigma_+ = \sigma_-/\tilde{\sigma}$ is analytic in the strip and bounded by Lemma 7.21. Its derivatives on the line $\operatorname{Im} z = -\eta$ decay: $\log \sigma_-$ is the Cauchy integral of $\log \tilde{\sigma}^{1/2}$, whose j -th derivative at z is $O(\log(e + |z|)/|z|^j)$ (split the integral at $|\tau - x| = |z|/2$ and use $\frac{1}{2} \log \log$ growth outside and the bound $|(\log \tilde{\sigma})'| \leq C/(|x| \log |x|)$ inside), and $\tilde{\sigma}^{(j)} = O(|x|^{-j})$. The improper integral defining k_+ converges after two integrations by parts, since $(1/\sigma_+)' = O(t^{-2})$ on the line; shifting the contour to $\operatorname{Im} t = -\eta$ (the vertical connectors vanish as $|\operatorname{Re} t| \rightarrow \infty$ because $1/\sigma_+$ is bounded and $e^{-it\xi}$ is) gives $k_+(\xi) = \frac{e^{-\eta\xi}}{2\pi} \int e^{-ix\xi} \sigma_+(x - i\eta)^{-1} dx$, and $j+2$ integrations by parts on that line, with the derivative bounds just stated, give $|k_+^{(j)}(\xi)| \leq C_{\delta} e^{-\eta\xi}$ for $\xi \geq \delta$. That k_+ is real follows from $\sigma_+(-t) = \overline{\sigma_+(t)}$, and that it is supported on $[0, \infty)$ from the analyticity and boundedness of $1/\sigma_+$ in the upper half-plane (Paley–Wiener). $\square$

Proposition 7.23 (exact Wiener–Hopf representation of a prime-free critical point). *Let $f \in L^2(-a, a)$ be odd and a critical point of $W_\infty - 2P^2$: $K_\infty f = \lambda f + 2Ps$ on $(-a, a)$, $P = \langle f, s \rangle$, with $\lambda > \Psi_\infty(0)$, and let $\sigma, t_\lambda, \tilde{\sigma}, \sigma_\pm$ be as in Lemma 7.22. With $g_{\text{out}} = K_\infty f$ outside the window define on the line*

$$\tilde{F} = e^{-ita} F, \quad \tilde{G}_+ = \int_0^\infty g_{\text{out}}(a+u) e^{itu} du, \quad \tilde{H} = e^{-ita} \left(2P\Sigma + \int_{y < -a} g_{\text{out}}(y) e^{ity} dy \right),$$

and $\hat{H}_- = \tilde{H} + t_\lambda \left[\frac{\tilde{H}(t) - \tilde{H}(t_\lambda)}{t - t_\lambda} - \frac{\tilde{H}(t) - \tilde{H}(-t_\lambda)}{t + t_\lambda} \right]$. Then $\hat{H}_-/\sigma_+ \in H^1(\mathbb{R})$ and

$$\tilde{F} = \frac{P_- [\hat{H}_-/\sigma_+]}{\sigma_-} \quad \text{on } \mathbb{R}, \quad (14)$$

where $P_\pm = \frac{1}{2}(I \pm i\mathcal{H})$ are the Riesz projections onto the boundary values of H^2 of the upper and lower half-planes.

Proof. The equation. As in the proof of Proposition 7.9, $K_\infty f = \lambda f + 2Ps + g_{\text{out}}$ on the line, so $\sigma F = 2P\Sigma + G$ with $G = \widehat{g_{\text{out}}}$; multiplying by e^{-ita} and splitting G at the window, $\sigma \tilde{F} = \tilde{H} + \tilde{G}_+$. Since $J_\Gamma(s) = \frac{1}{2|s|} + O(1)$ near 0 and $J_\Gamma(s) = O(e^{-|s|/2})$, $g_{\text{out}}(a+u)$ is $-\frac{1}{2}$ times the Hilbert transform of f at $a+u$ plus a bounded, exponentially decaying function, so $g_{\text{out}} \in L^1 \cap L^2$ outside the window, with $yg_{\text{out}} \in L^2$. Hence: $\tilde{F}(z) = \int_{-2a}^0 f(a+u) e^{izu} du$ is entire and bounded by $\|f\|_1$ in the closed lower half-plane; $\tilde{G}_+$ is analytic in the upper half-plane, continuous on its closure, bounded by $\|g_{\text{out}}\|_1$, and tends to zero at infinity there (Riemann–Lebesgue for the family $g_{\text{out}}(a+\cdot)e^{-yu}$, $y \geq 0$, which is compact in L^1); $\tilde{H}$ is analytic in the lower half-plane, continuous on its closure, bounded, and its restriction to the line is in $H^1(\mathbb{R})$, being the transform of a function h with $h, yh \in L^2$.

The real zeros. $\sigma(\pm t_\lambda) = 0$ forces $(\tilde{H} + \tilde{G}_+)(\pm t_\lambda) = 0$. Multiply the equation by $(t^2 + t_\lambda^2)/(t^2 - t_\lambda^2) = 1 + t_\lambda \left[\frac{1}{t - t_\lambda} - \frac{1}{t + t_\lambda} \right]$ and write each product $t_\lambda \tilde{H}(t)/(t \mp t_\lambda)$ as a difference quotient plus $t_\lambda \tilde{H}(\pm t_\lambda)/(t \mp t_\lambda)$, likewise for $\tilde{G}_+$: the terms with real poles cancel, and $\tilde{\sigma} \tilde{F} = \hat{H}_- + \hat{G}_+$ with $\hat{G}_+$ built from $\tilde{G}_+$ as $\hat{H}_-$ from $\tilde{H}$. The difference quotients are analytic where $\tilde{H}$ (resp. $\tilde{G}_+$) is, continuous up to the line, bounded, and $O(1/t)$ with derivative $O(t^{-2})$; so $\hat{H}_-$ and $\hat{G}_+$ have the properties listed for $\tilde{H}$ and $\tilde{G}_+$.

Liouville. Put $\varphi = \hat{H}_-/\sigma_+$; it is in $H^1(\mathbb{R})$ because $1/\sigma_+$ is bounded with bounded derivative (Lemma 7.21). Then $P_\pm \varphi \in H^2$ of the respective half-planes have boundary values in $H^1(\mathbb{R}) \subset C(\mathbb{R}) \cap L^\infty$, are the Poisson integrals of these boundary values, hence bounded and continuous on the closed half-planes, and $|P_\pm \varphi(x \pm iy)| \leq \|\varphi\|_2 (4\pi y)^{-1/2}$. Define $E = \sigma_- \tilde{F} - P_- \varphi$ on the closed lower half-plane and $E = \hat{G}_+/\sigma_+ + P_+ \varphi$ on the closed upper one. On the line the two agree, since their difference is $(\tilde{\sigma} \tilde{F} - \hat{H}_- - \hat{G}_+)/\sigma_+ = 0$. So E is continuous on $\mathbb{C}$ and analytic off the line, hence entire; it is bounded by $C(\log(e + |z|))^{1/2}$ (below by the bounds on σ_- , $\tilde{F}$ and $P_- \varphi$; above by those on $\hat{G}_+$, $1/\sigma_+$ and $P_+ \varphi$), so it is constant by Cauchy's estimate, and $E(iy) \rightarrow 0$ as $y \rightarrow \infty$. Hence $E \equiv 0$, which is (14). $\square$

Corollary 7.24 (the form of the edge). *In the setting of Proposition 7.23, fix $b > 0$ and collect the far edge in*

$$\Phi_-(t) = \int_0^\infty g_{\text{out}}(-a-v) e^{-itv} dv - \frac{2iP \sinh(a/2)}{t - ib},$$

so that $e^{-2ita} \Phi_-$ is the transform of the part $\text{far}(u) = (g_{\text{out}}(a+u) + 2P \sinh(a/2) e^{b(u+2a)}) \mathbf{1}_{u < -2a}$ of the source of $\tilde{H}$. Then, exactly on the line,

$$\tilde{F}(t) = \frac{A(t)}{t \sigma_-(t)} + \frac{e^{-2ita} \Phi_-(t)}{\tilde{\sigma}(t)}, \quad A(t) = t P_- \left[\frac{\hat{H}_- - e^{-2ita} \Phi_-}{\sigma_+} \right](t) - t P_+ \left[\frac{e^{-2ita} \Phi_-}{\sigma_+} \right](t), \quad (15)$$

with $A(t) = A_0 + O(\log t/t)$, A_0 purely imaginary, and $A(-t) = -\overline{A(t)}$. Moreover

$$F(t) = \frac{A(t)e^{ita}}{t\sigma_-(t)} - \frac{\overline{A(t)}e^{-ita}}{t\sigma_+(t)} + O(t^{-2}(\log t)^{-1/2}), \quad S(t) = \operatorname{Im} \frac{A(t)e^{ita}}{t\sigma_-(t)} + O(t^{-2}(\log t)^{-1/2}), \quad (16)$$

so that, writing $A(t) = -i|A_0|e^{-i\kappa(t)/t}(1+\varepsilon(t))$ with κ real, $S(t) = -|A_0|\cos(ta+\theta(t)-\kappa(t)/t)/(t\tilde{\sigma}(t)^{1/2}) + O(t^{-2}(\log t)^{-1/2}) + O(\varepsilon/t)$, $\kappa(t) = O(\log t)$ and $\varepsilon(t) = O(1/t)$.

Proof. (15) is (14) with $P_-[e^{-2ita}\Phi_-/\sigma_+] = e^{-2ita}\Phi_-/\sigma_+ - P_+[e^{-2ita}\Phi_-/\sigma_+]$ and $\sigma_+\sigma_- = \tilde{\sigma}$; the function $e^{-2ita}\Phi_-/\sigma_+$ is in L^2 since $\text{far} \in L^1 \cap L^2$. The symmetry $A(-t) = -\overline{A(t)}$ follows from $\tilde{F}(-t) = \tilde{F}(t)$, $\Phi_-(-t) = \overline{\Phi_-(t)}$, $\sigma_-(-t) = \overline{\sigma_-(t)}$ and the parity of $\tilde{\sigma}$.

The near part. Integrating $2P\Sigma$ by parts, $\hat{H}_- - e^{-2ita}\Phi_- = A'/(t-ib) + \Psi_2$ with $A' = -2iP \sinh(a/2) - t_\lambda(\tilde{H}(t_\lambda) - \tilde{H}(-t_\lambda))$, purely imaginary because $\tilde{H}(-t) = \overline{\tilde{H}(t)}$, and Ψ_2 analytic in the lower half-plane, $O(t^{-2})$ on the line. Now $P_-[1/((t-ib)\sigma_+)] = 1/((t-ib)\sigma_+(ib))$ exactly, because $[1/\sigma_+(t) - 1/\sigma_+(ib)]/(t-ib)$ is in H^2 of the upper half-plane; and $\sigma_+(ib) > 0$ (the outer function of an even weight is real and positive on the imaginary axis). Split the source of Ψ_2 with a smooth partition of unity into a part supported in $(-a, 0]$ and a part supported in $u < -a/2$. The first is smooth on $(-a, 0]$ and continuous at 0 with a kink (the step of the polar term at the edge has been removed), so its transform ψ satisfies $\psi = O(t^{-2})$, $\psi' = O(t^{-3})$, and for such ψ one has $tP_-[\psi/\sigma_+](t) = -\frac{i}{2\pi} \int \psi/\sigma_+ + O(\log t/t)$: indeed $P_- = \frac{1}{2}(I - i\mathcal{H})$, $t\psi = O(1/t)$, and $t\mathcal{H}[\psi/\sigma_+](t) = \frac{1}{\pi} \int \psi/\sigma_+ + \frac{1}{\pi} \int \frac{\psi(\tau)\tau}{\sigma_+(\tau)(t-\tau)} d\tau$, where the last integral is $O(\log t/t)$ on $|\tau| < t/2$, $O(1/t)$ on $|\tau - t| < t/2$ (the principal value of $[m(\tau) - m(t)]/(t-\tau)$ with $m = \tau\psi/\sigma_+$, $m' = O(\tau^{-2})$) and $O(1/t)$ on $|\tau| > 3t/2$. The integral $\int \psi/\sigma_+$ is real, as $\psi(-\tau) = \overline{\psi(\tau)}$ and $\sigma_+(-\tau) = \overline{\sigma_+(\tau)}$.

The far parts. Every remaining piece of $\hat{H}_-$ (the far source of Φ_- and the second part of the source of Ψ_2) is the transform $\tilde{\Phi}$ of a real function $\text{src} \in L^1 \cap L^2$ supported in $u \leq -\delta$ for some $\delta > 0$. By Lemma 7.22, $\tilde{\Phi}/\sigma_+$ is the transform of $\text{src} * k_+$, and $P_+[\tilde{\Phi}/\sigma_+]$ is the transform of its restriction to $u > 0$, $r(u) = \int_{u' \leq -\delta} \text{src}(u')k_+(u-u') du'$, which uses k_+ only on $[\delta, \infty)$: r is real, C^2 on $[0, \infty)$, and $r, r', r'' \in L^1$. Integrating by parts, $\int_0^\infty r(u)e^{itu} du = ir(0^+)/t - r'(0^+)/t^2 + o(t^{-2})$, so $-tP_+[\tilde{\Phi}/\sigma_+](t) = -ir(0^+) + O(1/t)$ with $r(0^+)$ real; and $tP_-[\tilde{\Phi}/\sigma_+] = t\tilde{\Phi}/\sigma_+ - tP_+[\tilde{\Phi}/\sigma_+]$, where $t\tilde{\Phi}/\sigma_+ = O(1/t)$ for the piece of Ψ_2 ($\tilde{\Phi} = O(t^{-2})$ there). Collecting, $A(t) = A_0 + O(\log t/t)$ with $A_0 = A'/\sigma_+(ib) - \frac{i}{2\pi} \int \psi/\sigma_+ - i \sum r(0^+)$ purely imaginary.

The mirror. From $\sigma\tilde{F} = \tilde{H} + \tilde{G}_+$ and (15), $\tilde{G}_+ = \sigma\tilde{F} - \tilde{H} = \frac{\sigma}{\tilde{\sigma}} \frac{\sigma_+ A}{t} + \frac{\sigma}{\tilde{\sigma}} e^{-2ita}\Phi_- - 2P\tilde{\Sigma} - e^{-2ita}(\Phi_- + \frac{2iP \sinh(a/2)}{t-ib})$, where $\tilde{\Sigma} = e^{-ita}\Sigma = -i \sinh(a/2)(1 + e^{-2ita})/t + O(t^{-2})$ and $\sigma/\tilde{\sigma} = 1 - 2t_\lambda^2/(t^2 + t_\lambda^2)$; the terms in e^{-2ita} cancel up to $O(t^{-2})$ (Φ_- is bounded), and $\tilde{G}_+(t) = \sigma_+(t)A(t)/t + 2iP \sinh(a/2)/t + O(t^{-2}(\log t)^{1/2})$. Since f is odd, $g_{\text{out}}(-a-v) = -g_{\text{out}}(a+v)$, so $\Phi_- = -\tilde{G}_+ - 2iP \sinh(a/2)/(t-ib) = -\sigma_- \overline{A}/t + O(t^{-2}(\log t)^{1/2})$, and the second term of (15) is $-e^{-2ita}\overline{A(t)}/(t\sigma_+(t)) + O(t^{-2}(\log t)^{-1/2})$. Multiplying by e^{ita} gives (16); $F = 2iS$ is purely imaginary, and the two terms are conjugate up to sign, so S is the imaginary part of the first. The last statement is $A(-t) = -\overline{A(t)}$ read on $A(t) - A_0 = \beta(t)/t$: $\beta(-t) = \overline{\beta(t)}$, so the even part of β is real, a phase to first order, while its odd part, which is $O(1/t)$ by the expansions above, is imaginary. $\square$

Corollary 7.24 is Conjecture 7.19(i) except for three things it does not decide: the value $|A_0|^2 = -\lambda'/2$, which is the boundary law (Propositions 8.3 and 8.4 prove it under (H_∞) , Corollary 8.5); the constancy of the phase $\kappa(t)$, for which the bound is $O(\log t)$ while the data of Computation 7.20 give a constant to 10^{-4} ; and the rate, $O(\log t/t)$ against the measured t^{-2} . With an entry inside the window the same holds with the effective symbol of Proposition 8.8, and part (ii) of the conjecture, the echo form, is Corollary 8.9(iii); at every support with a finite echo set the effective symbol is

the Schur complement of the echo tree, Proposition 8.14, and the echo form is Corollary 8.15(iii). Its content for the tail law is this: $\int_{|t|>T} |F|^2 \tilde{\sigma} dt = 8|A_0|^2 \int_T^\infty \cos^2(ta + \theta - \kappa/t) dt/t^2 (1 + o(1)) = 4|A_0|^2/T (1 + o(1))$, so for every prime-free critical point the tail energy of Corollary 7.13 equals $2|A_0|^2/(\pi T)$ at every height beyond the onset, and the tail law $-\lambda'/(\pi T)$ of Computation 7.14 is the single identity $2|A_0|^2 = -\lambda'$: with $|A_0| = C$, the boundary law. Item 2 of the program is therefore reduced to the flux identity, and to the echo form with primes.

Heuristic 7.25 (why the effective height is tied to the horizon). Below the horizon the zeros are sparser than the a/π degrees of freedom per unit height of a function of type a , and the minimizer vanishes at all of them: their share of the floor is 10^{-5} , 10^{-7} , 10^{-13} at $a = 0.6, 0.8, 1.0$. Above it, the excess of the zero density over the Nyquist density is $\log(t/T^*)/2\pi$, the same function of t/T^* for every a ; so, after scaling heights by the horizon, the minimizer faces the same problem at every support, which is why T_{eff}/T^* should tend to a constant, and why the profiles $\vartheta(T/T^*)$ should converge as a grows. The only way for a function of type a to be small at more points than it has freedom for is to steepen at the edge (the edge law and the Hadamard formula), and the edge amplitude feeds the tail at every height at once, at the rate $C^2/(\pi T)$; the minimizer balances the gain from hiding at one more zero just above the horizon against that cost, and the balance falls at a fixed multiple of T^* . The spare freedom below the horizon is worth $T^*/2\pi$ zeros, and the accumulated excess above it, $\int_{T^*}^T \log(t/T^*) dt/2\pi$, exhausts it at $T = eT^*$; the measured median heights are below that and rising with a . Two things enter that no counting supplies: positivity of every term, which is RH, and the absence of gaps between zeros wider than the resolution π/a above a few horizons, in which a shape could park its tail. The explanation is therefore conditional on where the zeros are; it says what Conjecture 7.1 asserts about them, not why the prime side must produce it.

The count in the heuristic has an exact form.

Lemma 7.26 (exhaustion height). *The number of zeros in $[-T, T]$ equals the Shannon number $2aT/\pi$ of the window $[-a, a]$ at bandwidth T , up to $O(\log T)$, exactly at $T = eT^*(a)$.*

Proof. $2N(T) = \frac{T}{\pi} (\log \frac{T}{2\pi} - 1) + O(\log T)$, and $\log \frac{T}{2\pi} - 1 = 2a$ at $T = 2\pi e^{2a+1} = eT^*$. $\square$

Below eT^* a function of type a has more freedom than there are zeros to avoid; above it, fewer. So the minimizer can be expected to hide from the zeros up to about eT^* and to be the pure edge tail beyond, which is the onset of the tail law; and its floor to sit in the zeros around eT^* . The data agree: the median height of the floor's mass is 2.39, 2.32, 2.65, 2.67, 2.74 horizons at $a = 0.6, \dots, 1.5$ (Computation 7.6), and $\vartheta(eT^*)$ read from the same tables is 0.43, 0.45, 0.47, 0.49, 0.50, while the law is within 10–15% at that height. Conjecture 7.1 then takes its sharpest conditional form.

Proposition 7.27 (sampling above the exhaustion height implies Conjecture 7.1). *Assume RH, let $\kappa \geq 1$, and suppose that for the minimizer f at every support $a \geq a_0$*

$$(S_\kappa) \quad \int_{|t|>\kappa T^*} |F(t)|^2 \rho(t) dt \leq M \sum_{|\gamma|>\kappa T^*} |F(\gamma)|^2 \quad (\text{the zeros above } \kappa T^* \text{ sample the tail of } F \text{ with constant } M), \text{ and}$$

$$(E_\kappa) \quad \int_{|t|>\kappa T^*} |F(t)|^2 \rho(t) dt \geq \frac{2C(a)^2}{\pi \kappa T^* M'} \quad (\text{the edge overlap of Lemma 7.11 holds at height } \kappa T^* \text{ within the factor } M'),$$

and that the boundary law $\lambda' = -2C^2$ holds. Then $\lambda' \geq -\pi \kappa M M' T^ \lambda$ on $[a_0, \infty)$: Conjecture 7.1 with $c = \pi \kappa M M'$.*

Proof. $\lambda = \sum_{\gamma} |F(\gamma)|^2 \geq \sum_{|\gamma| > \kappa T^*} |F(\gamma)|^2 \geq M^{-1} \int_{|t| > \kappa T^*} |F|^2 \rho \geq \frac{2C^2}{\pi \kappa M M' T^*}$, so $-\lambda' = 2C^2 \leq \pi \kappa M M' T^* \lambda$. $\square$

Both hypotheses are measured. (S_{κ}) with $\kappa = e$: the density integrals and the zero sums of Computation 7.6 agree to 7% at five horizons and to 1% at twenty, so $M \approx 1$; (E_e) : $T\vartheta(T)/T_{\text{eff}}$ at eT^* is 1.10–1.15 at every support, so $M' \approx 1$ with the inequality in the favourable direction. The measured constant, $c_0 = \Phi'/T^*$ between 3.3 and 3.9, is $\pi e \vartheta(eT^*)$ times that factor, against the bound $\pi e \approx 8.5$ of the proposition. What (S_{κ}) asks of the zeros is a sampling inequality on the half-line above eT^* for the high-frequency part of functions of type a with a logarithmic edge, with density $\rho \geq (2a+1)/2\pi$ against the Nyquist density a/π ; what (E_{κ}) asks of the minimizer is that its edge layer be no narrower than $1/(eT^*)$, i.e. that it hide from no zero above the exhaustion height. The first is a statement about the zeros alone, the second about the minimizer alone, and the count of Lemma 7.26 is the reason both should hold at a fixed multiple of the horizon.

The entries have a zero-side reading in the same picture. The tail of the floor carries $\cos^2(\gamma a) = \frac{1}{2}(1 + \cos 2\gamma a)$ over the zeros, so it contains the average of $\cos(2\gamma a)$ over the tail-weighted zeros, and by the explicit formula the zeros' rotations align exactly when $2a = \log n$.

Computation 7.28 (the zeros' rotation at the entries). `rh_alignment.py`, first 6000 zeros: $R(a) = \sum_{\gamma} \cos(2\gamma a) w(\gamma) / \sum_{\gamma} w(\gamma)$ with the tail weight $w = \gamma^{-2} \log^{-1}(\gamma/2\pi)$, on $0.3 \leq a \leq 1.3$ in steps of 0.0005. Its mean absolute value is 0.27 and its 95th percentile 0.53; at the entries of the primes it is -0.53 (a_2), -0.62 (a_3 , the minimum over the whole range, at $a = 0.5495$), -0.57 (a_5), -0.56 (a_7), -0.44 (a_{11}), each minimum within 0.0005 of a_n ; at the prime powers 4, 8, 9, whose weights $\Lambda(n)n^{-1/2} = 0.35, 0.25, 0.37$ are below the noise of this truncation, it is $+0.10, -0.08, +0.22$. Around a_3 the curve is a cusp: with the zeros above $3T^*$ only and weight γ^{-2} , R is $-0.097, -0.158, -0.226, -0.161, -0.103$ at offsets $-0.004, -0.002, 0, +0.002, +0.004$, linear flanks meeting at the entry. A cusp in the tail mass is a jump in λ' : this is the kink of Proposition 4.1 seen from the zeros, and with the edge law in place of a jump it is the soft kink of (5). The relay, the soft kink, the sawtooth of Computation 4.4 and the tail law are one mechanism: the zeros rotate with the window, align at the primes, and the alignment is felt through the tail that the edge feeds.

The sampling hypothesis (S_{κ}) has an exact form: the explicit formula turns the defect of the zeros against their density into a prime sum, and the prime sum lives on the entries, where the autocorrelation of the minimizer is singular.

Proposition 7.29 (the sampling defect is the prime sum at the entries). *Assume RH. Let $f \in L^2(-a, a)$ be odd with $P = \langle f, s \rangle$, and for $T > 0$ let $\chi_T(t) = 1 - e^{-(t/T)^{2m}}$ with m even ($m = 2$ unless said otherwise), an entire even high-pass at height T , and*

$$\check{h}_T(x) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T(t) e^{itx} dt,$$

the autocorrelation of f with its structure below the scale $1/T$ removed. Then

$$\sum_{\gamma} |F(\gamma)|^2 \chi_T(\gamma) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T(t) \Psi_{\infty}(t) dt - 2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \check{h}_T(\log n) - 2P^2(1 - e^{-(2T)^{-2m}}), \quad (17)$$

Proof. Weil's explicit formula in the normalization of (1) reads, for an even h analytic in $|\text{Im } t| < \frac{1}{2} + \varepsilon$ and $O(|t|^{-1-\varepsilon})$ there, $\sum_{\gamma} h(\gamma) = \frac{1}{2\pi} \int h \Psi_{\infty} - \sum_n \Lambda(n) n^{-1/2} (\check{h}(\log n) + \check{h}(-\log n)) + h(i/2) +$

$h(-i/2)$, $\check{h}$ the inverse transform [1, 3]; for $h = |F|^2$ it is the identity $Q(f) = \sum_{\gamma} |F(\gamma)|^2$ of §7.1. Apply it to $h(t) = F(t)F(-t)\chi_T(t)$, which is entire and even, equals $|F|^2\chi_T$ on the line, and is $O(|t|^{-1-\varepsilon})$ in every strip, f having compact support and χ_T being bounded there (for fixed m , $\text{Re}(t/T)^{2m} > 0$ in a strip once $|\text{Re } t| \gg m$). For odd f , $F(\pm i/2) = \mp P$, so $h(\pm i/2) = -P^2\chi_T(i/2) = -P^2(1 - e^{-(2T)^{-2m}})$, $i^{2m} = 1$ for even m , and $\check{h}$ is even. $\square$

The first term on the right of (17) is the density integral of (S_{κ}) with the exact density $\Psi_{\infty}/2\pi$, and the last is negligible. So the sampling constant is $M_{\chi}(T) = 1 + (2\sum_n \Lambda(n)n^{-1/2}\check{h}_T(\log n) + O(P^2(2T)^{-2m}))/\sum_{\gamma} |F(\gamma)|^2\chi_T(\gamma)$: the zeros above a height sample the minimizer at their mean density up to the prime sum at the entries. For an entry m outside the window, $\check{h}_T(\log m) = (g * \check{\chi}_T)(\log m)$ is the autocorrelation smeared at scale $1/T$ to a point at distance $\log m - 2a$ from its support, which is negligible once $T(\log m - 2a)$ is large: this is the alignment of Computation 7.28, confined to a neighbourhood of width $1/T$ of each entry. For the entries inside the window the term is set by the echoes.

Corollary 7.30 (the analytic form of the sampling constant). *Under RH and the hypotheses of Proposition 8.8 (one entry n inside the window, $a_2 < a < a_3$), with $c = \Lambda(n)n^{-1/2}$,*

$$M_{\chi}(T) = 1 + 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T (1 + o(1)) = 1 + \frac{2c^2}{\sigma_{\infty}(T)^2} (1 + O(1/\log T)) \quad (T \rightarrow \infty),$$

$$\langle g \rangle_T = \frac{\int_0^{\infty} \chi_T g t^{-2} dt}{\int_0^{\infty} \chi_T t^{-2} dt} :$$

the defect of the sampling is twice the square of the echo's amplitude $\rho = c/\tilde{\sigma}_{\infty}$ of (12), averaged over the tail with the weight χ_T/t^2 of the edge, and tends to zero like $1/\log^2 T$. The average matters at the heights computed: $\langle \tilde{\sigma}_{\infty}^{-2} \rangle_T = \sigma_{\infty}(T)^{-2}(1 - 2/\sigma_{\infty}(T) + \dots)$ is 0.7 times $\sigma_{\infty}(T)^{-2}$ at $T = 10T^*$.

Proof. By Corollary 8.9(iii), $|F|^2 = \frac{4|A_0|^2}{t^2\tilde{\sigma}} [\cos \varphi + \rho \cos(\varphi - t \log n)]^2 (1 + o(1))$ with $\varphi = ta + \theta - \kappa/t$. In $\check{h}_T(\log n) = \frac{1}{2\pi} \int |F|^2 \chi_T \cos(t \log n) dt$ the cross term $2\rho \cos \varphi \cos(\varphi - t \log n) = \rho[\cos(t \log n) + \cos(2\varphi - t \log n)]$ gives, against $\cos(t \log n)$, the non-oscillating $\frac{1}{2}\rho$, the other products oscillating at the frequencies $2a$, $2a \pm \log n$ and $2a \pm 2 \log n$, all nonzero; so $\check{h}_T(\log n) = \frac{|A_0|^2}{\pi} \int_0^{\infty} \frac{\rho \chi_T}{t^2 \tilde{\sigma}} dt (1 + o(1))$ with $\rho/\tilde{\sigma} = c/\tilde{\sigma}_{\infty}^2 (1 + O(\tilde{\sigma}_{\infty}^{-2}))$, while the density integral is $\frac{1}{2\pi} \int \frac{4|A_0|^2 \cos^2 \varphi}{t^2 \tilde{\sigma}} \chi_T \Psi_{\infty} dt (1 + o(1)) = \frac{|A_0|^2}{\pi} \int_0^{\infty} \frac{\chi_T}{t^2} dt (1 + o(1))$, which is $|A_0|^2 I_4 / (\pi T)$ with $I_4 = \int_0^{\infty} (1 - e^{-s^4}) s^{-2} ds = \Gamma(3/4)$ for $m = 2$. The entries outside the window and the polar term are o of this, and (17) gives $M_{\chi} - 1 = 2c\check{h}_T(\log n)/(\text{density integral})(1 + o(1)) = 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T (1 + o(1))$; the expansion of the average is $\int_1^{\infty} du/(u^2(L + \log u)^2) = L^{-2}(1 - 2/L + O(L^{-2}))$ with $L = \log(T/2\pi) - \lambda$, for a cutoff sharp at T . $\square$

Computation 7.31 (the identity on the zeros). `rh_sampling_defect.py`, first 6000 zeros, logs in `data/`. On the minimizers with primes at $a = 0.4, 0.45, 0.5$ (one entry inside the window) and $a = 0.6$ (two entries, Computation 8.17), every term of (17) is evaluated at $T = \kappa T^*$ for $\kappa = 1, \dots, 20$, with the cutoffs $m = 2$ and $m = 4$: the zero sum (with the part above the last zero, 305–455 horizons, taken at the density), the density integral by quadrature on a grid of step 0.01 to 3000 and 0.05 to 1.2×10^5 , the prime sum over all $n < 400$, and the polar term. The identity holds at every height to 3×10^{-5} of the zero sum or better. The prime sum is the entries inside the window: at $a \leq 0.5$, $n = 2$ gives all of it to three digits from $\kappa = e$ on, and 92–97% at $\kappa = 1$, the rest the smearing to $n = 3$ at distance 0.1–0.3 outside. The sampling constant $M_{\chi} = D_{\chi}/Z_{\chi}$ with the cutoff $1 - e^{-(t/T)^8}$ against the averaged form $1 + 2c^2 \langle \tilde{\sigma}_{\infty}^{-2} \rangle_T$ of Corollary 7.30 and against $1 + 2c^2/\sigma_{\infty}(T)^2$:

κ	$a = 0.4$			$a = 0.45$			$a = 0.5$		
	M_χ	$\langle \cdot \rangle_T$	at T	M_χ	$\langle \cdot \rangle_T$	at T	M_χ	$\langle \cdot \rangle_T$	at T
e	1.068	1.083	1.151	1.085	1.076	1.133	1.128	1.070	1.120
6	1.044	1.047	1.072	1.046	1.044	1.066	1.051	1.041	1.062
10	1.034	1.035	1.050	1.033	1.033	1.047	1.032	1.031	1.044
20	1.024	1.024	1.034	1.023	1.023	1.032	1.021	1.022	1.030

From six horizons on the defect is the echo's, averaged over the tail, to a few parts in a thousand, and the value at T overestimates $M_\chi - 1$ by a factor 1.4: the average matters. Below six horizons the cutoff still reaches the heights where the minimizer hides from the zeros, the more so the smaller λ ($\lambda = 2 \times 10^{-4}$ at $a = 0.5$), and M_χ carries that. With the quartic cutoff (`log sampling_defect.log`) the leakage is larger and, between six and twenty horizons, happens to compensate the difference between the average and the value at T , so that M_χ there agrees with $1 + 2c^2/\sigma_\infty(T)^2$ to a few parts in a thousand; the sharper cutoff separates the two. The sharp constant $M(\kappa)$ of Computation 7.14, 1.01, 1.09, 1.02 at $\kappa = e$ for the three supports, sees the same defect at the sharp edge of the tail.

The arithmetic of Conjecture 7.1 is now in one place. By (17) the sampling defect at every support is $2 \sum_{\log n < 2a} \Lambda(n) n^{-1/2} \check{h}_T(\log n)$ plus the alignment at the entries within $1/T$ of the window's edge, and (S_κ) with a bounded M asks that this prime sum be at most a fixed fraction of the tail at $T = \kappa T^*$, uniformly in a : a statement about the autocorrelation of the minimizer at the entries, that is, about the echoes, and no longer about the zeros. For one entry it is Corollary 7.30; for a window containing many entries the echoes form a finite tree and $\sum_{\log n < 2a} \Lambda(n)^2/n \sim 2a^2$ by Mertens, so the defect need not tend to zero, but it is bounded as long as the echoes keep their size c_n/σ_∞ . No gap condition on the zeros enters.

The two hypotheses of Proposition 7.27 can be replaced by one, about the minimizer alone. By (17) with $\Psi = \Psi_\infty - \sum_n 2\Lambda(n) n^{-1/2} \cos(t \log n)$, the zero sum with the cutoff is $\frac{1}{2\pi} \int |F|^2 \chi_T \Psi$ minus the polar term: the prime sum at the entries is the oscillating part of the symbol. So what the zeros see is the tail of $|F|^2$ against the full symbol, and the sharp form of Computation 7.18, the local envelope of $t^2(\Psi - \lambda)|F|^2$, is the whole story.

Proposition 7.32 (the sharp form as the single hypothesis). *Assume RH. Let $\kappa \geq 1$, and for a support a put $T = \kappa T^*$ and $\chi_T^{(m)}(t) = 1 - e^{-(t/T)^{2m}}$ with m the even integer in $[\frac{3}{2}T - 2, \frac{3}{2}T)$. Suppose that for every support $a \geq a_0$*

$$(T_\kappa) \quad \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \chi_T^{(m)}(t) (\Psi(t) - \lambda) dt \geq \frac{-\lambda'(a)}{\pi M'' T} \quad (\text{the sharp tail law holds at height } \kappa T^* \text{ within the factor } M'').$$

Then $\lambda' \geq -\pi \kappa M'' T^ \lambda - \varepsilon(a)$ on $[a_0, \infty)$ with $0 \leq \varepsilon(a) \leq M'' e^{-8T^*(a)}$, and Conjecture 7.1 holds with $c = \pi \kappa M''$ up to this term, which the Grönwall argument of Section 7 absorbs.*

Proof. For this m the cutoff is bounded by $e^{1.01}$ in the strip $|\operatorname{Im} t| \leq \frac{1}{2} + \varepsilon$, $\varepsilon < 0.02$: for $|\operatorname{Re} t| \leq T$, $|(t/T)^{2m}| \leq (1 + (0.52/T)^2)^{3T/2}$, and for $|\operatorname{Re} t| \geq T$, $2m|\arg t| \leq 3(\frac{1}{2} + \varepsilon) < \pi/2$, so $\operatorname{Re}(t/T)^{2m} > 0$. Hence $h = F(t)F(-t)\chi_T^{(m)}$ satisfies the hypotheses of the explicit formula, and (17) holds with the polar term $2P^2(1 - e^{-(2T)^{-2m}})$, which lies in $[0, 2P^2(2T)^{-2m}]$ since m is even. Under RH, $\lambda = \sum_\gamma |F(\gamma)|^2 \geq \sum_\gamma |F(\gamma)|^2 \chi_T^{(m)}(\gamma) = \frac{1}{2\pi} \int |F|^2 \chi_T^{(m)}(\Psi - \lambda) + \lambda \frac{1}{2\pi} \int |F|^2 \chi_T^{(m)} - 2P^2(1 - e^{-(2T)^{-2m}})$, and $\lambda \geq 0$; so by (T_κ) , $-\lambda' \leq \pi M'' T \lambda + 2\pi M'' T P^2(2T)^{-2m}$. The last term is $\varepsilon(a)$: $P^2 \leq \|s\|^2 = \sinh a - a$, and $2m \log(2T) \geq (3\kappa T^* - 4) \log(2\kappa T^*)$ gives $\varepsilon \leq M'' e^{-8T^*}$ for $T^* \geq 2\pi e^{0.6}$, that is $a_0 \geq 0.3$. In the Grönwall argument, $\lambda(a) e^{\frac{\varepsilon}{2} T^*(a)} \geq \lambda(a_0) e^{\frac{\varepsilon}{2} T^*(a_0)} - \int_{a_0}^a \varepsilon e^{\frac{\varepsilon}{2} T^*}$, and the integral is bounded by $M'' \int e^{-(8-c/2)T^*}$, negligible against the first term for $c < 16$. $\square$

Under the hypotheses of Corollary 8.9 or 8.15, (T_κ) holds at a fixed support for κ large, with $M'' \rightarrow 1$: the sharp form is exact in the limit, and by Corollary 8.15(iv) it is, in the finite tree, an identity of the echo amplitudes. What Conjecture 7.1 asks is that the height from which the sharp form holds within a fixed factor be a bounded multiple of the horizon, uniformly in a . The measured onset is a few horizons at every support computed (Computations 7.18, 7.14, 7.6: 0.1% from five horizons on the prime-free critical points, a few per cent with the primes, including $a = 1.0$ and 1.25, where the echo set is infinite), and the bound $c = \pi e M''$ at $\kappa = e$ is again 8.5 against the observed 3.3–3.9. No sampling constant and no gap condition on the zeros remain in the hypothesis: the zeros enter only through the identity (17).

Heuristic 7.33 (prolate lower bound). The number of zeros in $[-T^*, T^*]$ is $2N(T^*) \approx (T^*/\pi)(2a-1)$, short of the Shannon number $2aT^*/\pi$ of $[-a, a] \times [-T^*, T^*]$ by T^*/π . By the Landau–Pollak–Slepian theory [17, 18], with the asymptotics of the prolate spectrum in the plunge region [19, 20] (about $(2/\pi^2) \log c$ eigenvalues between ε and $1 - \varepsilon$ for $c = aT^*$), a function in PW_a constrained to vanish on all zeros below the horizon can still keep all but $\eta \gtrsim \exp(-\pi T^*/(\log(aT^*) + O(1)))$ of its energy below the horizon; under RH this bounds λ from above, i.e. $\Phi \gtrsim \pi T^*/\log(aT^*)$, which is $0.7 T^*$ at $a = 1$. The observed value is $\Phi(1) = 59.5 = 1.28 T^*$ (Computation 6.1); the zeros are about twice as expensive a constraint set as the count alone predicts, and the difference is arithmetic. Conjecture 7.1 is the matching upper bound.

8 The boundary law

The derivative of the floor should be a boundary quantity. Since Q does not depend on a , $\lambda(a)$ is the infimum of a fixed form over the growing family $L^2(-a, a)$, and its derivative is a domain variation. For the Dirichlet fractional Laplacian, Hadamard’s formula expresses such a derivative through the boundary trace u/dist^s of the eigenfunction [11]; for the logarithmic Laplacian the boundary trace is the amplitude C of Computation 3.4. A Pohozaev identity for L_Δ with the local boundary term $\int_{\partial\Omega} u^2 \ln(\delta^{-2})(x \cdot \nu)$ was claimed in [10], which its authors withdrew on 26 April 2025 “due to a crucial error in Proof of Pohozaev identity”; to our knowledge no correct version exists. We therefore state the law as a conjecture and give the evidence.

Conjecture 8.1 (boundary law). *Assume (H_∞) . For a not an entry, with $C_\pm(a) = \lim_{\delta \rightarrow 0} \sqrt{\log(1/\delta)} |f(\pm(a - \delta))|$ the edge amplitudes of the minimizer (assumed to exist),*

$$\lambda'(a) = -(C_+(a)^2 + C_-(a)^2) = -2C(a)^2.$$

Equivalently, with Proposition 5.3, the Weil form on a window satisfies the Pohozaev-type identity $\mathcal{D}(f) = 2a C(a)^2$ on its minimizer. Proposition 8.3 below proves it under the edge law with two derivatives, and Propositions 8.4 and 8.8 supply that law from (H_∞) for the archimedean form and for $a_2 < a < a_3$; Computation 8.11 is the evidence at the other supports.

The law is a theorem under the edge law with derivatives, the hypothesis of Proposition 7.8: the boundary term of the Pohozaev identity is the overlap, at the scale $1/T$, of the leak of the dilation generator through the low-pass with the singular force outside the edge, and the logarithms cancel there as they did in Lemma 7.11. We first sharpen Lemma 7.12.

Lemma 8.2 (the edge force to second order). *Under the hypothesis of Proposition 7.8, the remainder $\rho(\delta) = g_{\text{out}}(a + \delta) + CL(\delta)^{1/2} - g_{\text{reg}}(a)$ of Lemma 7.12 satisfies $\rho(\delta) = o(L(\delta)^{-1/2})$ and $\rho'(\delta) = o(\delta^{-1} L(\delta)^{-3/2})$ as $\delta \rightarrow 0^+$.*

Proof. By the formula in the proof of Lemma 7.12, $g_{\text{out}}(a + \delta) = -\int_0^{2a} f(a - u)k(u + \delta) du + (K_P f)(a + \delta)$, and the prime part is smooth near the edge (it is made of copies of the interior of f). Write $k(x) = \frac{1}{2x} + k_0(x)$ with k_0 smooth and bounded; the part with k_0 is C^1 in δ . In the singular part put $f(a - u) = (\varphi + \psi)(u)$ for $u < \delta_0$ and substitute $u = \delta w$:

$$-\frac{1}{2} \int_0^{\delta_0} \frac{\varphi(u)}{u + \delta} du = -\frac{C}{2} \int_0^{\delta_0/\delta} \frac{(L(\delta) - \log w)^{-1/2}}{1 + w} dw,$$

and $\int_0^W \frac{(L - \log w)^{-1/2}}{1 + w} dw = \int_1^W \frac{(L - \log w)^{-1/2}}{w} dw + \int_0^\infty \left[\frac{1}{1+w} - \frac{\mathbf{1}_{w>1}}{w} \right] (L - \log w)^{-1/2} dw + O(L^{-1/2}/W)$. The first integral is $2L^{1/2} - 2L(\delta_0)^{1/2}$; in the second the bracket integrates to zero, so the expansion $(L - \log w)^{-1/2} = L^{-1/2} \left(1 + \frac{\log w}{2L} + O(L^{-2} \log^2 w) \right)$ leaves $O(L^{-3/2})$. Hence the φ -part is $-CL(\delta)^{1/2} + \text{const} + O(L^{-3/2})$, and the ψ -part, $-\frac{1}{2} \int \psi(u) du / (u + \delta)$, is $o\left(\int_\delta^{\delta_0} L(u)^{-3/2} du / u\right) + o(L^{-3/2}) = o(L(\delta)^{-1/2})$. Differentiating under the integral, $\frac{d}{d\delta} \left[-\frac{1}{2} \int \frac{(\varphi + \psi)(u)}{u + \delta} du \right] = \frac{1}{2} \int \frac{(\varphi + \psi)(u)}{(u + \delta)^2} du$, and the same substitution gives $\frac{C}{2} \delta^{-1} L^{-1/2} \int_0^\infty \frac{(1 - \log w/L)^{-1/2}}{(1 + w)^2} dw + o(\delta^{-1} L^{-3/2})$, where $\int_0^\infty \frac{\log w}{(1 + w)^2} dw = 0$ by $w \mapsto 1/w$; so the derivative is $\frac{C}{2} \delta^{-1} L^{-1/2} (1 + O(L^{-2})) + o(\delta^{-1} L^{-3/2})$, which is $\frac{d}{d\delta} [-CL^{1/2}] + o(\delta^{-1} L^{-3/2})$. $\square$

Proposition 8.3 (edge law implies boundary law). *Let f be the minimizer at a support a that is not an entry, and suppose it satisfies the hypothesis of Proposition 7.8, the edge law with two derivatives. Then, with B_T as in Proposition 7.15,*

$$\lim_{T \rightarrow \infty} B_T = aC^2, \quad \text{hence} \quad \mathcal{D}(f) = 2aC^2 \quad \text{and} \quad \lambda'(a) = -2C^2$$

at every point of differentiability of λ (Proposition 5.3): Conjecture 8.1 holds. More precisely, in the model $\psi \equiv 0$,

$$B_T = aC^2 J_1(L_T) - aC g_{\text{reg}}(a) L_T^{-1/2} (1 + O(1/L_T)) + O(1/T),$$

$$J_1(L) = -\frac{2}{\pi} \int_0^\infty \int_0^\infty \left[\frac{L + \log(1/v)}{L + \log(1/s)} \right]^{1/2} k'(s + v) ds dv = 1 + O(L^{-2}),$$

with $k(w) = \sin w/w$, $L_T = L(1/T)$ and $g_{\text{reg}}(a) = 2P \sinh(a/2)$.

Proof. One edge. g_{out} and $(xf)'$ are odd, so $B_T = 2B_T^+$ with $B_T^+ = \int_{\delta>0} ((xf)')_T(a + \delta) g_{\text{out}}(a + \delta) d\delta$.

The leak. Near the right edge write $f(a - \delta) = u(\delta) = (\varphi + \psi)(\delta)\chi(\delta)$ as in the proof of Proposition 7.8, so that $(xf)'(a - \delta) = -au'(\delta) + b(\delta)$ with $b = u + \delta u' = O(L^{-1/2})$ bounded; the rest of $(xf)'$ is C^2 with compact support in $(-a, a)$ except at the left edge, where it has the mirror singularity. For $\delta > 0$, $((xf)')_T(a + \delta) = \int (xf)'(y) D_T(a + \delta - y) dy$ with $D_T(s) = \sin(Ts)/(\pi s)$. The part of $(xf)'$ supported at distance at least $c > 0$ from the right edge has transform $O(t^{-1} \log^{-1/2} t)$ (Step 1 of that proof), and its leak at $a + \delta$ is the tail $\int_{|t|>T}$ of the inversion integral against $e^{-it(a+\delta-y)}$, $|a + \delta - y| \geq c$, which is $O(1/(cT \log^{1/2} T))$ uniformly in δ after one integration by parts; paired with $g_{\text{out}} \in L^1$ it tends to zero. The part b leaks like f itself, $O(L_T^{-1/2} \min(1, 1/(T\delta)))$, and its pairing with $-CL^{1/2}$ is $O(1/T)$ by Lemma 7.11, with $g_{\text{reg}} + \rho$ it is $O(L_T^{-1/2} \log T/T)$. The main part is $-au'$, whose leak is, by one integration by parts ($u(0) = 0$, u cut off smoothly at δ_0),

$$-a \int_0^{\delta_0} u'(\delta') D_T(\delta + \delta') d\delta' = a \ell'_T(\delta), \quad \ell_T(\delta) = \int_0^{\delta_0} u(\delta') D_T(\delta + \delta') d\delta',$$

the derivative of the leak of f of Lemma 7.11.

The pairing. Write $g_{\text{out}}(a + \delta) = -CL(\delta)^{1/2} + g_{\text{reg}}(a) + \rho(\delta)$ by Lemmas 7.12 and 8.2. (i) With the singular force, $a \int \ell'_T(\delta) (-CL(\delta)^{1/2}) d\delta = -aC \iint u(\delta') L(\delta)^{1/2} \partial_\delta D_T(\delta + \delta') d\delta' d\delta$. For $u = \varphi\chi$ the substitution $s = T\delta'$, $v = T\delta$ gives $L(\delta') = L_T + \log(1/s)$, $L(\delta) = L_T + \log(1/v)$, $\partial_\delta D_T(\delta + \delta') = T^2 k'(s+v)/\pi$, $d\delta' d\delta = ds dv/T^2$, and the double integral becomes $\frac{aC^2}{2} J_1(L_T)$, taken in the Abel sense and cut where the model ends, as in Lemma 7.11. Here $J_1(\infty) = -\frac{2}{\pi} \int_0^\infty \int_0^\infty k'(s+v) ds dv = \frac{2}{\pi} \int_0^\infty k = 1$, and the term of order $1/L$ vanishes for the reason given there: $k'(s+v)$ is symmetric in (s, v) and the logarithm of the ratio antisymmetric. Numerically $J_1(L) = 1.029, 1.016, 1.012, 1.008, 1.006, 1.004$ for $L = 3, \dots, 8$, about $1 + 0.27/L^2$ (`rh_sum_rule.py j1`). The ψ -part of u contributes $o(1)$: with $|\psi'(\delta')| \leq \varepsilon(\delta')\delta'^{-1}L^{-3/2}$, $\varepsilon \rightarrow 0$, the same substitution bounds it by $o(1)$ times $\int_0^\infty (L + \log(1/s))^{-3/2} ds/s = 2L^{-1/2}$, which balances the $L^{1/2}$ of the force. (ii) With the regular value, $a g_{\text{reg}} \int_0^\infty \ell'_T(\delta) d\delta = -a g_{\text{reg}} \ell_T(0^+) = -\frac{aC}{2} g_{\text{reg}} L_T^{-1/2} (1 + O(1/L_T))$, since $\ell_T(0^+) = \frac{C}{\pi} \int_0^\infty (L_T + \log(1/s))^{-1/2} \frac{\sin s}{s} ds + o(L_T^{-1/2})$. (iii) With the remainder, $a \int \ell'_T \rho = -a \int \ell_T \rho'$ (as $\rho(0^+) = 0$ and $\ell_T(\infty) = 0$), and $|\ell_T(\delta)| \leq CL_T^{-1/2} \min(1, 1/(T\delta))$ with $\rho' = o(\delta^{-1}L^{-3/2})$ gives $o(L_T^{-1})$.

Collecting, $B_T^+ = \frac{aC^2}{2} J_1(L_T) - \frac{aC}{2} g_{\text{reg}} L_T^{-1/2} (1 + O(1/L_T)) + o(1)$, so $B_T \rightarrow aC^2$; the rest is Proposition 7.15, which gives $\mathcal{D} = 2B_\infty$, and Proposition 5.3, which gives $-a\lambda' = \mathcal{D}$. The regular value is $g_{\text{reg}}(a) = 2P \sinh(a/2)$ by Lemma 7.12. $\square$

The rate is visible in the data of Computation 7.17: on the prime-free critical point at $a = 0.5$ ($C = 1.0615$, $\beta = -1.469$ fitted on the edge nodes, against $C = 1.0626$ and $\beta = -1.400$ of Proposition 8.4 below, the fitted β absorbing the $O(L^{-2})$ correction over the fit's window; $g_{\text{reg}} = 0.0679$), $aC^2 - B_T = 0.040, 0.014, 0.0135, 0.0125$ at 3, 5, 10, 20 horizons, where the leading term $aC g_{\text{reg}} L_T^{-1/2}$ is 0.027 to 0.019; at these heights L_T is only 1.8 to 3.7 and the expansion is not yet quantitative, but the sign and the order are those of the law. In Fourier terms the same term is the polar cross term of the edge, $-\frac{1}{\pi} \int_T^\infty 4aP \sinh(a/2) |A_0| \sin \theta(t) dt / (t\tilde{\sigma}^{1/2})$ with $\sin \theta \approx -\pi/(4 \log t)$, and $C = |A_0|$.

For the prime-free form the edge law with its two derivatives, and with its constants, follows from the representation.

Proposition 8.4 (the edge law from the representation). *Assume (H_∞) , and let f be the minimizer of $W_\infty - 2P^2$ at a support a with $\lambda = \lambda_\infty(a) > \Psi_\infty(0)$, A_0 its edge amplitude (Corollary 7.24). Then f satisfies the hypothesis of Proposition 7.8 with*

$$C = |A_0|, \quad \beta = -\gamma - \log(2\pi a) - \lambda,$$

γ Euler's constant, and with $\psi^{(k)}(\delta) = O(\delta^{-k} L(\delta)^{-5/2})$ for $k \leq 2$. Equivalently, with $\sigma_\infty = \Psi_\infty - \lambda$,

$$f(a - \delta) = |A_0| (\sigma_\infty(1/\delta) - \gamma)^{-1/2} (1 + O(L^{-2})) :$$

the profile of the edge is the inverse square root of the symbol at frequency $1/\delta$, shifted by Euler's constant.

Proof. The model. For $s > 0$ and χ smooth, 1 on $[0, \delta_0/2]$ and 0 beyond $\delta_0 < 1$, put $\Phi_s(t) = \int_0^{\delta_0} (\log(1/\delta))^{-s} \chi(\delta) e^{-it\delta} d\delta$. From the Mellin representation $(\log(1/\delta))^{-s} = \Gamma(s)^{-1} \int_0^\infty x^{s-1} \delta^x dx$ and $\int_0^\infty \delta^x e^{-it\delta} d\delta = \Gamma(1+x)(it)^{-1-x} = \frac{-i}{t} \Gamma(1+x) e^{-x\Lambda}$ with $\Lambda = \log t + i\pi/2$ ($t > 0$), the Mellin-transform method for Fourier integrals with a logarithmic endpoint singularity [13, Ch. 4] gives, as $t \rightarrow +\infty$,

$$\Phi_s(t) = \frac{-i}{t} \Lambda^{-s} \sum_{k \leq N} \frac{\Gamma(s+k)}{\Gamma(s) k!} \Gamma^{(k)}(1) \Lambda^{-k} + O(t^{-1} |\Lambda|^{-s-N}),$$

an expansion that may be differentiated in t term by term. For $s = \frac{1}{2}$, with $\Gamma'(1) = -\gamma$ and $\Gamma''(1) = \gamma^2 + \pi^2/6$,

$$\Phi_{1/2}(t) = \frac{-i}{t} \Lambda^{-1/2} \left[1 - \frac{\gamma}{2\Lambda} + \frac{3(\gamma^2 + \pi^2/6)}{8\Lambda^2} + O(\Lambda^{-3}) \right] = \frac{-i}{t} (\Lambda + \gamma)^{-1/2} \left[1 + \frac{\pi^2}{16\Lambda^2} + O(\Lambda^{-3}) \right],$$

and with $M = \log t + \gamma$ one has $(\Lambda + \gamma)^{-1/2} = (M + i\pi/2)^{-1/2} = M^{-1/2} e^{-i\pi/(4M)} (1 - \frac{\pi^2}{16M^2} + O(M^{-3}))$, so that

$$\Phi_{1/2}(t) = \frac{-i}{t} M^{-1/2} e^{-i\pi/(4M)} (1 + O(M^{-3})) :$$

the real corrections of order M^{-2} cancel. Scaling δ by $e^{\beta'}$, the transform of $C(\log(e^{\beta'}/\delta))^{-1/2} \chi(\delta)$ is $\frac{-iC}{t} M_{\beta'}^{-1/2} e^{-i\pi/(4M_{\beta'})} (1 + O(M^{-3}))$ with $M_{\beta'} = \log t + \beta' + \gamma$.

The factor. By Stirling's formula and the rational factor, $\tilde{\sigma}(t) = \ell(t)(1 + O(t^{-2}))$ with $\ell(t) = \log(t/2\pi) - \lambda$; and substituting $\tau = tx$ in (10), $\theta(t) = \frac{1}{\pi} \int_0^\infty \frac{u(tx) - u(t)}{1-x^2} dx$ with $u = \log \tilde{\sigma}$. For $tx \geq \tau_0$, $u(tx) - u(t) = \log(1 + \log x/\ell(t)) + O(t^{-2})$; expanding in powers of $\log x/\ell$, with $\int_0^\infty \frac{(\log x)^k}{1-x^2} dx = 0$ for even k (by $x \mapsto 1/x$) and $= -\pi^2/4, -\pi^4/8$ for $k = 1, 3$, while the range $x < \tau_0/t$ and the tails $|\log x| > \ell$ contribute $O(\log \log t/t)$,

$$\theta(t) = -\frac{\pi}{4\ell} - \frac{\pi^3}{24\ell^3} + O(\ell^{-5}) + O\left(\frac{\log \log t}{t}\right),$$

and the same representation differentiated under the integral gives $\theta^{(j)}$, $j \leq 3$, with the expansion differentiated term by term. Hence, differentially,

$$\frac{A_0}{t\sigma_-(t)} = \frac{-i|A_0|}{t} \ell^{-1/2} e^{-i\pi/(4\ell)} \left(1 + O(\ell^{-3}) + O\left(\frac{\log \log t}{t}\right) \right),$$

where $A_0 = -i|A_0|$ for a minimizer positive near the right edge.

Matching. Choose β' with $\beta' + \gamma = -\log 2\pi - \lambda$, so that $M_{\beta'} = \ell$: the transform $\hat{\varphi}$ of the model $\varphi(\delta) = |A_0|(\log(a/\delta) + \beta)^{-1/2} \chi(\delta)$, $\beta = \beta' - \log a = -\gamma - \log(2\pi a) - \lambda$, agrees with $A_0/(t\sigma_-)$ to relative order ℓ^{-3} . By Corollary 7.24,

$$D(t) := \tilde{F}(t) - \hat{\varphi}(t) = \frac{A(t) - A_0}{t\sigma_-(t)} + \left[\frac{A_0}{t\sigma_-(t)} - \hat{\varphi}(t) \right] + \frac{e^{-2ita}\Phi_-(t)}{\tilde{\sigma}(t)}.$$

(i) $A(t) - A_0 = O(\log t/t)$ with $(A - A_0)^{(j)} = O(\log t/t^{1+j})$, $j \leq 3$: its pieces in the proof of Corollary 7.24 are a rational function, t times the Riesz projection of a smooth function with derivatives $O(t^{-2-j})$, and t times the transform of a C^∞ function on $u > 0$ with all derivatives integrable, and in each case the derivative of $tP_\pm[\cdot]$ minus its limit loses one power of t (for the projection, by the Hilbert-transform estimate of that proof applied to the derivatives); so the first piece is $O(t^{-2} \log t)$ with derivatives $O(t^{-2-j} \log t)$. (ii) The bracket is $O(t^{-1} \ell^{-7/2})$ with derivatives $O(t^{-1-j} \ell^{-7/2})$, $j \leq 3$. (iii) The third piece is the transform of $\text{far} * (1/\tilde{\sigma})$ with far supported in $u \leq -2a$, and the kernel of $1/\tilde{\sigma}$ is C^∞ and exponentially decaying away from the origin ($\tilde{\sigma}$ is zero-free in a strip around the line, as in Lemma 7.22), so its inverse transform is C^∞ on $(-2a + \varepsilon, 0]$.

The remainder. For $D \in C^3$ with $D^{(j)}(t) = O(t^{-1-j} \Lambda(t))$, $j \leq 3$, Λ slowly varying, the inverse transform ψ satisfies $\psi^{(k)}(\delta) = O(\delta^{-k} \Lambda(1/\delta))$ for $k = 1, 2$: split $\int (it)^k D e^{it\delta} dt$ at $|t| = 1/\delta$, bound the inner part by $\int_{1/\delta}^{1/\delta} t^{k-1} \Lambda$, and integrate the outer part by parts three times, the boundary terms being $(t^k D)^{(m)}(1/\delta) \delta^{-m-1}$, $m \leq 2$, and the integral $\delta^{-3} \int_{1/\delta}^\infty |(t^k D)^{(m)}|$. Applied to (ii) with

$\Lambda = \ell^{-7/2}$, to (i) with $\Lambda = t^{-1} \log t$ (where the same splitting gives $O(\delta^{1-k} \log^2(1/\delta))$), and with (iii) smooth, the inverse transform $\psi = f(a - \delta) - \varphi(\delta)$ of D satisfies $\psi^{(k)} = O(\delta^{-k} L^{-7/2}) + O(\delta^{1-k} \log^2(1/\delta))$ for $k = 1, 2$. Since $f(a^-) = 0$ by Theorem 3.2 and $\varphi(0^+) = 0$, $\psi(\delta) = \int_0^\delta \psi' = O(L(\delta)^{-5/2})$. Finally f is C^∞ inside the window, as $(K_\infty - \lambda)f = 2Ps$ there and the symbol is elliptic at infinity. This is the hypothesis of Proposition 7.8 with the stated C , β and remainder; the second form of the statement is $\log(a/\delta) + \beta = \sigma_\infty(1/\delta) - \gamma + O(\delta^2)$. $\square$

Corollary 8.5 (the boundary law and the tail law for the prime-free form). *Assume (H_∞) . For the minimizer of $W_\infty - 2P^2$ at a support a with $\lambda_\infty(a) > \Psi_\infty(0)$, at every point of differentiability of λ_∞ ,*

$$\lambda'_\infty(a) = -2|A_0|^2, \quad \lim_{T \rightarrow \infty} T \cdot \frac{1}{2\pi} \int_{|t| > T} |F(t)|^2 (\Psi_\infty(t) - \lambda) dt = \frac{2|A_0|^2}{\pi} = -\frac{\lambda'_\infty(a)}{\pi} :$$

the analytic tail law of Computation 7.14 is a theorem, and Conjecture 7.19(i) holds with $|A|^2 = -\lambda'/2$.

Proof. Propositions 5.3, 7.15 and 8.3 hold for the prime-free form with the same proofs (K_∞ , Ψ_∞ and $W_\infty - 2P^2$ in place of K , Ψ and Q ; the entries play no role). Proposition 8.4 supplies the hypothesis of Proposition 8.3 with $C = |A_0|$, which gives the first identity, and the tail energy is $2|A_0|^2/(\pi T)(1 + o(1))$ by Corollary 7.24, as computed after it. $\square$

The boundedness hypothesis entered the proposition at one point, $f(a^-) = 0$ from Theorem 3.2; the representation makes it a conclusion.

Proposition 8.6 (the prime-free minimizer is bounded). *Let $f \in L^2(-a, a)$ be an odd critical point of $W_\infty - 2P^2$ with $\lambda > \Psi_\infty(0)$ (in particular the minimizer, Lemma 1.4). Then $f \in C^\infty(-a, a)$, f is continuous on $\mathbb{R}$ with $f(\pm a) = 0$, and f satisfies the edge law of Proposition 8.4 with its constants: (H_∞) holds for the prime-free form.*

Proof. Propositions 7.23 and Corollary 7.24 assume only $f \in L^2$, and in the proof of Proposition 8.4 the hypothesis (H_∞) is used once, through $f(a^-) = 0$, to fix $\psi(0^+) = 0$. Without it: ψ' is integrable near 0 ($\psi' = O(\delta^{-1} L^{-7/2})$), so $c = \psi(0^+)$ exists and $f(a - \delta) \rightarrow c$ as $\delta \rightarrow 0^+$. If $c \neq 0$, f has a jump of size c at the edge, and $\tilde{F}(t) = \int_{-2a}^0 f(a+v)e^{itv} dv = \frac{c}{it} + o(1/t)$ as $t \rightarrow \infty$: indeed $f(a+v) - c \rightarrow 0$ as $v \rightarrow 0^-$ with integrable derivative near 0, so the transform of $(f(a+v) - c)\chi(v)$ is $(it)^{-1} \int(\dots)'e^{itv} = o(1/t)$ by the Riemann–Lebesgue lemma, the transform of $c\chi(v)$ is $c/(it) + O(t^{-2})$, and the part away from the edge is treated at the far edge in the same way, where the mirror jump $-c$ gives $-ce^{-2ita}/(it) + o(1/t)$. But Corollary 7.24 gives $\tilde{F} = A(t)/(t\sigma_-) + e^{-2ita}\Phi_-/\tilde{\sigma}$ with A bounded, $|\sigma_-| \asymp (\log t)^{1/2}$, and $\Phi_- = O((\log t)^{1/2}/t)$ (the far force $-CL^{1/2}$ has a transform of that order), so $\tilde{F} = O(1/(t\sqrt{\log t})) = o(1/t)$, and $c/(it) - ce^{-2ita}/(it) = o(1/t)$ forces $c = 0$. So $f(a^-) = 0$, $f = \varphi + \psi$ is bounded near the edges, and the edge law holds with $\psi(0^+) = 0$. Inside the window $(K_\infty - \lambda)f = 2Ps$, and for a cutoff $\chi \in C_c^\infty(-a, a)$, $(K_\infty - \lambda)(\chi f) = 2P\chi s + [K_\infty, \chi]f$, where the commutator has the symbol $\chi'(x)\partial_t \Psi_\infty + \dots$ of order -1 ($\partial_t \Psi_\infty = O(1/t)$), so $[K_\infty, \chi]f \in H^1$ for $f \in L^2$; since $\widehat{\chi f}$ is entire and σ_∞ has simple real zeros, $\widehat{\chi f} = (2P\chi s + [K_\infty, \chi]f)/\sigma_\infty$ gives $\chi f \in H^1$ with a logarithm to spare, and iterating with cutoffs $\chi_k \prec \chi_{k-1}$, the commutator term gaining one derivative at each step from the regularity already obtained (the part of f outside the support of χ_{k-1} contributing a smooth function, the kernel being smooth off the diagonal), $f \in C^\infty(-a, a)$. Continuity on $\mathbb{R}$ follows, with $f(\pm a) = 0$. $\square$

With entries the same argument runs through Proposition 8.18, which assumes only $f \in L^2$, and through Theorem 9.12, whose one use of (H_∞) is again $f(a^-) = 0$ in Step 3: the representation gives $\tilde{F} = A_a m_- / (t \sigma_{\infty, -}) + e^{-2ita} \Phi_- / \tilde{\sigma}_a = O(1/(t\sqrt{\log t}))$, so the edge value $e(0^+) + \psi(0^+)$ vanishes, and the minimizer is bounded, by the edge profile, the echo part and the smooth interior. At every support this rests on the steps of that theorem listed as sketch density; below a_∞ , and for the prime-free form, it is complete. Where (H_∞) is assumed in Sections 3–9, it is therefore a hypothesis only above a_∞ , and there only until those steps are written.

Computation 8.7 (the constants of the edge). `rh_edge_constants.py`, log in `data/`. On the prime-free edge-FEM critical points at $a = 0.3, 0.4, 0.5, 0.6$ ($\lambda = 0.2226, -0.0780, -0.3222, -0.5359$; $\lambda' = -3.429, -2.666, -2.258, -2.039$ from the dilation engine) the ratio of the profile $f(a - \delta)$ to the prediction $(-\lambda'/2)^{1/2} (\log(a/\delta) + \beta)^{-1/2}$ of Proposition 8.4 and Corollary 8.5, with $\beta = -\gamma - \log(2\pi a) - \lambda = -1.434, -1.421, -1.400, -1.368$ and nothing fitted, is 1.001–1.003 at every node between $\delta = 10^{-10}$ and 10^{-5} at all four supports; it rises to 1.01 at $\delta = 10^{-3}$, where L is only 6, and to 1.004 at the last nodes near 10^{-12} . On the mesh refined below 10^{-6} (Computation 7.20) the ratio at $a = 0.5$ is 1.0003, 1.0005, 1.0009 at $\delta = 10^{-10}, 10^{-8}, 10^{-6}$, so the residual is the discretization. The amplitude C of Computation 3.4, fitted there, is $(-\lambda'/2)^{1/2}$, and β is the symbol's: $\log(a/\delta) + \beta = \sigma_\infty(1/\delta) - \gamma$.

The entries inside the window are a finite perturbation of this structure, and for one entry everything goes through with the same constants.

Proposition 8.8 (the minimizer with an entry inside the window). *Assume (H_∞) . Let f be the minimizer at a support a with exactly one entry n inside the window and $a < d < 2a$, $d = \log n$ (that is, $a_2 < a < a_3$, $n = 2$), $c = \Lambda(n)n^{-1/2}$, $\lambda > \Psi_\infty(0)$. Let $\tilde{\sigma}_\infty$ be $\Psi_\infty - \lambda$ with its real zero divided out and k the kernel of $1/\tilde{\sigma}_\infty$, and put $L = (-a, a - d)$, $M = (a - d, d - a)$, $R = (d - a, a)$, $f_X = f \mathbf{1}_X$. Then*

$$f(a+u) = c \left(k * f_R(a + \cdot + d) \right)(u) + r(u), \quad r \in C^1 \text{ near } u = -d, \quad r''(u) = O(|u+d|^{-1} (\log 1/|u+d|)^{-5/2}), \quad (18)$$

the echo of the right edge at $a - d$; and, with $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - c^2/\tilde{\sigma}_\infty$, $\tilde{\sigma}_{\text{eff}}$ its positive part after dividing out the finitely many real zeros and $\sigma_{\text{eff}, \pm}$ its factors (Lemma 7.21),

$$\tilde{F} = \frac{P_- [\hat{H}'_- / \sigma_{\text{eff}, +}]}{\sigma_{\text{eff}, -}} \quad \text{on } \mathbb{R}, \quad (19)$$

where $\hat{H}'_-$ is in $H^1(\mathbb{R})$, analytic in the lower half-plane, and is the transform of a source made of the polar step and a smooth part near the edge, and of sources supported at distance at least $\delta_0 = \min(d, 2a - d)$ to the left of it.

Proof. On the window $Kf = \lambda f + 2Ps$ and $K = K_\infty + K_P$ with $K_P f = -\sum_m c_m (f(\cdot - \log m) + f(\cdot + \log m))$; the shifts by $\log m > 2a$ move the window off itself, and for $m = n$, $f(y - d) = f_L(y - d)$ on R , $f(y + d) = f_R(y + d)$ on L , both vanish on M . Hence on the whole line

$$K_\infty f = \lambda f + 2Ps \mathbf{1}_{[-a, a]} + c f_L(\cdot - d) + c f_R(\cdot + d) + g_\infty \mathbf{1}_{|y| > a}, \quad g_\infty = K_\infty f \text{ outside,}$$

and after transforming and multiplying by e^{-ita} , $\sigma_\infty \tilde{F} = \tilde{H} + \tilde{G}_+$ with $\tilde{G}_+$ the archimedean force to the right of a and $\tilde{H}$ the transform of the “ $-$ ” sources: the polar term, the archimedean force to the left of $-a$, and the two copies, supported in $u \in (d - 2a, 0)$ and $u \in (-2a, -d)$ respectively

($u = y - a$). All sources are in $L^1 \cap L^2$ with $uh \in L^2$, so Proposition 7.23 applies verbatim: $\tilde{F} = P_-[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-}$ with $\hat{H}_-$ the sources corrected at the real zero.

The echo. The copy on L has transform $ce^{-itd}\tilde{F}_R$, $\tilde{F}_R = e^{-ita}\widehat{f_R}$, and its contribution to $\tilde{F}$ is $P_-[ce^{-itd}\tilde{F}_R/\sigma_{\infty,+}]/\sigma_{\infty,-} = ce^{-itd}\tilde{F}_R/\tilde{\sigma}_\infty - P_+[ce^{-itd}\tilde{F}_R/\sigma_{\infty,+}]/\sigma_{\infty,-}$. The first term is the transform of $ck * f_R(a + \cdot + d)$; the second is the transform of a function smooth on $u < 0$: by Lemma 7.22 the kernel of $1/\sigma_{\infty,+}$ is smooth on $[d, \infty)$, so $P_+[\cdot]$ is the transform of a smooth function on $u > 0$, and $1/\sigma_{\infty,-}$ then spreads it to $u < 0$ through its kernel evaluated away from the origin. The remaining terms of $\tilde{F}$ are the polar step, the far force (smooth near $-d$), the copy on R (supported in $u \in (d - 2a, 0)$, whose contribution is smooth away from its support's endpoints), and the rational corrections of Proposition 7.23, which are weighted primitives of the sources, $-ie^{\pm it\lambda u} \int_{-\infty}^u h(u')e^{\mp it\lambda u'} du'$: one derivative smoother than the sources, with the same singular support and a step at the edge. All are smooth near $u = -d$ except the correction of the copy on L , which is C^1 there with the second derivative of the primitive of the edge profile, $O(|u + d|^{-1}L^{-3/2})$, and $O(|u + d|^{-1}L^{-5/2})$ after the convolution with k , which gains a logarithm; this is (18), with k smooth away from the origin and $f_R(a + \cdot + d)$ supported in $u \leq -d$.

The return. The copy on R near the edge is $cf_L(a + u - d) = cf(a + (u - d))$ for $u \in (d - 2a, 0)$, and by (18) at $u - d$, $f(a + (u - d)) = c(k * f_R(a + \cdot))(u) + r(u - d)$; so the copy on R equals $c^2(k * f_R(a + \cdot))(u)\mathbf{1}_{u < 0}$ plus $cr(u - d)\mathbf{1}_{u < 0}$, a function C^1 up to the edge with a step there (value $cr(-d)$) and a transform $O(t^{-2}(\log t)^{-3/2})$ beyond the step, plus, at its left end $u = d - 2a$, the copy of the left edge. Its transform is therefore $c^2P_-[\tilde{F}_R/\tilde{\sigma}_\infty]$ plus a step plus a far-type source. Since $\tilde{F}_R = \tilde{F} - \tilde{F}_L - \tilde{F}_M$ and f_L, f_M are supported at distance at least $2a - d$ from the edge, $c^2P_-[\tilde{F}_R/\tilde{\sigma}_\infty] = c^2P_-[\tilde{F}/\tilde{\sigma}_\infty] - c^2P_-[(\tilde{F}_L + \tilde{F}_M)/\tilde{\sigma}_\infty]$, the second term the transform of a source smooth near the edge. Write $c^2P_-[\tilde{F}/\tilde{\sigma}_\infty] = c^2\tilde{F}/\tilde{\sigma}_\infty - c^2P_+[\tilde{F}/\tilde{\sigma}_\infty]$ and move the first term to the left of $\tilde{\sigma}_\infty\tilde{F} = \hat{H}_- + \hat{G}_+$: this is $\sigma_{\text{eff}}\tilde{F} = \hat{H}'_- + \hat{G}'_+$ with $\hat{H}'_- = \hat{H}_- - c^2P_-[\tilde{F}/\tilde{\sigma}_\infty]$ and $\hat{G}'_+ = \hat{G}_+ + c^2P_+[\tilde{F}/\tilde{\sigma}_\infty]$, a “−” and a “+” function with the properties required in the proof of Proposition 7.23 ($1/\tilde{\sigma}_\infty$ has an integrable kernel, so $f * k \in L^1$). The symbol σ_{eff} is real, even, real-analytic, $\sim \log t$, with finitely many real zeros; dividing them out and applying the Liouville argument of that proof gives (19), and $\hat{H}'_-$ has the stated structure. $\square$

Corollary 8.9 (the edge, the echo and the tail law with an entry inside the window). *Under the hypotheses of Proposition 8.8: (i) the edge of the transform has the form of Corollary 7.24 with σ_{eff} in place of σ , $A(t) \rightarrow A_0$ purely imaginary; (ii) f satisfies the hypothesis of Proposition 7.8 at the edges, with the same constants as in Proposition 8.4, $C = |A_0|$ and $\beta = -\gamma - \log(2\pi a) - \lambda$, and is smooth on the window except at the edges and at the two echo points $\pm(a - d)$, where it is continuous with the singularity of $k * (\text{edge profile})$; (iii) the echo form (12) holds, with $\rho = c/\tilde{\sigma}_\infty$; (iv) $\lambda'(a) = -2|A_0|^2$ at every point of differentiability (Conjecture 8.1 at these supports); (v) under RH, $\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\lambda'(a)/\pi$: Conjecture 7.7 holds for $a_2 < a < a_3$.*

Proof. (i) is the proof of Corollary 7.24 applied to (19): the near sources of $\hat{H}'_-$ are the steps $-2iP \sinh(a/2) - icf(a - d)$ and a smooth part, the far sources sit at distance at least δ_0 , and Lemma 7.22 holds for σ_{eff} (a smooth symbol, $\sigma_{\text{eff}} = \ell - c^2/\ell + O(t^{-2})$, zero-free in a strip once its real zeros are divided out), so the far feedback is again $-ir(0^+) + O(1/t)$ and the symmetry argument gives A_0 purely imaginary. (ii) The proof of Proposition 8.4 goes through with σ_{eff} : $\theta_{\text{eff}} = \theta_\infty + \frac{1}{2}\mathcal{H}[\log(1 - c^2/\ell^2)] = -\pi/(4\ell) + O(\ell^{-3})$, since the added term is even and $O(\ell^{-2})$ with the expansion of that proof giving $\pi c^2/(2\ell^3)$ at the first order, and $\tilde{\sigma}_{\text{eff}}^{-1/2} = \ell^{-1/2}(1 + c^2/(2\ell^2) + O(\ell^{-3}))$; so the edge term matches the model $|A_0|(\log(a/\delta) + \beta)^{-1/2}$ with the same β up to a relative $c^2/(2\ell^2)$, whose inverse transform is $\frac{1}{2}c^2|A_0|(L + \beta)^{-5/2}(1 + O(1/L))$ by the Mellin expansion with $s = \frac{5}{2}$, inside the remainder class; the far-type sources contribute smooth functions near the edge. The

regularity off the edges is interior regularity on L , M , R (the forcing $cf_L(\cdot - d)$ on R is smooth inside R), and the structure at $\pm(a - d)$ is (18) and its mirror. (iii) The copy on L contributes $ce^{-itd}\tilde{F}_R/\tilde{\sigma}_\infty$ to $\tilde{F}$ up to smooth feedback, by the proof above, and $\tilde{F}_R = A(t)/(t\sigma_{\text{eff},-}) + \dots$ by (i); with $F = 2i\text{Im}(F_R + F_e)$ this is (12). (iv) Lemma 8.2 holds with the copies, which are continuous across the edge with the cusp of the echo, $O(L^{-3/2})$ with derivative $O(\delta^{-1}L^{-5/2})$, inside the remainder class; and in the proof of Proposition 8.3 the singularities of $(xf)'$ at the echo points, at distance at least δ_0 from the edge, leak like the left edge does. (v) Proposition 7.8 applies with one change in its Step 1: f_0 now carries the two echo cusps, whose transforms are $O(t^{-1}L^{-3/2})$ with the phases $e^{\pm it(a-d)}$; their squares are $O(t^{-2}L^{-3})$, and their cross terms with the edges oscillate at the frequencies d and $2a - d$, both nonzero, which Step 2 handles exactly as it handles $\cos(2ta)$. Hence $\lim T \sum_{|\gamma|>T} |F(\gamma)|^2 = 2|A_0|^2/\pi = -\lambda'/\pi$ by (iv). $\square$

Computation 8.10 (the constants of the edge with the prime inside the window). `rh_edge_constants.py`, log in `data/`. On the minimizers with primes at $a = 0.4, 0.45, 0.5$ ($\lambda = 0.01471, 0.002405, 0.000194$; $\lambda' = -0.4659, -0.1013, -0.01124$ from the dilation engine; the entry 2 inside the window, $0.347 < a < 0.549$) the ratio of $f(a - \delta)$ to $(-\lambda'/2)^{1/2}(\log(a/\delta) + \beta)^{-1/2}$ with $\beta = -\gamma - \log(2\pi a) - \lambda$ ($-1.514, -1.619, -1.722$), nothing fitted, is 1.000–1.002 at every node between $\delta = 10^{-12}$ and 10^{-6} at the three supports, 1.004–1.008 at 10^{-4} , and 1.03–1.05 at 10^{-3} , where L is 6 and the relative correction $c^2/(2L^2)$ of the symbol, 0.3%, is joined by the corrections of the asymptotic regime. The alternative $\beta - 2c$, which a first-order return of the echo would produce, is off by 2–4% over the whole range. So the prime inside the window changes neither constant: the edge amplitude is $(-\lambda'/2)^{1/2}$ and the shift is the symbol's.

Computation 8.11 (the boundary law). C from the edge FEM (two-stage mesh to $10^{-12}a$, fit of $1/f^2$ against $\log(a/\delta)$ on $14 \leq t \leq 20$, residual $\leq 5 \times 10^{-5}$, mesh bias 10^{-3} to 10^{-2} in C) and λ'_{48} from `rh_dilation.py` (Hellmann–Feynman, ball arithmetic, 48 modes; the truncation error of λ'_{48} is below 10^{-4} at these supports by the K -stability of Computation 4.4); data in `boundary_law.csv` and `data/`, reproduced by `rh_hadamard.py`:

a	primes	λ (FEM)	C	β	λ'_{48}	$r = -\lambda'/2C^2$
0.3	no	+0.2225566	1.30891	−1.49	−3.42881	1.0007
0.4	no	−0.0780502	1.15425	−1.48	−2.66640	1.0007
0.5	no	−0.3222425	1.06222	−1.45	−2.25815	1.0007
0.6	no	−0.5358860	1.00932	−1.42	−2.03880	1.0007
0.4	yes	1.470510×10^{-2}	0.48266	−1.57	−0.465947	1.0001
0.45	yes	2.404578×10^{-3}	0.22494	−1.67	−0.101283	1.0009
0.5	yes	1.936801×10^{-4}	0.07493	−1.78	-1.123556×10^{-2}	1.0005
0.549	yes	1.491447×10^{-5}	0.02412	−1.87	-1.143529×10^{-3}	0.983

Seven cases give $r = 1$ to better than 10^{-3} , across supports where the prime-free floor is positive and negative and with the primes present; the prime-free offset 7×10^{-4} is the mesh bias of C on that family, and the full-form offsets $(1, 9, 5 \times 10^{-4})$ are within the stated bias. The last case measures C at $a_3 = 0.54931$ and λ' at 0.549, inside the sawtooth where Φ' changes by about 9% per 0.001 (Computation 4.4); its 1.7% is within the combined error of the amplitude (10^{-2} in C , hence 2% in C^2) and the offset in a . The evidence is confined to $a \leq 0.6$ (heights below 22), far from the regime of Conjecture 7.1; extending it needs a multiprecision edge engine.

An exactly solvable case: for one half of the Dirichlet form of L_Δ on $(-a, a)$, the scaling identity $L_\Delta(u(\cdot/a)) = (L_\Delta u)(\cdot/a) - 2\log a u(\cdot/a)$ gives $\lambda(a) = \lambda(1) - \log a$, so the law predicts

$C(a)^2 = 1/(2a)$. The prime-free form differs from this by the bounded corrections (the polar term and $\operatorname{Re} \psi - \log |t|$); its amplitudes give $2aC^2 = 1.028, 1.066, 1.129, 1.223$ at $a = 0.3, \dots, 0.6$ and its derivatives $-a\lambda' = 1.029, 1.067, 1.129, 1.223$, both near 1 and equal to each other, which is the content of the table restated, not an independent test.

Remark 8.12 (consequence of Conjecture 8.1). $\Phi'(a) = 2C(a)^2/\lambda(a)$, and Conjecture 7.1 becomes

$$C(a)^2 \leq \frac{c}{2} T^*(a) \lambda(a) \quad (a \geq a_0) :$$

the edge amplitude of the minimizer is bounded by its energy times the horizon. This is the form in which the conjecture is a statement about one function at one place. The relay of Computation 4.4 reads: as a approaches an entry from below, λ falls toward the deleted form's failure point while C stays of the same order, and $\Phi' = 2C^2/\lambda$ rises until the entry rescues the floor. Under the Riemann Hypothesis, $C(a)$ is the boundary trace of the band-limited function that best hides from the zeros above the horizon.

8.1 The echo tree

Proposition 8.8 treats one entry and one echo. At a general support the entries copy the window onto itself in several ways, the echoes have echoes, and a copy of an echo can return to the edge along several chains. The structure is combinatorial, and it is finite as long as the walks that the entries generate inside the window are confined.

Let $D_a = \{\log m : \Lambda(m) \neq 0, \log m < 2a\}$ be the entries inside the window and $c_d = \Lambda(m)m^{-1/2}$ for $d = \log m$. A *walk* is a finite sequence of points of $[-a, a]$ whose consecutive points differ by $\pm d$ with $d \in D_a$. The *echo set* S is the set of points visited by walks starting at a or $-a$, and $S^\circ = S \cap (-a, a)$; $S = -S$. The walks are *confined* at a , (F_a) , if the set of points visited by walks starting anywhere in $[-a, a]$ is finite. Write $S = \{s_0 = -a < s_1 < \dots < s_N = a\}$, $I_j = (s_{j-1}, s_j)$, $f_j = f\mathbf{1}_{I_j}$. On S° let $C_{pq} = c_{|p-q|}$ if $|p-q| \in D_a$ and 0 otherwise (the weighted adjacency of the echo points), $b_p = c_{a-p}$ if $a-p \in D_a$ and 0 otherwise, $\bar{b}_p = b_{-p}$ (the couplings to the two edges), and let C_a be the matrix on $S^\circ \cup \{a\}$ obtained from C by adjoining the row and column b ; C and C_a are real symmetric.

Lemma 8.13 (the entries move the pieces onto pieces). *For $d \in D_a$ and each j , each of $I_j + d$ and $I_j - d$ is either disjoint from $(-a, a)$ or equal to some $I_{j'}$.*

Proof. If $s_{j-1} + d \geq a$, the interval $I_j + d$ lies to the right of the window. Otherwise $s_{j-1} + d \in [-a, a]$ is one step from a point of S , so it belongs to S ; if also $s_j + d \leq a$, then $s_j + d \in S$ for the same reason, and no point q of S lies strictly between them, since $q - d$ would be a point of S in I_j ; so $I_j + d = I_{j'}$. The remaining case $s_{j-1} + d < a < s_j + d$ is impossible: then $a - d \in I_j$, while $a - d \in S$ because $d < 2a$. The same holds for $-d$. $\square$

Proposition 8.14 (the echo tree). *Assume (H_∞) . Let f be the minimizer at a support a that is not an entry, with $\lambda > \Psi_\infty(0)$ and the walks confined, (F_a) . Let $\tilde{\sigma}_\infty$ and k be as in Proposition 8.8, and for $p \in S$ let $\Phi_p(t) = \int \chi(v) f(p+v) e^{itv} dv$, with one fixed $\chi \in C_c^\infty$ equal to 1 near 0 and supported in $(-\delta_1, \delta_1)$, δ_1 half the least gap of S ; for $p = \pm a$ this is the transform of f near the edge, and $\Phi_{-a} = -\Phi_a$. Then:*

(i) $f \in C^\infty((-a, a) \setminus S^\circ)$.

(ii) For $p \in S^\circ$, near $v = 0$,

$$f(p+v) = \sum_{q \in S, |p-q| \in D_a} c_{|p-q|} (k * f(q + \cdot))(v) + r_p(v), \quad (20)$$

where $f(q + \cdot)$ is the window function shifted by q and r_p is C^1 near 0 with $r_p''(v) = O(|v|^{-1}(\log 1/|v|)^{-5/2})$. In Fourier form,

$$(\tilde{\sigma}_\infty(t)I - C)\Phi_{S^\circ}(t) = b\Phi_a(t) + \bar{b}\Phi_{-a}(t) + \Psi(t), \quad \Psi(t) = O(t^{-2}(\log t)^{-3/2}). \quad (21)$$

(iii) Let $R = b^T(\tilde{\sigma}_\infty I - C)^{-1}b$ and $\bar{R} = \bar{b}^T(\tilde{\sigma}_\infty I - C)^{-1}\bar{b}$, the returns of the right edge to itself and to the left edge through the tree, and

$$\sigma_{\text{eff}} = \tilde{\sigma}_\infty - R = \frac{\det(\tilde{\sigma}_\infty I - C_a)}{\det(\tilde{\sigma}_\infty I - C)}, \quad (22)$$

a rational function of $\tilde{\sigma}_\infty(t)$ whose real zeros and poles are the heights at which $\tilde{\sigma}_\infty$ equals an eigenvalue of C_a , respectively of C : finitely many, none above the height where $\tilde{\sigma}_\infty > \|C_a\|$, and $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2/\tilde{\sigma}_\infty + O(\tilde{\sigma}_\infty^{-2})$. With $\tilde{\sigma}_{\text{eff}} > 0$ the symbol with its real zeros and poles divided out by rational factors and $\sigma_{\text{eff},\pm}$ its factors (Lemma 7.21),

$$\tilde{F} = \frac{P_-[\hat{H}_-''/\sigma_{\text{eff},+}]}{\sigma_{\text{eff},-}} \quad \text{on } \mathbb{R}, \quad (23)$$

where $\hat{H}_-'' \in H^1(\mathbb{R})$ is analytic in the lower half-plane and is the transform of a source made of steps at the edge, a part smooth near the edge, a part whose transform is $O(t^{-2}(\log t)^{-3/2})$, the mirror chains $P_-[\bar{R}\Phi_{-a}]$, and sources supported at distance at least $\delta_0 = a - \max S^\circ$ to the left of the edge.

Proof. The equation on the line. As in the proof of Proposition 8.8, with $g_\infty = K_\infty f$ outside the window,

$$(K_\infty - \lambda)f = 2Ps\mathbf{1}_{[-a,a]} + \sum_{d \in D_a} c_d \sum_j (f_j(\cdot - d) + f_j(\cdot + d))\mathbf{1}_{(-a,a)} + g_\infty \mathbf{1}_{|y|>a},$$

and by Lemma 8.13 every copy $f_j(\cdot \mp d)\mathbf{1}_{(-a,a)}$ is either zero or the whole shifted piece, landing on some $I_{j'}$. All sources are in $L^1 \cap L^2$ with $yh \in L^2$, so Proposition 7.23 applies at the right edge with $\hat{H}_-$ the sources to the left of a corrected at $\pm t_\lambda$: $\tilde{F} = P_-[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-} = \hat{H}_-/\tilde{\sigma}_\infty - P_+[\hat{H}_-/\sigma_{\infty,+}]/\sigma_{\infty,-}$, and on the window the second term is the transform of a smooth function (Lemma 7.22: $P_+[\cdot]$ is the transform of a function on $u > 0$, and the kernel of $1/\sigma_{\infty,-}$ is smooth away from the origin). So on $(-a,a)$, $f = k * \hat{h}_- + (\text{smooth})$, where $\hat{h}_-$ is the source with its corrections, which are weighted primitives of the sources, $-ie^{\pm it_\lambda u} \int_{-\infty}^u h(u')e^{\mp it_\lambda u'} du'$ on $u < 0$: one derivative smoother than h , with the same singular support, and a step at the edge. Since k is smooth away from the origin, $k * g$ is smooth near a point where g is smooth, for $g \in L^1$.

Regularity, (i). Let $p \notin S$ and let $O(p)$ be the set of points visited by walks from p : finite by (F_a) and disjoint from S , since a walk from p to $\pm a$ reversed is a walk from $\pm a$ to p . The sources near $q \in O(p)$ are the polar term, smooth inside the window, the copies landing on intervals adjacent to q , which are $c_d f(q \mp d + \cdot)$ with $q \mp d \in O(p)$ (a copy landing on an interval not adjacent to q is supported away from q), the far force, and the corrections; so the vector $(f(q + \cdot))_{q \in O(p)}$ solves near 0 the system $(K_\infty - \lambda)f(q + \cdot) = \sum_{q' \in O(p), |q-q'| \in D_a} c_{|q-q'|}(1 + \mathcal{M})f(q' + \cdot) + (\text{smooth})$, with $\mathcal{M}$ the primitive of the corrections, an operator of order -1 . The commutator of K_∞ with a cutoff has a bounded kernel $(J_\Gamma(s)(\chi(y) - \chi(y - s)))$ is bounded near $s = 0$, so the usual bootstrap, one derivative per step on a shrinking neighbourhood and for all $q \in O(p)$ at once, gives $f \in C^\infty$ near p .

The local relations, (ii). For $p \in S^\circ$ the same list of sources near p holds with $q \in S$: the copies landing on the two intervals adjacent to p are $c_{|p-q|}f(q + \cdot)$ for $q = p \mp d \in S$ (one-sided when $q = \pm a$), the other copies, the polar term, the far force and the far corrections are smooth near p , and the corrections of the adjacent copies are C^1 at p with the second derivative of the primitive of an echo, $O(|v|^{-1}L^{-3/2})$ before and $O(|v|^{-1}L^{-5/2})$ after the convolution with k , which gains a logarithm; this is (20). Transforming (20) against χ : $\chi \cdot (k * f(q + \cdot))$ has the transform $\Phi_q/\tilde{\sigma}_\infty$ up to the transform of a smooth compactly supported function (the error of the localization, $\chi(k * (\chi f(q + \cdot))) - k * (\chi f(q + \cdot))$, is smooth since k is smooth away from the origin), and χr_p has a transform $O(t^{-2}(\log t)^{-3/2})$; multiplying by $\tilde{\sigma}_\infty$ gives (21), where Ψ collects $\tilde{\sigma}_\infty \widehat{\chi r_p}$ and the rapidly decaying errors.

The edge, (iii). The sources landing on $I_N = (s_{N-1}, a)$ are $c_d f_{j(d)}(\cdot - d)$ with $I_{j(d)} = I_N - d$, that is, $c_d f(a - d + u)$ for $u \in (s_{N-1} - a, 0)$, for each $d \in D_a$; their transforms are $c_d P_-[\Phi_{a-d}]$ up to far sources (the cut at $s_{N-1} - a$, at distance δ_0) and smooth ones. By (21), wherever $\tilde{\sigma}_\infty(t)I - C$ is invertible, $\sum_d c_d \Phi_{a-d} = b^T \Phi_{S^\circ} = R \Phi_a + \bar{R} \Phi_{-a} + b^T (\tilde{\sigma}_\infty I - C)^{-1} \Psi$. Here $\Phi_a = \tilde{F} +$ (far source) by the choice of χ , and $P_-[R\tilde{F}] = R\tilde{F} - P_+[R\tilde{F}]$; moving $R\tilde{F}$ to the left of $\tilde{\sigma}_\infty \tilde{F} = \hat{H}_- + \hat{G}_+$ gives $\sigma_{\text{eff}} \tilde{F} = \hat{H}'_- + \hat{G}'_+$ with $\hat{H}'_- = \hat{H}_- - R\tilde{F} + P_-[R \Phi_a + \bar{R} \Phi_{-a} + b^T (\tilde{\sigma}_\infty I - C)^{-1} \Psi]$ (the copies on I_N) and $\hat{G}'_+ = \hat{G}_+ + P_+[R\tilde{F}]$. The identity (22) is the Schur complement of the entry (a, a) of $\tilde{\sigma}_\infty I - C_a$, and since $\tilde{\sigma}_\infty$ is real-analytic with $\tilde{\sigma}_\infty \rightarrow \infty$, its level sets at the eigenvalues of C and C_a are finite. At a real zero $\pm t_j$ of σ_{eff} the right-hand side vanishes, and the rational correction of Proposition 7.23 applies; at a real pole $\pm s_k$ (an eigenvalue of C reached by $\tilde{\sigma}_\infty$: a standing wave of the interior tree) both $\sigma_{\text{eff}} \tilde{F}$ and $\hat{H}'_-$ have a simple pole, with equal residues, and dividing by the rational factor $(t^2 + s_k^2)/(t^2 - s_k^2)$ of $\tilde{\sigma}_{\text{eff}}/\sigma_{\text{eff}}$ removes them; the Liouville function of the proof of Proposition 7.23 is then entire off $\pm s_k$ with at most simple poles there, hence a rational function of the form $\sum_k c_k^\pm/(z \mp s_k)$ by its growth, and the $c_k^\pm$ are fixed by the regularity of $\tilde{F}$ at $\pm s_k$; these terms are transforms of one-sided exponentials, smooth near the edge up to a step. Collecting, $\hat{H}''_-$ is $\hat{H}'_-$ with these rational corrections, and its sources are as stated: the polar step and the steps $c_d r_{a-d}(0^-)$ of the smooth parts of the copies, smooth parts, the corrections of order $t^{-2}(\log t)^{-3/2}$, the mirror chains, and the far sources. Lemma 7.21 applies to $\tilde{\sigma}_{\text{eff}}$, which is smooth, positive, and $\tilde{\sigma}_\infty(1 + O(\tilde{\sigma}_\infty^{-2}))$ at infinity with bounded log-derivative, and Lemma 7.22 to its factors, since σ_{eff} is analytic in the strip of $\tilde{\sigma}_\infty$ with $R = O(1/\log |t|)$ there; the Liouville argument of Proposition 7.23 then gives (23). $\square$

Corollary 8.15 (the edge, the echoes and the tail law at a support with a finite tree). *Under the hypotheses of Proposition 8.14, with $\rho = (\tilde{\sigma}_\infty I - C)^{-1}b$ (the echo amplitudes) and $E(t) = 1 + \sum_{p \in S^\circ} \rho_p(t) e^{-it(a-p)}$:*

- (i) *the edge of the transform has the form of Corollary 7.24 with σ_{eff} in place of σ , $A(t) \rightarrow A_0$ purely imaginary;*
- (ii) *f satisfies the edge law with derivatives with the constants of Proposition 8.4, $C = |A_0|$ and $\beta = -\gamma - \log(2\pi a) - \lambda$, and is smooth on the window except at the edges and at the points of S° , where it has the cusps of (20);*
- (iii) *the echo form:*

$$S(t) = -\frac{|A_0| |E(t)| \cos(ta + \theta_{\text{eff}}(t) + \arg E(t) - \kappa(t)/t)}{t \tilde{\sigma}_{\text{eff}}(t)^{1/2}} + O(t^{-2}(\log t)^{-3/2}),$$

with θ_{eff} the Hilbert phase of $\tilde{\sigma}_{\text{eff}}$ and $\kappa(t) = O(\log t)$ real; the chains of the tree are the terms

of the Neumann series $\rho = \sum_{\ell \geq 0} C^\ell b / \tilde{\sigma}_\infty^{\ell+1}$, convergent where $\tilde{\sigma}_\infty > \|C\|$, and R is a finite continued fraction along a chain;

(iv) the sharp form is an identity of the tree: with $\sigma_a = \tilde{\sigma}_\infty - \sum_{d \in D_a} 2c_d \cos(\tau d)$ and t frozen in $\tilde{\sigma}_\infty$ and ρ ,

$$\langle \sigma_a(\tau) |E(\tau)|^2 \rangle_\tau = \sigma_{\text{eff}}(t), \quad \langle \sigma_a(\tau) e^{2i\tau a} E(\tau)^2 \rangle_\tau = -\bar{R}(t)$$

(means over $\tau \in \mathbb{R}$); hence the local envelope of $t^2(\Psi - \lambda)|F(t)|^2$ over the oscillations is $2|A_0|^2(1 - \bar{R}/\tilde{\sigma}_{\text{eff}} + O(\theta_{\text{eff}}^2))(1 + o(1))$, and $\bar{R} \equiv 0$ when the trees of the two edges are disjoint;

(v) $\lambda'(a) = -2|A_0|^2$ at every point of differentiability, and under RH, $\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\lambda'(a)/\pi$: Conjecture 7.7 holds at a ;

(vi) under RH, when the trees of the two edges are disjoint, the sampling constant of Corollary 7.30 is

$$M_\chi(T) - 1 = \frac{R + \tilde{\sigma}_\infty(|\rho|^2 - 1)}{\sigma_{\text{eff}}} \Big|_{t=T} (1 + o(1)) = \frac{2|b|^2}{\tilde{\sigma}_\infty(T)^2} (1 + O(1/\tilde{\sigma}_\infty)), \quad |\rho|^2 := 1 + \sum_p \rho_p^2,$$

$$\text{and } |b|^2 = \sum_{\log m < 2a} \Lambda(m)^2/m.$$

Proof. (i), (ii), (v) are the proofs of Corollary 8.9(i), (ii), (iv), (v) with σ_{eff} of (22): the near sources are steps and smooth parts, the far feedback is $-ir(0^+) + O(1/t)$, the corrections of order $t^{-2}(\log t)^{-3/2}$ contribute $O(t^{-1}(\log t)^{-3/2})$ to $A(t)$, inside $A_0 + O(\log t/t)$; $\tilde{\sigma}_{\text{eff}}^{-1/2} = \ell^{-1/2}(1 + |b|^2/(2\ell^2) + O(\ell^{-3}))$ and $\theta_{\text{eff}} = -\pi/(4\ell) + O(\ell^{-3})$, so the profile matches the model with the same β up to a relative $|b|^2/(2\ell^2)$, inside the remainder class; the cusps at the finitely many points of S° , at distance at least δ_0 from the edges, have transforms $O(t^{-1}L^{-3/2})$ with the phases e^{itp} , their cross terms with the edges oscillate at the nonzero frequencies $a \mp p$, and Step 2 of Proposition 7.8 handles them as it handles $\cos(2ta)$. (iii) By (21), $\Phi_{S^\circ} = \rho \Phi_a + \bar{\rho} \Phi_{-a} + O(t^{-2}(\log t)^{-3/2})$ with $\bar{\rho} = (\tilde{\sigma}_\infty I - C)^{-1} \bar{b}$, $\bar{\rho}_p = \rho_{-p}$ by the symmetry $S = -S$, and $F = \sum_{p \in S} e^{itp} \Phi_p$ up to the transform of a smooth function; so $F = e^{ita} \Phi_a E(t) + e^{-ita} \Phi_{-a} \overline{E(t)} + \dots = 2i \operatorname{Im}(e^{ita} \Phi_a E) + \dots$, and $\Phi_a = \tilde{F} + (\text{far}) = A(t)/(t\sigma_{\text{eff},-}) + \dots$ by (i). The Neumann series and the continued fraction are the expansion of the resolvent and the Schur complement of a path. (iv) Write $E = \sum_{p \in S^\circ \cup \{a\}} e_p e^{-i\tau(a-p)}$ with $e_a = 1$, $e_p = \rho_p$. The mean of $|E|^2$ is $|\rho|^2$ and the mean of $2 \cos(\tau d) |E|^2$ is $\sum_{|p-q|=d} e_p e_q$, so the mean of $\sigma_a |E|^2$ is $\tilde{\sigma}_\infty |\rho|^2 - e^T C_a e = \tilde{\sigma}_\infty |\rho|^2 - 2b^T \rho - \rho^T C \rho$, and $\rho^T C \rho = \tilde{\sigma}_\infty(|\rho|^2 - 1) - b^T \rho$ by $(\tilde{\sigma}_\infty - C)\rho = b$; the sum is $\tilde{\sigma}_\infty - b^T \rho = \sigma_{\text{eff}}$. For the mirror term, $e^{2i\tau a} E^2 = \sum_{p,q} e_p e_q e^{i\tau(p+q)}$, whose mean against σ_a is $\tilde{\sigma}_\infty \sum_{p \in S^\circ} \rho_p \rho_{-p} - \sum_p e_p (C_S e)_{-p}$, with C_S the adjacency on S and $e_{-a} = 0$; by $(\tilde{\sigma}_\infty - C)\rho = b$, $(C_S e)_{-p} = \tilde{\sigma}_\infty \rho_{-p}$ for $p \in S^\circ$ and $(C_S e)_{-a} = \bar{b}^T \rho = \bar{R}$, which leaves $-\bar{R}$. With $|F|^2 = 2|\Phi_a|^2 |E|^2 - 2 \operatorname{Re}(e^{2ita} \Phi_a^2 E^2)$ and $\Phi_a^2 = -|A_0|^2 e^{2i\theta_{\text{eff}} - 2i\kappa/t} / (t^2 \tilde{\sigma}_{\text{eff}})(1 + o(1))$, the envelope follows; $\bar{R} \equiv 0$ if no chain of C joins the two trees. (vi) In (17), $\frac{1}{2\pi} \int |F|^2 \chi_T \Psi_\infty = (\tilde{\sigma}_\infty + \lambda) |\rho|^2 I_T (1 + o(1))$ and $2 \sum_n \Lambda(n) n^{-1/2} \check{h}_T(\log n) = e^T C_a e I_T (1 + o(1))$ with $I_T = \frac{1}{2\pi} \int 2|\Phi_a|^2 \chi_T$, the oscillating terms averaging out and the mirror cross terms having no mean when the trees are disjoint; so $M_\chi - 1 = e^T C_a e / (\tilde{\sigma}_\infty |\rho|^2 - e^T C_a e)(1 + o(1))$, and $e^T C_a e = R + \tilde{\sigma}_\infty(|\rho|^2 - 1)$ by the computation in (iv). $\square$

Computation 8.16 (the echo set). `rh_echo_tree.py`, log in `data/`. The echo set S and the orbits of walks from 200 interior points, with a cap of 5000 points, for $0.35 \leq a \leq 0.84$:

a	entries	$ S^\circ $	chain	orbit	$\ C\ $	$\ C_a\ $	trees of $\pm a$
0.35–0.53	2	2	1	2	0	0.490	disjoint
0.55–0.68	2, 3	6	2	4	0.490	0.901	disjoint
0.70–0.75	2, 3, 4	12	3	7	0.901	1.172	disjoint
0.775–0.80	2, 3, 4	22	6	12	1.172	1.219	disjoint
0.825	2, 3, 4, 5	148	30	75	1.471	1.471	disjoint
0.8375	2, 3, 4, 5	426	49				
0.8422	2, 3, 4, 5	3434	482				
0.8430	2, 3, 4, 5	8854	1572				

(chain: the longest shortest chain from the edge; orbit: the largest orbit of a walk from an interior point.) The walks are confined up to $a_\infty = 0.8430$ (between 0.84297 and 0.84336) and not beyond: with the entries 2, 3, 4, 5 in a window of length $2a > 1.686$ the net moves $\log 3 - \log 2$, $\log 4 - \log 3$, $\log 5 - \log 4$ can be composed indefinitely without leaving the window, and the echo set becomes dense. Below a_∞ the tree is finite; its size grows from 2 points for one entry, through 6 (one echo of an echo, $a - \log 3 + \log 2$) for $a_3 < a < a_4$, to thousands near a_∞ . At every support of the table the trees of the two edges are disjoint, so $\tilde{R} \equiv 0$ and the sharp form of Corollary 8.15(iv) holds without the mirror correction. The resonances: $\min \tilde{\sigma}_\infty = 1.003$ (at $t = 6.6$, just above $t_\lambda = 6.29$, for $\lambda \approx 0$), so σ_{eff} has no real zeros or poles for $a < a_4$ ($\|C_a\| = 0.90$), real zeros but no poles for $0.70 < a < 0.77$, and both beyond; all of them lie below the horizon ($\tilde{\sigma}_\infty(T^*) = 2a - \lambda$). Where the walks deconfine, the symbol of the window is negative on sets reaching far above the horizon (the last sign change at $33T^*$ for $a = 1.0$ and beyond $150T^*$ for $a = 1.25$ in Computation 7.20): the infinite tree has a band, and the finite tree does not.

Computation 8.17 (the tree at $a = 0.6$: the second echo). Edge FEM at $a = 0.6$ with the entries 2 and 3 (486 nodes, the mesh of Computation 8.11, $\lambda = 5.9459 \times 10^{-7}$; $\lambda_{64} = 5.9583 \times 10^{-7}$ and $\lambda'_{64} = -4.1545 \times 10^{-5}$ from the dilation engine; $T^* = 20.86$). The tree: $S^\circ = \{\pm 0.0931, \pm 0.1945, \pm 0.4986\}$, the right edge's chain $a \rightarrow a - \log 3 = -0.4986 \rightarrow a - \log 3 + \log 2 = 0.1945$ and its branch $a \rightarrow a - \log 2 = -0.0931$, $b = (c_3, c_2, 0)$ on $(-0.4986, -0.0931, 0.1945)$, C with the single link c_2 between -0.4986 and 0.1945 ; so $R = c_2^2/\tilde{\sigma}_\infty + c_3^2/(\tilde{\sigma}_\infty - c_2^2/\tilde{\sigma}_\infty)$ and the second echo has the amplitude $\rho_{0.1945} = c_2 c_3 / (\tilde{\sigma}_\infty^2 - c_2^2)$: 0.040, 0.026, 0.018 at 5, 10, 20 T^* . (i) *The constants*. With nothing fitted, $f/[(-\lambda'/2)^{1/2}(L + \beta)^{-1/2}]$, $\beta = -1.904$, is 1.0003–1.003 on $10^{-12} < \delta < 10^{-6}$ and 1.013 at 10^{-4} : the two entries change neither constant. (ii) *The nulls and the pointwise form*, 463 nulls on 3.5–120 T^* : with the tree's phase $\theta_{\text{eff}} + \arg E$ the residual $r + \vartheta - \kappa/t$ has rms 0.0114 on 5–80 T^* (300 nulls), against 0.0162 for the first generation alone ($\rho_p = b_p/\tilde{\sigma}_\infty$, $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2/\tilde{\sigma}_\infty$), 0.036 for the window's symbol and 0.41 for no shift; at 5, 7, 10, 15, 20, 30 T^* the tree leaves $-0.002, -0.006, -0.006, -0.004, -0.003, -0.0002$ where the first generation leaves $-0.039, +0.019, +0.010, +0.017, -0.011, +0.008$. The relative L^2 distance of S from the tree form over one period, $|A| = (-\lambda'/2)^{1/2}$, is 0.038, 0.0088, 0.0034 at 5, 10, 20 T^* , against 0.053, 0.028, 0.020 for the first generation and 0.20, 0.097, 0.055 for the window's symbol: the first generation's error is the missing second echo, 0.04, 0.026, 0.018. Beyond 35 T^* the interior mesh ($1/\text{spacing} \approx 34T^*$) limits S and all forms degrade alike (0.011 at 50 T^* , 0.11 at 100). (iii) *The sampling defect*. The identity (17) holds at every height to 10^{-5} of the zero sum. With $\lambda = 6 \times 10^{-7}$ the quartic cutoff is too soft, its weight $(t/T)^4$ below the horizon letting through the density integral there (which is 10^{-3} at $\kappa = 1$ against a zero sum of 6×10^{-7}), so the cutoff $1 - e^{-(t/T)^8}$ is used (Proposition 7.29 with $m = 4$). Its M_χ against the tree's prediction of Corollary 8.15(vi) averaged over the tail, $1 + \langle e^T C_a e / \tilde{\sigma}_\infty \rangle_T / \langle \sigma_{\text{eff}} / \tilde{\sigma}_\infty \rangle_T$, is 1.110/1.106, 1.076/1.079, 1.050/1.057 at $\kappa = 6, 10, 20$ (first order $2|b|^2 \langle \tilde{\sigma}_\infty^{-2} \rangle_T$: 1.098, 1.075, 1.054; the value at T instead of the average, 1.160, 1.113, 1.077, is off by a factor 0.7 in $M_\chi - 1$); at $\kappa = e$ the cutoff still lets through the density integral below the horizon,

and $M_\chi = 1.68$ against 1.19. The sharp constant $M(\kappa)$ of Computation 7.14 is 1.14, 1.12, 1.064, 1.049 at $\kappa = e, 6, 10, 20$.

8.2 Beyond the finite tree: the window's symbol and the band

Above a_∞ the echo set is dense and the tree has no finite matrix. The representation that survives is the one with the symbol of the window, which has only two sources, the polar term and the force outside the far edge; the copies are inside the symbol. It is exact at every support, and what it does not give is the asymptotics of its edge amplitude below the band of the echo lattice, where the symbol changes sign.

Proposition 8.18 (the window's symbol at every support). *Assume (H_∞) . Let f be the minimizer at a support a that is not an entry, $\sigma_a = \Psi_a - \lambda$ with $\Psi_a = \Psi_\infty - \sum_{d \in D_a} 2c_d \cos(td)$ the symbol of the window, $\pm t_1, \dots, \pm t_N$ its real zeros, finitely many, all below the last sign change t_s , $\tilde{\sigma}_a = \sigma_a \prod_j (t^2 + t_j^2)/(t^2 - t_j^2) > 0$, and $g_{\text{out}} = K_a f$ outside the window, K_a the operator with symbol Ψ_a . Then:*

- (i) *Lemma 7.21 holds for $\tilde{\sigma}_a$ with the bounds on θ and θ' replaced by $O(\log \log(e + |t|))$: $\sigma_{a,\pm} = \tilde{\sigma}_a^{1/2} e^{\pm i\theta_a}$ are the boundary values of the outer functions, $c \leq |\sigma_{a,\pm}(z)| \leq C(\log(e + |z|))^{1/2}$, and $1/\sigma_{a,\pm}$ are bounded on the line with derivative $O(\log \log(e + |t|))$.*
- (ii) *With $\tilde{H}$ the transform of $e^{-ita}(2Ps\mathbf{1}_{[-a,a]} + g_{\text{out}}\mathbf{1}_{y < -a})$ and $\hat{H}_-$ its correction at the $\pm t_j$ as in Proposition 7.23,*

$$\tilde{F} = \frac{P_-[\hat{H}_-/\sigma_{a,+}]}{\sigma_{a,-}} \quad \text{on } \mathbb{R}, \quad (24)$$

and, with Φ_- as in Corollary 7.24 built from g_{out} , $\tilde{F} = A_a(t)/(t\sigma_{a,-}) + e^{-2ita}\Phi_-/\tilde{\sigma}_a$ with A_a given by the formula of (15) with $\sigma_{a,\pm}$, $A_a(-t) = -\overline{A_a(t)}$.

- (iii) *At a support with a finite tree, (24) and (23) represent the same $\tilde{F}$: the echoes of the right edge inside the window are the first terms of the expansion of $1/\sigma_{a,-}$ in the lattice frequencies, and the terms of that expansion which would place an echo beyond the far edge are cancelled by the far source.*

Proof. σ_a is real-analytic, even, and tends to $+\infty$, so its real zeros are finitely many and lie below t_s ; $\tilde{\sigma}_a$ is even, positive, C^2 , bounded below, and $O(\log t)$. (i) In the proof of Lemma 7.21 the hypothesis $(\log \tilde{\sigma})' \in L^2$ serves only to bound $\theta' = \frac{1}{2}\mathcal{H}[(\log \tilde{\sigma})']$. Write $\log \tilde{\sigma}_a = \log \tilde{\sigma}_\infty + u$ with $\tilde{\sigma}_\infty$ the archimedean symbol with its real zero divided out; u is C^2 with u', u'' bounded ($\tilde{\sigma}_a \geq c_1$ and σ'_a, σ''_a are bounded), and $u = O(\log \log(e + |t|))$ since $c_1 \leq \tilde{\sigma}_a \leq C\tilde{\sigma}_\infty$. The Hilbert transform of u' is bounded on $|\tau - t| < 1$ by $\sup |u''|$, and on $|\tau - t| > 1$ an integration by parts gives $u(t \pm 1) + \int_{|\tau-t|>1} u(\tau)(t - \tau)^{-2} d\tau = O(\log \log(e + |t|))$; the same integration by parts in (10) bounds $\mathcal{H}[u]$. The modulus bounds of the lemma use only $\log \tilde{\sigma}_a$, and hold. (ii) On the window $K_a f = \lambda f + 2Ps$, since the copies $f(\cdot \mp \log m)$ with $\log m > 2a$ vanish there; so on the line $K_a f = \lambda f \mathbf{1}_{[-a,a]} + 2Ps \mathbf{1}_{[-a,a]} + g_{\text{out}} \mathbf{1}_{y > a}$, where $g_{\text{out}} = K_\infty f + \sum_{d \in D_a} c_d (\text{copies landing outside})$ is in $L^1 \cap L^2$ with $yg_{\text{out}} \in L^2$, and $\sigma_a \tilde{F} = \tilde{H} + \tilde{G}_+$. The proof of Proposition 7.23 applies with $\sigma_{a,\pm}$: the rational correction at the N pairs of zeros, the H^1 property of $\hat{H}_-/\sigma_{a,+}$ (from (i)), and the Liouville argument with the bounds of (i). The edge form is (15) read with $\sigma_{a,\pm}$, and the symmetry is as before. (iii) Both representations are exact for the same function, and $\tilde{F}$ is supported in $(-2a, 0)$; for one entry, with $\tilde{\sigma}_\infty$ frozen at s the factorization $s - 2c \cos \tau = \sigma_{\text{fp}}(1 - \rho_{\text{fp}} e^{i\tau})(1 - \rho_{\text{fp}} e^{-i\tau})$, $\sigma_{\text{fp}} + c^2/\sigma_{\text{fp}} = s$, $\rho_{\text{fp}} = c/\sigma_{\text{fp}}$, is exact for $s \geq 2c$, and $1/\sigma_{a,-} = \sigma_{\text{fp}}^{-1/2} \sum_k \rho_{\text{fp}}^k e^{-ik\tau}$ is the infinite chain of echoes at $a - kd$, of which the window holds $k = 0, 1$. $\square$

What (ii) does not give is $A_a(t) \rightarrow A_0$. In the proof of Corollary 7.24 the far feedback $-tP_+[e^{-2ita}\Phi_-/\sigma_+]$ was $-ir(0^+) + O(1/t)$ because the kernel of $1/\sigma_+$ is smooth beyond the origin (Lemma 7.22); the kernel of $1/\sigma_{a,+}$ is singular at the points $\sum_d n_d d$ of the lattice, because $\log \sigma_{a,+}$ contains all the harmonics of the entries, and the feedback of the far edge through it is the family of its echoes landing beyond the right edge, an almost-periodic function of t whose amplitudes decay only like inverse powers of $\log t$. In the finite tree this feedback is $\bar{R}\Phi_{-a}$, and it is absent when the trees of the two edges are disjoint; in the window's symbol it is always present, together with the rational corrections of the N zeros, which are $O(\sum_j t_j^2/t^2)$ and, with N zeros up to t_s , are not small below $t \approx N^{1/2}t_s$. So (24) is exact everywhere and asymptotic only far above the last sign change. Section 9.2 proves that $A_a(t) \rightarrow A_0$ at every support that is not an entry, the mirror chains being the leading correction, and derives the edge law and the boundary law from it; the finite tree is what makes the asymptotics start at the horizon.

Lemma 8.19 (the periodic limit of the window's symbol). *Let ν be the law of*

$$X = \sum_{p \leq e^{2a}} \sum_{k \geq 1, p^k < e^{2a}} 2c_{p^k} \cos(k\tau_p)$$

with independent uniform phases τ_p , one per prime, so that $\text{supp } \nu = [-X_{\max}, X_{\max}]$ with $X_{\max} = \sum_{d \in D_a} 2c_d$. For every continuous φ and every s ,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \varphi\left(s - \sum_{d \in D_a} 2c_d \cos(td)\right) dt = \int \varphi(s - x) d\nu(x);$$

in particular the mean of $\log |s - \sum_d 2c_d \cos(td)|$ over t is the logarithmic potential $U_\nu(s) = \int \log |s - x| d\nu(x)$, a continuous function of s . Put $G(s) = e^{U_\nu(s)}$, the geometric mean of the window's symbol at the frozen height s . Then

$$G(s) = s - \sum_{d \in D_a} \frac{c_d^2}{s} + O(s^{-3}) \quad (s \rightarrow \infty),$$

equal to the effective symbol of the finite tree to this order; and for one entry

$$G(s) = \frac{1}{2}(s + \sqrt{s^2 - 4c^2}) \quad (s \geq 2c), \quad G(s) = c \quad (|s| \leq 2c),$$

the former being the fixed point of the infinite continued fraction $\sigma \mapsto s - c^2/\sigma$.

Proof. The numbers $\log p$ for distinct primes are linearly independent over $\mathbb{Q}$, so $(t \log p \bmod 2\pi)_p$ is equidistributed on the torus (Weyl), the prime powers riding on the phases of their primes; this gives the limit. The law of $\sum_k 2c_{p^k} \cos(k\tau_p)$ for one prime is the push-forward of $d\tau/2\pi$ by a trigonometric polynomial, absolutely continuous with inverse-square-root singularities, and ν is the convolution of these laws over the primes; so $\log |\cdot|$ is locally integrable against ν and U_ν is continuous. For $s > X_{\max}$, $U_\nu(s) = \log s + \mathbb{E} \log(1 - X/s) = \log s - \mathbb{E}[X^2]/(2s^2) + O(s^{-3})$ with $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = \sum_d 2c_d^2$. For one entry ν is the arcsine law on $[-2c, 2c]$, whose logarithmic potential is $\log \frac{1}{2}(|s| + \sqrt{s^2 - 4c^2})$ outside and $\log c$ inside the support. $\square$

The band of the echo lattice is $\{t : \tilde{\sigma}_\infty(t) < X_{\max}\}$, the heights at which the periodic limit of the window's symbol changes sign: up to $47T^*$ at $a = 1$, $412T^*$ at $a = 1.25$ and $2 \times 10^4 T^*$ at $a = 1.5$, while the actual last sign changes are at 33, 259 and $397T^*$ (Computation 8.20). Below a_∞ the band is empty or lies below the horizon ($X_{\max} = 2.25$ at $a = 0.6$, $\tilde{\sigma}_\infty(T^*) = 2a$), which is the

other face of the confinement: a finite tree has resonances, not a band. In the band the geometric mean of the symbol is $G(\tilde{\sigma}_\infty)$, finite and smooth, so the outer factor of the window's symbol has modulus $G^{1/2}$ on average, and $|F|^2$ cannot follow $1/|\sigma_a|$ where $\sigma_a < 0$ without the envelope of $t^2\sigma_a|F|^2$ losing twice the measure of the negative set. Whether the minimizer loses it, or hides from the negative set as it hides from the zeros, cannot be read from the K -mode minimizers of Computation 7.6, the only ones computed in the band, which above a few horizons carry the edge alone (Computation 9.16); what Computation 8.20 measures on them is the envelope, which they carry correctly, and Proposition 9.7 answers the pointwise question above the band top.

Computation 8.20 (the sharp form in the band). `rh_band.py`, logs `band_a*.log` in `data/`. On the K -mode minimizers of Computation 7.6 at $a = 1.0, 1.25, 1.5$ ($K = 80, 140, 230$; $\lambda = 1.5 \times 10^{-26}$, 3.4×10^{-51} , 9.2×10^{-93} ; λ' from the dilation engine; F in closed form through spherical Bessel functions at 80 digits, reproducing the zero sums of Computation 7.6 to all digits), the local envelope of $t^2(\Psi - \lambda)|F|^2$ over the window of width $4T^*$ centred at κT^* , divided by $-\lambda'$; the fraction of the window on which the window's symbol σ_a is negative; and the part of the envelope carried by the negative set:

κ	$a = 1.0$			$a = 1.25$			$a = 1.5$		
	envelope	$\sigma_a < 0$	on it	envelope	$\sigma_a < 0$	on it	envelope	$\sigma_a < 0$	on it
3	1.708	0.049	-0.172	2.151	0.060	-0.217	3.156	0.084	-1.40
5	1.071	0.026	-0.018	1.216	0.045	-0.016	1.084	0.061	-0.025
7	1.046	0.019	-0.004	1.049	0.031	-0.008	1.027	0.048	-0.015
10	1.001	0.011	-0.000	1.012	0.019	-0.004	1.010	0.034	-0.009
15	0.987	0.008	-0.001	1.013	0.012	-0.001	0.998	0.026	-0.005
20	0.968	0.005	-0.000	0.996	0.010	-0.001	0.997	0.018	-0.003
30	0.989	0	0	0.997	0.007	-0.000	0.976	0.012	-0.002
50	1.041	0	0	0.990	0.003	-0.000	0.994	0.008	-0.001

From five horizons on the envelope is $-\lambda'$ within 2–8%, and within 1–3% from seven, at all three supports, inside the region where the symbol of the window changes sign (its last sign change is at 33, 259 and $397T^*$). The negative set occupies 2.6–6% of the heights at $5T^*$ and carries -0.02 of the envelope, and the correlation coefficient of $t^2|F|^2$ with $1/|\sigma_a|$ over the window is below 0.04 in modulus from $5T^*$ on; but these are properties of the polynomial, not of the minimizer. Above $(2K - 1)/a$, which is 3.4, 2.9, $2.4T^*$ here, the K -mode transform is the edge's alone (Computation 9.16), and it cannot show the echo structure that the minimizer has (Corollary 8.15(iii), and the FEM of Computation 8.17). What the polynomial carries correctly is the envelope, a mean, to the few per cent of $\sigma_{\text{eff}}/\tilde{\sigma}_\infty - 1$. The pointwise statement that holds for the minimizer is Proposition 9.7: above the band top $|F|^2 \propto 1/\sigma_a$ to first order in c_d/s , the first-generation echoes being the first terms of the expansion of $1/\sigma_{a,-}$, while inside the band the law is carried by the mean (Corollary 8.15(iv)). The envelope against the archimedean symbol alone, $\langle t^2(\Psi_\infty - \lambda)|F|^2 \rangle / (-\lambda')$, is 1.24, 1.35, 1.18 at $5T^*$ and 1.05, 1.02, 1.01 at $10T^*$: the prime sum at the entries, the sampling defect of Proposition 7.29, is 15–30% of the tail at five horizons and a few per cent at ten. Below five horizons the windows reach the mass of the floor (medians 2.4 – $2.7T^*$ in Computation 7.6) and the envelope is not a tail quantity. The K -mode minimizers resolve the edge to $\delta \approx e^{-2}a/K^2$, heights of order 10^3T^* , and their tails were validated on the zeros in Computation 7.6.

The band that matters is not the periodic limit's. The walks are confined to the window even when the echo set is dense: from a point p only the steps $\pm d$ with $p \pm d \in [-a, a]$ are allowed, and from most points these are a small fraction of the $2|D_a|$ steps of the lattice. The echo graph is the quasi-periodic structure cut out of $\mathbb{Z}^{\pi(e^{2a})}$ by the window, and its adjacency is a far smaller operator than the periodic limit's.

Proposition 8.21 (the confined lattice). *Let a be any support that is not an entry, S its echo set, finite or not, and C the weighted adjacency on S° , $C_{pq} = c_{|p-q|}$ for $|p-q| \in D_a$. Let $D_{\max} = \max_{p \in S^\circ} \sum_{q \sim p} C_{pq}$ be the largest weighted degree. Then C is a bounded self-adjoint operator on $\ell^2(S^\circ)$ with*

$$\rho(C) = \|C\| \leq D_{\max} \leq \sum_{d \in D_a, d \leq a} 2c_d + \sum_{d \in D_a, a < d < 2a} c_d < X_{\max} = \sum_{d \in D_a} 2c_d,$$

and $\rho(C)$ is the supremum of the top eigenvalues of the adjacencies of the finite subgraphs. In the other direction, let Σ be the full echo lattice of the window, the set of all points $\pm a + \sum_p m_p \log p \in [-a, a]$ ($m_p \in \mathbb{Z}$, p prime, $p < e^{2a}$), of which S is the component of $\pm a$ under the allowed steps, with the same adjacency; then

$$\rho(C_\Sigma) \geq \bar{\rho}(a) := \sum_{d \in D_a} 2c_d \left(1 - \frac{d}{2a}\right), \quad \bar{\rho}(a) = \frac{4e^a - 4a - 4}{a} (1 + o(1)) \quad (a \rightarrow \infty),$$

the mean weighted degree of the lattice. The formal echo expansion of Corollary 8.15(iii), $\rho = \sum_{\ell \geq 0} C^\ell b / \tilde{\sigma}_\infty^{\ell+1}$, converges in $\ell^2(S^\circ)$ at every height where $\tilde{\sigma}_\infty(t) > \rho(C)$, and the return $R = b^T (\tilde{\sigma}_\infty - C)^{-1} b$ and the effective symbol $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - R$ are real-analytic functions of t there, with $\sigma_{\text{eff}} = \tilde{\sigma}_\infty - |b|^2 / \tilde{\sigma}_\infty + O(\tilde{\sigma}_\infty^{-2})$. The band of the confined lattice is $\{t : \tilde{\sigma}_\infty(t) \leq \rho(C)\}$; it ends at $t_{\text{band}} = 2\pi e^{\rho(C)+\lambda} (1 + o(1))$.

Proof. A symmetric matrix with non-negative entries is bounded by its largest row sum (Schur's test), which is $D_{\max}$. A step $\pm d$ from p counts in the degree of p only if $p \pm d \in [-a, a]$; both directions are possible only if $d \leq a - |p| \leq a$, so an entry $d > a$ contributes at most c_d and an entry $d \leq a$ at most $2c_d$, which is the middle bound, and it is strictly below $X_{\max}$ because some entry lies in $(a, 2a)$ at every support $a \geq a_2$. For a bounded self-adjoint operator with non-negative entries the norm is the supremum of $\langle v, Cv \rangle$ over finitely supported unit vectors, which is the supremum of the top eigenvalues of the finite truncations, increasing with the truncation. For the lower bound take the points of Σ with exponent vectors in the cube $|m_p| \leq L$; by Weyl's equidistribution their projections $\pm a + \sum_p m_p \log p$ are uniformly distributed modulo any interval, so a fixed fraction of the $(2L+1)^\pi$ vectors (π the number of primes) lands in the window with projections uniformly distributed in $[-a, a]$, and all but $O(L^{\pi-1})$ of their allowed steps stay in the cube; the Rayleigh quotient of the normalized indicator of this set is the mean over it of the weighted degree, and the weighted degree of a point at p is $\sum_d c_d (\mathbf{1}_{p+d \leq a} + \mathbf{1}_{p-d \geq -a})$, whose mean over p uniform in $[-a, a]$ is $\sum_d 2c_d (1 - d/2a)$. The asymptotic form is the prime number theorem in the form $\sum_{\log m < X} \Lambda(m) m^{-1/2} (X - \log m) = 4e^{X/2} - 2X - 4 + o(e^{X/2})$. The Neumann series converges in operator norm for $\tilde{\sigma}_\infty > \|C\|$, and the resolvent is analytic in $\tilde{\sigma}_\infty$ there, hence in t , with the expansion of its first terms. $\square$

What the proposition does not do is extend Proposition 8.14 to a dense echo set: the local relations (20) were proved with the other singular points at a positive distance, and the representation with the effective symbol needs them. Beyond a_∞ the exact statement is Proposition 8.18, the echo expansion is formal, and the band of the confined lattice is where it ceases to converge. It is nevertheless the quantity to compare with the onset: the tail law of the zeros was measured from about five horizons at $a = 1.0, 1.25, 1.5$ (Computation 7.6), where the periodic band reaches 47, 412 and 2×10^4 horizons, and Computation 8.22 finds the confined band ending at one to three.

Computation 8.22 (the band of the confined lattice). `rh_lattice_band.py`, logs `lattice_band*.log` in `data/`. The echo set by breadth-first search from $\pm a$ in the exponent-vector representation, truncated to the first N points by chain length, and the top eigenvalue ρ_N of the truncated adjacency by power iteration; $D_{\max}$ over the points found; the mean degree $\bar{\rho}(a)$ of Proposition 8.21; and the band tops in units of the horizon, $t_{\text{band}}/T^* = e^{\rho-2a}$:

a	entries	ρ_{500}	ρ_{2000}	ρ_{8000}	ρ_{40000}	$\bar{\rho}(a)$	$D_{\max}$	$X_{\max}$	confined t_{band}/T^*	periodic
0.8	3	1.219 (all 24 points)				1.05	1.47	2.94	0.68	4
0.9	4	1.659	1.659	1.659		1.41	2.19	4.38	0.87	13
1.0	5	1.920	1.936	1.941		1.75	2.93	5.85	0.94	47
1.1	7	2.275	2.293	2.298		2.15	3.54	7.08	1.10	131
1.25	8	2.663	2.773	2.862	2.912	2.80	4.26	8.52	≥ 1.5	412
1.5	12	3.209	3.446	3.622	3.855	4.09	6.51	13.0	≥ 2.4	2×10^4

The chains reached by 40000 points are 13 links long at $a = 1.25$ and 7 at $a = 1.5$, where ρ_N is still rising; the mean degree $\bar{\rho}$ is a lower bound for the lattice, which the truncations approach from below at $a = 1.5$ and exceed by 10% at $a \leq 1.25$, so $\rho(C)$ is within a few per cent of $\bar{\rho}$ at these supports: the confined lattice is nearly regular. Its spectral radius is a third of the periodic band top and two thirds of the largest degree. The band of the confined lattice ends at 0.9, 1.5 and about 3.0 horizons at $a = 1.0, 1.25, 1.5$ (the last from $\bar{\rho}$), against the periodic band top at 47, 412 and 2×10^4 horizons and the last sign change of the window's symbol at 33, 259 and 397. The onset of the tail law of the zeros is five horizons at the three supports (Computation 7.6: $T\vartheta/T_{\text{eff}} = 1.03, 1.04, 1.015$ at $5T^*$) and of the sharp form five to seven (Computation 8.20): two to five times the confined band top, as it is a few horizons above the horizon for the finite trees, whose band lies below it.

The lattice's band against the edge's. The lower bound $\bar{\rho}(a) \sim 4e^a/a$ grows exponentially while $\tilde{\sigma}_{\infty}(\kappa T^*) = 2a + \log \kappa$ grows linearly, so the band of the confined lattice outruns the horizon: $e^{\bar{\rho}(a)-2a}$ is 3.0, 34 and 2041 at $a = 1.5, 2$ and 2.5 . If the onset of the sharp form followed the spectral radius, the effective height of the zeros would outrun the horizon from $a \approx 2$ on and the route of Proposition 7.32 to Conjecture 7.1 would need κ growing like $e^{\bar{\rho}(a)-2a}$. It does not follow it: the edge is coupled to the lattice through b alone, and the spectral measure of C at b sits at the bottom of the spectrum (Proposition 8.23, Computation 8.24), with its upper tail ending within a few horizons at every support up to $a = 2.5$. The spectral radius is the band of the worst coupling, not of the edge's. The decisive measurement remains a minimizer at $a = 2$ ($T^* = 343$, about 500 modes at 1500 bits), where the edge's band predicts the onset of the law at seven to ten horizons rather than five.

The edge does not see the whole lattice. It is coupled to it through b alone, the first-generation echoes, and what enters the effective symbol is the return $R = b^T(\tilde{\sigma}_{\infty} - C)^{-1}b$: a function of the spectral measure of C at b , which may have little weight near the top of the spectrum.

Proposition 8.23 (the band the edge sees). *Let C be the adjacency of the confined lattice on $\ell^2(S^\circ)$ and μ_b its spectral measure at b , $\int \varphi d\mu_b = \langle b, \varphi(C)b \rangle$, a positive measure of mass $|b|^2$ supported in $[-\rho(C), \rho(C)]$. Then for real s outside $\text{supp } \mu_b$,*

$$R(s) = \int \frac{d\mu_b(\nu)}{s - \nu}, \quad \sigma_{\text{eff}}(s) = s - R(s),$$

R is real-analytic and decreasing on $(\text{supp } \mu_b, \infty)$ with $R(s) = |b|^2/s + \langle b, Cb \rangle/s^2 + O(s^{-3})$, and σ_{eff} is increasing there. For s inside the support, $R(s \pm i0) = \text{p.v.} \int d\mu_b(\nu)/(s - \nu) \mp i\pi\mu'_b(s)$ wherever μ_b has a density: the effective symbol with limiting absorption is complex, the imaginary

part being the rate at which the edge radiates into the lattice. The moments $\int \nu^k d\mu_b = \langle b, C^k b \rangle$ count the weighted closed walks of length $k + 2$ through the edge, so the truncation of the lattice to the points within g steps of the edge reproduces the moments of μ_b of order up to $2g - 2$ exactly. The band the edge sees is $\{t : \tilde{\sigma}_\infty(t) \in \text{supp } \mu_b\}$, contained in the band of the confined lattice; it reaches the top of its own component's band (Proposition 9.26), and what ends within a few horizons is its mass, bounded by the moments.

Proof. The spectral theorem for the bounded self-adjoint C , the resolvent identity $\langle b, (s - C)^{-1} b \rangle = \int (s - \nu)^{-1} d\mu_b$ for $s \notin \sigma(C)$, and the Sokhotski–Plemelj formula for the boundary values of a Stieltjes transform. A walk of length k from a first-generation point to a first-generation point, weighted by its steps and by b at both ends, is a closed walk of length $k + 2$ through the edge, and all such walks of length $k \leq 2g - 2$ stay within g steps of the edge. $\square$

Computation 8.24 (the spectral measure at the edge). `rh_spectral_measure.py`, logs in `data/`. The spectral measure μ_b of the confined lattice at the edge's coupling, by 80 Lanczos steps from b on the lattice truncated to 150000 points (the moments of μ_b are exact to order $2g - 2$, g the depth reached: 58, 18, 9, 6, 5, 4, 4 links at $a = 1.0, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5$), checked against the dense eigenproblem on the first 6000 points. The weight of μ_b above $s = \tilde{\sigma}_\infty(\kappa T^*)$ as a fraction of $|b|^2$, the heights at which it drops below 5% and 1%, and σ_{eff}/s at five and ten horizons against the first order $1 - |b|^2/s^2$:

a	$ b ^2$	top of μ_b	weight above $\tilde{\sigma}_\infty(\kappa T^*)$				tail ends (T^*)		σ_{eff}/s at 5, 10 T^*	
			$\kappa = 1$	2	3	5	5%	1%	lattice	first order
1.0	1.82	1.95	0	0	0	0	0.70	0.78	0.844, 0.894	0.860, 0.902
1.25	2.54	2.93	0.031	0	0	0	0.94	1.30	0.817, 0.873	0.850, 0.890
1.5	4.00	3.94	0.040	0.004	0	0	0.94	1.65	0.761, 0.832	0.812, 0.858
1.75	5.37	4.84	0.059	0.010	0.007	0	1.13	2.06	0.704, 0.801	0.795, 0.841
2.0	7.07	5.52	0.047	0.017	0.013	0	1.04	4.6	0.52, 0.767	0.775, 0.822
2.25	9.13	5.83	0.063	0.029	0.026	0	1.47	3.8	0.55, 0.731	0.755, 0.803
2.5	11.2	6.52	0.056	0.028	0.020	0	1.40	4.6	0.28, 0.701	0.745, 0.791

The measure the edge sees sits at the bottom of the spectrum: its mean $\langle b, Cb \rangle / |b|^2$ is 0.09–0.25 (two first-generation points are adjacent only when their entries differ by an entry), its width $(\langle b, C^2 b \rangle / |b|^2)^{1/2}$ is 1.0–2.3, and its weight above the horizon's height $\tilde{\sigma}_\infty(T^*) = 2a$ is at most 6% at every support. Its 1% quantile, the band the edge sees, ends at 0.8, 1.3, 1.7, 2.1 horizons at $a = 1.0$ –1.75, where the depth is adequate, and at 4–5 horizons at $a = 2$ –2.5, where the truncation reaches four or five links and the extreme tail is underestimated (the top node moves up with the depth: $4.34 \rightarrow 5.52$ at $a = 2$ between 6000 and 150000 points, while the 5% quantile moves from 1.4 to $1.0 T^*$ and is stable); the top of the lattice's own spectrum is $\rho(C) \geq \bar{\rho}(a)$, at 34 and 2000 horizons for $a = 2$ and 2.5, and the edge's measure reaches the top of its own component's band (Proposition 9.26), which the truncations put at 5.52, 5.83, 6.52 for $a = 2, 2.25, 2.5$, within 2% of five horizons and still rising, with a weight they cannot resolve. What the exact moments bound is the mass: by Chebyshev with the highest exact even moment (orders 14, 14, 14, 10, 8, 6, 6), the weight above five horizons is at most $2 \times 10^{-6}, 10^{-4}, 10^{-3}, 0.8\%, 2.2\%, 4.7\%, 5.6\%$ at $a = 1.0, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5$; the Lanczos measure's vanishing weight above s at $a \geq 1.5$ is the truncation's, its top node lying below s . At five horizons σ_{eff}/s is 0.84, 0.82, 0.76, 0.70 at $a = 1.0$ –1.75 (a few per cent below the first order, the longer chains) and 0.5 at $a \geq 2$, where s is just above the resolved top of the support and the return is large: the onset there is predicted a little above five horizons, at seven to ten, where σ_{eff}/s is 0.72–0.77. So the band that the sharp form has to clear is the one the edge sees, whose mass stays within a few horizons as far as $a = 2.5$, and the prediction of Computation 8.22, drawn from the spectral radius, is withdrawn: the spectral

radius bounds the band of every coupling, and the edge's coupling sits far below it. What remains predicted for $a = 2$ is a mild recession of the onset, from five horizons to about seven to ten, and the sampling defect σ_{eff}/s growing from 0.84 to 0.7 at ten horizons.

9 The reduction to one inequality

This section states the main theorem and names the gap. The theorems of Sections 5, 8 and 9.2 reduce the Hypothesis to one inequality about one function, every other link being a theorem under the boundedness hypothesis (H_∞) at every support that is not an entry; the inequality is as hard as the Hypothesis, because it fails wherever the floor is negative, and the section gives the form in which it would have to be attacked, which is a statement about multiplicative relations among the integers below e^{2a} . It is where the paper rests.

Theorem 9.1 (the reduction). *Assume (H_∞). Let $a_Z = 0.8$ be the support to which the floor is certified positive (Theorem 1.3), and suppose that for some $c > 0$ and every support $a \geq a_Z$ that is not an entry,*

$$2|A_0(a)|^2 \leq cT^*(a)\lambda(a), \quad (25)$$

where $A_0(a)$ is the edge amplitude of the minimizer, the constant term of the amplitude $A(t)$ in the representation of its transform with the symbol of the window (Proposition 8.18). Then Conjecture 7.1 holds with this c and the Riemann Hypothesis follows (Proposition 7.2). The link is the boundary law $\lambda'(a) = -2|A_0(a)|^2$: a theorem at every support that is not an entry below $a_\infty = 0.843$ (Corollary 8.15(v)) and at almost every support in general (Theorem 9.12, under the Diophantine condition of Lemma 9.6(iii)); it is verified at every support computed (Computation 8.11).

Proof. Corollary 8.15(v) gives $\lambda' = -2|A_0|^2$ at every point of differentiability below a_∞ , with A_0 the amplitude of the representation with the effective symbol, which is the amplitude of the representation with the window's symbol by Proposition 8.18(iii); Theorem 9.12(iv) gives it at almost every support. So (25) reads $\lambda' \geq -cT^*\lambda$ almost everywhere on $[a_Z, \infty)$, which is Conjecture 7.1 with $a_0 = a_Z$, given the absolute continuity of λ : Lemma 1.4 gives continuity and monotonicity only, and absolute continuity is Proposition 5.6 under (H_{BV}). Without (H_{BV}) the same conclusion would follow from a lower bound on the Dini derivative: Proposition 5.3 with Proposition 8.3 gives $aD_-\lambda(a) \geq -\mathcal{D}(f) = -2a|A_0(a)|^2$ at every support that is not an entry, and a continuous non-increasing function whose lower Dini derivative is bounded below by $-M$ off a countable set is Lipschitz with constant M ; what this needs is a bound on $|A_0(a)|$ locally uniform in a , which the source norms of Corollary 7.24 should supply from $\|f\|_2 = 1$ alone. That is Lemma 9.14 and Proposition 9.15 in Section 9.2, under the local boundedness of the height $T_0(a)$ there. $\square$

Three remarks on the inequality.

It is one inequality, and everything else is proved. The chain is: boundary law (a theorem at every support, Theorem 9.12) $\Rightarrow$ (25) is the conjecture; the analytic tail law (Corollary 8.15(v) and Theorem 9.12(iii), a theorem at every support, no zeros involved) makes $2|A_0|^2$ the limit of $\pi T \cdot \frac{1}{2\pi} \int_{|t|>T} |F|^2(\Psi - \lambda)$, so (25) says that the energy of the minimizer against the shifted symbol above κ horizons is at most $c\lambda/(\pi\kappa)$ once the law has set in; and the zeros enter nowhere. Under RH this energy is the zero sum above κT^* (Proposition 7.29), which is at most λ because every term is a square: that is the content of Propositions 7.27 and 7.32, and it is where RH is used. Without RH the zero sum is $\sum_\rho F(\gamma_\rho)F(-\gamma_\rho)$ with the pairs off the line contributing $2\operatorname{Re} F(\gamma)\overline{F(\bar{\gamma})}$, of either sign, and the bound by λ is lost.

It contains the Hypothesis. If RH fails, the floor is negative from some support on (the odd-sector Weil criterion on windows, Section 1), while $2|A_0|^2 \geq 0$; so (25) fails there. A proof of (25) must therefore use the minimality of f in a way that excludes a negative floor, and no argument that treats λ as a given number can succeed. The quantity on the left is a boundary trace and the one on the right an energy; for the Dirichlet Laplacian the corresponding inequality, $u'(a)^2 \leq C\lambda$ for the ground state, is the Rellich–Pohozaev identity, and it is unconditional because the identity’s remainder is a sum of squares. The dilation identity of Section 5 is the Pohozaev identity of the Weil form, $a\lambda' = D_\infty(f) + D_P(f) + 2P^2 + 2PM$; what is needed is a sign for its remainder against $T^*\lambda$, and the obstruction is the prime part D_P , whose sign is not fixed.

A bootstrap in height does not replace it. The zeros are known to lie on the line up to a height H (3×10^{12} at the time of writing), and in outline the standard reasoning runs: the explicit formula gives, unconditionally, $\lambda = \sum_{|\gamma| \leq H} |F(\gamma)|^2 + \sum_{|\gamma| > H} F(\gamma)F(-\gamma)$, the second sum is of the order of e^a times the tail energy above H (F is evaluated in the strip $|\text{Im}| \leq \frac{1}{2}$, where $|F(t + iy)|$ is the transform of $f(u)e^{-yu}$), that is $e^a|A_0|^2/H$ by the analytic tail law; so $\lambda > 0$ follows from the first sum exceeding this, which with $2|A_0|^2 = -\lambda' \approx 3.8T^*\lambda$ holds when $e^aT^* \ll H$, that is $a \lesssim \frac{1}{3} \log(H/2\pi) \approx 9$. Positivity of the floor up to $a \approx 9$ is the known consequence of the verified zeros; the inequality (25) is needed for all a , and the gain per step of any such bootstrap is $O(1/T^*)$, which does not sum (the memo `RH_TOP3_PROOF_ARCHITECTURE.md` in [24], Section 3).

The prime-side form. At a support with a finite tree the amplitude is explicit in the Wiener–Hopf data: with φ_1 the solution of the eigen-equation with unit polar source and λ as a parameter, $f = P\varphi_1$, the normalization fixes $P^2 = 1/\|\varphi_1\|^2$, the secular equation $\langle \varphi_1, s \rangle = 1$ fixes $\lambda(a)$ as the lowest root, and $A_0 = P A_0^{(1)}(\lambda, a)$ with $A_0^{(1)}$ the amplitude of Corollary 7.24 for the unit source. So (25) is $2|A_0^{(1)}|^2 \leq cT^*\lambda\|\varphi_1\|^2$ at the root of the secular equation, a statement in which only the archimedean symbol and the entries below e^{2a} appear, through σ_{eff} and its factors. Beyond a_∞ the same holds with the window’s symbol, and the only structure of the entries that enters the effective symbol is the spectral measure of the echo lattice at the edge (Proposition 8.23), whose moments are

$$\int \nu^k d\mu_b(\nu) = \langle b, C^k b \rangle = \sum \prod_{i=1}^{k+2} \frac{\Lambda(m_i)}{\sqrt{m_i}}, \quad (26)$$

the sum over the closed walks of length $k+2$ through the edge: sequences of prime powers $m_i < e^{2a}$ and signs ε_i with $\prod m_i^{\varepsilon_i} = 1$ whose partial products stay in the window $[1, e^{2a}]$ when started at its end. The onset of the tail law, and with it the constant in (25), is set by the upper tail of μ_b (Computation 8.24); its uniformity in a is the statement that these multiplicative relations, weighted by $\Lambda(m)m^{-1/2}$, do not accumulate at the top of the spectrum, and Computation 9.2 measures their growth. This is the point at which the problem has become arithmetic, and it is where the paper stops.

Computation 9.2 (the moments of the edge’s measure). `rh_spectral_measure.py (moments)`, `log spectral_moments.log` in `data/`. The moments $m_k = \langle b, C^k b \rangle / |b|^2$ of the edge’s measure from the Gauss quadrature of Computation 8.24, exact for $k \leq 2g - 2$ with g the depth (58, 18, 9, 6, 5, 4, 4 links at $a = 1.0, \dots, 2.5$); m_1 checked against the direct sum over the ordered pairs of entries whose ratio is an entry ($4/2, 8/2, 8/4, 9/3, 25/5, 27/3, 27/9, \dots$), $m_1|b|^2 = 2 \sum c_m c_{m'} c_{m/m'}$, to all digits; the height of five horizons $\tilde{\sigma}_\infty(5T^*) = 2a + \log 5$; and the Chebyshev bound m_8/s^8 on the weight of μ_b above $s = 2a + \log 5$:

a	$ b ^2$	m_1	m_2	$m_4^{1/4}$	$m_6^{1/6}$	$m_8^{1/8}$	top of μ_b	$2a$	$2a + \log 5$	m_8/s^8
1.0	1.82	0.091	0.94	1.12	1.22	1.29	1.95	2.0	3.61	3×10^{-4}
1.25	2.54	0.247	1.65	1.57	1.75	1.89	2.93	2.5	4.11	0.002
1.5	4.00	0.188	2.65	1.98	2.23	2.41	3.94	3.0	4.61	0.006
1.75	5.37	0.255	3.85	2.41	2.73	2.97	4.84	3.5	5.11	0.013
2.0	7.07	0.236	5.15	2.81	3.19	3.49	5.52	4.0	5.61	0.022
2.25	9.13	0.199	6.86	3.24	3.68	(3.99)	5.83	4.5	6.11	(0.033)
2.5	11.2	0.198	8.71	3.66	4.09	(4.40)	6.52	5.0	6.61	(0.039)

(Parentheses: beyond the exact order at that depth.) The first moment is small because a ratio of two prime powers is rarely a prime power: m_1 is 0.1–0.25 at every support, while the second, the mean weighted degree of the first-generation points, grows like e^a . The even roots $m_k^{1/k}$ increase slowly towards the top of the support and sit, at $k = 8$, a constant 2.1–2.3 below the height of five horizons at every support, a factor nine in t ; the Chebyshev bound with the highest exact even moment puts the weight of μ_b above five horizons below 2×10^{-6} at $a = 1$, 10^{-3} at 1.5, 2.2% at 2 and 5.6% at 2.5, from the moments alone (Computation 8.24, Proposition 9.26). What the uniformity of the onset asks is that this persist: that the closed walks through the edge of length k , the multiplicative relations among the integers below e^{2a} with partial products in the window, weighted by $\prod \Lambda(m_i) m_i^{-1/2}$, have $m_k^{1/k} \leq 2a + \log \kappa - \delta$ for some κ, δ and a k growing with a . Since m_k grows exponentially in a at fixed k while $(2a + \log \kappa)^k$ grows polynomially, the order k must grow with a , and the statement is about long relations, of length comparable to a .

9.1 The plane

The third leg of the triangulation of Section 7.1, the zeros, was taken on the line. Unconditionally the zeros live in the strip, and the floor is a quadratic functional on the plane: with F odd and entire, $F(z)F(-z) = -F(z)^2 = 4S(z)^2$ for every complex z , $S(z) = \int_0^a f(u) \sin(zu) du$, and the explicit formula reads, for every odd f supported in $[-a, a]$ and without any hypothesis,

$$Q(f) = \sum_{\rho} q(\gamma_{\rho}), \quad q(z) = 4S(z)^2, \quad \rho = \frac{1}{2} + i\gamma_{\rho}, \quad (27)$$

the sum over all nontrivial zeros with multiplicity, $\gamma_{\rho} = \gamma_0 - i\delta$ for a zero $\rho = \frac{1}{2} + \delta + i\gamma_0$ off the line. On the line $q \geq 0$. Off it, q is complex; a zero and its three companions $\rho, 1 - \rho, \bar{\rho}, 1 - \bar{\rho}$, that is $\pm\gamma_0 \pm i\delta$, contribute $4 \operatorname{Re} q(\gamma_0 - i\delta) = 16 \operatorname{Re} S(\gamma_0 - i\delta)^2$, of either sign; and to second order in the displacement, $q(x + iy) = 4(S(x) + iyS'(x))^2 + O(y^3)$ has real part $4S(x)^2 - 4y^2 S'(x)^2$, negative wherever $y|S'| > |S|$, which is near the nulls of S , where the zeros are. The Hypothesis is the statement that all the terms of (27) are squares; the following theorem says what the floor does when they are not.

Theorem 9.3 (the floor reads the plane). *Let $\delta_{\max} = \sup_{\rho} (\operatorname{Re} \rho - \frac{1}{2}) \in [0, \frac{1}{2})$, and write $\log^+(-\lambda) = 0$ where $\lambda \geq 0$. Then:*

- (i) $\lambda(a) \geq 0$ for all a if and only if $\delta_{\max} = 0$, that is, if and only if the Riemann Hypothesis holds (Weil's criterion on the odd sector [3]).
- (ii) Assume (H_{∞}) and that the total variation of the minimizer is at most exponential in a . Then $\limsup_{a \rightarrow \infty} \log^+(-\lambda(a))/2a \leq \delta_{\max}$.
- (iii) If the supremum is attained, at a zero $\rho_0 = \frac{1}{2} + \delta_{\max} + i\gamma_0$ with $\delta_{\max} > 0$ (as it is when only finitely many zeros lie off the line), then $\lambda(a) \leq -e^{2\delta_{\max}a}/(2\delta_{\max})(1 + o(1))$, so that $\lim \log(-\lambda(a))/2a = \delta_{\max}$.

Proof. (i) is Weil's criterion: on the line every term of (27) is a square, and an odd test function with $Q < 0$ exists when a zero is off the line; the odd sector suffices because the test functions of the criterion can be taken odd (a zero and its mirror $1 - \bar{\rho}$ have the same $|\gamma|$).

(ii) Let $V(a) \leq e^{Ca}$ be the total variation of f , so that $|F(x + iy)| \leq V(a)e^{a|y|/|x|}$, and $\int |F(x + iy)|^2 dx \leq 2\pi e^{2a|y|}$ (Plancherel–Pólya). By Gallagher's inequality for sums over the zeros of a function of exponential type a , $\sum_{|\gamma_\rho| \leq T} |F(\gamma_\rho)|^2 \leq C(a + \log T) \sup_{|y| \leq \delta_{\max}} \int |F(x + iy)|^2 dx \leq C(a + \log T)e^{2\delta_{\max}a}$, and the zeros above $T = e^{(2C+1)a}$ contribute at most $\sum_{|\gamma| > T} V^2 e^{2\delta_{\max}a} / \gamma^2 \leq e^{2\delta_{\max}a} e^{2Ca} \log T / T \leq e^{2\delta_{\max}a}$. So $\lambda = Q(f) = \sum_\rho q(\gamma_\rho) \geq -Ca^2 e^{2\delta_{\max}a}$.

(iii) Take $f(u) = \sinh(\delta_{\max}u) \cos(\gamma_0 u) \phi(u/a)$ on $[-a, a]$, odd, with ϕ a fixed smooth even cutoff, $\phi = 1$ on $[-1 + \eta, 1 - \eta]$, and write $\delta_0 = \delta_{\max}$. The term of the quartet of ρ_0 is $-4 \operatorname{Re} w^2$ with $w = F(\gamma_0 - i\delta_0) = \int f(u) e^{\delta_0 u} e^{i\gamma_0 u} du$, whose real part is $\int \sinh^2(\delta_0 u) \cos^2(\gamma_0 u) \phi(u/a) du = c_\phi e^{2\delta_0 a} (1 + o(1))$ with $c_\phi = \frac{1}{2} \int_0^\infty e^{-2\delta_0 v} \phi(1 - v/a) dv \cdot (1 + o(1))$ and imaginary part smaller by $O(\delta_0/\gamma_0)$; $\|f\|^2 = c_\phi e^{2\delta_0 a} (1 + o(1))$ with the same c_ϕ ($\int \sinh^2 \cos^2 \phi$ as well). Every other zero has $\delta_j \leq \delta_0$. Those within distance 1 of $\pm\gamma_0$ number at most $C \log \gamma_0$, a fixed number as $a \rightarrow \infty$, and $|F(\gamma_j - i\delta_j)| \leq \int |f| e^{\delta_j u} \leq c'_\phi e^{(\delta_0 + \delta_j)a} \cdot (a|\gamma_j - \gamma_0|)^{-N}$ for those at distance at least $1/a$, by the smoothness of the cutoff, while a zero at distance less than $1/a$ of γ_0 with $\delta_j = \delta_0$ has $w_j = w(1 + O(1/a))$ and its term has the sign of the main one; the zeros at distance between 1 and T contribute at most $C(a + \log T)e^{4\delta_0 a} a^{-N}$ by the same decay and the density of zeros, and those above $T = e^a$ at most $e^{(1+2\delta_0+2\delta_0)a} \log T / T$ by $|F(x + iy)| \leq V e^{a|y|/|x|}$ with $V \leq C e^{\delta_0 a} \gamma_0 a$. The zeros on the line contribute at most $\sum_\gamma |F(\gamma)|^2 \leq C(a + \log T) \cdot 2\pi \|f\|^2$ below T (Gallagher) and the same tail bound above. Dividing $Q(f)$ by $\|f\|^2$: $\lambda(a) \leq -4c_\phi e^{2\delta_0 a} (1 + o(1)) + O(a^2)$, and letting $\eta \rightarrow 0$, $c_\phi \rightarrow 1/(8\delta_0)$. $\square$

Corollary 9.4 (the edge reads the plane). *Assume (H_∞) ; the boundary law $\lambda' = -2|A_0|^2$ holds at almost every support (Theorem 9.12). If the Hypothesis fails with the supremum attained, the edge amplitude of the minimizer grows exponentially, $\lim_{\frac{1}{2a}} \log |A_0(a)|^2 = \delta_{\max}$; if it holds, together with Conjecture 7.1, it decays faster than any exponential, $|A_0(a)|^2 \leq \frac{c}{2} T^*(a) \lambda(a) \leq \frac{c}{2} T^*(a) \lambda(a_Z) e^{-\frac{c}{2}(T^*(a) - T^*(a_Z))}$. There is no third behaviour: either the edge of the minimizer reads the distance of the zeros from the line, or it reads the horizon.*

Proof. $-\lambda' = 2|A_0|^2$ with $\lambda = -e^{2\delta_{\max}a + o(a)}$ by Theorem 9.3(ii), (iii) gives the first statement, since $2 \int_{a_1}^a |A_0|^2 = \lambda(a_1) - \lambda(a)$; the second is Theorem 9.1 and Proposition 7.2. $\square$

Three consequences for the program. First, every unconditional theorem of this paper holds on both branches, since none of them uses the sign of the floor: the Wiener–Hopf representation, the edge law with its constants ($\beta = -\gamma - \log(2\pi a) - \lambda$ grows like $e^{2\delta_{\max}a}$ on the negative branch, so the profile is flat down to $\delta \approx e^{-|\lambda|}$ and the minimizer approaches its edge with what is, at every visible scale, a jump: the function $\sinh(\delta_0 u) \cos(\gamma_0 u)$ that tracks the quartet), the boundary law and the analytic tail law. This is why they cannot decide the Hypothesis by themselves, and why the inequality (25) is exactly the statement that the negative branch does not occur. Second, a proof must find a quantity that the two branches force apart *before* the floor changes sign, at supports where $\lambda > 0$ still but the quartet already shapes the minimizer; the natural candidate is the quadratic functional on the plane itself, whose second-order response to a displacement of the zeros is $-16y^2 \sum_{\gamma > 0} S'(\gamma)^2$ at the zeros below the horizon, where the minimizer's transform vanishes with a large slope: Computation 9.5 measures this stiffness and finds the floor annulled by displacements of the first zeros that shrink like $\lambda^{1/2}$, that is like $e^{-cT^*/2}$. Third, the triangulation itself: the floor is the matrices (an upper bound at every truncation), the symbol (exact), and the zeros in the plane (exact); its agreement on the computed zeros tests the identity (27), not the

Hypothesis, since those zeros are verified to lie on the line; what the plane adds is the rate, which turns the Hypothesis into a statement about one number, the exponential rate of the negative part of the floor, that the unconditional structure of the minimizer carries to its edge.

Computation 9.5 (the floor on the plane). `rh_plane.py`, `log_plane.log` in `data/`. On the K -mode minimizers at $a = 0.6, 0.8, 1.0$ (Computation 7.6; F at complex heights by the Bessel series and the recurrence at 60 digits), with the first 6000 zeros. (i) *The stiffness*. A quartet at $\pm\gamma \pm iy$ in place of the pair $\pm\gamma$ changes the floor by $-16y^2 S'(\gamma)^2 + O(y^4)$ at fixed f ; the stiffness $\kappa_2 = \sum_{\gamma>0} 16S'(\gamma)^2$ and the displacement $y_c = (\lambda/\kappa_2)^{1/2}$ at which a uniform displacement of the zeros annuls the floor to second order:

a	λ	κ_2	κ_2/λ	y_c	share of κ_2 below T^*
0.6	5.97×10^{-7}	4.83×10^{-3}	8.1×10^3	1.1×10^{-2}	0.986 (one zero)
0.8	1.57×10^{-14}	2.05×10^{-3}	1.3×10^{11}	2.8×10^{-6}	1.000
1.0	1.48×10^{-26}	1.23×10^{-3}	8.3×10^{22}	3.5×10^{-12}	1.000

The stiffness is of order 10^{-3} and sits entirely on the zeros below the horizon, where the minimizer's transform vanishes with a slope of order one; so y_c shrinks like $\lambda^{1/2}$, that is like $e^{-cT^*/2}$: the floor at support a is a certificate that the zeros below the horizon lie within $y_c(a)$ of the line, and the certificate sharpens doubly exponentially with the support, which is the resolution e^{-cT^*} of Section 7 read on the plane. (ii) *The growth off the line*. The ratio $r(y) = \langle |q(x + iy)| \rangle / \langle q(x) \rangle$ over windows of one horizon, against the growth $\cosh(2ya)$ of the edge's cosine of frequency a :

a	window	$y = 0.05$	0.1	0.2	0.3	0.45
0.6	$\cosh(2ya)$	1.0018	1.0072	1.0289	1.0655	1.1494
	$[T^*, 2T^*]$	1.0037	1.0150	1.0602	1.1363	1.3115
	$[5T^*, 6T^*]$	1.0021	1.0084	1.0338	1.0766	1.1748
	$[10T^*, 11T^*]$	1.0018	1.0073	1.0294	1.0666	1.1520
0.8	$\cosh(2ya)$	1.0032	1.0128	1.0516	1.1174	1.2706
	$[T^*, 2T^*]$	1.0040	1.0161	1.0649	1.1482	1.3450
	$[5T^*, 6T^*]$	1.0032	1.0127	1.0512	1.1165	1.2687
	$[10T^*, 11T^*]$	1.0032	1.0129	1.0518	1.1179	1.2717
1.0	$\cosh(2ya)$	1.0050	1.0201	1.0811	1.1855	1.4331
	$[T^*, 2T^*]$	1.0026	1.0105	1.0423	1.0965	1.2245
	$[5T^*, 6T^*]$	1.0050	1.0200	1.0809	1.1852	1.4326
	$[10T^*, 11T^*]$	1.0051	1.0205	1.0827	1.1893	1.4422

Above the horizon the growth of the transform into the strip is the edge's to three or four digits at every support and every displacement up to 0.45: the plane reads the edge. In the mass region it departs from it, upward at $a = 0.6, 0.8$ (the transform hides from the zeros there, with nulls denser than the edge's cosine, so $|S'|$ is large where $|S|$ is small) and downward at $a = 1.0$ (the frequency content of $[T^*, 2T^*]$ then comes from the interior of the window). (iii) *The sign*. The fraction of heights at which $\operatorname{Re} q(x + iy) < 0$ is 0.02–0.05 at $y = 0.1$, 0.07–0.15 at 0.2, 0.12–0.19 at 0.3 and 0.19–0.27 at 0.45, nearly independent of the height and of the support: a zero displaced by y lowers the floor at that fraction of the positions, around the nulls of S , and raises it elsewhere; the minimizer on the negative branch of Theorem 9.3 is the function that moves its nulls to make the displaced zeros' terms negative.

9.2 The boundary law at every support

Theorem 9.1 carries one link that is a theorem only below a_∞ : the boundary law $\lambda' = -2|A_0|^2$, which Corollary 8.15(v) proves from the edge law, and the edge law from the representation of the

transform with a symbol whose factors have smooth kernels away from the origin (Lemma 7.22). Beyond a_∞ the exact representation is the one with the window's symbol (Proposition 8.18), whose factors have kernels singular on the lattice generated by the entries, dense in the window. This section proves the edge law, and with it the boundary law and the analytic tail law, at every support that is not an entry and satisfies a Diophantine condition on $2a$ relative to the lattice of the entries, which holds at almost every support and at every resonant one (Lemma 9.6(iii)). Two facts carry the proof. Above its periodic band the window's symbol has an absolutely convergent expansion in the frequencies of the lattice, with coefficients indexed by the walks of the echo tree and summable with their lengths (Proposition 9.7); and a walk of k steps cannot return to within $\frac{1}{2}e^{-2ak}$ of its start, because $\log(M/M') \neq 0$ for integers $M \neq M'$ below e^{2ak} is at least $1/(2\min(M, M'))$ (Lemma 9.6). So the echoes that accumulate at an edge are long chains, and their total weight within ε of the edge is a power of ε . The mirror chains, the walks from the far edge across the window, are controlled in the same way at every support, the chain that lands exactly on the near edge at a resonant support included; they are the leading correction to the edge amplitude $A_a(t)$ of the window's representation, and Computation 9.16 finds them in the data at their predicted frequencies.

Lemma 9.6 (the lattice of the entries). *Let Λ_a be the additive group generated by D_a , that is $\Lambda_a = \{\log(M/M') : M, M' \in \mathbb{N} \text{ with all prime factors below } e^{2a}\}$, dense in $\mathbb{R}$ for $a > \frac{1}{2}\log 3$. A word of length k is a sequence $w = (\varepsilon_1 d_1, \dots, \varepsilon_k d_k)$ with $\varepsilon_i = \pm 1$, $d_i \in D_a$; its point is $u_w = \sum_i \varepsilon_i d_i \in \Lambda_a$ and its weight $c_w = \prod_i c_{d_i}$, so that $\sum_{|w|=k} c_w = X^k$ with $X = \sum_{d \in D_a} 2c_d$ the top of the periodic band (Lemma 8.19).*

- (i) *If $u_w \neq 0$ then $|u_w| > \frac{1}{2}e^{-2a|w|}$.*
- (ii) *Call a resonant if $2a \in \Lambda_a$; the resonant supports are countable. At a non-resonant support, $k_a(\varepsilon) = \min\{|w| : |u_w - 2a| < \varepsilon\}$ is finite for every $\varepsilon > 0$ and tends to infinity as $\varepsilon \rightarrow 0$, and $d_a = \min_{|w| \leq 2} |u_w - 2a| > 0$. At a resonant support $2a = u_{w_0}$ for a word with $|w_0| \geq 2$, and $|u_w - 2a| > \frac{1}{2}e^{-2a(|w|+|w_0|)}$ for every word with $u_w \neq 2a$.*
- (iii) *Say that a satisfies the Diophantine condition (D_a) if there are $c_D > 0$ and $C_D \geq 0$ with*

$$|u_w - 2a| \geq c_D e^{-C_D|w|} \quad \text{for every word } w \text{ with } u_w \neq 2a.$$

Every resonant support satisfies (D_a) with $c_D = \frac{1}{2}e^{-2a|w_0|}$, $C_D = 2a$, by (ii). For every $C_D > \log(2|D_a|)$ the set of supports in an interval between consecutive entries that violate (D_a) with that C_D (for every c_D) has measure zero: so almost every support satisfies (D_a) , and the exceptional set is contained in a set of measure zero that is the union, over the intervals between entries, of the supports $2a$ approximable to within $e^{-C_D k}$ by lattice points of word length k for infinitely many k .

Proof. (i) $u_w = \log(M/M')$ with $M = \prod_{\varepsilon_i=1} m_i$ and $M' = \prod_{\varepsilon_i=-1} m_i$ ($d_i = \log m_i$), positive integers below $e^{2a|w|}$; if $u_w \neq 0$ then $M \neq M'$ and $|u_w| \geq \log(1+1/\min(M, M')) \geq 1/(2\min(M, M')) > \frac{1}{2}e^{-2a|w|}$. (ii) Λ_a is countable. There are finitely many words of each length, so at a non-resonant support $\min_{|w| \leq k} |u_w - 2a| > 0$ for every k , which is $k_a(\varepsilon) \rightarrow \infty$ and $d_a > 0$, while $k_a(\varepsilon) < \infty$ because Λ_a is dense. A single step is shorter than $2a$, the entries being below e^{2a} , so a word with $u_{w_0} = 2a$ has length at least 2; and for $u_w \neq 2a = u_{w_0}$ the concatenation of w with the reverse of w_0 is a word of length $|w| + |w_0|$ with the point $u_w - 2a \neq 0$, to which (i) applies. (iii) On an interval I between consecutive entries $|D_a| =: n$ is constant and the words of length k number at most $(2n)^k$. The set $B_k = \{a \in I : |u_w - 2a| < e^{-C_D k} \text{ for some } |w| = k \text{ with } u_w \neq 2a\}$ has

measure at most $(2n)^k \cdot e^{-C_D k}$, and $\sum_k (2ne^{-C_D})^k < \infty$ for $C_D > \log(2n)$; by the Borel–Cantelli lemma almost every $a \in I$ lies in finitely many B_k , and for such a the finitely many exceptions are handled by a smaller c_D , the distances $|u_w - 2a|$ with $u_w \neq 2a$ and $|w| \leq k_0$ being finitely many positive numbers. The resonant case is (ii). $\square$

Proposition 9.7 (the window’s factors above the band). *Let $s > X$ and $\sigma_s(\tau) = s - \sum_{d \in D_a} 2c_d \cos(\tau d)$, the window’s symbol at the frozen height s . Then:*

(i) $\log \sigma_s(\tau) = \log s + \sum_{u \in \Lambda_a} \beta_u(s) e^{i\tau u}$, absolutely convergent, with

$$\beta_u(s) = - \sum_{k \geq 1} \frac{1}{k} \sum_{|w|=k, u_w=u} \frac{c_w}{s^k} \leq 0, \quad \beta_{-u} = \beta_u, \quad \sum_u |\beta_u(s)| = -\log(1 - X/s),$$

and $\beta_0(s) = \log G(s) - \log s$, G the geometric mean of Lemma 8.19.

(ii) $\sigma_{s,\pm}(\tau) = G(s)^{1/2} \exp(\sum_{\pm u > 0} \beta_u(s) e^{i\tau u})$ are the Wiener–Hopf factors of σ_s : boundary values of bounded, zero-free analytic functions in the upper and lower half-planes with bounded inverses, $\sigma_{s,+}\sigma_{s,-} = \sigma_s$ on the line, $\sigma_{s,-} = \overline{\sigma_{s,+}}$, and $\arg \sigma_{s,+}(\tau) = \sum_{u > 0} \beta_u(s) \sin(\tau u)$, an odd function vanishing at 0.

(iii) $1/\sigma_{s,+} = G(s)^{-1/2} \sum_{u \geq 0} \rho_u(s) e^{i\tau u}$ with $\rho_0 = 1$, $\rho_u(s) \geq 0$, and $\sum_{u > 0} \rho_u(s) \leq (1 - X/s)^{-1} - 1$, with $\rho_u(s) = O(s^{-k(u)})$, $k(u)$ the least length of a word with $u_w = u$; the terms of the expansion that arise from words of total length k have total weight at most $(X/s)^k$, hence

$$\sum_{0 < u < \varepsilon} \rho_u(s) \leq \frac{(2\varepsilon)^{\alpha(s)}}{1 - X/s}, \quad \alpha(s) = \frac{\log(s/X)}{2a},$$

and near $2a$: at a non-resonant support $\sum_{|u-2a| < \varepsilon} \rho_u(s) \leq (X/s)^{k_a(\varepsilon)}/(1 - X/s)$, which is $O(s^{-3})$ for $\varepsilon < d_a$ and $O(s^{-N})$ for every N as $\varepsilon \rightarrow 0$, while at a resonant support, $2a = u_{w_0}$, the words landing exactly on $2a$ have total weight $\rho_{2a}(s) \leq (X/s)^{|w_0|}/(1 - X/s) = O(s^{-2})$ and the others within ε at most $(s/X)^{|w_0|} (2\varepsilon)^{\alpha(s)}/(1 - X/s)$ (Lemma 9.6(ii)). The same holds for $1/\sigma_{s,-}$ with $u \leq 0$.

(iv) $(s/G(s))^{1/2} = 1 + |b|^2/(2s^2) + O(s^{-4})$ with $|b|^2 = \sum_{d \in D_a} c_d^2$, and $\rho_{\pm d}(s) = c_d/s + O(s^{-2})$ for $d \in D_a$: the first terms of the expansion are the first-generation echoes of Corollary 8.15, and the correction to the modulus of the factor is that of the effective symbol, $\tilde{\sigma}_{\text{eff}}^{-1/2} = \tilde{\sigma}_{\infty}^{-1/2}(1 + |b|^2/(2\tilde{\sigma}_{\infty}^2) + \dots)$.

Proof. (i) With $x_s(\tau) = \sum_d (c_d/s)(e^{i\tau d} + e^{-i\tau d})$, $|x_s| \leq X/s < 1$, so $\log \sigma_s = \log s + \log(1 - x_s) = \log s - \sum_{k \geq 1} x_s^k/k$, and $x_s^k = \sum_{|w|=k} c_w s^{-k} e^{i\tau u_w}$; the double series converges absolutely, with sum of moduli $\sum_k (X/s)^k/k = -\log(1 - X/s)$, and regrouping by u gives the expansion, the symmetry from $w \mapsto -w$, and the sign. Its constant term is the mean of $\log \sigma_s$ over τ , which is $U_\nu(s) = \log G(s)$ by Lemma 8.19. (ii) For $\text{Im } z \geq 0$ and $u > 0$, $|e^{izu}| \leq 1$, so $B_+(z) = \sum_{u > 0} \beta_u(s) e^{izu}$ converges absolutely and uniformly on the closed upper half-plane, is analytic inside and continuous up to the line, with $|B_+| \leq -\log(1 - X/s)$; hence $\sigma_{s,+} = G^{1/2} e^{B_+}$ is analytic, bounded, zero-free, with $|\sigma_{s,+}| \geq G^{1/2}(1 - X/s)$; likewise $\sigma_{s,-} = G^{1/2} e^{B_-}$ below. On the line $B_+ + B_- = \sum_{u \neq 0} \beta_u e^{i\tau u} = \log \sigma_s - \log G$, so $\sigma_{s,+}\sigma_{s,-} = \sigma_s$, and $B_- = \overline{B_+}$ since $\beta_{-u} = \beta_u$ is real; the argument is $\text{Im } B_+$. (iii) $1/\sigma_{s,+} = G^{-1/2} e^{-B_+} = G^{-1/2} \sum_{n \geq 0} (-B_+)^n/n!$, where $-B_+ = \sum_{u > 0} |\beta_u| e^{i\tau u}$ has nonnegative coefficients; so the regrouped coefficients ρ_u are nonnegative, $\rho_0 = 1$ (no product of frequencies

$u > 0$ vanishes), and $\sum_{u>0} \rho_u = e^{\sum_{u>0} |\beta_u|} - 1 \leq (1 - X/s)^{-1} - 1$. Mark the length of the words by a variable z : replacing c_w by $c_w z^{|w|}$ throughout, every coefficient of the resulting power series in z is bounded by the corresponding coefficient of $\exp(\sum_k (Xz/s)^k/k) = (1 - Xz/s)^{-1}$, which is $(X/s)^k$ at z^k . A term contributing to ρ_u is a product $\beta_{u_1} \cdots \beta_{u_n}$ with $u_1 + \cdots + u_n = u$, each β_{u_i} a sum over words w_i with $u_{w_i} = u_i$, and the concatenation $w_1 \cdots w_n$ is a word of length $k = \sum |w_i|$ with $u_w = u$; if $0 < u < \varepsilon$ then $k > \log(1/2\varepsilon)/(2a)$ by Lemma 9.6(i), and $\sum_{k>k_\varepsilon} (X/s)^k \leq (X/s)^{k_\varepsilon}/(1 - X/s) = (2\varepsilon)^{\alpha(s)}/(1 - X/s)$. Near $2a$: at a non-resonant support a word within ε of $2a$ has length at least $k_a(\varepsilon)$, which is at least 3 for $\varepsilon < d_a$; at a resonant one, Lemma 9.6(ii) gives $|w| > \log(1/2\varepsilon)/2a - |w_0|$ for the words with $0 < |u_w - 2a| < \varepsilon$; the same geometric sums give the bounds. (iv) is Lemma 8.19, $G = s - |b|^2/s + O(s^{-3})$, and the first terms of the series: $\beta_{\pm d} = -c_d/s + O(s^{-2})$ (the one-letter words; the two-letter words with $u_w = d$ are $O(s^{-2})$), so $\rho_{\pm d} = |\beta_{\pm d}| + O(s^{-2})$. $\square$

In the frozen picture the kernel of $1/\sigma_{s,+}$ is the discrete measure $G^{-1/2} \sum_{u \geq 0} \rho_u \delta_u$ on the positive part of the lattice: the chain expansion of Proposition 8.18(iii), now at every support and with the weights of all the chains, summable above the band. Its mass near the origin is a power of the distance, and its mass near $2a$ is $O(s^{-3})$ at every support, the resonant ones putting an $O(s^{-2})$ atom at $2a$ itself. In the actual symbol $s = \tilde{\sigma}_\infty(t)$ is slowly varying, the band top $t_{\text{band}} = 2\pi e^{X+\lambda}$ is a fixed height, and the edge law is a statement about $t \rightarrow \infty$, so every support is eventually above its band; the following theorem turns this into the edge law by the method of Corollary 7.24 and Proposition 8.4.

The proof of the theorem below is written so that every estimate is a lemma with its constants. Three lemmas carry the analytic part: the kernel of a symbol with finitely many derivatives, the Hilbert transform of a modulated symbol, and the regularity of the frozen coefficients β_u along the archimedean symbol. Throughout, $\check{g}(\eta) = \frac{1}{2\pi} \int g(t) e^{-it\eta} dt$ is the inverse transform, taken in the sense of tempered distributions and, on $\eta \neq 0$, as the oscillatory integral obtained by integrating by parts.

Lemma 9.8 (the kernel of a symbol with finitely many derivatives). *Let $p \in \{0, 2\}$, $j \geq 0$, and let $g \in C^{j+2}(\mathbb{R})$ satisfy $|g^{(i)}(t)| \leq M(1 + |t|)^{-p-i}$ for $0 \leq i \leq j+2$. Then $\check{g}$ is a C^j function on $\mathbb{R} \setminus \{0\}$, and for $0 \leq m \leq j$ and $\eta \neq 0$:*

- (i) if $p = 0$: $|\check{g}^{(m)}(\eta)| \leq c_m M |\eta|^{-1-m} \min(1, |\eta|^{-1})$, with $c_m = \frac{1}{\pi} (\frac{1}{m+1} + \sum_{k=0}^{m+2} d_{k,m})$ and $d_{k,m} = \sum_{i \leq \min(k,m)} \binom{k}{i} \frac{m!}{(m-i)!}$;
- (ii) if $p = 2$: for $0 < |\eta| \leq 1$, $|\check{g}(\eta)| \leq c'_0 M$, $|\check{g}'(\eta)| \leq c'_1 M(1 + \log(1/|\eta|))$, and $|\check{g}^{(m)}(\eta)| \leq c'_m M |\eta|^{1-m}$ for $2 \leq m \leq j$, with $c'_m = \frac{1}{\pi} (2 + \sum_{k=0}^{m+2} d_{k,m})$; for $|\eta| \geq 1$ the bounds of (i) hold, an order -2 symbol being of order 0 with the same M .

Proof. For $\eta \neq 0$ put $T = 1/|\eta|$ and write $2\pi \check{g}^{(m)}(\eta) = \int (-it)^m g(t) e^{-it\eta} dt$ as the sum over $|t| < T$ and $|t| > T$, the second part integrated by parts $m+2$ times on each half-line; the boundary terms at infinity vanish because $|\partial^k(t^m g)(t)| \leq M d_{k,m} (1 + |t|)^{m-p-k} \rightarrow 0$ for $k \geq m+1$, and $|\partial^k(t^m g)(t)| \leq M d_{k,m} (1 + |t|)^{m-p-k}$ follows from Leibniz's rule. For $p = 0$ and $|\eta| \leq 1$ ($T \geq 1$): the inner part is at most $2MT^{m+1}/(m+1)$; the boundary terms at $\pm T$ are at most $2 \sum_{k \leq m+1} M d_{k,m} T^{m-k} |\eta|^{-k-1} = 2M |\eta|^{-m-1} \sum_{k \leq m+1} d_{k,m}$; the remaining integral is at most $2|\eta|^{-m-2} \int_T^\infty M d_{m+2,m} t^{-2} dt = 2M |\eta|^{-m-1} d_{m+2,m}$. Dividing by 2π gives $c_m M |\eta|^{-1-m}$. For $|\eta| \geq 1$ integrate by parts $m+2$ times over the whole line: $2\pi |\check{g}^{(m)}(\eta)| \leq |\eta|^{-m-2} \int |\partial^{m+2}(t^m g)| \leq |\eta|^{-m-2} M d_{m+2,m} \int (1 + |t|)^{-2} dt = 2M d_{m+2,m} |\eta|^{-m-2}$, which is at most $2\pi c_m M |\eta|^{-m-2}$. For $p = 2$ the same splitting with the extra factor $(1 + |t|)^{-2}$ gives, for $m \geq 2$, an inner part at most

$2MT^{m-1}/(m-1)$ and the same boundary and remainder bounds with T^{m-p-k} in place of T^{m-k} , hence $|\eta|^{1-m}$; for $m=0$ the inner part is at most $2M$ and the rest is bounded by $2M \sum_k d_{k,0} |\eta|^{\dots} \leq 2M \sum_k d_{k,0}$ for $|\eta| \leq 1$; for $m=1$ the inner part is at most $2M \int_0^T (1+t)^{-1} dt \leq 2M(1+\log T)$. The bounds for $|\eta| \geq 1$ are as in (i). That $\check{g}$ is C^j on $\eta \neq 0$ and that the oscillatory integral is the distribution $\check{g}$ there is the standard fact that $t^m g$ is a tempered function whose transform on $\eta \neq 0$ is given by the integrated-by-parts formula. $\square$

Lemma 9.9 (the Hilbert transform of a modulated symbol). *Let $u \neq 0$, $j \geq 0$, and let $g \in C^{j+3}(\mathbb{R})$ satisfy $|g^{(i)}(t)| \leq M(1+|t|)^{-i}$ for $i \leq j+3$. Then $h = g e^{i \cdot u}$ has a Hilbert transform $\mathcal{H}[h] = \mathcal{F}[-i \operatorname{sign}(\xi) \check{h}]$, and*

$$\mathcal{H}[h](t) = -i \operatorname{sign}(u) g(t) e^{itu} + E(t), \quad E(t) = -2i \operatorname{sign}(u) \int_{\operatorname{sign} \xi = -\operatorname{sign} u} \check{g}(\xi - u) e^{it\xi} d\xi,$$

where $E \in C^j(\mathbb{R})$ and, for $0 \leq k \leq j$ and $t \neq 0$,

$$|E^{(k)}(t)| \leq \frac{e_k M}{|u| |t|^{k+1}}, \quad e_k = 2 \left(k! c_0 + \sum_{m=1}^{k+1} \binom{k+1}{m} \frac{k!}{(m-1)!} (c_m + c_m) \right),$$

with the c_m of Lemma 9.8(i). In particular $\sum_u |E_u^{(k)}(t)| \leq e_k |t|^{-k-1} \sum_u M_u / |u|$ for a family g_u with constants M_u .

Proof. $\check{h}(\xi) = \check{g}(\xi - u)$, and $-i \operatorname{sign}(\xi) \check{h} = -i \operatorname{sign}(u) \check{h} - i(\operatorname{sign}(\xi) - \operatorname{sign}(u)) \check{h}$; the first term transforms to $-i \operatorname{sign}(u) h$, and the second is supported where $\operatorname{sign} \xi = -\operatorname{sign} u$, where $\operatorname{sign}(\xi) - \operatorname{sign}(u) = -2 \operatorname{sign}(u)$ and $|\xi - u| \geq |u|$. On that half-line $\check{g}(\xi - u)$ is C^j with $|\check{g}^{(m)}(\xi - u)| \leq c_m M |\xi - u|^{-1-m} \min(1, |\xi - u|^{-1})$ by Lemma 9.8, so E is an absolutely convergent integral, C^j by differentiation under the integral sign ($|\xi|^k |\check{g}(\xi - u)| \leq c_0 M |\xi - u|^{k-2}$ is integrable at infinity for $k \leq j$ once $j+2$ derivatives of g are controlled; we assumed $j+3$). Integrate $E^{(k)}(t) = -2i \operatorname{sign}(u) \int (i\xi)^k \check{g}(\xi - u) e^{it\xi} d\xi$ by parts $k+1$ times over the half-line. The boundary terms at $\xi = 0$ are $\partial^m [(i\xi)^k \check{g}(\xi - u)]_{\xi=0} / (it)^{m+1}$, $m \leq k$, which vanish for $m < k$ and equal $i^k k! \check{g}(-u) / (it)^{k+1}$ for $m = k$, of modulus at most $k! c_0 M |u|^{-1} |t|^{-k-1}$; the boundary terms at infinity vanish. The remaining integral is $(it)^{-k-1} \int \partial^{k+1} [(i\xi)^k \check{g}(\xi - u)] e^{it\xi} d\xi$ with $\partial^{k+1} [\xi^k \check{g}(\xi - u)] = \sum_{m=1}^{k+1} \binom{k+1}{m} \frac{k!}{(m-1)!} \xi^{m-1} \check{g}^{(m)}(\xi - u)$, and on the half-line $|\xi|^{m-1} |\check{g}^{(m)}(\xi - u)| \leq c_m M |\xi - u|^{m-1} |\xi - u|^{-1-m} \min(1, |\xi - u|^{-1}) \leq c_m M |\xi - u|^{-2}$, whose integral over $|\xi - u| \geq |u|$ is at most $2c_m M / |u|$. Collecting gives e_k (the factor 2 in front is the 2 of $-2i \operatorname{sign}(u)$; the two copies of c_m are a safe overcount of the half-line integral). $\square$

Lemma 9.10 (the frozen coefficients along the archimedean symbol). *Let t_λ be the real zero of σ_∞ and $s(t) = \tilde{\sigma}_\infty(t)$. There is an absolute height t_S such that for $t \geq \max(t_S, 4t_\lambda)$ the function $\tilde{\sigma}_\infty$ is analytic on the disc $|z - t| \leq t/2$ with $|\tilde{\sigma}_\infty(z) - s(t)| \leq 2$ there. Consequently, for $s_* > X$ and $t_1 \geq \max(t_S, 4t_\lambda)$ with $s(2t_1) - 2 \geq s_*$, and for every $u \in \Lambda_a$: the function $\beta_u(\tilde{\sigma}_\infty(t))$ is C^∞ on $t \geq 2t_1$ with*

$$\left| \frac{d^i}{dt^i} \beta_u(\tilde{\sigma}_\infty(t)) \right| \leq i! (2/t)^i |\beta_u|(s_*) \quad (t \geq 2t_1, i \geq 0),$$

where $|\beta_u|(s) = \sum_k \frac{1}{k} \sum_{|w|=k, u_w=u} c_w s^{-k}$ is decreasing in s ; and with χ even, smooth, 0 on $|t| \leq 2t_1$, 1 on $|t| \geq 4t_1$, $|\chi^{(i)}| \leq \kappa_i t_1^{-i}$, the function $g_u = \chi \beta_u(\tilde{\sigma}_\infty(\cdot))$ satisfies $|g_u^{(i)}(t)| \leq M_u (1+|t|)^{-i}$ for $i \leq 6$ with $M_u = \mu |\beta_u|(s_*)$, $\mu = \max_{i \leq 6} \sum_{l \leq i} \binom{i}{l} \kappa_l 4^l (i-l)! 2^{i-l} (1+1/(2t_1))^i$. The same holds for β_0 and for the coefficients $\rho_u(s) = \sum_k r_{u,k} s^{-k}$ of Proposition 9.7(iii), which are power series in $1/s$ with nonnegative coefficients, with $\rho_u(s_*)$ in place of $|\beta_u|(s_*)$; moreover $\rho_u(s) \leq \rho_u(s_*) s_*/s$

for $s \geq s_*$ and $u \neq 0$, since every term has $k \geq 1$. And $g_0 = \chi\beta_0(\tilde{\sigma}_\infty(\cdot))$ is even and real with $\beta_0(s) = -|b|^2 s^{-2} + O(s^{-4})$; its Hilbert transform is real, even in the sense $\mathcal{H}[g_0](-t) = -\mathcal{H}[g_0](t)$, and

$$\mathcal{H}[g_0](t) = -\frac{\pi}{2} \frac{d}{d \log t} [\beta_0(\tilde{\sigma}_\infty(t))] + O((\log t)^{-5}) + O\left(\frac{t_1 \log t_1}{t}\right) = -\frac{\pi|b|^2}{\ell(t)^3} + O(\ell^{-5}) + O\left(\frac{t_1 \log t_1}{t}\right),$$

with $\ell(t) = \log(t/2\pi) - \lambda$, and the same expansion differentiated term by term for $j \leq 3$ derivatives.

Proof. For z in the disc, $\operatorname{Re} z \geq t/2 > 0$, so $w_\pm = \frac{1}{4} \pm iz/2$ have $\operatorname{Im} w_+ = \operatorname{Re} z/2 \geq t/4$ and $\operatorname{Im} w_- \leq -t/4$: both lie in $|\arg w| \leq 3\pi/4$ once $t \geq 2$, where Stirling's formula $\psi(w) = \log w - \frac{1}{2w} + r(w)$, $|r(w)| \leq \frac{1}{8|w|^2 \cos^2(\arg w/2)}$ [15, 5.11.ii] applies; hence $\Psi_\infty(z) = \log(z/2\pi) + \rho(z)$ with $|\rho(z)| \leq 4/t$ for $t \geq t_S$, an absolute constant, and the rational factor $(z^2 + t_\lambda^2)/(z^2 - t_\lambda^2)$ is analytic on the disc and within $4t_\lambda^2/t^2 \cdot \log$ -size of 1 once $t \geq 4t_\lambda$ (its poles $\pm t_\lambda$ lie outside the disc, and $\sigma_\infty(\pm t_\lambda) = 0$ removes the real ones). With $|\log(z/t)| \leq \log 2 + \pi/6 < 1.3$ on the disc, $|\tilde{\sigma}_\infty(z) - s(t)| \leq 1.3 + 4/t + O(t_\lambda^2 \log t/t^2) \leq 2$ for $t \geq t_S$. Then $|\tilde{\sigma}_\infty(z)| \geq s(t) - 2 \geq s(2t_1) - 2 \geq s_* > X$ on the disc for $t \geq 2t_1$ (s is increasing), so $\beta_u \circ \tilde{\sigma}_\infty$ is analytic there, bounded by $|\beta_u|(s_*)$ since every term of its series has modulus at most $c_w/(k|\tilde{\sigma}_\infty(z)|^k)$; Cauchy's estimate on the disc of radius $t/2$ gives the derivative bound. The bound for g_u is Leibniz's rule with $|\chi^{(l)}| \leq \kappa_l t_1^{-l} \leq \kappa_l (4/t)^l$ on $|t| \leq 4t_1$ and $(2/t)^i \leq 2^i (1 + 1/(2t_1))^i (1+t)^{-i}$ for $t \geq 2t_1$. For the Hilbert transform of g_0 , substitute $\tau = tx$ in $\mathcal{H}[g_0](t) = \frac{1}{\pi} \int_0^\infty \frac{2t(g_0(\tau) - g_0(t))}{t^2 - \tau^2} d\tau$ (valid for even g_0) and expand $g_0(tx) - g_0(t)$ in powers of $\log x$ as in the proof of Proposition 8.4, $\int_0^\infty \frac{(\log x)^k}{1-x^2} dx$ vanishing for even k and equal to $-\pi^2/4$, $-\pi^4/8$ for $k = 1, 3$: the terms with $k = 1, 3$ give $-\frac{\pi}{2} \frac{d}{d \log t} g_0 - \frac{\pi^3}{24} \frac{d^3}{d(\log t)^3} g_0$, the second of order ℓ^{-5} and the range $x < 4t_1/t$, where $\chi \neq 1$, contributes $O(t_1 \log t_1/t)$ (the integrand is bounded by $\sup |g_0| \cdot 2t/(t^2 - \tau^2)$ there). The derivatives are the same representation differentiated under the integral. $\square$

Lemma 9.11 (inversion of a remainder). *Let $\Lambda \geq 0$ be non-increasing on $[T_0, \infty)$ with $\Lambda(t) \leq \Lambda_0 \Lambda(2t)$ (doubling), and let D be bounded on $|t| \leq T_0$, C^3 on $|t| \geq T_0$, with $|D^{(j)}(t)| \leq D_0 |t|^{-1-j} \Lambda(|t|)$ for $j \leq 3$ and $D(-t) = \overline{D(t)}$. Then the inverse transform $\psi = \check{D}$ is a real C^2 function on $\mathbb{R} \setminus \{0\}$, and for $0 < |x| \leq 1/T_0$ and $k = 1, 2$,*

$$|\psi^{(k)}(x)| \leq \frac{D_0 \Lambda(1/|x|)}{\pi |x|^k} \left(\frac{\Lambda_0}{1 - 2^{-k}} + \frac{2^{k+1} \Lambda_0}{3 - k} + 6 \cdot 2^k \right) + \frac{\|D\|_{L^\infty(|t| \leq T_0)} T_0^{k+1}}{\pi(k+1)}.$$

If moreover $\Lambda(t) \leq \Lambda_1 (\log t)^{-5/2}$, then $\psi^{(k)}(x) = O(|x|^{-k} L^{-5/2})$: ψ is in the remainder class of Proposition 8.4.

Proof. $2\pi\psi^{(k)}(x) = \int (-it)^k D(t) e^{-itx} dt$ as an oscillatory integral. Split at $|t| = 1/|x| =: T$. The inner part is at most $\int_{|t| < T_0} |t|^k \|D\|_\infty + 2 \int_{T_0}^T t^{k-1} D_0 \Lambda(t) dt$, and with the doubling property $\int_{T_0}^T t^{k-1} \Lambda \leq \sum_{n \geq 0} \int_{T_0 2^{-n-1}}^{T_0 2^{-n}} t^{k-1} \Lambda \leq \Lambda(T) \sum_n \Lambda_0^{n+1} \dots$; more simply, $\Lambda(t) \leq \Lambda_0 \Lambda(2t) \leq \dots$ gives $\Lambda(T 2^{-n}) \leq \Lambda_0^n \Lambda(T)$ and $\int_{T_0 2^{-n-1}}^{T_0 2^{-n}} t^{k-1} dt \leq (T 2^{-n})^k / k$, so the sum is at most $\Lambda(T) T^k \sum_n \Lambda_0^n 2^{-nk} / k$; we use the cruder bound $\int_{T_0}^T t^{k-1} \Lambda \leq \Lambda_0 \Lambda(T) T^k / (1 - 2^{-k})$, valid when $\Lambda_0 \leq 2^k$, which is the case for the two functions used below ($\Lambda = 1$ and $\Lambda = (\log t)^{-5/2}$ with $\Lambda_0 = 1$; and $\Lambda = t^{-1/2}$ with $\Lambda_0 = \sqrt{2}$). The outer part is integrated by parts three times on each half-line: the boundary terms are $|(t^k D)^{(m)}(T)| |x|^{-m-1}$, $m \leq 2$, each at most $2^k D_0 T^{k-1-m} \Lambda(T) |x|^{-m-1} = 2^k D_0 \Lambda(T) |x|^{-k}$ ($(t^k D)^{(m)}$ is a sum of at most 2^m terms $t^{k-i} D^{(m-i)}$, each bounded by $D_0 t^{k-1-m} \Lambda$), six of them in all; the remaining integral is at most $|x|^{-3} \cdot 2 \int_T^\infty |(t^k D)'''| \leq |x|^{-3} \cdot 2 \cdot 2^3 D_0 \int_T^\infty t^{k-4} \Lambda \leq$

$|x|^{-3} \cdot 16D_0\Lambda(T)T^{k-3}/(3-k)$, using that Λ is non-increasing. Collecting and dividing by 2π gives the bound (the constant in front of $2^{k+1}\Lambda_0/(3-k)$ is a safe overcount). The integrals converge absolutely after the integrations by parts, so ψ is C^2 on $x \neq 0$, and real since $D(-t) = \overline{D(t)}$. $\square$

Theorem 9.12 (the edge law and the boundary law at every support). *Assume (H_∞) . Let a be a support that is not an entry and satisfies the Diophantine condition (D_a) of Lemma 9.6(iii) with constants c_D, C_D (almost every support, and every resonant support), f the minimizer at a , $\sigma_a, \tilde{\sigma}_a, \sigma_{a,\pm}, A_a(t)$ as in Proposition 8.18, $s = s(t) = \tilde{\sigma}_\infty(t)$, $w = \tilde{\sigma}_a/\tilde{\sigma}_\infty$, $C_w = \sup_{\mathbb{R}} |\log w|$, $C_R = \sum_j t_j^2$ over the real zeros of σ_a , and let $s_* > X$ with $\log(s_*/X) > 2\max(2a, C_D)$ (so $\alpha(s_*) = \log(s_*/X)/2a > 2$ and $\alpha_D = \log(s_*/X)/C_D > 2$) and t_1 be as in Lemma 9.10, with $2t_1$ above the real zeros of σ_a . Then:*

(i) (the factors) For $|t| \geq 8t_1$, with $m_\pm = \sigma_{\infty,\pm}/\sigma_{a,\pm}$,

$$m_\pm(t) = \left(\frac{s}{G(s)}\right)^{1/2} e^{\mp \frac{i}{2}\mathcal{H}[g_0](t)} \sum_{\pm u \geq 0} \rho_u(s) e^{itu} (1 + \varepsilon_\pm(t)), \quad |\varepsilon_\pm^{(j)}(t)| \leq \frac{\varepsilon_j t_1}{|t|^{1+j}} \quad (j \leq 3), \quad (28)$$

where $\mathcal{H}[g_0] = -\pi|b|^2\ell^{-3} + O(\ell^{-5})$ is the real slowly varying phase of Lemma 9.10 and ε_j depends only on a, s_*, C_w, C_R and t_1 , explicitly in the proof.

(ii) (the amplitude) There is a height $T_2(a)$ such that on $|t| \geq T_2$

$$A_a(t) = A_0 + M_a(t) + R_1(t) + R_0(t), \quad M_a(t) = \sum_{\nu \in \Lambda_a \cup (\Lambda_a - 2a), \nu > 0} m_\nu(t) e^{it\nu},$$

with A_0 purely imaginary; M_a the mirror chains, the walks from the far edge across the window, with slowly varying coefficients of total weight $\sum_\nu \|m_\nu\| \leq 4\eta(s_*)|A_0| + O(1/T_2)$, $\eta(s) = (X/s)^{k_a}(1 - X/s)^{-2}$, $k_a \geq 2$ the least length of a word exceeding $2a$, and first order $m_{u-2a} = \overline{A_0}e^{-2i\theta_\infty}\gamma_u(s)$ for $u > 2a$; $R_1^{(j)} = O(t^{-1-j}(\log t)^j)$ for $j \leq 3$; and R_0 slowly varying, $O(s^{-2})$ with derivatives gaining, which at a resonant support carries the atom of the chain landing on the edge (Proposition 9.18).

(iii) (the edge) With $\varphi(\delta) = |A_0|(\log(a/\delta) + \beta)^{-1/2}$, $\beta = -\gamma - \log(2\pi a) - \lambda$, and $L = \log(a/\delta) + \beta$, on $0 < \delta < \delta_0$

$$f(a - \delta) = \varphi(\delta) + e(\delta) + \psi(\delta), \quad \psi^{(k)}(\delta) = O(\delta^{-k}L^{-5/2}) \quad (k = 1, 2), \quad e(0^+) + \psi(0^+) = 0,$$

where the echo part e is the inverse transform, at position $-\delta$, of

$$\hat{e}(t) = \frac{A_0 (s/G)^{1/2} e^{\frac{i}{2}\mathcal{H}[g_0]}}{t \sigma_{\infty,-}(t)} \sum_{u \in \Lambda_a, u < 0} \rho_u(s) e^{itu} (1 + \varepsilon_-(t)) :$$

a sum over the lattice points $u < 0$ of cusps of the type of (20) at distance $|u|$ from the edge, with weights $\rho_u(s)$ whose total within ε of the edge is at most $(2\varepsilon)^{\alpha(s)}/(1 - X/s)$ (Proposition 9.7(iii)), together with the cusps of the mirror chains at the positions $\nu + u$ and $u - 2a - \nu$ inside the window, whose weights within ε of the edge sum to $O(\varepsilon^{\alpha_D})$ by (D_a) ; e is continuous with $e(\delta) - e(0^+) = O(\delta)$. Hence the edge law holds pointwise with the constants of Proposition 8.4,

$$f(a - \delta) = \varphi(\delta)(1 + O(L^{-2})),$$

while the two-derivative form holds for $\varphi + \psi$ and fails for e exactly at the echo points, where f has the cusps of its echoes.

(iv) (the laws) $\lambda'(a) = -2|A_0|^2$ at every point of differentiability of λ , and under RH

$$\lim_{T \rightarrow \infty} T \sum_{|\gamma| > T} |F(\gamma)|^2 = -\lambda'(a)/\pi.$$

Since λ is absolutely continuous (Lemma 1.4, Proposition 5.6) and the entries are countable, $\lambda(a_2) - \lambda(a_1) = -2 \int_{a_1}^{a_2} |A_0(a)|^2 da$ for all $a_1 < a_2$.

Proof. Throughout, $w = \tilde{\sigma}_a/\tilde{\sigma}_\infty = (1-x)R$ with $x = \sum_d 2c_d \cos(td)/\tilde{\sigma}_\infty$ and $R = \prod_j (t^2 + t_j^2)/(t^2 - t_j^2)$ the rational factor of the real zeros, and $m_\pm = \sigma_{\infty, \pm}/\sigma_{a, \pm} = \exp(-P_\pm[\log w])$, bounded above and below together with $1/m_\pm$ (Proposition 8.18(i)), analytic in the respective half-planes; $1/\sigma_{a, \pm} = m_\pm/\sigma_{\infty, \pm}$, and the kernels of m_+ , m_- are supported on $u \geq 0$, $u \leq 0$. The Hilbert transform is taken through the kernel, as in Lemma 9.9; on the functions below it agrees with the boundary values of the Cauchy integrals that define $\sigma_{a, \pm}$ in Lemma 7.21.

Step 1: the factors on the line, (28). Let χ be the cutoff of Lemma 9.10. On $|t| \geq 2t_1$, $s(t) \geq s(2t_1) > X$, so Proposition 9.7(i) gives $\log w = \sum_{u \in \Lambda_a} \beta_u(s) e^{itu} + \log R$, absolutely and uniformly convergent; hence on the whole line

$$\log w = \sum_{u \in \Lambda_a} g_u e^{itu} + r_0, \quad g_u = \chi \beta_u(\tilde{\sigma}_\infty(\cdot)), \quad r_0 = (1 - \chi) \log w + \chi \log R,$$

with $\sum_u \sup |g_u| \leq -\log(1 - X/s(2t_1))$. By Lemma 9.10, $|g_u^{(i)}| \leq M_u(1 + |t|)^{-i}$, $i \leq 6$, $M_u = \mu|\beta_u|(s_*)$, so Lemma 9.9 applies to each $g_u e^{i \cdot u}$, $u \neq 0$, with $j = 3$: $P_\pm[g_u e^{i \cdot u}] = \frac{1}{2} g_u e^{itu} \pm \frac{i}{2} \mathcal{H}[g_u e^{i \cdot u}] = \frac{1}{2} (1 \pm \text{sign } u) g_u e^{itu} \pm \frac{i}{2} E_u$, and $P_\pm[g_0] = \frac{1}{2} g_0 \pm \frac{i}{2} \mathcal{H}[g_0]$. The interchange of $P_\pm$ with the sum over u is dominated convergence on the kernels, $\sum_u M_u < \infty$. Therefore

$$-P_\pm[\log w] = -\frac{1}{2} g_0 \mp \frac{i}{2} \mathcal{H}[g_0] - \sum_{\pm u > 0} g_u e^{itu} + e_\pm, \quad e_\pm = \mp \frac{i}{2} \sum_{u \neq 0} E_u - P_\pm[r_0].$$

On $|t| \geq 8t_1$ one has $g_u = \beta_u(s)$, $e^{-g_0/2} = (s/G(s))^{1/2}$ (Proposition 9.7(i), $\beta_0 = \log G - \log s$) and $\exp(-\sum_{\pm u > 0} \beta_u(s) e^{itu}) = \sum_{\pm u \geq 0} \rho_u(s) e^{itu}$ (Proposition 9.7(iii)), which is (28) with $1 + \varepsilon_\pm = e^{e_\pm}$. The bounds on $e_\pm$ for $|t| \geq 8t_1$, $j \leq 3$:

- (a) $|\sum_{u \neq 0} E_u^{(j)}(t)| \leq e_j |t|^{-1-j} \sum_{u \neq 0} M_u/|u| \leq B_j |t|^{-1-j}$, $B_j = 2e_j \mu(-\log(1 - X e^{2a}/s_*))$, since $|u_w| > \frac{1}{2} e^{-2a|w|}$ (Lemma 9.6(i)) gives $\sum_{u \neq 0} |\beta_u|(s_*)/|u| \leq \sum_k \frac{1}{k} \sum_{|w|=k} c_w s_*^{-k} 2e^{2ak} = -2\log(1 - X e^{2a}/s_*)$; this is where $\alpha(s_*) > 1$ is used.
- (b) $r_0 = \chi \log R$ on $|t| \geq 8t_1$, and for $|t| \geq 2t_1$ (above the real zeros) $\log R = \sum_j \log(1 + 2t_j^2/(t^2 - t_j^2))$ satisfies $|(\log R)^{(j)}| \leq r_j C_R (1 + |t|)^{-2-j}$ with absolute constants r_j ; so $|P_\pm[r_0]^{(j)}| \leq \frac{1}{2} r_j C_R |t|^{-2-j} + \frac{1}{2} |\mathcal{H}[r_0]^{(j)}|$.
- (c) $\mathcal{H}[(1 - \chi) \log w]^{(j)}(t) = \frac{j!}{\pi} \int \frac{(1 - \chi) \log w(\tau)}{(t - \tau)^{j+1}} d\tau$ with $|\tau| \leq 4t_1 \leq |t|/2$, so its modulus is at most $\frac{j!}{\pi} 8t_1 C_w (2/|t|)^{j+1}$.
- (d) $\chi \log R$ is a symbol of order -2 with $M = r C_R$, $r = \max_{j \leq 6} r_j \max_l \kappa_l$ (Leibniz with χ), so by Lemma 9.8(ii) its kernel $\tilde{r}$ is C^3 on $\xi \neq 0$ with the stated bounds, and $\mathcal{H}[\chi \log R]^{(j)}(t) = \int (-i \text{sign } \xi) (i\xi)^j \tilde{r}(\xi) e^{it\xi} d\xi$; integrating by parts $j + 1$ times on each half-line, the boundary

terms at 0 vanish because $\partial^m[\xi^j \tilde{r}] \rightarrow 0$ as $\xi \rightarrow 0$ for $m \leq j$ (each term is $\xi^{j-i} \tilde{r}^{(m-i)}$ with $|\tilde{r}^{(m-i)}| \leq c' M |\xi|^{1-(m-i)}(1 + \log(1/|\xi|))$), and

$$|\mathcal{H}[\chi \log R]^{(j)}(t)| \leq |t|^{-1-j} \int |\partial^{j+1}[\xi^j \tilde{r}]| d\xi \leq h_j r C_R |t|^{-1-j}, \quad h_j = \sum_{m=1}^{j+1} \binom{j+1}{m} \frac{j!}{(m-1)!} (4c'_m + c_m)$$

(the two integrals, near and far, bounded as in the proof of Lemma 9.9).

Collecting, $|e_{\pm}^{(j)}(t)| \leq \varepsilon'_j t_1 |t|^{-1-j}$ on $|t| \geq 8t_1$ with $\varepsilon'_j = \frac{1}{2}(B_j + r_j C_R + j! 2^{j+4} C_w / \pi + h_j r C_R)$ (we used $t_1 \geq 1$ and $|t| \geq 8t_1$ to absorb the $|t|^{-2-j}$ term). Since $|e_{\pm}| \leq \varepsilon'_0 / 8 \leq 1$ once t_1 is also taken with $\varepsilon'_0 \leq 8$, which only enlarges t_1 (the constants B_j depend on s_* and a alone, C_w , C_R on a alone), Faà di Bruno's formula for $e^{e_{\pm}} - 1$ gives $|\varepsilon_{\pm}^{(j)}| \leq \varepsilon_j t_1 |t|^{-1-j}$ with $\varepsilon_j = e \mathcal{B}_j(\varepsilon'_0, \dots, \varepsilon'_j)$, $\mathcal{B}_j$ the complete Bell polynomial ($\varepsilon_0 = e \varepsilon'_0$). This is (i).

Step 2: the amplitude, (ii). This is Proposition 9.18: on $|t| \geq T_2(a)$, $A_a = A_0 + M_a + R_1 + R_0$, with the mirror chains M_a at the positive frequencies of $\mathcal{N} = \Lambda_a \cup (\Lambda_a - 2a)$, of total weight $4\eta(s_*)|A_0| + O(1/T_2)$ and first order $\overline{A_0} e^{-2i\theta_{\infty}} \gamma_u(s) e^{it(u-2a)}$, R_1 with derivatives $O(t^{-1-j}(\log t)^j)$ and R_0 slowly varying, $O(s^{-2})$ with derivatives gaining, carrying the resonant atom when there is one. The Diophantine condition enters through the Hilbert errors of the far feedback, $\sum_u |\gamma_u|/|u-2a-\nu|$, and nowhere else; the symmetry $A_a(-t) = -\overline{A_a(t)}$ is Proposition 8.18(ii).

Step 3: the edge, (iii). By Proposition 8.18(ii) and (i), on $|t| \geq 8t_1$,

$$\tilde{F} = \frac{A_a m_-}{t\sigma_{\infty,-}} + \frac{e^{-2ita}\Phi_-}{\tilde{\sigma}_a}, \quad m_- = m_-^{(0)}(1 + \varepsilon_-), \quad m_-^{(0)} = \left(\frac{s}{G}\right)^{1/2} e^{\frac{i}{2}\mathcal{H}[g_0]} \sum_{u \leq 0} \rho_u(s) e^{itu}.$$

Write $\hat{\varphi}$ for the transform of the model $\varphi\chi_0$ of Proposition 8.4 (χ_0 its cutoff), $\hat{e} = \sum_{u < 0} \hat{e}_u$ with $\hat{e}_u = \rho_u(s) e^{itu} A_0 (s/G)^{1/2} e^{\frac{i}{2}\mathcal{H}[g_0]} (1 + \varepsilon_-) / (t\sigma_{\infty,-})$ (the $u < 0$ terms, with their share of ε_-), and decompose

$$\begin{aligned} \tilde{F} - \hat{\varphi} - \hat{e} &= D_1 + D_2 + D_3 + D_4, \\ D_1 &= \frac{A_0 (s/G)^{1/2} e^{\frac{i}{2}\mathcal{H}[g_0]}}{t\sigma_{\infty,-}} - \hat{\varphi}, \quad D_2 = \frac{A_0 (s/G)^{1/2} e^{\frac{i}{2}\mathcal{H}[g_0]} \varepsilon_-}{t\sigma_{\infty,-}}, \\ D_3 &= \frac{(A_a - A_0) m_-}{t\sigma_{\infty,-}}, \quad D_4 = \frac{e^{-2ita}\Phi_-}{\tilde{\sigma}_a}, \end{aligned}$$

all four extended by the exact difference on $|t| < 8t_1$, where they are bounded (every function involved is). Each D_i is inverted by Lemma 9.11, with the inverse transform read at the position $-\delta$.

D_1 : $(s/G)^{1/2} = 1 + |b|^2/(2s^2) + O(s^{-4})$ (Proposition 9.7(iv)) and $e^{\frac{i}{2}\mathcal{H}[g_0]} = 1 - \frac{i\pi|b|^2}{2\ell^3} + O(\ell^{-5})$ (Lemma 9.10), both with derivatives gaining t^{-j} (Cauchy's estimate on the disc of Lemma 9.10 for the first, the differentiated expansion for the second); and $A_0/(t\sigma_{\infty,-}) - \hat{\varphi} = O(t^{-1}\ell^{-7/2})$ with derivatives $O(t^{-1-j}\ell^{-7/2})$, $j \leq 3$ (the factor and matching paragraphs of Proposition 8.4). So $|D_1^{(j)}| \leq D_0 |t|^{-1-j}\ell^{-5/2}$ and Lemma 9.11 with $\Lambda = \ell^{-5/2}$ gives $\psi_1^{(k)} = O(\delta^{-k} L^{-5/2})$, $k = 1, 2$: this is the term $|b|^2/(2\ell^2)$ of Corollary 8.15(ii), and the exponent 5/2 of the theorem is its exponent.

D_2 : by (i), $|\varepsilon_-^{(j)}| \leq \varepsilon_j t_1 |t|^{-1-j}$, so $|D_2^{(j)}| \leq D'_0 t_1 |t|^{-2-j}\ell^{-1/2}$ and Lemma 9.11 with $\Lambda(t) = t_1 t^{-1}\ell(t)^{-1/2}$ (doubling, $\Lambda_0 = 2$) gives $\psi_2^{(k)} = O(t_1 \delta^{1-k})$: bounded for $k = 1$, $O(\delta^{-1})$ for $k = 2$, inside the class.

The echo part: for $u < 0$ the function $\hat{e}_u$ is e^{itu} times $D_u = \rho_u(s)A_0(s/G)^{1/2}e^{\frac{i}{2}\mathcal{H}[g_0]}(1 + \varepsilon_-)/(t\sigma_{\infty,-})$, and by Lemma 9.10 ($\rho_u(s) \leq \rho_u(s_*)s_*/s$ with the Cauchy bounds for the derivatives) $|D_u^{(j)}| \leq D_0''\rho_u(s_*)s_*|t|^{-1-j}\ell^{-3/2}$, $j \leq 3$; Lemma 9.11 with $\Lambda = \ell^{-3/2}$ gives a real C^2 function $\tilde{e}_u$ on $x \neq 0$ with $|\tilde{e}_u^{(k)}(x)| \leq C\rho_u(s_*)s_*|x|^{-k}L(|x|)^{-3/2}$, $k = 1, 2$, and $e_u(\delta) = \tilde{e}_u(\delta - |u|)$ is the cusp of weight ρ_u at distance $|u|$ from the edge; e_u is continuous and bounded by $\|D_u\|_1/2\pi = O(\rho_u(s_*)s_*)$ since $D_u = O(t^{-1}\ell^{-3/2})$ is integrable, and $\int_0^\delta |e'_u| \leq C\rho_u(s_*)s_* \min(1, 2\delta/|u|)$ (for $|u| \geq 2\delta$ the derivative is bounded by $C\rho_u(s_*) \cdot 2/|u|$ on $(0, \delta)$; for $|u| < 2\delta$ the integral of $|x|^{-1}L^{-3/2}$ over $|x| \leq 3\delta$ is at most $4L(3\delta)^{-1/2} \leq 4$). Summing, $e = \sum_{u < 0} e_u$ converges uniformly, is continuous, and

$$|e(\delta) - e(0^+)| \leq Cs_* \left(2\delta \sum_{u \neq 0} \frac{\rho_u(s_*)}{|u|} + \sum_{0 < |u| < 2\delta} \rho_u(s_*) \right) \leq Cs_* \left(\frac{4\delta}{1 - X/s_*} \sum_k \left(\frac{Xe^{2a}}{s_*} \right)^k + \frac{(4\delta)^{\alpha(s_*)}}{1 - X/s_*} \right) = O(\delta),$$

by Lemma 9.6(i) and Proposition 9.7(iii), with $\alpha(s_*) > 1$. This is the structure of the echo part stated in (iii).

D_3 and D_4 . By Step 2, $D_3 = (M_a + R_1 + R_0)m_-/(t\sigma_{\infty,-})$ with $m_- = \sum_{u \leq 0} \mu_u(s)e^{itu}(1 + \varepsilon_-)$: a sum over the positions $p = \nu + u$ ($\nu \in \mathcal{N}$, $\nu \geq 0$, $u \leq 0$) of e^{itp} times a symbol of order -1 with the factor $\ell^{-1/2}$ and derivatives gaining up to the logarithms of $\mathcal{X}$, of weight $w_p \leq \|m_\nu\|_{\mathcal{X}}\rho_u(s_*)s_*$. The terms with $p = 0$ are in the remainder class by Lemma 9.11 (R_1 with $\Lambda = t^{-1}(\log t)^4\ell^{-1/2}$, R_0 with $\Lambda = s^{-2}\ell^{-1/2} = O(\ell^{-5/2})$). A term with $p > 0$ sits beyond the edge: its inverse transform at the positions $-\delta \in (-\delta_0, 0)$ is the function of Lemma 9.11 evaluated at distance at least p from its singularity, C^2 there with derivatives at most Cw_pp^{-k} , and $\sum_p w_pp^{-2} < \infty$ because $p \in \Lambda_a \cup (\Lambda_a - 2a)$ is bounded below by $\frac{1}{2}e^{-2ak}$ or $c_De^{-C_Dk}$ for a word of length k (Lemma 9.6) while the weights of length k sum to $O((X/s_*)^k)$, and $\log(s_*/X) > 2\max(2a, C_D)$. A term with $p < 0$ is a cusp at distance $|p|$ from the edge, of the type of e_u , and belongs to the echo part; those within ε of the edge are the mirror chains landing within ε of the near edge, whose words have length at least $\log(c_D/\varepsilon)/C_D$ by (D_a), so their weights sum to $O(\varepsilon^{\alpha_D})$ with $\alpha_D = \log(s_*/X)/C_D > 2$. $D_4 = e^{-2ita}\Phi_-/\tilde{\sigma}_a$: by Lemma 9.17, $D_4 = -e^{-2ita}\overline{A}_a m_+/(t\sigma_{\infty,+}) + e^{-2ita}\Xi m_+ m_-/\tilde{\sigma}_\infty$. The first term is the mirror image of D_3 : with $\overline{A}_a = \overline{A}_0 + \overline{M}_a + \overline{R}$ and $m_+ = \sum_{u \geq 0} \mu_u(s)e^{itu}(1 + \varepsilon_+)$ it is a sum over the positions $p = u - 2a - \nu$ of copies of the mirrored profile, the far edge itself at $p = -2a$, the mirror chains inside the window at $p < 0$ (cusps of the echo part, those within ε of the near edge of weight $O(\varepsilon^{\alpha_D})$), the terms with $p > 0$ smooth from inside with the summable bounds, and $p = 0$ the resonant atom. The Ξ term is an explicit symbol of order -2 times $m_+ m_-/\tilde{\sigma}_\infty$, a sum over the positions $u + u' - 2a$ and $u + u'$ ($u \geq 0 \geq u'$) of symbols of order -2 with the factor ℓ^{-1} , each inverting by Lemma 9.11 ($\Lambda = t^{-1}\ell^{-1}$) to a continuous function with ψ' bounded and $\psi'' = O(|x|^{-1}L^{-1})$ at its position: inside the class at positive distance, weak cusps of the echo part at negative positions, with the same summability. So $\psi = \psi_1 + \psi_2 + \psi_3 + \psi_4$ satisfies $\psi^{(k)} = O(\delta^{-k}L^{-5/2})$, $k = 1, 2$, and the echo part $e = \sum_p e_p$ runs over the positions $p \in (-\delta_0, 0)$ of the forms u (the echoes of the right edge), $\nu + u$ and $u - 2a - \nu$ (the mirror chains), with weights w_p , $\sum w_p < \infty$ and $\sum_{|p| < \varepsilon} w_p \leq W\varepsilon^{\alpha'}$, $\alpha' = \min(\alpha(s_*), \alpha_D) > 2$; the bound $e(\delta) - e(0^+) = O(\delta)$ is proved as for the e_u , with $\sum_p w_p/|p| < \infty$ from the same lower bounds on $|p|$.

Finally $f(a^-) = 0$ (Theorem 3.2) and $\varphi(0^+) = 0$ give $e(0^+) + \psi(0^+) = 0$, hence $f(a - \delta) - \varphi(\delta) = [e(\delta) - e(0^+)] + \int_0^\delta \psi' = O(\delta) + O(L^{-5/2})$, which is the pointwise law $\varphi(1 + O(L^{-2}))$. This is (iii).

Step 4: the laws, (iv). Proposition 8.3 is applied with $u = (\varphi + e + \psi)\chi$ near the right edge; φ and ψ are as there, and the echo part contributes nothing to B_∞ by Lemma 9.19, whose hypotheses are (iii): the cusps e_p with their weights, the transforms' bounds from the symbols that produced them, and $\sum_p w_p(\log(1/|p|))^{3/2} < \infty$ because $\log(1/|p|) \leq \max(2a, C_D)|w| + \log(2/c_D)$ for the word of p while $\sum_k (X/s_*)^k k^{3/2} < \infty$. So $B_\infty = a|A_0|^2$, hence $\mathcal{D}(f) = 2a|A_0|^2$ and $\lambda' = -2|A_0|^2$

at every point of differentiability, by Propositions 7.15 and 5.3. Proposition 7.8 is replaced by Lemma 9.20: by (i), (ii) and Lemma 9.17, $\tilde{F} = A_a/(t\sigma_{a,-}) - e^{-2ita}\overline{A_a}/(t\sigma_{a,+}) + e^{-2ita}\Xi/\tilde{\sigma}_a$, so $|F|^2 = 4|A_a|^2 \sin^2(ta + \theta_a - \arg(iA_a))/(t^2\tilde{\sigma}_a) + O(t^{-3} \log t)$; expanding $|A_a|^2 = |A_0|^2 + 2 \operatorname{Re} \overline{A_0}(M_a + R) + O(\eta^2)$, $1/\tilde{\sigma}_a = \tilde{\sigma}_\infty^{-1} R^{-1}(1-x)^{-1}$ with $(1-x)^{-1} = \sum_k x^k$ a lattice sum of frozen coefficients, and $e^{2i\theta_a} = e^{2i\theta_\infty} e^{2i\theta_s} (1 + \varepsilon')$ with $e^{2i\theta_s}$ a lattice sum, every term is of the form of the lemma: the non-oscillating part is $4|A_0|^2(t^2\tilde{\sigma}_\infty)^{-1} \cdot \frac{1}{2}(1 - \cos(2ta + 2\theta_\infty))(1 + q)$ with q the mean of the lattice sums, $O(1/\log t)$, and the oscillating terms have the frequencies $\nu \in (\Lambda_a \cup (\pm 2a + \Lambda_a) \cup (\pm 4a + \Lambda_a)) \setminus \{0\}$ with slowly varying coefficients whose weights, products of the frozen weights, satisfy $\sum w_\nu(1 + \nu^2) < \infty$ ($|\nu| \leq 4a + 2ak$ for total word length k); frequency zero occurs only through a word landing on $\pm 2a$ or $\pm 4a$, at a resonant support, with a weight $O(s^{-2})$ that is part of q . The lemma needs no lower bound on the frequencies. Hence $\lim T \sum_{|\gamma| > T} |F(\gamma)|^2 = 2|A_0|^2/\pi = -\lambda'/\pi$ under RH. The last statement of the theorem integrates λ' over the supports that are not entries. $\square$

Verification notes. The proof above was rewritten so that every estimate of Step 1 is a lemma with its constants, and the rewriting changed three things in the statement. First, the error of the factors is $O(t_1/t)$, not $O(t^{-1/2})$: the earlier version split the lattice sum at $|u| = t^{-1/2}$ under $\alpha(s_1) \geq 1$; the kernel form of the Hilbert transform (Lemmas 9.8 and 9.9) gives the error $e_k M_u/(|u|t^{k+1})$ for each frequency with all derivatives gaining, and the sum converges exactly when $\alpha(s_*) > 1$. Second, the frequency $u = 0$ of $\log w$, the constant term $\log(G/s)$, has a Hilbert transform that is not small in $1/t$ but slowly varying, $-\pi|b|^2/\ell^3$: it is a real phase of the factors, omitted in the earlier statement of (28), and it enters the edge profile as a relative correction $O(\ell^{-3})$, inside the remainder class. Third, the edge law with two derivatives in the pointwise sense of Proposition 7.8 does not hold at a support where the echo lattice is dense: the minimizer has at every lattice point $u < 0$ the cusp of its echo, with the derivative $\rho_u|x|^{-1}L^{-3/2}$ at distance x from it, and the lattice accumulates at the edge; the correct statement separates the echo part e , whose weights within ε of the edge sum to $O(\varepsilon^\alpha)$, and the pointwise law $f(a - \delta) = \varphi(\delta)(1 + O(L^{-2}))$ survives because $e(\delta) - e(0^+) = O(\delta)$. The three points that the first pass left at the density of a research sketch are now Proposition 9.18 (the far-source feedback as a fixed point, uniform over the lattice under (D_a)), the per-position treatment of D_3 and D_4 in Step 3 (no cancellation between the mirror terms and the far source is needed: the exact transform vanishes outside the window by itself, and near the edge every term is controlled individually), and Lemmas 9.19 and 9.20. What that pass cost is the Diophantine condition: the earlier version claimed the rate at every non-entry support, but the frozen expansion of the far feedback converges absolutely only when $\sum_u |\gamma_u|/|u - 2a| < \infty$, which (D_a) guarantees and which can fail on a null set of supports where $2a$ is approximable by lattice points faster than any exponential in the word length; at those supports the conclusions are not claimed. Two points remain at sketch density: the statement that the Hilbert transform of a symbol of order 0 is a symbol up to one logarithm per derivative, used in Proposition 9.18 for the frequency-zero terms with its one-line proof; and the local boundedness in a of the heights $T_0(a)$, $T_2(a)$, used by Proposition 9.15. The earlier list, for the record: (1) in Step 2, the uniformity over the lattice points $v - 2a \rightarrow 0^+$ of the far-source feedback, which produces the mirror amplitudes m_v and the $O(t^{-1/2})$; (2) in Step 3, the derivative bounds with the gain t^{-j} for $A_a - A_0$, and the cancellation between the mirror terms that would land outside the window and the far source D_4 through the kernel of $1/\tilde{\sigma}_a$; (3) in Step 4, the two extensions, of the flux pairing of Proposition 8.3 and of the zero sums of Proposition 7.8, to the echo part, which are written above with their estimates but not as separate lemmas. Items (1) and (2) are statements about the kernel of $1/\sigma_{a,+}$ on $(0, \varepsilon)$, where Lemma 9.8 and Proposition 9.7(iii) give the ingredients; item (3) is bookkeeping of the type of Corollary 8.15. Below a_∞ , where the lattice

is finite, none of the three arises and the theorem is Corollary 8.15(ii), (v). Two hypotheses can be removed by the same representation, and this is the next step after the three items. (H_∞) enters the chain Proposition 7.23, Corollary 7.24, Proposition 8.4 and the theorem above at one point only, $f(a^-) = 0$ from Theorem 3.2, used to fix the constant of integration of ψ ; but ψ' is integrable, so $\psi(0^+)$ exists, a nonzero value would put a term $c/(it)$ into $\tilde{F}$, and the representation gives $\tilde{F} = O(1/(t\sqrt{\log t})) = o(1/t)$, so the value is zero: the edge profile is bounded, the interior is C^∞ by the commutator bootstrap ($[K_\infty, \chi]$ is of order -1 , the symbol's derivative being $1/t$), and the minimizer is bounded, a corollary rather than a hypothesis. (H_{BV}) enters only through the absolute continuity of λ in Theorem 9.1, and the Dini-derivative argument stated in its proof replaces it once $|A_0(a)|$ is bounded locally uniformly in a .

Corollary 9.13 (the reduction, unconditional on the boundary law). *Assume (H_∞) and the inequality (25) at almost every support $a \geq a_Z$. Then Conjecture 7.1 holds with the constant c of (25) and the Riemann Hypothesis follows.*

Proof. Theorem 9.12(iv) gives $\lambda' = -2|A_0|^2$ at almost every $a \geq a_Z$, so (25) reads $\lambda' \geq -cT^*\lambda$ almost everywhere on $[a_Z, \infty)$; with the absolute continuity of λ this is Conjecture 7.1 with $a_0 = a_Z$, and Proposition 7.2 applies. $\square$

The reduction uses the absolute continuity of the floor, which Lemma 1.4 does not give and Proposition 5.6 gives under (H_{BV}) . The boundary law supplies it without that hypothesis, through the Dini derivatives of Proposition 5.3, once the edge amplitude is bounded locally uniformly in the support; and the representation bounds the amplitude by the energy of the tail.

Lemma 9.14 (the edge amplitude is bounded by the tail energy). *Let a be a support that is not an entry and f the minimizer, with $\|f\|_2 = 1$, and suppose Theorem 9.12(i)–(ii) hold at a . Let $T_0(a)$ be a height above the last sign change t_s of σ_a such that*

$$\int_{T_0}^{\infty} |F(t)|^2 \sigma_a(t) dt \geq \frac{|A_0(a)|^2}{T_0},$$

which exists because the tail integral is $2|A_0|^2 T^{-1}(1 + o(1))$ by Theorem 9.12(ii) and the mirror structure of Corollary 7.24 read with the window's symbol. Then

$$|A_0(a)|^2 \leq \pi T_0(a) \left(2P(f)^2 + \max_{|t| \leq T_0} |\sigma_a(t)| \right) \leq \pi T_0(a) \left(2(\sinh a - a) + \max_{|t| \leq T_0} |\sigma_a| \right).$$

In particular $|A_0(a)|$ is bounded on every compact set of supports on which $T_0(a)$ is bounded.

Proof. On the window $K_a f = Kf$, the copies of f shifted by more than $2a$ not meeting it, so the eigen-equation gives $\frac{1}{2\pi} \int |F|^2 (\Psi_a - \lambda) dt = \langle f, K_a f \rangle - \lambda \|f\|^2 = 2P^2$ (Proposition 7.9 at $T = 0$). Split at $|t| = T_0$. On $|t| < T_0$, $|\frac{1}{2\pi} \int_{|t| < T_0} |F|^2 \sigma_a| \leq \max_{|t| \leq T_0} |\sigma_a| \cdot \frac{1}{2\pi} \int |F|^2 = \max_{|t| \leq T_0} |\sigma_a|$. On $|t| > T_0 > t_s$ the integrand is positive and, by parity and the hypothesis, $\frac{1}{2\pi} \int_{|t| > T_0} |F|^2 \sigma_a = \frac{1}{\pi} \int_{T_0}^{\infty} |F|^2 \sigma_a \geq \frac{|A_0|^2}{\pi T_0}$. Hence $|A_0|^2/(\pi T_0) \leq 2P^2 + \max_{|t| \leq T_0} |\sigma_a|$, and $P^2 \leq \|s\|^2 \|f\|^2 = \sinh a - a$ by Cauchy–Schwarz. $\square$

Proposition 9.15 (the floor is Lipschitz, from the boundary law). *Assume (H_∞) and Theorem 9.12 at every support of $[a_1, a_2]$ that is not an entry, and that $T_0(a)$ of Lemma 9.14 is bounded on $[a_1, a_2]$ by T_0^* . Then λ is Lipschitz on $[a_1, a_2]$, with constant $M = 2\pi T_0^* (2(\sinh a_2 - a_2) + \max_{a \in [a_1, a_2]} \max_{|t| \leq T_0^*} |\sigma_a(t)|)$, hence absolutely continuous there, without (H_{BV}) .*

Proof. At a support $a \in (a_1, a_2)$ that is not an entry, Proposition 5.3 gives $a D_- \lambda(a) \geq -\mathcal{D}(f)$ for the minimizer f : its hypothesis, that g_f is differentiable at the entries, holds because f is continuous with $f' \in L^1$, the edge profile having the derivative $O(\delta^{-1} L^{-3/2})$, the echo cusps the integrable derivatives of Theorem 9.12(iii) with summable weights, and the interior being smooth. Proposition 8.3, applied as in Theorem 9.12(iv), gives $\mathcal{D}(f) = 2a|A_0(a)|^2$; so $D_- \lambda(a) \geq -2|A_0(a)|^2 \geq -M$ by Lemma 9.14. The entries are countable and λ is continuous and non-increasing (Lemma 1.4); a continuous function whose lower left Dini derivative is at least $-M$ at every point outside a countable set satisfies $g(y) - g(x) \geq -M(y - x)$ for $x < y$ [14, Ch. VI, §4], applied to $g(x) = \lambda(x) + Mx$, which is then non-decreasing; with the monotonicity of λ , $|\lambda(y) - \lambda(x)| \leq M(y - x)$. $\square$

With Proposition 9.15 the hypothesis (H_{BV}) leaves the main theorem, at the price of the local boundedness of $T_0(a)$: a statement about the constants of Theorem 9.12 as functions of the support between consecutive entries, where t_1 , C_w , C_R and s_* vary continuously with the symbol, the two one-sided symbols at an entry both being admissible. This is recorded here, not proved.

Three remarks. First, the theorem is about λ' , not about the onset: the asymptotics it establishes begin where the chain expansion converges with room, $s > X e^{2a}$, which is a height $2\pi e^{X e^{2a}}$, astronomically far above the horizon for $a \geq 1$; the measured onset of the tail law, a few horizons at every support computed, is governed by the band the edge sees (Proposition 8.23), and Computation 9.30 finds that it is not uniform in a : the bulk of the edge's measure recedes like e^{ca} horizons, $0.4 < c < 1$. Second, below a_∞ the window's representation and the tree's represent the same function, and the theorem recovers Corollary 8.15(ii) by a different route: the tree's effective symbol absorbs the echoes exactly from the horizon, the window's symbol does so only asymptotically, through the mirror chains and the slowly varying factor $(s/G)^{1/2}$, and the two agree on the constants because $\rho_0 = 1$ and $(s/G)^{1/2} - 1 = O(s^{-2})$. Third, the resonant supports, $2a \in \Lambda_a$: a chain from the far edge lands exactly on the near edge, with the weight $\rho_{2a}(L) = O(L^{-2})$ of a word of length at least two; it is a relative correction of that order to the edge amplitude, inside the remainder class, and the Diophantine condition holds there automatically (Lemma 9.6(ii)), so the theorem holds at every resonant support. The supports it leaves out are the entries, where λ has its soft kinks, and the null set of supports violating (D_a) , where $2a$ is approximable by lattice points faster than any exponential in the word length.

Computation 9.16 (the mirror chains, and what the K -mode minimizers resolve). `rh_mirror_chains.py` and `rh_band_phase.py`, logs `mirror_chains_a0.6.log` and `band_phase_a*.log` in `data/`. The K -mode minimizer is a polynomial of degree $2K - 1$ on $[-a, a]$: it resolves the edge to $\delta \approx a/(2K - 1)^2$, heights of order $(2K - 1)^2/a$, but the interior cusps only to heights of order $(2K - 1)/a$. (i) At $a = 0.6$, against the edge FEM of Computation 8.17 (interior resolved to $34 T^*$), the K -mode transform ($K = 48$, degree 95, $(2K - 1)/a = 7.6 T^*$) agrees to 0.07%, 0.3%, 0.6% at 2, 3, $5 T^*$ and departs at $7 T^*$ (2.9%) to 15–20% at 10 – $30 T^*$. (ii) On 81 nulls at 10 – $30 T^*$, the residual $r + \theta_\infty - \kappa/t$ regressed on $(\alpha_d \sin td + \beta_d \cos td)/\tilde{\sigma}_\infty$: the FEM gives $\alpha_2 = 0.502$, $\alpha_3 = 0.635$, $|\beta_d| \leq 0.005$ against the echo form's $c_2 = 0.490$, $c_3 = 0.634$ (rms $0.139 \rightarrow 0.022$); the polynomial gives $\alpha_2 = -0.003$, $\alpha_3 = 0.16$ (rms $0.055 \rightarrow 0.048$). The K -mode minimizers carry the edge alone above a few horizons, and above $(2K - 1)/a$ (3.4, 2.9, $2.4 T^*$ at $a = 1.0, 1.25, 1.5$) their transforms say nothing about the pointwise structure of the minimizer; the envelope, a mean, they get right to the few per cent of $\sigma_{\text{eff}}/\tilde{\sigma}_\infty - 1$. On 3112 nulls at 3 – $400 T^*$ of the $K = 64$ minimizer at $a = 0.8$ ($(2K - 1)/a = 5.1 T^*$, $(2K - 1)^2/a = 650 T^*$) the archimedean phase fits to rms 0.008–0.017 from $10 T^*$ on, with $|\alpha_d| \leq 0.0011$ for the three entries, and the window's symbol phase to 0.10–0.16; the local amplitude $\langle 4t^2 \tilde{\sigma}_a S^2 \rangle / (-\lambda')$ over one period is 1.04 ± 0.04 at 47 – $100 T^*$ and 0.95 ± 0.03 at 200 – $400 T^*$; on 5755 nulls of the $K = 80$ minimizer at $a = 1.0$ (edge resolved to $540 T^*$) the archimedean phase fits to rms 0.010–0.017 from 30 to $400 T^*$,

while on $10\text{--}30 T^*$ the residual is 0.31 and fits no echo form: between $(2K - 1)/a$ and a few times that height the polynomial's interior error still reaches its transform. The $K = 200$ minimizer at $a = 1.0$ (`kmode_K200/coeffs_1.0.json`; $\lambda_{200} = 1.4361 \times 10^{-26}$, $a\lambda' = -2.4201 \times 10^{-24}$, against 1.4820×10^{-26} and -2.5169×10^{-24} at $K = 80$; interior resolved to $(2K - 1)/a = 8.6 T^*$) on the same 63 nulls at $3\text{--}8 T^*$ (`band_phase_a1.0_K200_3-8.log`): the residual against the archimedean phase is 0.56, the regression on the five entries gives $\alpha_2 = 0.38$, $\alpha_3 = 0.51$, $\alpha_4 = 0.11$, $\alpha_5 = 0.41$, $\alpha_7 = 0.32$ against $c_d = 0.49, 0.63, 0.35, 0.72, 0.74$, all now with the sign of the echo form and 60–80% of its size where $K = 80$ gave signs at random, but the residual falls only to 0.54: at $a = 1$ the band of the window's symbol, with 87 real zeros up to $33 T^*$, dominates these heights, and the echo form is not the right description there; what the $K = 200$ run adds is that the interior structure the polynomial resolves does carry the entries' frequencies with the predicted signs. This is the edge of the minimizer, at heights where the sampling identity of Proposition 7.29 and the tail law were measured on these functions; it is not their interior. (iii) The window's symbol on the FEM nulls, 114 at $5\text{--}34 T^*$: $\tilde{\sigma}_a$ has 11 real zeros below $2.2 T^*$ and its band top is $2.9 T^*$ ($X = 2.249$). The residual with the phase θ_a of $\tilde{\sigma}_a$ has rms 0.0365 (tree form 0.0009, archimedean phase 0.145); regressed on $(p \sin + q \cos)(t(v - 2a))/\tilde{\sigma}_\infty^2$ over the three two-step mirror words $v = d + d' > 2a$ ($\log 4, \log 6, \log 9$; $v - 2a = 0.186, 0.592, 0.997$), it drops to 0.0186 with the amplitudes 0.190, 0.514, 0.348 against the chain weights $\sum c_d c_{d'} = 0.240, 0.622, 0.402$, a common factor 0.8; by height band the residual falls from 0.046 to 0.030 ($5\text{--}10$ to $20\text{--}34 T^*$) and after the fit from 0.033 to 0.013, while $(X/s)^2$ falls from 0.49 to 0.25 and $(X/s)^3$, the three-step chains left out, from 0.34 to 0.13. The same regression on the tree's residual finds nothing (0.002, 0.001, 0.002).

9.3 The amplitude as a fixed point, and the two laws

The proof of Theorem 9.12 used, in its Steps 2 and 4, the following statements.

Lemma 9.17 (the mirror identity with the window's symbol). *In the setting of Proposition 8.18, with Φ_- built from g_{out} as in Corollary 7.24,*

$$\Phi_-(t) = -\frac{\sigma_{a,-}(t) \overline{A_a(t)}}{t} + \Xi(t), \quad \Xi(t) = \frac{2iP \sinh(a/2)}{t} - \frac{2iP \sinh(a/2)}{t - ib} + \Xi_1(t),$$

where Ξ_1 is a sum of explicit rational functions of t (from $\tilde{\Sigma}$ and the corrections at the N pairs of zeros) with $|\Xi_1^{(j)}(t)| \leq \underline{C}_\Xi(1 + N) t_s^2 |t|^{-2-j}$ for $|t| \geq 2t_s$, $j \leq 3$; so Ξ is $O(t^{-2})$ with all derivatives gaining, and $\Xi(-t) = \overline{\Xi(t)}$.

Proof. This is the mirror paragraph of Corollary 7.24 with $\sigma_a, \tilde{\sigma}_a, \sigma_{a,\pm}$: from $\sigma_a \tilde{F} = \tilde{H} + \tilde{G}_+$ and $\tilde{F} = A_a/(t\sigma_{a,-}) + e^{-2ita}\Phi_-/\tilde{\sigma}_a$, $\tilde{G}_+ = \frac{\sigma_a}{\tilde{\sigma}_a} \frac{\sigma_{a,+} A_a}{t} + \frac{\sigma_a}{\tilde{\sigma}_a} e^{-2ita}\Phi_- - 2P\tilde{\Sigma} - e^{-2ita}(\Phi_- + \frac{2iP \sinh(a/2)}{t - ib})$, where $\sigma_a/\tilde{\sigma}_a = \prod_j (t^2 - t_j^2)/(t^2 + t_j^2) = 1 - \sum_j 2t_j^2/(t^2 + t_j^2) + \dots$ is 1 plus an explicit rational function $O(Nt_s^2/t^2)$ and $\tilde{\Sigma} = e^{-ita}\Sigma = -i \sinh(a/2)(1 + e^{-2ita})/t + (\text{explicit}, O(t^{-2}))$ is exact; the terms in e^{-2ita} are $e^{-2ita}[(\sigma_a/\tilde{\sigma}_a - 1)\Phi_- + 2iP \sinh(a/2)(1/t - 1/(t - ib))]$, explicit and $O(t^{-2})$ since Φ_- is bounded. Oddness of f gives $g_{\text{out}}(-a - v) = -g_{\text{out}}(a + v)$, hence $\Phi_- = -\overline{\tilde{G}_+} - 2iP \sinh(a/2)/(t - ib)$; substituting $\tilde{G}_+$ and using $\overline{\sigma_{a,+}} = \sigma_{a,-}$ on the line gives the identity, every term of Ξ being one of the explicit rational functions named, whose derivatives are bounded as stated. (The term $e^{-2ita}[\dots]$ of $\tilde{G}_+$ becomes $e^{2ita}[\dots]$ in Ξ_1 after conjugation; it enters below through $e^{-2ita}\Xi/\tilde{\sigma}_a$ as a non-oscillating explicit $O(t^{-2})$ symbol, and the bound on $\Xi_1^{(j)}$ is understood for its coefficient.) $\square$

The amplitude A_a satisfies an equation in which the far edge feeds back through the window's factors, and the equation is a contraction above a height that depends on the support. Fix $T \geq 8t_1$

and let $\mathcal{X}_T$ be the space of C^4 functions on $|t| \geq T$ with

$$\|B\|_{\mathcal{X}} = \max_{i \leq 4} \sup_{|t| \geq T} \frac{|t|^i}{(\log |t|)^i} |B^{(i)}(t)| < \infty,$$

the slowly varying functions (the logarithmic weight is there because the Hilbert transform of a symbol of order 0 is a symbol up to one logarithm per derivative: in the kernel form $\mathcal{H}[g]^{(j)} = \int (-i \operatorname{sign} \xi) (i\xi)^j \tilde{g} e^{it\xi} d\xi$ the j -th integration by parts meets $\int_{1/t}^1 |\tilde{g}| \leq c_0 M \log t$, and no more; we use this as stated, its proof being the one just given), and let $\mathcal{Y}_T$ be the space of sums $B = \sum_{\nu \in \mathcal{N}} b_{\nu}(t) e^{it\nu}$ over the frequency set $\mathcal{N} = \Lambda_a \cup (\Lambda_a - 2a)$ with $b_{\nu} \in \mathcal{X}_T$ and $\|B\|_{\mathcal{Y}} = \sum_{\nu} \|b_{\nu}\|_{\mathcal{X}} < \infty$. Under (D_a) and Lemma 9.6(i), every nonzero frequency of $\mathcal{N}$ is bounded away from zero in terms of the word lengths: $|\nu| \geq \frac{1}{2} e^{-2ak}$ for $\nu = u_w \in \Lambda_a \setminus \{0\}$ and $|\nu| \geq c_D e^{-C_D k}$ for $\nu = u_w - 2a \neq 0$, $k = |w|$.

Proposition 9.18 (the amplitude as a fixed point). *In the setting of Theorem 9.12, with (D_a) and $\log(s_*/X) > 2 \max(2a, C_D)$, there is a height $T_2(a) \geq 8t_1$, explicit in the constants of Lemmas 9.9, 9.10 and 9.17, such that on $|t| \geq T_2$:*

- (i) $A_a = N + \mathcal{T}A_a$, where $\mathcal{T}B = P_+[e^{-2ita}(\sigma_{a,-}/\sigma_{a,+})\bar{B}]$ and $N = tP_-[(\hat{H}_- - e^{-2ita}\Phi_-)/\sigma_{a,+}] - tP_+[e^{-2ita}\Xi/\sigma_{a,+}]$ is built from the near sources and the explicit Ξ of Lemma 9.17;
- (ii) $N \in \mathcal{Y}_{T_2}$, $N = A_0^{\text{near}} + N_1$ with A_0^{near} a purely imaginary constant and $\|N_1\|_{\mathcal{Y}} \leq C_N/T_2$; $\mathcal{T}$ maps $\mathcal{Y}_{T_2}$ to itself with $\|\mathcal{T}\| \leq \frac{1}{2}$; and $\mathcal{T}(A_0^{\text{near}}) = \overline{A_0^{\text{near}}} e^{-2i\theta_{\infty}} \sum_{u>2a} \gamma_u(s) e^{it(u-2a)} + E_0$ with $\|E_0\|_{\mathcal{Y}} \leq C_E/T_2$, where $e^{-2i\theta_s} = \sum_{u \in \Lambda_a} \gamma_u(s) e^{itu}$ is the frozen phase of Proposition 9.7(ii), $\sum_u |\gamma_u|(s) \leq (1 - X/s)^{-2}$, and $\sum_{u>2a} |\gamma_u|(s) \leq \eta(s) := (X/s)^{k_a} (1 - X/s)^{-2}$;
- (iii) consequently $A_a = (I - \mathcal{T})^{-1}N = \sum_{n \geq 0} \mathcal{T}^n N$ in $\mathcal{Y}_{T_2}$, and

$$A_a(t) = A_0 + M_a(t) + R(t), \quad M_a = \sum_{\nu \in \mathcal{N}, \nu > 0} m_{\nu}(t) e^{it\nu}, \quad \sum_{\nu} \|m_{\nu}\|_{\mathcal{X}} \leq 4\eta(s_*)|A_0| + C_M/T_2,$$

with $m_{u-2a} = \overline{A_0} e^{-2i\theta_{\infty}} \gamma_u(s) + O(\eta^2)$ for $u > 2a$, A_0 purely imaginary, and R slowly varying, $R = R_1 + R_0$ with $\|R_1\|_{\mathcal{X}} \leq C_R/T_2$ (so $R_1^{(j)} = O(t^{-1-j})$) and R_0 the frequency-zero part of the series beyond the constant, $|R_0^{(j)}(t)| \leq C_0 t^{-j} s^{-2}$, which at a resonant support carries the atom $\rho_{2a}(s) = O(s^{-|w_0|})$.

Proof. (i) By Proposition 8.18(ii), $A_a = tP_-[(\hat{H}_- - e^{-2ita}\Phi_-)/\sigma_{a,+}] - tP_+[e^{-2ita}\Phi_-/\sigma_{a,+}]$, and Lemma 9.17 gives $-tP_+[e^{-2ita}\Phi_-/\sigma_{a,+}] = P_+[e^{-2ita}(\sigma_{a,-}/\sigma_{a,+})\bar{A}_a] - tP_+[e^{-2ita}\Xi/\sigma_{a,+}]$.

The factor. On $|t| \geq 8t_1$, $\sigma_{a,-}/\sigma_{a,+} = (m_+/m_-)e^{-2i\theta_{\infty}}$ and, by (28), $m_+/m_- = e^{-i\mathcal{H}[g_0]} (\sum_{u \geq 0} \rho_u e^{itu}) / (\sum_{u \leq 0} (1 + \varepsilon_+)/(1 + \varepsilon_-))$; the quotient of the two lattice sums is $\sigma_{s,-}/\sigma_{s,+} = e^{-2i\theta_s}$ (Proposition 9.7(ii)), whose expansion $\sum_u \gamma_u(s) e^{itu}$ is the product of the expansions of $1/\sigma_{s,+}$ and $\sigma_{s,-}$, each with non-negative coefficients summing to at most $(1 - X/s)^{-1}$, so $\sum |\gamma_u| \leq (1 - X/s)^{-2}$, with γ_u a power series in $1/s$ with the Cauchy bounds of Lemma 9.10, and $\sum_{u>2a} |\gamma_u| \leq (X/s)^{k_a} (1 - X/s)^{-2}$ since a word landing beyond $2a$ has length at least k_a . The factor $(1 + \varepsilon_+)/(1 + \varepsilon_-) e^{-i\mathcal{H}[g_0]} = 1 + \varepsilon'_{\pm}$ is slowly varying with $\|\varepsilon'\|_{\mathcal{X}} \leq C/T$ for the ε part and $O(s^{-3})$ with derivatives gaining for the phase, by (28) and Lemma 9.10.

The operator on one term. For $B = b(t) e^{it\nu}$ with $b \in \mathcal{X}_T$, $e^{-2ita}(\sigma_{a,-}/\sigma_{a,+})\bar{B} = \sum_u \gamma_u e^{it(u-2a-\nu)} g_u$ with $g_u = e^{-2i\theta_{\infty}}(1 + \varepsilon')\bar{b}$, a symbol with $|g_u^{(i)}| \leq C\|b\|_{\mathcal{X}}(1 + |t|)^{-i}$ for $i \leq 4$ (the derivatives of θ_{∞} gain, Proposition 8.4). Lemma 9.9 with $w = u - 2a - \nu$ gives, for $w \neq 0$, $P_+[e^{itw}\gamma_u g_u] =$

$\mathbf{1}_{w>0}\gamma_u g_u e^{itw} + \frac{i}{2}E_{u,\nu}$ with $\|E_{u,\nu}\|_{\mathcal{X}} \leq e_3 C \|b\|_{\mathcal{X}} \sup_{s \geq s_*} |\gamma_u|/(|w|T)$ (the derivative bounds of the lemma are exactly the $\mathcal{X}$ -norm), and for $w = 0$, $P_+[\gamma_u g_u] = \frac{1}{2}\gamma_u g_u + \frac{i}{2}\mathcal{H}[\gamma_u g_u]$, slowly varying with $\mathcal{X}$ -norm at most $C|\gamma_u|\|b\|_{\mathcal{X}}$ (the Hilbert transform of a symbol is a symbol of the same class, by Lemma 9.8 and the argument of Lemma 9.10). The frequencies that occur are $u - 2a - \nu$: for $\nu \in \Lambda_a$ they lie in $\Lambda_a - 2a$, for $\nu \in \Lambda_a - 2a$ in Λ_a , so $\mathcal{T}$ maps $\mathcal{Y}_T$ into itself, and the surviving terms, $w > 0$, have $u > 2a + \nu$: when $\nu \geq 0$ this forces $u > 2a$ and the weight $\eta(s)$; the frequency-zero terms occur only when $u = 2a + \nu$, which for $\nu \in \Lambda_a$ is a word landing on $2a$ (only at a resonant support, weight $O(s^{-|w_0|})$) and for $\nu = u' - 2a$ means $u = u'$ (weight $|\gamma_{u'}|$, which is $O(\eta)$ when $\nu > 0$). Summing the Hilbert errors with the lower bounds on $|w|$: $\sum_u |\gamma_u|(s_*)/|u - 2a - \nu| \leq \sum_k (X/s_*)^k (1 - X/s_*)^{-2} \max(2e^{2ak}, c_D^{-1}e^{C_D k}) \cdot (\dots)$, finite by the choice $\log(s_*/X) > 2\max(2a, C_D)$ (the words of length k contribute total weight at most $(X/s_*)^k (1 - X/s_*)^{-2}$, by the marking argument of Proposition 9.7(iii)). Hence, for $B \in \mathcal{Y}_T$ whose frequencies are all nonnegative,

$$\|\mathcal{T}B\|_{\mathcal{Y}} \leq \left(\eta(s_*)C_1 + \frac{C_2}{T} + \rho_{2a}(s_*)C_3 \right) \|B\|_{\mathcal{Y}},$$

and T_2 is chosen so that the bracket is at most $\frac{1}{2}$ (every term tends to zero as $T \rightarrow \infty$, since s_* may be taken as $s(2t_1) - 2$ with $t_1 \rightarrow \infty$). The series $\sum \mathcal{T}^n N$ has nonnegative frequencies at every order because N does and $\mathcal{T}$ keeps only $w \geq 0$ up to the Hilbert errors, which are slowly varying, frequency zero.

The inhomogeneity. N has the near sources of Corollary 7.24, read through $1/\sigma_{a,+} = m_+/\sigma_{\infty,+}$ and (28): $tP_-[A'/((t - ib)\sigma_{a,+})]$ splits into the $u = 0$ term, $A'/((t - ib)\sigma_{a,+}(ib)) \cdot t$ exactly as before up to the slowly varying factor $(s/G)^{1/2}e^{-\frac{i}{2}\mathcal{H}[g_0]}(1 + \varepsilon_+)$, and the terms $u > 0$, $tP_-[e^{itu}\mu_u(1 + \varepsilon_+)A'/((t - ib)\sigma_{\infty,+})]$, which are t times Hilbert errors of symbols of order -1 at the positive frequency u : by the integration by parts of Lemma 9.9 applied to a symbol of order -1 , each is a constant $2\check{g}_u(-u)$ (purely imaginary, the source being real and the kernel real) plus a term of $\mathcal{X}$ -norm $O(\rho_u(s_*)/(uT))$, and $\sum_u \rho_u(s_*)/u < \infty$ by Lemma 9.6(i). The smooth near source Ψ_2^{near} (order -2) and the term $-tP_+[e^{-2ita}\Xi/\sigma_{a,+}]$ (order -2 , frequencies $u - 2a$ and u) are handled identically, the $w < 0$ terms giving constants $2\check{g}(-w)$ bounded by $c'_0 M$ of Lemma 9.8(ii), summable, and the rest $\mathcal{X}$ -small; the rational corrections at the N pairs of zeros are explicit and $\mathcal{X}$ -small. Collecting, $N = A_0^{\text{near}} + N_1$ as stated, the constants being purely imaginary by the symmetry $A_a(-t) = -\overline{A_a(t)}$ applied term by term.

(iii) The Neumann series converges in $\mathcal{Y}_{T_2}$ with $\|A_a - N\|_{\mathcal{Y}} \leq 2\|\mathcal{T}N\|_{\mathcal{Y}}$. Its first-order term is $\mathcal{T}(A_0^{\text{near}}) + \mathcal{T}N_1$, the former computed in (ii); the frequency-zero part of the whole series is a slowly varying function whose constant part, together with A_0^{near} , is A_0 , and whose non-constant part R_0 collects the $u = 2a + \nu$ terms (weights $O(\eta^2)$ or the resonant atom), the factors $(s/G)^{1/2} - 1$, $e^{\mp \frac{i}{2}\mathcal{H}[g_0]} - 1$, which are $O(s^{-2})$ with derivatives gaining; the positive frequencies form M_a with the stated bound, the first order explicit; and R_1 collects the $\mathcal{X}$ -small terms. That A_0 is purely imaginary is the symmetry. $\square$

The two laws of part (iv) are proved by Propositions 8.3 and 7.8, whose hypotheses the echo part does not satisfy pointwise; the two lemmas below are the extensions those proofs need.

Lemma 9.19 (the echo part in the flux pairing). *Let $e = \sum_p e_p$ be a sum of cusps at positions $p \in (-\delta_0, 0)$ with weights $w_p \geq 0$, $\sum_p w_p < \infty$, each e_p continuous, C^2 off $\delta = |p|$, with $|e_p^{(k)}(\delta)| \leq w_p|\delta - |p||^{-k}L(|\delta - |p||)^{-3/2}$ for $k = 1, 2$, and whose transforms (placed at $a + p$) are $e^{it(a+p)}D_p(t)$ with $|D_p^{(j)}| \leq w_p|t|^{-1-j}\ell^{-3/2}$ for $j \leq 3$. Suppose $\sum_p w_p(\log(1/|p|))^{3/2} < \infty$ and $\sum_{|p|<\varepsilon} w_p \leq W\varepsilon^\alpha$*

for some $\alpha > 1$. Then the contribution of e to the pairing B_T^+ of Proposition 8.3, $\int_0^{\delta_0} ((xe)')_T(a + \delta) g_{\text{out}}(a + \delta) d\delta$, is $O(T^{-1} \ell_T^{-3/2} \sum_p w_p (1 + \log(1/|p|))^{3/2} + W L_T^{1/2} T^{-\alpha})$, and tends to zero as $T \rightarrow \infty$.

Proof. Since $(xe_p)'$ vanishes outside the window, its low-pass at $a + \delta$ is minus its high-pass, $-\frac{1}{2\pi} \int_{|t|>T} \widehat{(xe_p)'}(t) e^{-it(a+\delta)} dt$, and $\widehat{(xe_p)'} = -t[i(a+p)D_p + D_p'] e^{it(a+p)} = h_p(t) e^{it(a+p)}$ with $|h_p^{(j)}| \leq C_a w_p \ell_T^{-3/2} |t|^{-j}$, $j \leq 2$. Two integrations by parts in $\int_{|t|>T} h_p e^{-it(|p|+\delta)} dt$ give the bound $C_a w_p \ell_T^{-3/2} [(T(|p| + \delta))^{-1} + (T(|p| + \delta))^{-2}]$, and the trivial bound $\|(xe_p)'\|_1 \cdot T/\pi \leq C_a w_p T$ holds always (e'_p is integrable with $\|e'_p\|_1 \leq 4w_p$). So the leak at $a + \delta$ is at most $C_a w_p \min(T, \ell_T^{-3/2}/(T(|p| + \delta)))$. Paired with $|g_{\text{out}}(a + \delta)| \leq CL(\delta)^{1/2} + C'$ on $(0, \delta_0)$: the region $|p| + \delta < 1/T$ exists only for $|p| < 1/T$ and contributes at most $C_a w_p T \cdot T^{-1} L(1/T)^{1/2} \cdot C$; the rest contributes at most $C_a w_p \ell_T^{-3/2} T^{-1} \int_0^{\delta_0} L(\delta)^{1/2} (|p| + \delta)^{-1} d\delta \leq C_a w_p \ell_T^{-3/2} T^{-1} (L(|p|)^{1/2} + \frac{2}{3} L(|p|)^{3/2})$. Summing over p with the two hypotheses gives the bound; both terms tend to zero, the first because $\alpha > 1$ beats $L_T^{1/2}$. $\square$

Lemma 9.20 (the zero sums with the lattice phases). *Assume RH. Let F be the transform of a real odd f supported in $[-a, a]$ and suppose that for $t \geq T_0$*

$$|F(t)|^2 = \frac{4|A_0|^2}{t^2 \tilde{\sigma}_\infty(t)} \left[\frac{1}{2} (1 - \cos(2ta + 2\theta_\infty(t))) \right] (1 + q(t)) + \sum_{\nu \in \mathcal{N}'} c_\nu(t) e^{it\nu} + O(t^{-2} (\log t)^{-3/2}),$$

where q is real and slowly varying with $q(t) = O(1/\log t)$, $q' = O(1/t)$, the frequency set $\mathcal{N}' \subset (\Lambda_a \cup (2a + \Lambda_a) \cup (-2a + \Lambda_a)) \setminus \{0\}$ is countable with the coefficients c_ν slowly varying, $|c_\nu| \leq w_\nu t^{-2} (\log t)^{-1}$, $|c'_\nu| \leq w_\nu t^{-3}$, $|c''_\nu| \leq w_\nu t^{-4}$, and $\sum_\nu w_\nu (1 + \nu^2) < \infty$; no lower bound on $|\nu|$ is assumed. Then $\lim_{T \rightarrow \infty} T \sum_{|\gamma|>T} |F(\gamma)|^2 = 2|A_0|^2/\pi$.

Proof. Step 2 of the proof of Proposition 7.8 applies to each term. The non-oscillating term is $2|A_0|^2 w(t)(1 + q)$ with $w = t^{-2} \tilde{\sigma}_\infty^{-1}$, and $\sum_{\gamma>T} w(\gamma)(1 + q(\gamma)) = \frac{1}{2\pi T} (1 + o(1))$ by the computation there, q being absorbed into the $O(1/\log t)$; the term $\cos(2ta + 2\theta_\infty)$ is the oscillatory sum of that proof with the phase $2\theta_\infty = O(1/\log t)$, whose derivatives gain, so the integrations by parts against $\sin(2ta)/2a$ go through with $G'' = O(a^2 w)$ unchanged. For a frequency $\nu \neq 0$ the term $G_\nu = c_\nu e^{it\nu}$ gives, as there, $\int_T^\infty G_\nu \rho dt = O(w_\nu T^{-2} (\log T)^{-1} \rho(T)/|\nu|)$ by one integration by parts when $|\nu| \geq 1/T$, and the trivial bound $\int_T^\infty |c_\nu| \rho = O(w_\nu/T)$ always, so this integral is $O(w_\nu T^{-1} \min(1, 1/(|\nu|T)))$, whose sum over ν is $o(1/T)$ by dominated convergence, each term being $o(w_\nu/T)$ because $\nu \neq 0$; $\int G_\nu R' = o(w_\nu/T)$, and $|\int_T^\infty G_\nu dS| \leq |G_\nu(T)S(T)| + |G'_\nu(T)S_1(T)| + \int_T^\infty |S_1 G''_\nu|$ with $|G''_\nu| \leq w_\nu (\nu^2 + 2|\nu|/T + 1/T^2) T^{-2} (\log T)^{-1}$ on $t \geq T$, each $O(w_\nu (1 + \nu^2)/(T(\log \log T)^2))$ by Littlewood's bounds; summing over ν with $\sum w_\nu (1 + \nu^2) < \infty$ gives $o(1/T)$ for the whole oscillatory part. The remainder $O(t^{-2} (\log t)^{-3/2})$ sums to $o(1/T)$ as in that proof. $\square$

9.4 The global and the local

Two objects in the companion work are global. The complete Weil form $W(q, q)$ of [22] is the form of this paper on all of $C_c^\infty(\mathbb{R})$, every support at once, and its positivity is the Hypothesis; the atlas organizes it through the Suzuki metric and its defect E_N over the first N zeros, and proves local support theorems, the latest being the Checkpoint 20 certificate on the odd sector up to $a = 3/8$. The Möbius energy of [23],

$$R(N) = \sum_{m, n \leq N} \frac{\mu(m)\mu(n)}{\max(m, n)} = \int_1^N \frac{M(t)^2}{t^2} dt + \frac{M(N)^2}{N}, \quad M(t) = \sum_{n \leq t} \mu(n),$$

is the Green energy of the Möbius history to N , the norm of $\sum_{n \leq N} \mu(n) n^{-s}$ on the critical line in $L^2(dt/(\frac{1}{4} + t^2))$, and its growth is the Hypothesis: $R(N) \ll_\varepsilon N^\varepsilon$ if and only if RH, and a zero of real part $\beta_0 > \frac{1}{2}$ forces $R(N) > N^{2\beta_0-1-\delta}$ infinitely often [23, Theorem 12.6]. The floor $\lambda(a)$ is local, one window at a time. This section records what connects them, and in what sense the local statements of this paper tolerate windows that close late.

Proposition 9.21 (the window form in the atlas). *The odd arithmetic operator of [22, CP19 (0.5)], $A_a^- = C_a - \log(2\pi) - \Pi_a - K_{\alpha,a} - 2|s_a\rangle\langle s_a|$ on $L_{\text{odd}}^2(-a, a)$, is the operator of the form Q of Section 1: its archimedean part is Weil's kernel $\kappa(t) = e^{-t/2}/(1 - e^{-2t})$ in the form $C_a - \log(2\pi) - K_{\alpha,a}$, its prime part the shifts $\Pi_a q = \sum_{\log n < 2a} b_n[q(\cdot + \log n) + q(\cdot - \log n)]$ with $b_n = \Lambda(n)/\sqrt{n} = c_n$, and its polar part $-2|s_a\rangle\langle s_a|$ with $s_a = \sinh(x/2)$; so $Q_a^-(q) = \langle q, A_a^- q \rangle = Q(q)$ on the odd window, and the Checkpoint 20 inequality $Q_a^-(q) > \frac{5278}{10^6} \|q\|_2^2$ for $0 < a \leq \frac{3}{8}$ [22, CP20 (0.1)] is the statement*

$$\lambda(a) \geq 5.278 \times 10^{-3} \quad (0 < a \leq \tfrac{3}{8}),$$

a certified lower bound on the floor, with an exact rational certificate. Against it, the 40-mode minimum is an upper bound: $5.28 \times 10^{-3} \leq \lambda(\frac{3}{8}) \leq \lambda_{40}(\frac{3}{8}) = 3.14 \times 10^{-2}$ (`floor_grid_K40.csv`), and $\lambda_{40} = 6.5 \times 10^{-2}$ at $a = 0.35$, 9.8×10^{-3} at 0.4125.

Proof. The identification is term by term: the Weil distribution's archimedean part has the kernel κ with the local singularity $1/(2|t|)$, which is the kernel of $\log|D|$, so that $\log|D| - \log(2\pi) - K_\alpha$, $\alpha = \kappa - 1/(2|t|)$, with the atlas's finite part and gamma correction, is the operator with the multiplier $\text{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi$ of Section 1; the prime shifts are the terms $-2c_n \cos(t \log n)$ of Ψ in the position representation, and the polar term is $-2P(f)^2$. The bounds are those of the two documents. $\square$

The number both sides read is $\delta_{\max} = \sup_\rho (\text{Re } \rho - \frac{1}{2})$.

Theorem 9.22 (two readings of $\delta_{\max}$). *Write $\log^+ x = \max(\log x, 0)$.*

$$(i) \text{ Unconditionally, } \limsup_{N \rightarrow \infty} \frac{\log^+ R(N)}{2 \log N} = \delta_{\max}.$$

$$(ii) \text{ Under the hypotheses of Theorem 9.3(ii) and (iii), } \limsup_{a \rightarrow \infty} \frac{\log^+ (-\lambda(a))}{2a} = \delta_{\max}.$$

So, in the common variable $a = \log N$, the global energy of the Möbius history to N and the local floor of the window of half-length $\log N$ grow at the same exponential rate, $2\delta_{\max}$, and both rates vanish if and only if the Hypothesis holds.

Proof. (ii) is Theorem 9.3. (i) If $\delta_{\max} = 0$ then $R(N) \ll_\varepsilon N^\varepsilon$ by [23, Theorem 12.6(iii)], so the limit is 0. If $\delta_{\max} > 0$, every zero of real part $\beta_0 > \frac{1}{2}$ gives $\limsup \log R(N)/\log N \geq 2\beta_0 - 1 - \delta$ for all $\delta > 0$ by [23, (47)], and the supremum over the zeros gives $\geq 2\delta_{\max}$. For the upper bound put $\Theta = \frac{1}{2} + \delta_{\max}$; the classical bound $M(x) \ll_\varepsilon x^{\Theta+\varepsilon}$ (Perron's formula with $1/\zeta(s) \ll |t|^\varepsilon$ on $\text{Re } s \geq \Theta + \varepsilon$, the Borel–Carathéodory argument of [5, §14.2] in the zero-free half-plane, as in [5, §14.25]) gives $R(N) \leq N^{-1} M(N)^2 + \sum_{k < N} M(k)^2 k^{-2} \ll_\varepsilon N^{2\Theta-1+\varepsilon} = N^{2\delta_{\max}+\varepsilon}$. $\square$

Two remarks. First, the detection scale. The window of half-length a sees a zero $\rho_0 = \frac{1}{2} + \delta_0 + i\gamma_0$ only when its quartet term outweighs the on-line zeros: in the proof of Theorem 9.3(iii) the test function $\sinh(\delta_0 u) \cos(\gamma_0 u) \phi(u/a)$ has $\lambda(a) \leq -4c_\phi e^{2\delta_0 a} (1 + o(1)) + O(a^2)$ with $c_\phi \rightarrow 1/(8\delta_0)$, so the floor is negative from a support $a_{\det} = O(\delta_0^{-1} \log(1/\delta_0))$ on, which is exponentially late in

$1/\delta_0$ in the variable $N = e^a$; and the Möbius energy exceeds its on-line growth $C_M \log N$ only once $N^{\delta_0} \gg |\rho_0 \zeta'(\rho_0)| \sqrt{C_M}$ [23, Remark 15.3], the same scale. The two readings close at the same time, and a floor that is positive, or a Möbius budget that is tame, up to any finite scale certifies nothing beyond it: this is the sense in which a window can take an incredibly long time to close, and it is the only sense in which the Hypothesis is harder than a finite computation. Second, the atlas's negative-witness transfer, $E_N \geq (1 + \delta/6S(q, q))^2 > 1$ for $N \geq N_0$ once an odd test q has $Q(q) = -\delta < 0$ [22, Current mathematical position], is made quantitative by Theorem 9.3: the witness at support a is the minimizer, with $\delta = -\lambda(a)\|f\|^2$ growing like $e^{2\delta_{\max}a}$; what the transfer to the Suzuki defect needs beyond this is the Suzuki norm $S(f, f)$ of the minimizer, which is not computed here.

The local statements of this paper tolerate late windows in a precise sense.

Proposition 9.23 (late windows are allowed). *Let $e \geq 0$ be locally integrable on $[a_Z, \infty)$. If*

$$\lambda'(a) \geq -(cT^*(a) + e(a))\lambda(a) \quad \text{for almost every } a \geq a_Z,$$

then $\lambda > 0$ on $[a_Z, \infty)$ and the Riemann Hypothesis holds. Under (H_∞) the hypothesis is the inequality $2|A_0(a)|^2 \leq (cT^(a) + e(a))\lambda(a)$ at almost every support (Theorem 9.12). Conversely, if the Hypothesis holds then $-\lambda'/\lambda$ is locally integrable on $[a_Z, \infty)$; so, given the absolute continuity of λ and $\lambda(a_Z) > 0$,*

$$RH \iff \lambda > 0 \text{ on } [a_Z, \infty) \iff -\lambda'/\lambda \in L^1_{\text{loc}}[a_Z, \infty).$$

Proof. Put $g = cT^* + e$ and $u(a) = \lambda(a) \exp \int_{a_Z}^a g$, locally absolutely continuous with $u' = e^{\int g}(\lambda' + g\lambda) \geq 0$ a.e., so u is non-decreasing and $\lambda(a) \geq \lambda(a_Z)e^{-\int_{a_Z}^a g} > 0$; Theorems 1.3 and 1.2 conclude as in Proposition 7.2. Conversely, if $\lambda > 0$ on $[a_Z, \infty)$ then $\log \lambda$ is absolutely continuous on every compact interval, λ being absolutely continuous and bounded below there, so $(\log \lambda)' = \lambda'/\lambda \in L^1_{\text{loc}}$; and if λ first vanishes at some $a_0 > a_Z$ then $\int_{a_Z}^a |\lambda'|/\lambda \geq \log \lambda(a_Z) - \log \lambda(a) \rightarrow \infty$ as $a \rightarrow a_0^-$, so $-\lambda'/\lambda$ is not integrable near a_0 . $\square$

The proposition says where the content of Conjecture 7.1 lies. Positivity of the floor for all a is equivalent to the local integrability of $-\lambda'/\lambda$ and says nothing about a rate; the conjecture, and the inequality (25), assert the rate cT^* , and a proof of that rate is what the structure of the minimizer might supply. Windows at which the rate is exceeded are permitted, as many as one likes, provided the excess is integrable: the entries, where the floor has its soft kinks and Theorem 9.12 does not apply, are countable and cost nothing; a support at which the onset of the tail law is late costs its excess $e(a)$ over the interval of lateness, and the integration pays it. What may not happen is a non-integrable excess, which is a sign change of the floor, which is a zero off the line.

Finally, the identification of Proposition 9.21 places this paper's results inside the atlas's objects: the floor of the odd window is the lowest point of the atlas's Q_a^- ; the edge law, the echo tree, the boundary law and the reduction describe the minimizer of that form at every support, where the atlas certifies its sign up to $a = 3/8$; and the all-support comparison that the atlas leaves open, the sign of the complete form on every window, is the inequality (25) in the form of Proposition 9.23.

9.5 Two transfers, and the blind spot

Section 9.4 left two connections open, the transfer of the floor's negative value to the atlas's Suzuki defect, and the relation of the atlas's paired-prime matrices to the echo tree; and it raised the question of blind spots: at which heights, and at what order in the distance from the line, does a window see a zero off the line. The two transfers are done here, each with the quantity it leaves uncontrolled named; the blind spot is a theorem, and its map is computed.

Proposition 9.24 (the witness in the Suzuki metric). *In the notation of [22, Appendix B and CP9, §B]: $S(f, g) = \langle L_0 f, L_0 g \rangle$ is Suzuki's metric, positive definite on the form domain, with $\|L_0 f\|_2 \leq \pi^{-1/2} M_A \|f'\|_1$ for $\text{supp } f \subseteq [-A, A]$ (B.7), M_A the local bound of Suzuki's feature; E_N is the Suzuki defect on the determining ladder of the first N zeros, non-decreasing in N , zero under RH; and the transfer (CP9 (D1)) takes a compact smooth odd q with $W(q, q) = -\eta < 0$ to $E_N \geq (1 + \eta/3S(q, q))^2$ for $N \geq N_0(q)$. Let a be a support with $\lambda(a) < 0$, f the minimizer, $\|f\| = 1$, $V_a = \|f'\|_1$ its total variation. Then for every $\varepsilon > 0$ there is N_0 with*

$$E_N \geq \left(1 + \frac{\pi(1 - \varepsilon)|\lambda(a)|}{3M_{a+\varepsilon}^2 V_a^2}\right)^2 \quad (N \geq N_0).$$

Under the hypotheses of Theorem 9.3(iii), $|\lambda(a)| \geq e^{2\delta_{\max} a}/(2\delta_{\max})(1 + o(1))$ at every large support, so the permanent floor of the defect grows with the support like $e^{2\delta_{\max} a}/(M_a^2 V_a^2)$. The two quantities not controlled here are Suzuki's constant M_a and the variation of the negative-branch minimizer, and the bound (B.7) is in any case far from sharp: under RH, $S(f, f) = W(f, f) = \lambda(a)$, of size e^{-cT^} , while $M_a^2 V_a^2/\pi$ is of order one, so the quantity to compute is $S(f, f)$ itself, through Suzuki's feature along the critical line; on the computed branch $V_a = 4 \sup |f|$ exactly, with $\sup |f| = 1.46, 1.42, 1.40, 1.38$ at $a = 0.6, 0.8, 1.0, 1.25$ (Computation 9.28).*

Proof. Let $f_\varepsilon = f * \phi_\varepsilon$ with ϕ_ε an even mollifier of width ε : odd, smooth, supported in $[-a - \varepsilon, a + \varepsilon]$, with $\|f'_\varepsilon\|_1 \leq \|f'\|_1$, and $f_\varepsilon \rightarrow f$ in the form norm $\int (1 + \log(2 + |\xi|)) |\hat{f}|^2$ (dominated convergence, f being in the form domain), so $Q(f_\varepsilon) \rightarrow \lambda(a)$ and $\|f_\varepsilon\| \rightarrow 1$. By (B.7), $S(f_\varepsilon, f_\varepsilon) \leq M_{a+\varepsilon}^2 V_a^2/\pi$. The transfer with $\eta = -Q(f_\varepsilon) \geq (1 - \varepsilon)|\lambda(a)|$ for ε small gives the bound. The growth is Theorem 9.3(iii); on the computed branch every minimizer is unimodal on each half-window (Section 7), so $V_a = 4 \sup |f|$. $\square$

Proposition 9.25 (the paired-prime matrices and the echo tree). *Let $\Pi_a q = \sum_{\log n < 2a} c_n [q(\cdot + \log n) + q(\cdot - \log n)]$ be the atlas's prime operator on the odd window [22, CP19 (3.3)], and $P_a = \sum_n c_n P_{a, \log n}$ its paired-prime matrix on the forced columns, $c^* P_a c = \langle W_a c, \Pi_a W_a c \rangle$.*

(i) *For every q in the odd form domain, $\langle q, \Pi_a q \rangle = 2 \sum_{\log n < 2a} c_n h_q(\log n)$ with $h_q(x) = \int q(y)q(y+x) dy$ the autocorrelation: the paired-prime form is the prime part of Q , $Q(q) = \frac{1}{2\pi} \int |\hat{q}|^2 \Psi_\infty - \langle q, \Pi_a q \rangle - 2P(q)^2$, and, with the cutoff χ_T of Proposition 7.29, its part above T is the sampling defect of the zeros above T against their density.*

(ii) *At a support with a finite tree, on the minimizer, the part of $\langle f, \Pi_a f \rangle$ carried by the frequencies above T is $(M_\chi(T) - 1)$ times the density integral, and by Corollary 8.15(vi)*

$$M_\chi(T) - 1 = \frac{\langle e^T C_a e / \tilde{\sigma}_\infty \rangle_T}{\langle \sigma_{\text{eff}} / \tilde{\sigma}_\infty \rangle_T} (1 + o(1)), \quad e = (1, \rho), \quad C_a = \begin{pmatrix} 0 & b^T \\ b & C \end{pmatrix},$$

the contraction of the tree's weighted adjacency, edges included, with the amplitude vector: $e^T C_a e = 2b^T \rho + \rho^T C \rho$.

(iii) *In the periodic model of Proposition 9.7, with $E(\tau) = \sum_{p \in S^\circ \cup \{a\}} e_p e^{-i\tau(a-p)}$, the mean over τ of $2 \cos(\tau d) |E(\tau)|^2$ is $\sum_{|p-q|=d} e_p e_q = e^T C^{(d)} e$, $C^{(d)}$ the adjacency of the step d on the echo set. So the paired-prime matrix of the entry n , compressed to the span of the edge profile and its echoes, is the adjacency of the step $\log n$, and P_a compressed to that span is the tree's weighted adjacency C_a : the atlas's "point of contact" is the echo tree.*

Proof. (i) is the prime term of (1) in the position representation, $-\frac{1}{2\pi} \int |\hat{q}|^2 2c_n \cos(t \log n) dt = -c_n \langle q, q(\cdot + \log n) + q(\cdot - \log n) \rangle$, and Proposition 7.29. (ii) is Corollary 8.15(vi) with its proof, where $R + \tilde{\sigma}_\infty(|\rho|^2 - 1) = e^T C_a e - \dots$ is the mean of $\sum_d 2c_d \cos(\tau d) |E|^2$. (iii) is the computation in the proof of Corollary 8.15(iv). $\square$

Proposition 9.26 (the band the edge sees: support and mass). *Let C be the adjacency of the interior lattice S° restricted to the component containing the first-generation points, μ_b its spectral measure at the edge's coupling b , a vector with finite support, and $m_k = \langle b, C^k b \rangle / |b|^2$.*

- (i) $\text{supp supp } \mu_b = \rho(C)$: the edge's band reaches the top of its component's band. At a non-resonant support the closures of the two edges are disjoint point sets (a common point would put $2a$ in Λ_a), so that component is the interior of the right edge's closure, one component in every computed case from $a = 0.9$ on, while just above a_∞ , at $a = 0.85$, the first generation still splits into two components within 10^5 points and chains of 9000 steps (`rh_lattice_connect.py`, `logs lattice_connect*.log`); its truncations give $\rho(C) \geq 1.94, 2.93, 3.94, 4.84, 5.52, 5.83, 6.52$ at $a = 1.0, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5$, still rising with the truncation at $a \geq 1.5$ and within 2% of $2a + \log 5$ at $a \geq 2$; and if the closure is the full lattice Σ of the window, as the density of the echo set suggests beyond a_∞ , Proposition 8.21 gives $\rho(C) \geq \bar{\rho}(a) = \sum_d 2c_d(1 - d/2a)$, which exceeds $2a + \log 5$ from $a = 1.75$ on and grows like $4e^a/a$: the edge's band then outruns the horizon.
- (ii) For every even k and every $s > 0$, $\mu_b([s, \infty)) \leq m_k |b|^2 / s^k$, and m_k is exact on the lattice truncated at depth g for $k \leq 2g - 2$ (Proposition 8.23).
- (iii) $m_1 = |b|^{-2} \sum_{p^j < e^{2a}} 2(j-1)(\log p)^3 p^{-j}$, exactly, with $|b|^2 = \sum_{p^j < e^{2a}} (\log p)^2 p^{-j} \sim 2a^2$; so $m_1 \rightarrow 0$ like $C_1/2a^2$, $C_1 = \sum_p 2(\log p)^3 / (p-1)^2 = 4.6315$: the mean of the edge's measure tends to zero while its support tends to infinity.
- (iv) The uniformity of the onset is therefore a statement about mass, not support: that $\mu_b([2a + \log \kappa, \infty)) \leq \eta |b|^2$ for some κ , some $\eta < 1$ and all a , which by (ii) follows from $m_k \leq \eta(2a + \log \kappa)^k$ for one even k and all a , and which no bound holding for all k can give, since $m_k^{1/k} \rightarrow \rho(C)$.
- (v) For $k = 2$ it holds with $\eta = \frac{5}{12} + o(1)$: exactly,

$$m_2 |b|^2 = \|Cb\|^2 = \sum_q \left(\sum_{(m,m') \rightarrow q} c_m c_{m'} \right)^2,$$

the inner sum over the ordered pairs of entries whose two-step walk from the edge lands at q , which is $a - \log(mm')$ when $mm' < e^{2a}$ (inward) or $a - \log(m/m')$ when $m' < m$ (outward), so that the walks are grouped by the rational mm' or m/m' ; and

$$m_2 \leq \frac{5}{12} (2a)^2 (1 + O(1/a)), \quad \text{hence} \quad \frac{\mu_b([2a + \log \kappa, \infty))}{|b|^2} \leq \frac{5}{12} \left(\frac{2a}{2a + \log \kappa} \right)^2 (1 + O(1/a))$$

for every $\kappa \geq 1$: the mass of the edge's measure above any fixed number of horizons is at most $\frac{5}{12} + o(1)$ of its total, uniformly in the support. Computed exactly, $m_2/(2a)^2$ rises from 0.23 at $a = 1$ to 0.39 at $a = 5$ (Computation 9.29).

Proof. (i) For a bounded self-adjoint operator with nonnegative matrix elements on a connected graph with positive weights, the local growth rate $r_x = \lim_n \langle \delta_x, C^{2n} \delta_x \rangle^{1/2n}$, the top of $|\text{supp } \mu_{\delta_x}|$,

is the same at every vertex: $(C^{2n+2m})_{xx} \geq (C^m)_{xy}^2 (C^{2n})_{yy}$, one term of a nonnegative sum, gives $r_x \geq r_y$; the union over x of the supports of the μ_{δ_x} is the spectrum, whose top is $\rho(C)$ for a nonnegative operator ($\langle v, Cv \rangle \leq \langle |v|, C|v| \rangle$); so $r_x = \rho(C)$ for all x , and $\langle b, C^{2n}b \rangle \geq b_x^2 (C^{2n})_{xx}$ at a point x with $b_x > 0$ gives $\sup \text{supp } \mu_b \geq \rho(C)$, the reverse being trivial. The two closures are disjoint at a non-resonant support since $a+u = -a+u'$ with $u, u' \in \Lambda_a$ gives $2a = u' - u \in \Lambda_a$; the truncated top eigenvalues are lower bounds for $\rho(C)$, the restriction of a nonnegative operator to a subgraph having the smaller norm; the bound $\rho(C_\Sigma) \geq \bar{\rho}(a)$ is Proposition 8.21, with $\bar{\rho}(1.75) = 5.65 > 5.11 = 2 \cdot 1.75 + \log 5$ (Computation 8.22). (ii) is Markov's inequality for $\nu^k \geq 0$ and Proposition 8.23. (iii) Two first-generation points $a - \log m, a - \log m'$ are adjacent when m/m' is an entry, which for prime powers forces $m = p^j, m' = p^i$ with $i < j$ (or the reverse) and the weight $c_{pj} c_{pi} c_{p^{j-i}} = (\log p)^3 p^{-j}$; summing over the ordered pairs gives $\langle b, Cb \rangle$, and $\sum_{j \geq 2} (j-1)p^{-j} = (p-1)^{-2}$; the denominator is Mertens' $\sum_{p \leq x} (\log p)^2 / p \sim \frac{1}{2}(\log x)^2$ at $x = e^{2a}$. The formula reproduces the computed m_1 at every support of Computation 9.2 to four digits. (iv) is (ii) and (i). (v) The landing points are as stated, two-step walks landing at the same point being those with the same rational. With N_q the number of walks landing at q , $(\sum_{(m,m') \rightarrow q} c_m c_{m'})^2 \leq N_q \sum_{(m,m') \rightarrow q} c_m^2 c_{m'}^2$; $N_q = 2$ at an inward point mm' with two distinct primes (the swap), $N_q = 1$ at an outward point m/m' with distinct primes (the pair is determined by the rational), and $N_q \leq 4a/\log 2$ at the points p^k of a single prime, whose walks have total squared weight at most $\sum_p (\log p)^4 / (p-1)^2 = O(1)$. So, with $B(y) = \sum_{\log m' \leq y} c_{m'}^2$,

$$m_2 |b|^2 \leq 2 \sum_m c_m^2 B(2a - \log m) + \sum_m c_m^2 B(\log m) + O(a) = \int_0^{2a} [2B(2a - y) + B(y)] dB(y) + O(a),$$

and Mertens' theorem, $B(y) = \frac{1}{2}y^2 + O(y)$, gives $\int_0^{2a} [(2a - y)^2 + \frac{1}{2}y^2] y dy (1 + O(1/a)) = \frac{10}{3}a^4 (1 + O(1/a))$, while $|b|^2 = B(2a) = 2a^2 (1 + O(1/a))$; (ii) with $k = 2$ gives the mass bound. $\square$

Theorem 9.27 (the blind spot). *Let f be odd, supported in $[-a, a]$, with $\|f\|_1 < \infty$, and $S(z) = \int_0^a f(u) \sin(zu) du$, entire of exponential type a with $|S^{(k)}(z)| \leq a^k \|f\|_1 e^{a|\text{Im } z|}$.*

(i) For real γ and δ ,

$$16 \text{Re } S(\gamma - i\delta)^2 = 16S(\gamma)^2 - 16\delta^2 (S'(\gamma)^2 + S(\gamma)S''(\gamma)) + R, \quad |R| \leq C a^4 \delta^4 \|f\|_1^2 e^{2a|\delta|},$$

with an absolute constant C : the quartet of a zero at distance δ from the line contributes, against a double zero at its height, a correction of second order in δ and even in δ . No functional $\sum_\rho q_f(\gamma_\rho)$ of the explicit formula, for any test function on any window, sees the distance of a zero from the line at first order; nor does the Möbius energy, in which a quartet enters $M(x)$ through $x^{1/2} \cosh(\delta \log x)$.

(ii) Let $\Lambda_Z(a) = \inf_{\|f\|=1} \sum_{\rho \in Z} q_f(\gamma_\rho)$ for a configuration Z in the strip, symmetric under $\rho \mapsto 1 - \rho$ and $\rho \mapsto \bar{\rho}$ with $\sum |\gamma_\rho|^{-2} < \infty$; $\Lambda_Z(a) = \lambda(a)$ when Z is the zeros of ζ . If Z has a double zero at height γ and Z_δ is Z with it replaced by the quartet at distance δ , then, with f a minimizer for Z ,

$$\Lambda_{Z_\delta}(a) \leq \Lambda_Z(a) - 16\delta^2 (S'(\gamma)^2 + S(\gamma)S''(\gamma)) + O(\delta^4) :$$

the floor is a function of the configuration with the single-zero sensitivity $\delta_c(\gamma) = (\lambda/16S'(\gamma)^2)^{1/2}$ at a null γ of the minimizer's transform.

(iii) Above the onset of the sharp form (Proposition 7.32), where $S = -|A_0| \cos(ta + \theta)/(t\tilde{\sigma}^{1/2})(1 + o(1))$ with $2|A_0|^2 = -\lambda'$,

$$\delta_c(\gamma)^2 \geq \frac{\gamma^2 \tilde{\sigma}(\gamma)}{8a^2} \cdot \frac{\lambda}{-\lambda'} (1 + o(1)) \geq \frac{\gamma^2 \tilde{\sigma}(\gamma)}{8a^2 c T^*} (1 + o(1))$$

under $-\lambda' \leq cT^*\lambda$, which exceeds $\frac{1}{4}$ at every support computed from the onset on: above the onset the window is blind, and below the horizon it resolves the zeros to $e^{-cT^*/2}$ (Computation 9.28).

Proof. (i) Expand $S(\gamma - i\delta) = \sum_k (-i\delta)^k S^{(k)}(\gamma)/k!$, absolutely convergent with the stated bounds; in $S(\gamma - i\delta)^2 = \sum_{k,l} (-i\delta)^{k+l} S^{(k)}S^{(l)}/(k!l!)$ the terms with $k+l$ odd are imaginary and drop from the real part, $k+l=0$ gives S^2 , $k+l=2$ gives $-\delta^2(S'^2 + SS'')$, and $k+l \geq 4$ is bounded by $\|f\|_1^2 \sum_{k+l \geq 4} (a\delta)^{k+l}/(k!l!) \leq C(a\delta)^4 \|f\|_1^2 e^{2a|\delta|}$. The quartet's total is $8[S(\gamma - i\delta)^2 + S(\gamma + i\delta)^2] = 16 \operatorname{Re} S(\gamma - i\delta)^2$ by $S(\bar{z}) = \overline{S(z)}$ and $S(-z) = -S(z)$, and a double zero at $\pm\gamma$ contributes $2 \cdot 2 \cdot 4S(\gamma)^2$. For $M(x)$, a zero ρ enters through $x^\rho/(\rho\zeta'(\rho))$ and its mirror through $x^{1-\bar{\rho}}$, whose moduli are $x^{1/2 \pm \delta}$. (ii) $\Lambda_{Z_\delta}(a) \leq \sum_{\rho \in Z_\delta} q_f(\gamma_\rho) = \Lambda_Z(a) + 16 \operatorname{Re} S(\gamma - i\delta)^2 - 16S(\gamma)^2$ and (i). (iii) $|S'| \leq a|A_0|/(\gamma\tilde{\sigma}^{1/2})(1 + o(1))$ from the sharp form, and $\lambda/16S'^2 \geq \gamma^2\tilde{\sigma}\lambda/(16a^2|A_0|^2) = \gamma^2\tilde{\sigma}\lambda/(8a^2(-\lambda'))$. $\square$

What this settles about blind spots. There is no first-order detector: every quantity built from the zeros through the explicit formula, the floor and the Möbius energy included, is even in the distance from the line, so a search for a quantity that sees an off-line zero at first order is over before it starts. The floor's sensitivity is second order with no suppression by the size of the floor: below the horizon the minimizer hides from the zeros, its transform crosses them with slopes $S'(\gamma)$ of order 10^{-3} to 10^{-2} at the first zeros while λ is e^{-cT^*} , and the sensitivity $\delta_c = (\lambda/16S'^2)^{1/2}$ is $e^{-cT^*/2}$, 3.5×10^{-12} at $a = 1$ for the first zero: no blind spot below the horizon, at any support. Above about one and a half horizons δ_c exceeds $\frac{1}{2}$ and the window is blind, at every support; the Möbius energy, with its weight $1/|\rho|^2$, is blind in the same place. The two readings of Theorem 9.22 therefore see the same zeros and miss the same zeros, and no functional of either kind sees the zeros above its scale. For the program this closes the search for a branch-separating quantity below the sign change: at supports where the floor is positive, every such quantity differs between the branches by a second-order term that the floor itself already carries with the sharpest coefficient, and the branch is decided only at the sign change. The Hypothesis is therefore not a matter of detection at any finite scale but of structure, which is the reduction of Theorem 9.1 in the form of Proposition 9.23.

Computation 9.28 (the blind spot). `rh_blind_spot.py`, `log blind_spot.log` in `data/`. On the K -mode minimizers (unit norm) at $a = 0.6, 0.8, 1.0, 1.25$ ($K = 48, 64, 80, 140$), the transform through spherical Bessel functions at 60 digits with the full-precision coefficients. (i) The second-order formula: at the first zero and at the zero nearest five horizons, the exact quartet difference $16 \operatorname{Re} S(\gamma - i\delta)^2 - 16S(\gamma)^2$ against $-16\delta^2(S'^2 + SS'')$ has ratio 1.0000 at $\delta = 10^{-3}$ and 10^{-2} and 0.996–1.005 at $\delta = 0.1$, at every support. (ii) The sensitivity $\delta_c(\gamma) = (\lambda/16S'(\gamma)^2)^{1/2}$, with the slopes of the transform at the zeros:

zero	$a = 0.6, T^* = 20.9$		$a = 0.8, T^* = 31.1$		$a = 1.0, T^* = 46.4$	
	$S'(\gamma)$	δ_c	$S'(\gamma)$	δ_c	$S'(\gamma)$	δ_c
$\gamma_1 = 14.13$	-1.7×10^{-2}	1.1×10^{-2}	-1.1×10^{-2}	2.8×10^{-6}	-8.7×10^{-3}	3.5×10^{-12}
$\gamma_2 = 21.02$	1.9×10^{-3}	0.10	6.3×10^{-4}	4.9×10^{-5}	3.3×10^{-4}	9.1×10^{-11}
$\gamma_3 = 25.01$	-6.8×10^{-4}	0.29	-1.2×10^{-4}	2.5×10^{-4}	-4.8×10^{-5}	6.3×10^{-10}
$\gamma_5 = 32.94$	-8.1×10^{-5}	2.4	-4.4×10^{-6}	7.1×10^{-3}	-7.7×10^{-7}	3.9×10^{-8}
$\gamma \approx T^*$	1.9×10^{-3}	0.10	1.1×10^{-5}	2.8×10^{-3}	-6.7×10^{-10}	4.5×10^{-5}
$\gamma \approx 2T^*$	-1.3×10^{-5}	15	-2.2×10^{-8}	1.4	1.5×10^{-14}	2.0
$\gamma \approx 3T^*$	2.0×10^{-6}	95	6.7×10^{-9}	4.6	5.2×10^{-15}	5.8
$\gamma \approx 5T^*$	-1.6×10^{-5}	12	-7.3×10^{-10}	43	2.3×10^{-15}	13
$\gamma \approx 10T^*$	-8.1×10^{-6}	24	6.1×10^{-10}	51	-2.8×10^{-16}	108

The sensitivity crosses $\frac{1}{2}$ between 1.2 and $1.5T^*$ at $a = 0.6$, between 1.1 and $2T^*$ at 0.8 , and between 1.0 and $2T^*$ at 1.0 ; above, it is 10–100 from five horizons on, the slopes being those of the edge’s cosine; below, it falls like $e^{-cT^*/2}$, from 10^{-2} at $a = 0.6$ to 10^{-12} at $a = 1.0$ for the first zero; at $a = 1.25$ the first zero has $\delta_c = 2 \times 10^{-24}$. (iii) The minimizer in position space: $\sup |f| = 1.462, 1.422, 1.400, 1.384$, attained at $u = 0.221, 0.226, 0.229, 0.232$ from the origin at the four supports, a position that does not move with the support, and the total variation is $4.00 \sup |f|$ in every case.

Computation 9.29 (the second and fourth moments of the edge’s measure to $a = 5$). `rh_edge_moments.py`, `log_edge_moments.log` in `data/`. The two-step and three-step walks from the edge grouped exactly by their rational landing point, over the entries below e^{2a} : $m_2|b|^2 = \|Cb\|^2$ and $m_4|b|^2 = \|C^2b\|^2$ as finite sums, with no truncation. At $a = 1.0$ – 2.5 the values reproduce every digit of the Lanczos moments of Computation 9.2. The Chebyshev bounds are on the mass of μ_b above five horizons, $s = 2a + \log 5$:

a	entries	$ b ^2$	m_2	$m_2/(2a)^2$	m_2/s^2	m_4	$m_4^{1/4}/ b $	m_4/s^4
1.0	5	1.822	0.937	0.234	0.072	1.58	0.83	0.009
1.5	12	4.003	2.650	0.294	0.125	15.4	0.99	0.034
2.0	24	7.073	5.154	0.322	0.164	62.2	1.06	0.063
2.5	47	11.159	8.708	0.348	0.199	179	1.10	0.094
3.0	98	16.711	12.929	0.359	0.223	399	1.09	0.119
3.5	209	23.145	18.007	0.368	0.243	778	1.10	0.142
4.0	465	30.578	23.980	0.375	0.260			
4.5	1068	38.992	30.841	0.381	0.274			
5.0	2532	48.466	38.521	0.385	0.286			

$m_2/(2a)^2$ rises towards the asymptotic bound $5/12 = 0.417$ of Proposition 9.26(v) and stays below it at every support; the Chebyshev bound with $k = 2$ on the mass above five horizons rises from 7% to 29% and tends to $5/12$. The fourth root $m_4^{1/4}/|b|$ settles at 1.10 from $a = 2.5$ on, so the fourth-moment bound tends to $(1.10/\sqrt{2})^4 = 0.36$, no better; and the roots $m_k^{1/k}/|b|$ increase with k (to 1.2 and 1.3 at $k = 6, 8$ where exact), so the moments cannot push the uniform bound below about a third. What they establish is that the edge’s measure keeps at least two thirds of its mass below any fixed number of horizons at every support; what they cannot see is how much smaller the mass above is (at $a \leq 1.25$, where the measure is resolved, it is 10^{-6} to 10^{-4}), which is a question about the shape of μ_b , not its moments.

Computation 9.30 (the shape of the edge’s measure, and the onset it predicts). From the exact moments of Computation 9.29 and the Lanczos moments of Computation 9.2 (`edge_moments.log`, `spectral_moments.log`). The normalized measure $\mu_b/|b|^2$ has mean $m_1 = 0.10$ – 0.25 and variance

$m_2 \rightarrow (5/12)(2a)^2$, a bulk of standard deviation $\sqrt{m_2} \rightarrow 1.29a$; the horizon $s = 2a$ sits at $z = 2a/\sqrt{m_2} = 2.07, 1.84, 1.76, 1.69, 1.67, 1.65$ standard deviations at $a = 1, 1.5, 2, 2.5, 3, 3.5$, tending to $(12/5)^{1/2} = 1.549$. The shape: the kurtosis m_4/m_2^2 is $1.80, 2.19, 2.34, 2.37, 2.39, 2.40$ at the same supports (semicircle 2, Gaussian 3), and $m_6/m_2^3 = 7.7, 7.6, 7.0$ at $a = 2, 2.25, 2.5$ (semicircle 5, Gaussian 15): the bulk is between the two, its tail heavier than a semicircle's. The mass above the horizon is therefore not small and does not tend to zero: a Gaussian bulk puts 3.9%, 4.5%, 4.8% above $2a$ at $a = 2, 2.5, 3$ and 6.1% in the limit, a semicircle 2.4%, 3.5%, 4.0% and 6.2%, and the Lanczos measure (depth 4–5) has 4.7%, 6.3%, 5.6% at $a = 2, 2.25, 2.5$ (Computation 8.24); the uniform bound $5/12$ of Proposition 9.26(v) is an upper bound for a quantity near 0.06. The height at which the mass above s falls to 1% recedes with the support: the Lanczos measure puts it at 4.6, 3.8, 4.6 horizons at $a = 2, 2.25, 2.5$; a Gaussian bulk would put it at $z = 2.33$, that is $s = 2a + (2.33\sqrt{m_2} - 2a)$, which is 3.6, 6.5, 10.6, 17.6 horizons at $a = 2, 2.5, 3, 3.5$ and grows like $e^{1.0a}$ in the limit, a semicircle at 1.3, 1.7, 2.1, 2.5 horizons, growing like $e^{0.41a}$. The one-percent onset of the sharp form, which Section 8.2 reads from the mass of μ_b above s , is therefore not uniform in the support: it recedes like e^{ca} horizons with c between 0.4 and 1, and the measured onsets of Computation 7.6, 4–5 horizons at $a = 2$ –2.5, sit between the two model shapes. Item 3 of the program, the uniform onset, is false as stated; the inequality's natural rate is then $cT^*(a)^{1+c'}$ rather than $cT^*(a)$, and Proposition 9.23 is the form that absorbs it, every such rate being integrable in a . What would settle the shape is a local law for μ_b in the bulk: the lattice is a product of one-dimensional chains over the primes cut by the confinement, its cycles come from $p^i p^j = p^{i+j}$ alone and carry $O(1)$ weight against $|b|^2 \sim 2a^2$ (the formula for m_1), so a cavity recursion should give the bulk to $O(a^{-2})$.

Proposition 9.31 (the law of the free lattice). *Let ν_a be the law of Lemma 8.19, $X = \sum_{p \leq e^{2a}} X_p$ with $X_p = \sum_{k: p^k < e^{2a}} 2c_{p^k} \cos(k\tau_p)$ and independent uniform phases. Then ν_a is the spectral measure, at a point, of the adjacency of the unconfined lattice $\mathbb{Z}^{\pi(e^{2a})}$ with the steps $\pm \log p^k$ of weight c_{p^k} : its moments are the numbers of closed walks weighted by the products of the weights, without the confinement to the window. Its variance is $\mathbb{E}X^2 = 2|b|^2 = 4a^2(1 + O(1/a))$, and ν_a is asymptotically Gaussian: $X/(2a)$ converges in law to $N(0, 1)$ as $a \rightarrow \infty$, with $\mathbb{E}X^4/(\mathbb{E}X^2)^2 \rightarrow 3$. The confined measure $\mu_b/|b|^2$ of Proposition 9.26 has variance $m_2 = \frac{5}{12}(2a)^2(1 + O(1/a))$, five twelfths of the free variance, and kurtosis 2.40 at $a = 3.5$ (Computation 9.30): the confinement to the window is what removes seven twelfths of the variance and bends the bulk from Gaussian towards semicircular, and a local law for μ_b is the continuum limit of the closed-walk count of Proposition 9.26(v) at every even order, of which the second is done.*

Proof. The unconfined lattice is the Cayley graph of $\mathbb{Z}^\pi$ with the generators ke_p of weight c_{p^k} ; its adjacency is the sum over p of commuting operators, each unitarily equivalent (Fourier series in the p -th coordinate) to multiplication by $\sum_k 2c_{p^k} \cos(k\tau_p)$ on $L^2(d\tau_p/2\pi)$, so the spectral measure at the origin is the law of the sum of the independent X_p , which is ν_a , and the moments are the closed-walk counts. $\mathbb{E}X_p = 0$, $\mathbb{E}X_p^2 = \sum_k 2c_{p^k}^2$, so $\mathbb{E}X^2 = 2 \sum_{d \in D_a} c_d^2 = 2|b|^2$, and $|b|^2 = \sum_{p^k < e^{2a}} (\log p)^2 p^{-k} = 2a^2(1 + O(1/a))$ by the prime number theorem in the form $\sum_{p \leq x} (\log p)^2/p = \frac{1}{2} \log^2 x + O(\log x)$. Lyapunov's condition: $|X_p| \leq \sum_k 2c_{p^k} \leq 2 \log p (p^{-1/2} + p^{-1} + \dots) \leq 6 \log p / \sqrt{p}$, so $\sum_p \mathbb{E}X_p^4 \leq 36 \sum_p (\log p)^4 p^{-2} < \infty$ while $(\mathbb{E}X^2)^2 \sim 16a^4 \rightarrow \infty$; hence $X/(\mathbb{E}X^2)^{1/2} \rightarrow N(0, 1)$ and the fourth moment ratio tends to 3 (the fourth moment of a sum of independent centred variables is $3(\sum \mathbb{E}X_p^2)^2 + \sum_p (\mathbb{E}X_p^4 - 3(\mathbb{E}X_p^2)^2)$, the correction being bounded). The confined variance is Proposition 9.26(v); the kurtosis is Computation 9.30. $\square$

Computation 9.32 (the even sector). `rh_weil_even.py`, `logs weil_even_K30.log` and `weil_even_K40.log` in `data/`. For even test functions the polar term of the explicit formula

has the opposite sign: with $c(y) = \cosh(y/2)$, $Q^{\text{even}}(f) = \langle f, K_a f \rangle + 2\langle f, c \rangle^2$, the operator K_a (archimedean part and prime shifts) being the same as in the odd sector; and off the diagonal the kernel of K_a is negative ($-J_\Gamma$, and $-c_n$ at the shifts), so on even functions folded to the half window K_a is an M-operator whose ground state is positive and overlaps $c > 0$. The Weil criterion on even windows, $Q^{\text{even}} \geq 0$ for all a , is then that K_a^{even} has at most one negative eigenvalue μ_1 and that the rank-one positive perturbation closes it, $1 + 2\langle c, K_a^{-1}c \rangle \leq 0$. On the even Legendre basis ($K = 40$ modes, degree 78, 400 bits; $K = 30$ to $a = 2$), K_a^{even} has exactly one negative eigenvalue at every support computed, $\mu_1 = -1.12, -1.54, -1.99, -2.42, -2.87$ at $a = 0.3, \dots, 0.7$ ($-4.31, -5.96, -7.92, -11.23$ at $a = 1, 1.3, 1.6, 2$), its ground state has no sign change on the Gauss nodes, and

a	0.3	0.4	0.5	0.6	0.7
$\mu_2(K_a^{\text{even}})$	0.727	0.236	1.19×10^{-2}	6.6×10^{-5}	5.6×10^{-8}
$1 + 2\langle c, K_a^{-1}c \rangle$	-7.39×10^{-3}	-1.54×10^{-4}	-7.4×10^{-7}	-1.2×10^{-9}	-3.0×10^{-13}
$\lambda_{40}^{\text{even}}(a) = \text{minspec}(K_a + 2cc^T)$	7.57×10^{-3}	1.81×10^{-4}	9.3×10^{-7}	1.6×10^{-9}	4.2×10^{-13}

The structure the review predicted is there: one negative direction, a positive ground state, and the Hypothesis on the even window reduced to the sign of one number. But the number sits at the edge of zero exactly as the odd floor does: μ_2 and λ^{even} collapse doubly exponentially with the support, the secular number tracks $-\lambda^{\text{even}}$, and from $a = 0.7$ on both are below the resolution of $K = 30$ modes at 200 bits; the even minimizer hides in the zeros as the odd one does. Beyond the table, at $K = 40$ and 400 bits, $\mu_2 = 5.4 \times 10^{-12}, 4.8 \times 10^{-17}, 1.7 \times 10^{-23}, 1.2 \times 10^{-35}, 2.4 \times 10^{-44}$ and $\lambda_{40}^{\text{even}} = 1.7 \times 10^{-17}, 4.4 \times 10^{-23}, 7.8 \times 10^{-30}, 1.5 \times 10^{-41}, 9.5 \times 10^{-49}$ at $a = 0.8, 0.9, 1, 1.25, 1.5$ (the odd floor is 1.4×10^{-26} at $a = 1$); these are K -mode upper bounds, not converged at this K , and the secular number is lost to double precision there. No sum of squares shows its face in the even sector either; what the sector offers is a positivity structure (the M-operator and its Perron ground state) that the odd sector lacks, and whether it can be used is open.

Where the paper rests

The main theorem is the reduction: under (H_∞) , the Riemann Hypothesis follows from the inequality (25) at almost every support, with any locally integrable excess, and every other link is proved. The inequality is not proved, and the paper does not claim it. Its remainder in the dilation identity is the prime part D_P , whose sign is not fixed; a proof would have to show that the minimality of f fixes it against $T^*\lambda$, and the arithmetic in which this would have to be done is that of the moments of the edge's measure, where the second moment is now bounded uniformly in the support and the finer structure is not. The converse, that the Hypothesis implies the inequality in the form of Proposition 9.23, is Proposition 7.27 up to the onset, and Computation 9.30 says what the onset is: a theorem of that form would make the inequality equivalent to the Hypothesis, the cleanest shape the reduction can take, and it is the reachable one.

10 Relation to the literature, and acknowledgement

Suzuki [4] organizes the Weil form on $[-a, a]$ as a family of self-adjoint operators and conjectures that an operator with the zeros as spectrum is their limit as $a \rightarrow \infty$; Conjecture 7.1 is a differential inequality for the ground state of that family. Zhu [6] certifies both parity sectors at $a = 0.8$; his Landau–Widom decay law is fitted to the even-sector floor, calibrated on $L \leq 1.2$ and validated at $L = 1.4, \dots, 2.0$, and his exploratory 950-mode value at $L = 1.19$, between 10^{-48} and 10^{-46} , is for the full space of test functions, i.e. the even floor; the odd floors of Section 6 interpolate to $\lambda(1.19) \approx$

3×10^{-44} , and the even floor is already three orders below the odd one at $a = 0.8$ (Theorem 1.3), so the two are consistent. Comparisons between his law and the odd Φ' are cross-sector and are labelled so. Connes–Consani [7] treat the range $a \leq \frac{1}{2} \log 2$ where only the archimedean place acts; the wall of Proposition 3.1 is that place acting alone at the edge for every a . The atlas [22] contains an exact-rational certificate of the odd floor to $a = 3/8$ with constant $1/200$, inside Zhu’s range, and the first application of Theorem 3.2 to this operator. The folds paper [21] computed the floors to $a = 1.1$, the deleted-form failure points and the symbol’s negative set, and described the relay as a sawtooth, which Computation 4.4 measures; the Möbius companion [23] treats the Green-energy side of the same program.

Computation 7.5 is the converse use of the explicit formula: the form was built from the primes and the archimedean symbol, and evaluating the minimizer on the actual zeros returns the floor and its derivative; with Computations 5.9 and the Fourier-side check the identity has now been measured from the matrices, from the symbol and from the zeros. An independent referee report on the first version of this paper identified the gaps corrected here: the identity had been asserted for the exact form on numerical grounds (now Theorem 5.2 for the K -mode form and Proposition 5.3 for the exact one), Conjecture 7.1 had been phrased in terms of Φ and so presupposed $\lambda > 0$, the abstract overstated several computations, the resolution threshold had been misattributed, the scaling identity carried a sign error, (5) lacked the constant $1 - \log 2$, and Computation 6.1 misdescribed its data; the report’s own recomputations of every table from the data files are gratefully acknowledged.

Reproducibility

All scripts and data are in [24]: `rh_weil_odd.py`, `rh_weil_arb.py` (python-flint 0.9.0), `rh_edge_fem.py` (numpy 2), `rh_soft_kink.py`, `rh_floor_grid.py`, `rh_arb_grid.py`, `rh_dilation.py` (python3 `rh_dilation.py` a K [bits] [noprimess]), `rh_hadamard.py`, `rh_fourier_check.py` (the Fourier-side check of Computation 5.9), `rh_bv_check.py`, `rh_zero_side.py` (Computation 7.5, with the zeros in `data/zeros_6000.txt`), `rh_tail_law.py` and `rh_alignment.py` (Computations 7.6 and 7.28), `rh_sum_rule.py` and `rh_cutoff_identity.py` (Computations 7.10, 7.17 and 7.18), `rh_wiener_hopf.py` (Computation 7.20), `rh_echo_tree.py` (Computation 8.16), `rh_sampling_defect.py` (Computation 7.31), `rh_band.py` (Computation 8.20), `rh_lattice_band.py` (Computation 8.22), `rh_spectral_measure.py` (Computations 8.24 and 9.2), `rh_plane.py` (Computation 9.5), `rh_mirror_chains.py` and `rh_band_phase.py` (Computation 9.16), `rh_blind_spot.py` (Computation 9.28), `rh_lattice_connect.py` and `rh_edge_moments.py`, `rh_weil_even.py` (Proposition 9.26, Computation 9.29), and the data `floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`, `boundary_law.csv`, `relay_scan.csv` and the folder `data/` (edge-FEM outputs and dilation logs behind Computation 8.11).

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